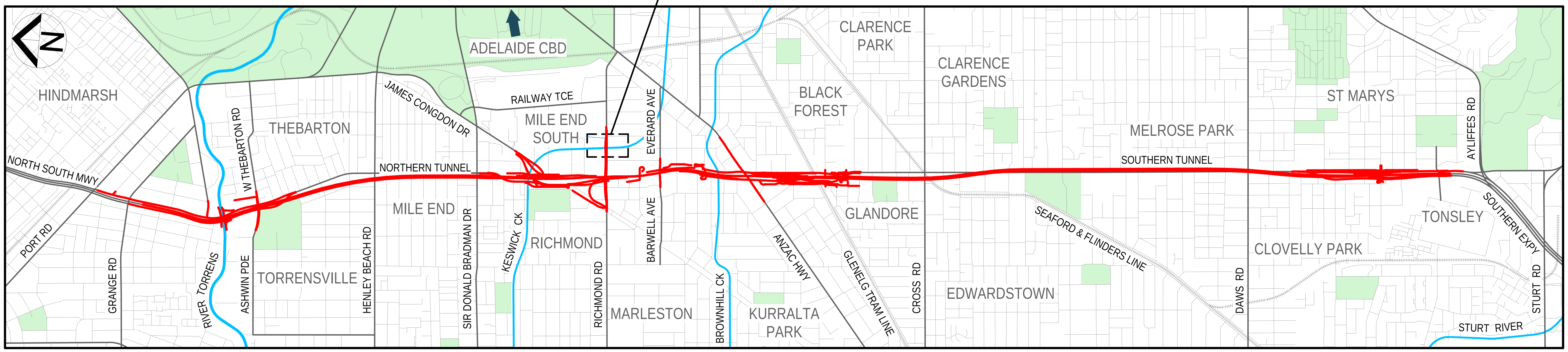
ROAD No. 6200
RICHMOND ROAD
RAILWAY TERRACE - SOUTH ROAD; MILE END SOUTH
BRIDGE OVER KESWICK CREEK

SITE OF WORKS



LOCALITY PLAN

SCALE 1 : 20000

PROJECT DOCUMENT REFERENCE
T2DPAA-T2D-C3S-BR-DRG-101001

FOR CONSTRUCTION

No.	AMENDMENT DESCRIPTION	BY	CHECK	ACCEPTANCE	DATE
0	ISSUED FOR CONSTRUCTION (KNet No: 24467656)	T2DA	T2DA	M. SHORT	28.03.26

INDEX SHEET REFERENCE: 8011 SHEET 2871002

T2D → TORRENS TO DARLINGTON

UNCONTROLLED COPY WHEN PRINTED | 100 MILLIMETRES ON ORIGINAL DRAWING | ALL DIMENSIONS ARE IN MILLIMETRES UNLESS SHOWN OTHERWISE

DESIGNER: A.SINGHAL / S.PODDAR
QUALIFICATION: MEng BEng
DATE: 25.03.26

REVIEWER: R.WOZNIAK
QUALIFICATION: FIEAust CPEng NER RPEV RPEQ MICE CEng
DATE: 25.03.26

INDEPENDENT DESIGN CERTIFIER (IF REQUIRED): T. JACKSON
QUALIFICATION: BEng (Hons), MEngSc
DATE: 27.03.26

SOUTH AUSTRALIA
Government of South Australia
Department for Infrastructure and Transport

PROJECT No.: 31921901	FILE No.: 2020/10577
DESIGN No.: #15551401	SURVEY No.: #15551401

PROJECT START ROAD RUNNING DISTANCE: NOT APPLICABLE
PROJECT END ROAD RUNNING DISTANCE: NOT APPLICABLE

SCALES: 400000 0 200000 400000 600000 800000
1:20000

ROAD No. 6200
RICHMOND ROAD
RAILWAY TERRACE - SOUTH ROAD; MILE END SOUTH
BRIDGE OVER KESWICK CREEK
TITLE AND LOCALITY

DESIGNED:	DRAFTED:	ACCEPTED FOR USE:	ACCEPTANCE FORM KNET No.:	DRAWING No.:	SHEET No.:	AMEND No.:
T2DA	T2DA	M. SHORT TITLE: DIRECTOR OF ENG. DATE: 28.03.26	24467656 IN ACCORDANCE WITH DP013	8011 SHEET LATITUDE -34.94188	2871001 SHEET LONGITUDE 138.57397	0

FILE NAME: Autodesk Docs://T2D - Torrens to Darlington /T2DPAA-T2D-C3S-BR-HSD-100302.rvt

DRAWING INDEX

PROJECT DOCUMENT REFERENCE
T2DPAA-T2D-C3S-BR-DRG-101002

FOR CONSTRUCTION

No.	AMENDMENT DESCRIPTION	BY	CHECK	ACCEPTANCE	DATE
0	ISSUED FOR CONSTRUCTION (KNet No: 24467656)	T2DA	T2DA	M. SHORT	28.03.26

INDEX SHEET REFERENCE: 8011 SHEET 2871002

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DATE: 27.03.26

Government of South Australia
Department for Infrastructure and Transport

PROJECT No.: 31921901	FILE No.: 2020/10577
DESIGN No.: #15551401	SURVEY No.: #15551401
PROJECT START ROAD RUNNING DISTANCE: NOT APPLICABLE	
PROJECT END ROAD RUNNING DISTANCE: NOT APPLICABLE	
SCALES: NOT TO SCALE	

ROAD No. 6200
RICHMOND ROAD
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DRAWING INDEX

DESIGNED:	DRAFTED:	ACCEPTED FOR USE:	ACCEPTANCE FORM KNET No.:	DRAWING No.:	SHEET No.:	AMEND No.:
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FILE NAME: Autodesk Docs://T2D - Torrens to Darlington /T2DPAA-T2D-C3S-BR-M3D-100302.rvt

GENERAL NOTES

G1. THESE DRAWINGS SHALL BE READ IN CONJUNCTION WITH ALL ENGINEERING DESIGN, THE CSCR, THE SPECIFICATIONS AND OTHER WRITTEN INSTRUCTIONS AS MAY BE ISSUED. WHEN REQUIREMENTS DIFFER BETWEEN SPECIFICATIONS, CONTRACT DOCUMENTS SHALL TAKE PRECEDENCE.

G2. UNLESS NOTED OTHERWISE:
- ALL DIMENSIONS ARE IN MILLIMETRES. ALL CHAINAGES ARE IN METRES.
- ALL REDUCED LEVELS ARE IN METRES.
- ALL SET OUT COORDINATES ARE TO GDA 2020 MAP GRID AUSTRALIA (MGA54).
- ALL LEVELS ARE TO AUSTRALIAN HEIGHT DATUM (AHD).

G3. ALL DIMENSIONS, LEVELS AND SET OUT COORDINATES SHALL BE VERIFIED BY THE CONTRACTOR PRIOR TO COMMENCING WORKS. THE CONTRACTOR SHALL REPORT ANY DISCREPANCIES AND SEEK FURTHER ADVICE FROM THE DESIGN ENGINEER. DIMENSIONS SHALL NOT BE OBTAINED BY SCALING FROM THE DESIGN DRAWINGS.

G4. WORKMANSHIP AND MATERIALS SHALL BE IN ACCORDANCE WITH THE PROJECT SPECIFICATION, CURRENT AUSTRALIAN STANDARDS INCLUDING ALL AMENDMENTS, AUSTRALIAN STANDARD CODES OF PRACTICE AND THE SPECIFICATIONS.

G5. THE CONTRACTOR SHALL OBTAIN ALL NECESSARY PERMITS AND APPROVALS FROM RELEVANT AUTHORITIES BEFORE COMMENCING THE WORK.

G6. THE NOMINATION OF PROPRIETARY ITEMS DOES NOT INDICATE AN EXCLUSIVE PREFERENCE. ALTERNATIVE EQUIVALENTS MAY BE SUBMITTED TO THE DESIGN ENGINEER AND RELEVANT AUTHORITIES FOR REVIEW AND APPROVAL. ALL PROPRIETARY ITEMS SHALL BE INSTALLED IN ACCORDANCE WITH THE MANUFACTURER'S SPECIFICATIONS AND RECOMMENDATIONS.

G7. THE CONTRACTOR IS RESPONSIBLE FOR MAINTAINING THE STABILITY OF THE STRUCTURE AND EXCAVATION UNTIL COMPLETION AND SHALL ENSURE NO PART BECOMES OVERSTRESSED. TEMPORARY WORKS DESIGN IS BY OTHERS. IF ANY TEMPORARY WORKS TO BE ATTACHED TO, SUPPORTING AND / OR SUPPORTED FROM THE PERMANENT STEELWORK INCLUDING WALKWAYS, SUPPORTS, RESTRAINTS AND FORMWORK, THE TEMPORARY WORK DESIGN IS TO BE SUBMITTED TO THE DESIGNER FOR APPROVAL.

G8. THE ANTICIPATED CONSTRUCTION SEQUENCE INDICATED ON THESE DESIGN DRAWINGS SHALL NOT BE VARIED WITHOUT PRIOR APPROVAL FROM THE DESIGN ENGINEER AND TEMPORARY WORKS DESIGN ENGINEER WHERE APPLICABLE.

G9. THE CONTRACTOR SHALL PRODUCE CONSTRUCTION PROCEDURES AND SAFE WORK METHOD STATEMENTS AS REQUIRED. IF ANY STRUCTURAL ELEMENT PRESENTS DIFFICULTY IN CONTRACTIBILITY OR SAFETY, THE MATTER SHALL BE REFERRED TO THE DESIGN ENGINEER FOR RESOLUTION BEFORE PROCEEDING WITH WORKS.

G10. THESE DRAWINGS SHALL NOT BE USED FOR COMMITTING TO MATERIAL ORDERS OR CONSTRUCTION UNTIL AUTHORISED AND ISSUED AS "FOR CONSTRUCTION".

EXISTING UTILITIES NOTES

U1. DUE ATTENTION AND CARE SHALL BE MAINTAINED WHILST CARRYING OUT CONSTRUCTION ACTIVITIES IN AREAS CONTAINING EXISTING UTILITIES.

U2. THE LOCATIONS OF ALL EXISTING UNDERGROUND UTILITIES IDENTIFIED ON THE DRAWINGS ARE INDICATIVE ONLY. ALL EXISTING UTILITIES LOCATIONS SHALL BE VERIFIED ON SITE AND ALL AFFECTED UTILITIES PROTECTED PRIOR TO COMMENCING WORK. THE CONTRACTOR SHALL SATISFY THEMSELVES THAT THE LOCATION OF ALL UTILITIES THAT MAY BE AFFECTED BY THE WORKS HAS BEEN CORRECTLY IDENTIFIED.

U3. PRIOR TO ANY EXCAVATION OR CONSTRUCTION ON THE SITE, THE CONTRACTOR SHALL CHECK WITH ALL RELEVANT AUTHORITIES AND OBTAIN ALL NECESSARY PERMITS AND BY SITE EXPLORATION TO DETERMINE THE LOCATION OF ANY EXISTING UTILITIES WHICH MAY AFFECT THE WORKS.

U4. THE CONTRACTOR SHALL CHECK THAT ANY CLEARANCES TO UTILITIES SPECIFIED ON THE DRAWINGS ARE ACHIEVED ON SITE.

U5. ALL EXCAVATIONS IN THE VICINITY OF KNOWN UTILITY LOCATIONS OR IN LOCATIONS WHERE THE EXACT UTILITY LOCATION HAS NOT BEEN ESTABLISHED, WORKS SHALL BE CARRIED OUT WITH APPROPRIATE MITIGATION MEASURES SUCH THAT NO DAMAGE TO THE UTILITY OCCURS.

U6. ALL EXCAVATIONS SHALL BE CARRIED OUT FOLLOWING THE REGULATIONS SET OUT BY EACH INDIVIDUAL UTILITY AUTHORITY. IT IS THE CONTRACTOR'S RESPONSIBILITY TO OBTAIN THESE REGULATIONS AND TO COMPLY WITH THEM. THE CONTRACTOR SHALL BE AWARE OF AND COMPLY WITH ALL UTILITY REGULATIONS AND STANDARDS IN RELATION TO THE USE OF MACHINERY AND EQUIPMENT IN THE VICINITY OF UTILITIES.

U7. THIS DESIGN COMPLIES WITH THE CLEARANCE REQUIREMENTS OF THE OFFICE OF THE TECHNICAL REGULATOR AND SERVICE AUTHORITIES. IF DURING CONSTRUCTION IT IS DETERMINED THAT THE CLEARANCE REQUIREMENTS CANNOT BE MET, WORK MUST IMMEDIATELY CEASE AND THE CONSTRUCTOR SITE REPRESENTATIVE NOTIFIED.

DESIGN LOADING NOTES

DL1. ELEMENT DESIGN LIFE:
- ALL STRUCTURAL ELEMENTS: 100 YEARS.
- JOINT SEALANT BETWEEN PRECAST BARRIERS TIME TO FIRST MAINTENANCE: 25 YEARS.

DL2. MATERIAL DENSITIES:
- CAST IN-SITU REINFORCED CONCRETE: 25.5 kN/m^3.
- PLAIN CONCRETE: 24 kN/m^3.
- STEEL: 77 kN/m^3.
- ASPHALT WEARING COURSE: 22 kN/m^3.
- FILL: 22 kN/m^3 COMPACTED EARTH FILL.
- PERMANENT FORMWORK: 0.92kN/m^2.

DL3. DESIGN LIVE LOADING:
- SM1600 BETWEEN TRAFFIC BARRIERS IN PERMANENT/ULTIMATE STAGE.
- PEDESTRIAN LOADING:
  - FOR LESS THAN OR EQUAL TO 10 m2 LOADED AREA: 5 kPa.
  - FOR GREATER THAN OR EQUAL TO 100 m2 LOADED AREA: 2 kPa.
  - FOR INTERMEDIATE VALUES, INTERPOLATE BETWEEN 5 kPa AND 2 kPa.

DL4. DIFFERENTIAL SETTLEMENT:
- POST-CONSTRUCTION MAXIMUM DIFFERENTIAL SETTLEMENT OF 5 mm BETWEEN ABUTMENTS AND CLSM BACKFILL.
- MAXIMUM DIFFERENTIAL SLS SETTLEMENT BETWEEN ABUTMENTS OF 10 mm IN ACCORDANCE WITH GEOTECHNICAL TECHNICAL NOTE T2DPAA-T2D-C3S-GE-TNT-100201.

DL5. EARTHQUAKE LOADING: AS 5100.2 AMENDMENT 2-2024:
- BRIDGE EARTHQUAKE DESIGN CATEGORY: BEDC-4
- DESIGN PERFORMANCE LEVEL: SERVICE (IMMEDIATE USE)
- EARTHQUAKE ANNUAL EXCEEDANCE PROBABILITY (P): 1/2000
- HAZARD FACTOR: 0.1
- PROBABILITY FACTOR, Kp: 1.7
- SITE SUBSOIL CLASS: Ce
- DESIGN DUCTILITY FACTOR: 1.25

DL6. WIND LOADING:
- WIND TERRAIN CATEGORY: 2.0
- WIND REGION: A5
- WIND VELOCITY ULS: 48m/s
- WIND VELOCITY SLS: 37m/s
- AVERAGE RECURRENCE INTERVAL (ARI) ULS = 2000 YEARS
- AVERAGE RECURRENCE INTERVAL (ARI) SLS = 20 YEARS

DL7. CONSTRUCTION LIVE LOAD:
- CONSTRUCTION LOAD ON BRIDGE DECK 1 kPa
- VEHICULAR SURCHARGE TO PILES ASSUMED 20 kPa
- WET CONCRETE DENSITY: 25.5 kN/m^3
- RETURN INTERVAL FOR WIND DURING CONSTRUCTION: 200YEARS IN ACCORDANCE WITH AS5100.2 TABLE 22.2.1

DL8. ROBUSTNESS DESIGN LOADING:
- PROTECTION SCREEN (MESH): 2 kN ULS LOAD OVER AN AREA OF 50 mm x 50 mm ON THE SCREEN AT ANY POINT.

DL9. MAINTENANCE PROVISION: 5 kPa LIVE LOAD OR 20kN POINT LOAD ON AN AREA OF 200 mm X 200 mm.

D10. THERMAL EFFECTS:
- THERMAL REGION = II
- MAX SHADE AIR TEMPERATURE = 47°C
- MIN SHADE AIR TEMPERATURE = -1°C
- MAX AVERAGE BRIDGE TEMPERATURE = 51°C
- MIN AVERAGE BRIDGE TEMPERATURE = 5°C

D11. FLOOD DATA:
- EFFECTS OF SCOURING AND FLOODING HAVE BEEN ASSESSED BASED ON THE BELOW TABLE
  - WITH CONCRETE LINING IN CHANNEL

RETURN PERIOD	VELOCITY (m/s)	FLOOD LEVEL (m)	SCOUR DEPTH (m)	DEBRIS MAT DEPTH (m)
1% AEP (1 IN 100 YR ARI)	2.10	19.04	0.00	2.12
0.05% AEP (1 IN 2000 YR ARI)	2.50	19.79	0.00	2.87

- WITH NO CONCRETE LINING IN CHANNEL

RETURN PERIOD	VELOCITY (m/s)	FLOOD LEVEL (m)	SCOUR DEPTH (m)	DEBRIS MAT DEPTH (m)
1% AEP (1 IN 100 YR ARI)	2.10	19.04	1.00	3.00
0.05% AEP (1 IN 2000 YR ARI)	2.50	19.79	2.10	3.00

- LOG IMPACT: 2000 kg MASS WITH STOPPING DISTANCE OF 75 mm
- BUOYANCY FORCE ON SUBMERGED DECK AREA

DESIGN LOADING NOTES (CONTINUED)

DL12. BARRIER COLLISION LOADING:
- BRIDGE BARRIER PERFORMANCE LEVEL: MEDIUM

DESIGN STANDARDS AND REFERENCES

DS1. T2D CSCR INCLUDING ALL APPLICABLE SPECIFICATIONS.

DS2. BRIDGE DESIGN: AS 5100 2017 INCLUDING AMENDMENT 1 AND AMENDMENT 2.

DS3. BRIDGE DESIGN REPORT (T2DPAA-T2D-C3S-BR-DRP-100201) INCLUDING GEOTECHNICAL TECHNICAL NOTE AND DURABILITY CHECKLIST APPENDICES.

DS4. DIT ENVIRONMENTAL AND HERITAGE TECHNICAL MANUAL FAUNA IMPACT ASSESSMENT GUIDELINE APPENDIX A.

DS5. DISABLED ACCESS AND DDA COMPLIANCE AS 1428.1.

DS6. DESIGN BASIS REPORT.

ABBREVIATIONS

CSCR	CONTRACT SCOPE AND CONTRACT REQUIREMENTS
DIT	DEPARTMENT FOR INFRASTRUCTURE AND TRANSPORT
HDG	HOT-DIP GALVANISED
NSOP	NOT SHOWN ON PLAN
SLS	SERVICEABILITY LIMIT STATE
SS	STAINLESS STEEL
ULS	ULTIMATE LIMIT STATE
UNO	UNLESS NOTED OTHERWISE
EJ	EXPANSION JOINT
CJ	CONSTRUCTION JOINT
EPS	EXPANDED POLYSTYRENE
INT	INTEGRAL
EB	ELASTOMERIC BEARING

EXCAVATION NOTES

E1. EARTHWORKS TO BE CARRIED OUT IN ACCORDANCE WITH THE SPECIFICATIONS.

E2. AN EARTHWORKS MANAGEMENT PLAN SHALL BE SUBMITTED FOR APPROVAL PRIOR TO CARRYING OUT EARTHWORKS. THIS PLAN SHALL INCLUDE AS A MINIMUM, BUT NOT LIMITED TO, THE ITEMS OUTLINED IN THE SPECIFICATIONS.

E3. THE CONTRACTOR SHALL NOT EXCAVATE IN EXCESS OF THE NOMINATED LEVELS AT EACH CONSTRUCTION STAGE. A TOLERANCE OF 300 mm OVER-EXCAVATION HAS BEEN CONSIDERED IN THE DESIGN IN ACCORDANCE WITH AS 5100.3. IN THE EVENT OF ANY OVER-EXCAVATION FROM THE DOCUMENTED DESIGN LEVELS, THE CONTRACTOR SHALL SWIFTLY UNDERTAKE THE FOLLOWING MEASURES:
- THE OVER-EXCAVATION SHALL BE CLEANED OF ANY LOOSE OR FRIABLE MATERIAL AND BACKFILLED USING GRADE 10 CONCRETE.
- THE CONTRACTOR SHALL IMMEDIATELY REPORT ANY OVER-EXCAVATION TO THE DESIGNER FOR REVIEW AND DETERMINATION OF ANY FURTHER REMEDIAL ACTION.

E4. TOPSOIL, UNSUITABLE MATERIAL CONTAINING LOOSE MATERIAL, RUBBISH AND EXISTING STRUCTURES (ORGANIC MATERIAL, PEAT, GRASS ROOTS, REFUSE AND BUILDING MATERIALS) OR DEBRIS AS PER THE SPECIFICATIONS SHALL BE REMOVED PRIOR TO PLACEMENT OF NEW FILL.

E5. DESIGN IS BASED ON COMPLETED INVESTIGATION RESULTS. REFER TO GEOTECHNICAL ASSESSMENT REPORT (T2DPAA-T2D-C3S-GE-TNT-100201).

E6. A GEOTECHNICAL REDUCTION FACTOR ϕg = 0.45 FOR BEARING, ϕg = 0.55 FOR SLIDING, OVERTURNING AND GLOBAL STABILITY HAS BEEN DETERMINED IN ACCORDANCE WITH AS 5100.3 AND IS IN LINE WITH THE GEOTECHNICAL DESIGN BASIS REPORT T2DPAA-T2D-PWA-GE-RPT-101801.

E7. UNO ENGINEERED FILL, WHERE REQUIRED, IS TO BE TYPE A FILL WITH MODIFIED COMPACTION OF 95% IN ACCORDANCE WITH THE SPECIFICATIONS AND A FRICTION ANGLE OF 34°. OTHER FILLS, WHERE REQUIRED, ARE TO BE TYPE B FILL WITH MODIFIED COMPACTION OF 90% IN ACCORDANCE WITH THE SPECIFICATIONS AND A FRICTION ANGLE OF 33°.

E8. CONTRACTOR TO IDENTIFY PLANT AND TECHNIQUES IN THE CONSTRUCTION DOCUMENTATION. BACKFILL EVENLY TO AVOID DIFFERENTIAL SOIL PRESSURES ON STRUCTURES, AND ONLY AFTER THE SPECIFIED CONCRETE STRENGTH IS ACHIEVED AND PERMANENT SUPPORTS INSTALLED.

PROJECT DOCUMENT REFERENCE
T2DPAA-T2D-C3S-BR-DRG-101005

FOR CONSTRUCTION

No.	AMENDMENT DESCRIPTION	BY	CHECK	ACCEPTANCE	DATE
0	ISSUED FOR CONSTRUCTION (KNet No: 24467656)	T2DA	T2DA	M. SHORT	28.03.26

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ROAD No. 6200
RICHMOND ROAD
RAILWAY TERRACE - SOUTH ROAD; MILE END SOUTH
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GENERAL NOTES - SHEET 01

DESIGNED: T2DA
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ACCEPTED FOR USE: M. SHORT
TITLE: DIRECTOR OF ENG.
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FORMWORK NOTES

F1. FORMWORK FOR CONCRETE SHALL BE IN ACCORDANCE WITH CONCRETE SPECIFICATION T2DPAA-T2D-PWA-DY-SPC-106201, THE SPECIFICATIONS AND AS 3610.

F2. THE DESIGN, CERTIFICATION AND PERFORMANCE OF FORMWORK AND FALSEWORK SYSTEMS SHALL BE THE RESPONSIBILITY OF THE CONTRACTOR. FORMWORK AND FALSEWORK SYSTEMS SHALL NOT BE DESIGNED TO RELY ON THE RESTRAINT OR SUPPORT FROM THE STRUCTURE WITHOUT PRIOR APPROVAL FROM THE DESIGN ENGINEER.

F3. FORMWORK SURFACE MATERIAL FOR EXPOSED SURFACES SHALL BE STEEL PLATE CONFORMING TO AS/NZS 1594 OR PLYWOOD CONFORMING TO AS/NZS 2271 UNO.

F4. UNLESS NOTED OTHERWISE, MINIMUM STRIPPING TIMES FOR CONCRETE SHALL BE IN ACCORDANCE WITH THE SPECIFICATIONS AND THE CONCRETE SPECIFICATION T2DPAA-T2D-PWA-DY-SPC-106201.

F5. DIMENSIONAL TOLERANCES SHALL COMPLY WITH THE SPECIFICATIONS.

F6. BEFORE PLACING CONCRETE, REMOVE ALL WATER, DUST, AND DEBRIS FROM THE FORMWORK.

CONCRETE NOTES

C1. ALL WORKMANSHIP AND MATERIALS SHALL BE IN ACCORDANCE WITH AS 5100, THE SPECIFICATIONS AND THE PROJECT CONCRETE SPECIFICATION T2D-PWA-DY-SPC-106201, EXCEPT WHERE VARIED BY THE PROJECT REQUIREMENTS.

C2. ALL CONCRETE SHALL BE CURED IN ACCORDANCE WITH THE CONCRETE SPECIFICATION ON T2DPAA-T2D-PWA-DY-SPC-106201 AND CIAZ09 'RECOMMENDED PRACTICE FOR CURING OF CONCRETE'.

C3. CONCRETE FINISH SHALL BE IN ACCORDANCE WITH THE SPECIFICATIONS AND AS FOLLOWS:
- FORMED CONCRETE SURFACES:
  - CLASS 1 - FOR EXPOSED AND TEXTURED SURFACES OF PRECAST ELEMENTS (PRECAST BRIDGE AND OFF-STRUCTURE BARRIERS)
  - CLASS 2 - FOR HIDDEN SURFACES OF PRECAST ELEMENTS AND CAST-IN-SITU SURFACES VISIBLE IN FINAL CONDITION (EXPOSED FACE OF ABUTMENT HEADSTOCK)
  - CLASS 3 - FOR CAST-IN-SITU SURFACES NOT VISIBLE IN FINAL CONDITION AND CONCRETE CAST AGAINST GROUND.
  - HIDDEN SURFACES, CLASS 4 (RETAINED FACE OF ABUTMENT HEADSTOCK)
- UNFORMED CONCRETE SURFACES:
  - VISIBLE SURFACES, FINISH U3
  - HIDDEN SURFACES, FINISH U2
- FOOTPATH SURFACES - FINISH U1
- CONCRETE SURFACE FOR FOOTPATHS SHALL BE LIGHT GREY EXPOSED AGGREGATE CONCRETE TO ACHIEVE AT LEAST CLASSIFICAITON P4 SLIP RESISTANCE, AS DEFINED IN TABLE 2 OF AS 4586. THIS SHALL BE READ IN CONJUCTION WITH HB198: 2014 GUIDE TO THE SPECIFICATION AND TESTING OF SLIP RESISTANCE OF PEDESTRIAN SURFACES.

C4. CONCRETE SHALL BE COMPACTED TO ENSURE A DENSE HOMOGENEOUS PRODUCT. THE INITIAL DISCHARGE FROM THE CONCRETE PUMP SHALL NOT BE USED UNTIL A CONSISTENT WORKABLE APPROVED MIX, IN ACCORDANCE WITH THE PROJECT REQUIREMENTS, IS DISCHARGED. SELF-COMPACTING CONCRETE SHALL ONLY BE USED FOR PRECAST ELEMENTS OR CAST IN-SITU PILES.

C5. NO HOLES, RECESSES, CHASES, OR EMBEDMENT OF PIPES AND CONDUITS OTHER THAN THOSE SHOWN ON THESE DESIGN DRAWINGS SHALL BE MADE IN CONCRETE MEMBERS WITHOUT PRIOR APPROVAL FROM THE DESIGN ENGINEER.

C6. CONSTRUCTION JOINTS ARE TO BE PROPERLY FORMED WHERE DETAILED ON THE DRAWINGS, OR AT LOCATIONS APPROVED BY THE DESIGN ENGINEER IN ACCORDANCE WITH SPECIFICATIONS. THE SURFACE ONTO WHICH A CONSTRUCTION JOINT IS CAST IS TO BE DELIBERATELY ROUGHENED.

C7. REINFORCEMENT CROSSING CONSTRUCTION JOINTS WHERE VISIBLE TO THE PUBLIC AND POSSIBLE FOR WATER INGRESS SHALL BE GALVANISED IN ACCORDANCE WITH SPECIFICATIONS.

C8. WHERE CONCRETE SURFACE HAS NOT BEEN PAINTED, ANTI-GRAFFITI COATING SHALL BE APPLIED TO ALL EXPOSED CONCRETE SURFACES THAT ARE ACCESSIBLE TO THE PUBLIC. THE COATING SHALL BE APPLIED TO ALL EXPOSED SURFACES THAT ARE WITHIN A MINIMUM HEIGHT OF 3m FROM ACCESSIBLE FOOTHOLDS. WHERE ANY PART OF THE STRUCTURAL ELEMENT REQUIRES ANTI-GRAFFITI COATING, THE TREATMENT SHALL BE APPLIED TO THE ENTIRE ELEMENT TO PREVENT DIFFERENTIAL VISUAL IMPACT. THE ANTI-GRAFFITI TREATMENTS MUST BE:
  a. APPROVED TO APAS - 1441/1, WHERE A PERMANENT CLEAR FINISH IS REQUIRED,
  b. APPROVED TO APAS - 1441/2, WHERE A COLOUR IS REQUIRED, AND
  c. COMPLY WITH THE TECHNICAL REQUIREMENTS SPECIFIED IN VICROADS STANDARD SECTION 685 ANTI-GRAFFITI PROTECTION AND GRAFFITI REMOVAL.

C9. A CURRENT ENVIRONMENTAL PRODUCT DECLARATION (EPD) IS REQUIRED AT THE TIME OF CONSTRUCTION FOR ALL CONCRETE MIX DESIGNS USED IN CONSTRUCTION OF THE BRIDGE.

C10. UNLESS SPECIFICALLY NOTED OTHERWISE ON THE DRAWING OR VARIED BY THE PROJECT REQUIREMENTS ALL CONCRETE EDGES HAVING A CONTAINED ANGLE LESS THAN 120° SHALL BE PROVIDED WITH 20 mm FILLETS OR CHAMFERS AS APPROPRIATE. MAINTAIN COVER TO REINFORCEMENT AT THESE DETAILS.

C11. THE CONTRACTOR SHALL CARRY OUT AN ASSESSMENT OF THE TEMPERATURE DIFFERENTIAL BETWEEN THE CENTRE AND SURFACE OF CONCRETE ELEMENTS AND LIMIT THE TEMPERATURE DIFFERENTIAL TO 25°C. THERMAL CONTROL PLANS MUST BE COMPLETED AND SUBMITTED TO THE PRINCIPAL PRIOR TO POURING OF RELEVANT CONCRETE STRUCTURES. REFER TO T2DPAA-T2D-PWA-DY-SPC-106201 CONCERETE SPECIFICATION FOR ADDITIONAL TEMPERATURE REQUIREMENTS.

CONCRETE NOTES (CONTINUED)

C12. THERMOCOUPLES THAT ARE CAPABLE OF MEASURING THE MAXIMUM TEMPERATURE AND DIFFERENTIAL TEMPERATURE SHALL BE LOCATED IN 50 MPa CONCRETE ELEMENTS THAT ARE GREATER THAN 500 mm IN THICKNESS. THERMOCOUPLE PAIRS SHALL BE ADDED AT A MINIMUM OF 20 m INTERVALS OR AS AGREED WITH THE SITE REPRESENTATIVE.

C13. ALL BLOCKOUTS SHALL BE FORMED. THE SURFACES OF BLOCKOUTS SHALL BE PREPARED AS PER THE REQUIREMENTS FOR A CONSTRUCTION JOINT.

C14. CORING OF HOLES IN CONCRETE MEMBERS IS PROHIBITED WITHOUT PRIOR APPROVAL FROM THE DESIGN ENGINEER.

C15. UNLESS NOTED OTHERWISE ALL GROUT FOR IN-SITU CONCRETE WORKS SHALL BE CONBEXTRA GP FREE-FLOW NON-SHRINK CEMENTITIOUS GROUT OR PRODUCTS HAVING EQUIVALENT FUNCTION OR PERFORMANCE GROUT STRENGTH SHALL BE A MINIMUM 40 MPa.

C16. PLACING OF CONCRETE IS TO BE CARRIED OUT IN ONE CONTINUOUS POUR WITHIN NOMINATED CONSTRUCTION JOINTS ON THE DRAWINGS UNO.

C17. CONCRETE COMPRESSIVE STRENGTH, EXPOSURE CLASSIFICATION AND NOMINAL COVER TO REINFORCEMENT SHALL COMPLY WITH THE FOLLOWING REQUIREMENTS UNO:

ELEMENT	EXPOSURE CLASSIFICATION	MINIMUM 28-DAY COMPRESSIVE STRENGTH (MPa)	NOMINAL COVER (mm)	SUPPLEMENTARY CEMENTITOUS MATERIAL % (MINIMUM / TARGET)
CFA AND CAST-IN-SITU BORED PILES	B1	50	40*	30% / 50%
CAST-IN-SITU ABUTMENT HEADSTOCKS	B1	40	45*	30% / 50%
CAST-IN-SITU OFF-STRUCTURE BARRIER PILE CAP	B1	50	40*	30% / 50%
CAST-IN-SITU DECK SLAB	B1	50	40	30% / 50%
PRECAST BARRIERS	B1	40	45	30% / 40%
OFF-STRUCTURE PRECAST BARRIERS	B1	50	30#	30% / 40%
CAST-IN-SITU INFILL CONCRETE	B1	32	50	30% / 50%
BLINDING CONCRETE	N/A	20	N/A	50% / 50%

* SURFACES CAST AGAINST GROUND TO HAVE 30 mm COVER ADDED AND SURFACES CAST AGAINST A DAMP-PROOF MEMBRANE OR BLINDING CONCRETE TO HAVE 10 mm ADDED.
# DENOTES RIGID FORMWORK AND INTENSE COMPACTION IN ACCORDANCE WITH AS 5100.5-2017 CL 4.14.3.3.

C18. CLSM BACKFILL MATERIAL SHALL BE IN ACCORDANCE TO THE SPECIFICATIONS FOR CONTROLLED LOW-STRENGTH MATERIAL (3-8 MPa).

C19. CONCRETE EMISSION LIMITS IN ACCORDANCE WITH THE SPECIFICATIONS ARE (ALL VALUES IN kg CO_2-e/m^3): 20MPa - 220, 32MPa - 272, 40MPa - 328 AND 50MPa - 405.

C20. NON-POTABLE WATER AND RECYCLED AGGREGATES SHALL BE USED WHERE RELEVANT FOR CONCRETE MIXES IN ACCORDANCE WITH THE CONCRETE SPECIFICATION T2DPAA-T2D-PWA-DY-SPC-106201.

C21. CAST IN-PLACE CONCRETE STITCH POUR MUST HAVE MAXIMUM SHRINKAGE STRAIN OF 600 $\mu\varepsilon$ AT 56 DAYS.

REINFORCEMENT

R1. ALL WORKMANSHIP AND MATERIALS SHALL BE IN ACCORDANCE WITH AS 5100, AS/NZS 4671 AND THE SPECIFICATIONS. MANUFACTURER AND PROCESSOR SHALL HOLD A VALID APPROVAL CERTIFICATION ISSUED BY THE AUSTRALIAN CERTIFICATION AUTHORITY FOR REINFORCING STEEL (ACRS).

R2. REINFORCEMENT SHOWN ON THE DRAWINGS IS REPRESENTED DIAGRAMMATICALLY AND NOT NECESSARILY SHOWN IN TRUE POSITION. FOR CLARITY, BAR LOCATIONS MAY BE EXAGGERATED. REINFORCING BARS SHOWN MAY REPRESENT MORE THAN ONE LENGTH AND / OR PROFILE.

R3. STEEL REINFORCEMENT SYMBOLS, GRADE AND DUCTILITY CLASS SHALL BE AS FOLLOWS:
- N HOT ROLLED DEFORMED BARS TO AS/NZS 4671 (GRADE 500 N)
- S HOT ROLLED DEFORMED BARS TO AS/NZS 4671 (GRADE 600 N)
- R STRUCTURAL GRADE PLAIN BARS TO AS/NZ 4671 (GRADE 250 N)
- SW WELDED MESH CLASS L TO AS/NZ 4671 (GRADE 500 MPa)
- RW RIBBED WIRE GRADE 500 RF TO AS 4671
- W HARD DRAWN STEEL REINFORCEMENT WIRE, GRADE 500 DUCTILITY CLASS L TO AS 4671
- TM HARD DRAWN STEEL TRENCH MESH, GRADE 500 DUCTILITY CLASS L TO AS 4671
- RL RECTANGULAR RIB MESH, GRADE 500 DUCTILITY CLASS L TO AS 4671
- SL SQUARE RIB MESH, GRADE 500 DUCTILITY CLASS L TO AS 4671
- GFRP GLASS FIBRE REINFORCED POLYMER BARS TO AC1 440-1R-06

R4. REINFORCEMENT SPACING NOT SHOWN SHALL BE TAKEN AS EQUAL.

REINFORCEMENT (CONTINUED)

R5. FITMENT AND REINFORCEMENT ANCHORAGE DETAILING INCLUDING HOOKS, BENDS AND COGS SHALL APPLY WITH AS 5100.5 UNLESS NOTED OTHERWISE. REBENDING OF REINFORCEMENT IS PROHIBITED.

R6. SPLICES IN REINFORCEMENT SHALL ONLY BE MADE IN THE POSITION SHOWN ON THESE DESIGN DRAWINGS UNLESS OTHERWISE APPROVED BY THE DESIGN ENGINEER. ALL SPLICES SHALL DEVELOP FULL STRENGTH IN ACCORDANCE WITH AS 5100.5 UNLESS NOTED OTHERWISE ON THE DESIGN DRAWINGS.

R7. ALL REINFORCEMENT SHALL BE SECURELY SUPPORTED IN ITS CORRECT POSITION DURING CONCRETING BY APPROVED BAR CHAIRS, SPACERS OR SUPPORT BARS.

R8. ALL REINFORCEMENT SHALL BE SECURELY TIED WITH WIRE TIES AND ALL TIE ENDS SHALL BE TURNED INTO THE MEMBER CLEAR OF THE COVER ZONE.

R9. APPROVAL FROM THE DESIGNER REQUIRED IN CASE OF ANY DEVIATION FROM DESIGN, DUE TO CUTTING OR BENDING OF REINFORCEMENT ON SITE.

R10. WELDING OF REINFORCEMENT IS ONLY PERMITTED WHERE SHOWN ON THE DESIGN DRAWINGS OR APPROVED OTHERWISE BY THE DESIGN ENGINEER.

R11. MECHANICAL SPLICES SHALL DEVELOP AT LEAST THE MINIMUM NOMINAL STRENGTH OF THE SMALLER BARS BEING SPLICED, WHEN TESTED IN ACCORDANCE WITH AS 1391.

R12. REINFORCEMENT ANCHORAGE DETAILING SHALL COMPLY WITH AS 5100.5. MINIMUM SPLICE AND DEVELOPMENT LENGTHS FOR DEFORMED BARS SHALL BE AS FOLLOWS UNO ON THE DESIGN DRAWINGS. LAP AND DEVELOPMENT LENGTHS ARE BASED ON A TYPICAL BAR SPACING OF 150 mm AND 45 mm COVER UNLESS OTHERWISE NOTED :

40 MPa COMPRESSIVE STRENGTH CONCRETE D500N								
BAR SIZE:	N12	N16	N20	N24	N28	N32	N36	N40
LAP LENGTH	400	600	750	1000	1250	1550	1850	2150
DEVELOPMENT	400	500	650	800	1050	1300	1500	1750

50 MPa COMPRESSIVE STRENGTH CONCRETE D500N								
BAR SIZE:	N12	N16	N20	N24	N28	N32	N36	N40
LAP LENGTH	400	600	700	900	1150	1400	1650	1950
DEVELOPMENT	350	500	600	750	950	1150	1400	1600

40 MPa COMPRESSIVE STRENGTH CONCRETE D600N								
BAR SIZE:	S11	S15	S18	S22	S26	S29	S33	S37
LAP LENGTH	500	700	800	1050	1350	1600	1950	2300
DEVELOPMENT	500	550	650	800	1050	1250	1500	1650

50 MPa COMPRESSIVE STRENGTH CONCRETE D600N								
BAR SIZE:	S11	S15	S18	S22	S26	S29	S33	S37
LAP LENGTH	400	600	700	950	1200	1400	1700	2050
DEVELOPMENT	400	550	650	800	950	1150	1350	1650

NOTE:
- LAPS IN REINFORCEMENT SHALL BE STAGGERED SO THAT NO MORE THAN 50% OF BARS ARE LAPPED IN ANY ONE CROSS SECTION. WHERE THIS IS NOT POSSIBLE THE MINIMUM LAP LENGTH SHALL BE INCREASED BY 25%.
- THE LAP LENGTH OF BUNDLES BARS SHALL BE INCREASED FROM THE VALUES SHOWN IN THE TABLE AS FOLLOWS:
  - 3 BAR BUNDLE - 20% INCREASE
  - 4 BAR BUNDLE - 33% INCREASE
- LAP LENGTHS SHALL BE INCREASED BY 1.5 x THE CLEAR SPACING BETWEEN BARS WHERE IT IS NOT POSSIBLE TO PROVIDE CONTACT BETWEEN LAPPED BARS AND THE CLEAR DISTANCE APART IS GREATER THAN 3 x THE BAR DIAMETER.
- WHEN SHOWN ON THE DRAWINGS, MECHANICAL SPLICES FOR REINFORCEMENT SHALL BE CAPABLE OF DEVELOPING A STRESS TENSION OR COMPRESSION NOT LESS THAN 125% OF THE FORCE WHICH IS DEVELOPED IN THE BAR AT NOMINAL YIELD. IN ADDITION, MECHANICAL SPLICES SHALL CONFORM TO THE REQUIREMENTS OF AS 5100.5.
- WHERE THE BAR SIZES AT A LAP VARY, THE LAP LENGTH SHALL BE BASED ON THE SIZE OF SMALLER BAR UNLESS NOTED OTHERWISE.
- FOR A HORIZONTAL BAR, WHERE THERE IS GREATER THAN 300 mm OF CONCRETE CAST BELOW THE BAR, THE DEVELOPMENT LENGTH SHALL BE INCREASED BY 30%.

PROJECT DOCUMENT REFERENCE
T2DPAA-T2D-C3S-BR-DRG-101006

FOR CONSTRUCTION

No.	AMENDMENT DESCRIPTION	BY	CHECK	ACCEPTANCE	DATE
0	ISSUED FOR CONSTRUCTION (KNet No: 24467656)	T2DA	T2DA	M. SHORT	28.03.26

INDEX SHEET REFERENCE: 8011 SHEET 2871002

T2D → TORRENS TO DARLINGTON

UNCONTROLLED COPY WHEN PRINTED ◄ 100 MILLIMETRES ON ORIGINAL DRAWING ► ALL DIMENSIONS ARE IN MILLIMETRES UNLESS SHOWN OTHERWISE

DESIGNER: A.SINGHAL / S.PODDAR
QUALIFICATION: MEng BEng
DATE: 25.03.26

REVIEWER: R.WOZNIAK
QUALIFICATION: FIEAust CPEng NER RPEV RPEQ MICE CEng
DATE: 25.03.26

INDEPENDENT DESIGN CERTIFIER (IF REQUIRED): T. JACKSON
QUALIFICATION: BEng (Hons), MEngSc
DATE: 27.03.26

Government of South Australia
Department for Infrastructure and Transport

PROJECT No.: 31921901 | FILE No.: 2020/10577
DESIGN No.: #15551401 | SURVEY No.: #15551401
PROJECT START ROAD RUNNING DISTANCE: NOT APPLICABLE
PROJECT END ROAD RUNNING DISTANCE: NOT APPLICABLE
SCALES: NOT TO SCALE

ROAD No. 6200
RICHMOND ROAD
RAILWAY TERRACE - SOUTH ROAD; MILE END SOUTH
BRIDGE OVER KESWICK CREEK
GENERAL NOTES - SHEET 02

DESIGNED: T2DA | DRAFTED: T2DA | ACCEPTED FOR USE: M. SHORT, TITLE: DIRECTOR OF ENG., DATE: 28.03.26 | ACCEPTANCE FORM KNET No.: 24467656 IN ACCORDANCE WITH DP013 | DRAWING No.: 8011 | SHEET No.: 2871006 | AMEND No.: 0

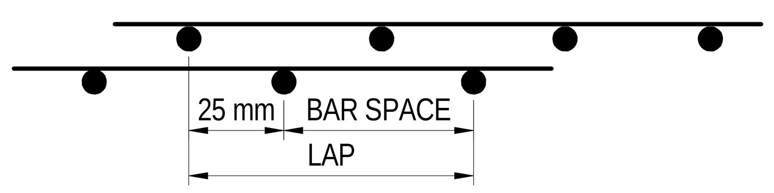
SHEET LATITUDE -34.94188 SHEET LONGITUDE 138.57397

FILE NAME: Autodesk Docs://T2D - Torrens to Darlington /T2DPAA-T2D-C3S-BR-M3D-100302.rvt

REINFORCEMENT (CONTINUED)

R13. INDIVIDUAL BARS WITHIN A BUNDLE SHALL BE TERMINATED/ SPLICED AT DIFFERENT POINTS AND STAGGERED BY AT LEAST 40 TIMES THE DIAMETER OF THE LARGER BAR UNO.

R14. MINIMUM MESH REINFORCEMENT LAP SHALL BE ONE BAR SPACING PLUS 25 mm.



R15. HEATING OF REINFORCEMENT IS NOT PERMITTED UNLESS SHOWN ON THE DESIGN DRAWINGS OR APPROVED OTHERWISE BY THE DESIGN ENGINEER.

R16. FIXING TOLERANCES FOR REINFORCEMENT SHALL COMPLY WITH THE SPECIFICATIONS AND AS 5100.5 UNO.

R17. IF REQUIRED, SPACING OF LIGS, LONGITUDINAL BARS OR OTHER REINFORCEMENT IS TO BE LOCALLY ADJUSTED TO AVOID CLASHES WITH REINFORCEMENT PROJECTED FROM CFA PILES OR OTHER CAST-IN ELEMENTS. MAXIMUM SPACING BETWEEN BARS SHALL NOT EXCEED 300 mm. MINIMUM SPACING BETWEEN BARS SHALL NOT BE LESS THAN 40 mm OR 1.5 x BAR DIA.

R18. HELICAL REINFORCEMENT TO BE SPLICED WITHIN ITS LENGTH EITHER BY WELDING OR MECHANICAL MEANS.

PRECAST CONCRETE NOTES

PC1. ALL PRECAST CONCRETE SHALL COMPLY WITH THE CONCRETE NOTES ON THESE DRAWINGS, AS 5100.5, THE SPECIFICATIONS AND PROJECT CONCRETE SPECIFICATION T2D-PWA-DY-SPC-106201.

PC2. THE CONTRACTOR SHALL REFER TO RELEVANT DRAWINGS TO ENSURE THAT ALL OPENINGS, RECESSES, FIXINGS AND FITTINGS SPECIFIED ON THE DRAWINGS ARE INCORPORATED INTO PRECAST ELEMENTS.

PC3. ALL FERRULES USED SHALL BE GALVANISED STEEL OR GRADE 316 STAINLESS STEEL AND FITTED ANCHORAGE BARS OF MINIMUM 10 mm DIAMETER. THE FERRULES SHALL DEVELOP THE FULL CAPACITY OF THE BOLT TO BE USED IN CONJUNCTION WITH THE FERRULE. A MINIMUM M20 DIAMETER FERRULE IS TO BE ADOPTED UNLESS NOTED OTHERWISE.

PC4. UNLESS NOTED OTHERWISE, FERRULES THAT WILL BE EXPOSED AFTER COMPLETION OF ERECTION ARE TO BE RECESSED 30 mm BELOW THE CONCRETE SURFACE AND ARE TO BE GROUTED ON COMPLETION WITH NON-SHRINK GROUT. BONDING AGENT TO BE APPLIED TO SURFACE PRIOR TO GROUTING.

PC5. ALL HOT DIP GALVANISED FERRULES AND FIXINGS SHALL BE IN ACCORDANCE WITH AS/NZS 1214. MINIMUM COATING THICKNESS IS 50 MICRONS.

PC6. GROUT FOR ALL PRECAST CONCRETE WORKS SHALL BE NON-SHRINK AND HAVE 28 DAYS CHARACTERISTIC STRENGTH OF 50 MPa. DETAILS OF THE PROPOSED GROUT TO BE SUBMITTED TO THE DESIGN ENGINEER FOR APPROVAL.

PC7. OILED SMOOTH STEEL FORMS OR APPROVED EQUIVALENT TO BE USED DURING CASTING.

LIFTING AND HANDLING OF PRECAST ELEMENTS NOTES

L1. THE ACTUAL MASS SHALL BE VERIFIED BY THE PRECAST ELEMENT FABRICATOR.

L2. LIFTING DEVICE, ANCHOR SIZE AND CAPACITY SHALL BE DESIGNED AND CERTIFIED BY THE PRECAST FABRICATOR.

L3. PRECAST ELEMENTS SHALL ONLY BE LIFTED BY THE LIFTING ANCHORS PROVIDED THE ANGLE BETWEEN THE SLINGS ALONG THE CENTERLINE SHALL NOT EXCEED 60°. PRECAST ELEMENTS SHALL ONLY BE LIFTED USING THE INSTALLED LIFTING ANCHOR.

L4. MINIMUM CONCRETE STRENGTH FOR LIFTING AND TRANSPORT OF PRECAST BARRIERS IS 20 MPa.

GALVANISING

G1. ITEMS ARE REQUIRED TO BE GALVANISED SHALL BE HOT-DIP GALVANISED IN ACCORDANCE WITH THE SPECIFICATIONS AND AS/NZS 4680 AND AS 1214 WHERE REQUIRED. TAP THREADED HOLES AFTER GALVANISING.

G2. STEEL SHALL BE PROVIDED WITH AIR VENT HOLES. THE LOCATION AND SIZE OF HOLES SHALL BE SHOWN ON SHOP DRAWINGS.

G3. ALL DAMAGE TO THE GALVANISING SHALL BE REPAIRED BY THE APPLICATION OF TWO COATS OF ORGANIC ZINC RICH EPOXY PRIMER AS APPROVED UNDER APAS-C-2916/1, TO A DRY FILM THICKNESS OF 100 um.

G4. ALL SHAPES EDGES AND BURRS, RESULTING FROM PUNCHING, CUTTING AND DRILLING, SHALL BE REMOVED PRIOR GALVANISING.

SPECIFICATIONS

DOCUMENT No.	DESCRIPTION	VERSION	DATE
RD-BF-C3	CONSTRUCTION OF CONCRETE SAFETY BARRIER SYSTEMS	0	31-Aug-23
RD-DK-D1	ROAD DRAINAGE DESIGN	1	30-Sep-24
PC-ST1	SUSTAINABILITY IN DESIGN AND CONSTRUCTION (T2D)	1	31-Jan-24
PR-LS-D1	LANDSCAPE AND URBAN DESIGN	1	30-Sep-24
ST-BF-C1	BEARINGS	0	31-Aug-23
ST-BF-C2	DECK EXPANSION JOINTS	0	31-Aug-23
ST-BF-C4	BRIDGEWORK SUNDRIES	0	31-Aug-23
ST-PI-C2	CAST-IN-PLACE CONCRETE PILES	0	31-Aug-23
ST-PI-C3	CONTINUOUS FLIGHT AUGER (CFA) PILES	0	31-Aug-23
ST-RE-C3	SUPPLY AND INSTALLATION OF GABIONS, RENO MATTRESSES AND MESH PANELS	0	31-Aug-23
ST-SC-C1	PRE-TENSIONED CONCRETE	0	31-Aug-23
ST-SC-C2	POST TENSIONED CONCRETE	0	31-Aug-23
ST-SC-C3	FIBRE REINFORCED POLYMER COMPOSITE STRENGTHENING OF CONCRETE STRUCTURES	0	31-Aug-23
ST-SC-C4	SPRAYED CONCRETE WORK	0	31-Aug-23
ST-SC-C6	FORMWORK	0	31-Aug-23
ST-SC-C7	PLACEMENT OF CONCRETE	1	30-Sep-24
ST-SC-S1	NORMAL CLASS CONCRETE	0	31-Aug-23
ST-SC-S2	GEOPOLYMER CONCRETE	0	31-Aug-23
ST-SC-S3	PRECAST CONCRETE UNITS	1	30-Sep-24
ST-SC-S4	LOW PRESSURE STEAM CURING OF PRECAST UNITS	1	31-Jan-24
ST-SC-S5	HEAT ACCELERATED CURING	0	31-Aug-23
ST-SC-S6	STEEL REINFORCEMENT	0	31-Aug-23
ST-SC-S7	SUPPLY OF CONCRETE	1	30-Sep-24
ST-SD-D1	DESIGN OF STRUCTURE	1	30-Sep-24
ST-SP-C1	EARTHWORKS FOR STRUCTURES	0	31-Aug-23
ST-SS-C1	TRANSPORTATION AND ERECTION OF STRUCTURAL MEMBERS	1	30-Sep-24
ST-SS-S1	FABRICATION OF STRUCTURAL STEELWORK	0	31-Aug-23
ST-SS-S2	PROTECTIVE TREATMENT OF STRUCTURAL STEELWORK	1	30-Sep-24
ST-SS-S3	GALVANISING	0	31-Aug-23
RD-EW-C1	EARTHWORKS	0	31-Aug-23
RD-EW-C4	CONTROLLED LOW STRENGTH MATERIAL	0	31-Aug-23
ST-RE-D2	DESIGN OF RETAINING WALLS	0	31-Aug-23
RD-EW-D1	DESIGN OF EARTHWORKS FOR ROADS	1	30-Sep-24

PROJECT DOCUMENT REFERENCE
T2DPAA-T2D-C3S-BR-DRG-101007

FOR CONSTRUCTION

No.	AMENDMENT DESCRIPTION	BY	CHECK	ACCEPTANCE	DATE
0	ISSUED FOR CONSTRUCTION (KNet No: 24467656)	T2DA	T2DA	M. SHORT	28.03.26

INDEX SHEET REFERENCE: 8011 SHEET 2871002

T2D → TORRENS TO DARLINGTON

DESIGNER: A.SINGHAL / S.PODDAR
QUALIFICATION: MEng BEng
DATE: 25.03.26

REVIEWER: R.WOZNIAK
QUALIFICATION: FIEAust CPEng NER RPEV RPEQ MICE CEng
DATE: 25.03.26

INDEPENDENT DESIGN CERTIFIER (IF REQUIRED): T. JACKSON
QUALIFICATION: BEng (Hons), MEngSc
DATE: 27.03.26

Government of South Australia
Department for Infrastructure and Transport

PROJECT No.: 31921901
FILE No.: 2020/10577
DESIGN No.: #15551401
SURVEY No.: #15551401
PROJECT START ROAD RUNNING DISTANCE: NOT APPLICABLE
PROJECT END ROAD RUNNING DISTANCE: NOT APPLICABLE
SCALES: NOT TO SCALE

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ROAD No. 6200
RICHMOND ROAD
RAILWAY TERRACE - SOUTH ROAD; MILE END SOUTH
BRIDGE OVER KESWICK CREEK
GENERAL NOTES - SHEET 03

DESIGNED: T2DA
DRAFTED: T2DA
ACCEPTED FOR USE: M. SHORT
TITLE: DIRECTOR OF ENG.
DATE: 28.03.26
ACCEPTANCE FORM KNET No.: 24467656
IN ACCORDANCE WITH DP013
DRAWING No.: 8011
SHEET No.: 2871007
AMEND No.: 0
SHEET LATITUDE -34.94188
SHEET LONGITUDE 138.57397

FILE NAME: Autodesk Docs://T2D - Torrens to Darlington /T2DPAA-T2D-C3S-BR-M3D-100302.rvt

INSTRUMENTATION AND MONITORING

IM1. FOR INSTRUMENTATION AND MONITORING REQUIREMENTS FOR BRIDGES, REFER TO THE INSTRUMENTATION AND MONITORING DESIGN BASIS REPORT, DOCUMENT NUMBER T2DPAA-T2D-PWA-GE-DRP-103801.

IM2. FOR INSTRUMENTATION AND MONITORING CONSTRUCTION SPECIFICATION, REFER TO T2DPAA-T2D-PWA-GE-SPC-100601.

IM3. INSTRUMENTATION AND MONITORING NOTES:

1. MONITORING RESULTS SHALL BE SUBJECT TO REVIEW BY THE CPST AS REQUIRED.
2. WHERE THERE IS NO MONITORING ARRAY IDENTIFIED ON THE INSTRUMENTATION AND MONITORING DRAWINGS, THE PREDICTED EFFECTS MAY BE REVIEWED WITHIN THE ZOI WITH ACTUAL SETTLEMENT RESULTS FROM InSAR MONITORING BY THE CPST. NOTE THAT InSAR MONITORING IS NOT REAL-TIME MONITORING WITH READINGS TYPICALLY TAKEN EVERY 12 DAYS.
3. MONITORING SPACING AND READING FREQUENCIES MAY BE ALTERED BY THE CPST WITH PRIOR NOTIFICATION TO THE CV, TO SUIT ENCOUNTERED GROUND CONDITIONS, AND AS A RESULT OF THE OBSERVED MONITORING PERFORMANCE OR ADJUSTED AS PART OF THE REVIEW AND INTERPRETATION PROCESS DURING CONSTRUCTION WITH CONSULTATION AND AGREEMENT FROM THE DESIGNER.
4. THE MONITORING FREQUENCY SHOWN IN T2DPAA-T2D-PWA-GE-SPC-100601, SKT-109004, GENERAL NOTES - SHEET 2, TABLE 3 MAY CHANGE AS A RESULT OF THE OBSERVED MONITORING PERFORMANCE OR ADJUSTED AS PART OF THE REVIEW AND INTERPRETATION PROCESS DURING CONSTRUCTION WITH CONSULTATION FROM THE CPST AND WITH PRIOR NOTIFICATION TO THE CV.
5. STABILISED READING REFERS TO MINIMUM MONTHLY READINGS TAKEN A MINIMUM OF THREE (3) TIMES AFTER CONSTRUCTION ACTIVITIES HAVE CEASED WITHIN THE ZOI WITH NO CHANGES OVER THE LONG-TERM BACKGROUND TREND. DETERMINATION OF A STABILISED MONITORING POINT AND CESSATION OF READINGS TO BE DETERMINED AND AGREED UPON BY CPST AND WITH PRIOR NOTIFICATION TO THE CV.
6. THESE PLANS MUST BE READ IN COLOUR WHEN PRINTED.
7. DE-COMMISSIONING OF INSTRUMENTS MAY OCCUR ONCE THE CESSATION OF MONITORING CONDITIONS IN T2DPAA-T2D-PWA-GE-SPC-100601, SKT-109004, GENERAL NOTES - SHEET 2 HAVE BEEN SATISFIED AND IN CONSULTATION WITH THE DESIGNER.
8. LOST OR DAMAGED INSTRUMENTS SHALL BE REPLACED AND RESURVEYED AS SOON AS PRACTICABLE, OR PRIOR TO THE NEXT SCHEDULED SURVEY, WHICHEVER IS SOONER; INCLUDING ANY BASELINE READINGS, OR AS REQUIRED BY CPST AND CV. AFTER THE REPLACEMENT OF DAMAGED INSTRUMENTS, THE MONITORING SHALL CONTINUE WITH REFERENCE TO THE PREVIOUS READINGS AND SHALL NOT BE RESET TO ZERO. THE CONSTRUCTION SITE REPRESENTATIVE MAY DECIDE NOT TO REPLACE DAMAGED OR LOST INSTRUMENTATION, DEPENDING ON THE LOCATION OF THE INSTRUMENT AND THE STATE OF CONSTRUCTION WORKS AND SUBJECT TO APPROVAL OF THE CPST AND CV TEAM. THE CONTRACTOR SHALL MAINTAIN ENOUGH INSTRUMENTS TO ENABLE ACTIVE INSTRUMENTS TO BE REPLACED.
9. DEFINITIONS:
   - CPST: CONSTRUCTION PHASE SERVICES GROUP FORMED BY SITE GEOTECHNICAL REPRESENTATIVE (SGR), THE DESIGNER AND CONSTRUCTION SITE REPRESENTATIVE (CSR) AS A MINIMUM WITH OTHER REPRESENTATIVES, AS REQUIRED (e.g. SUPERVISORS OR SUPERINTENDENTS, CONSTRUCTION MANAGER, SENIOR PROJECT ENGINEER, ETC.)
   - ZOI: ZONE OF INFLUENCE
   - InSAR: INTERFEROMETRIC SYNTHETIC APERTURE RADAR
   - CV: CONSTRUCTION VERIFIER
   - REFER TO INSTRUMENTATION AND MONITORING SPECIFICATION T2DPAA-T2D-PWA-GE- SPC-100601 FOR TERMS AND DEFINITIONS
10. THE NOMENCLATURE USED TO IDENTIFY INDIVIDUAL INSTRUMENTS IS "XXX-YYY-ZZZ-##" WHERE:
   - 'XXX' DENOTES INSTRUMENT TYPE E.G. INC = INCLINOMETER
   - 'YYY' DENOTES PROJECT ZONE e.g. 'S1S'
   - 'ZZZ' DENOTES STRUCTURE ID e.g. STSP = SOUTH TUNNEL SOUTH PORTAL
   - '##' DENOTES THE UNIQUE INSTRUMENT NUMBER

IM4. SUMMARY OF MONITORING AND INSTRUMENTATION SYMBOLS REQUIRED FOR BRIDGES

SYMBOL	DESCRIPTION
SSM ▲	SURFACE SETTLEMENT MARKERS

BRIDGE SETTLEMENT TRIGGER LEVELS (mm)				
LOCATION		AFTER COMPLETION OF BRIDGE CONSTRUCTION AND BEFORE LAYING A WEARING COURSE		
ABUTMENT	MARKER	GREEN	AMBER	RED
ABUTMENT A	SSM-C3S-BR06-01	2	4	6
ABUTMENT A	SSM-C3S-BR06-02	2	4	6
ABUTMENT A	SSM-C3S-BR06-03	2	4	6
ABUTMENT A	SSM-C3S-BR06-04	2	4	6
ABUTMENT A	SSM-C3S-BR06-05	2	4	6
ABUTMENT A	SSM-C3S-BR06-06	2	4	6
ABUTMENT A	SSM-C3S-BR06-07	2	4	6
ABUTMENT A	SSM-C3S-BR06-08	2	4	6
ABUTMENT A	SSM-C3S-BR06-09	2	4	6
ABUTMENT A	SSM-C3S-BR06-10	2	4	6
ABUTMENT A	SSM-C3S-BR06-11	2	4	6
ABUTMENT A	SSM-C3S-BR06-12	2	4	6

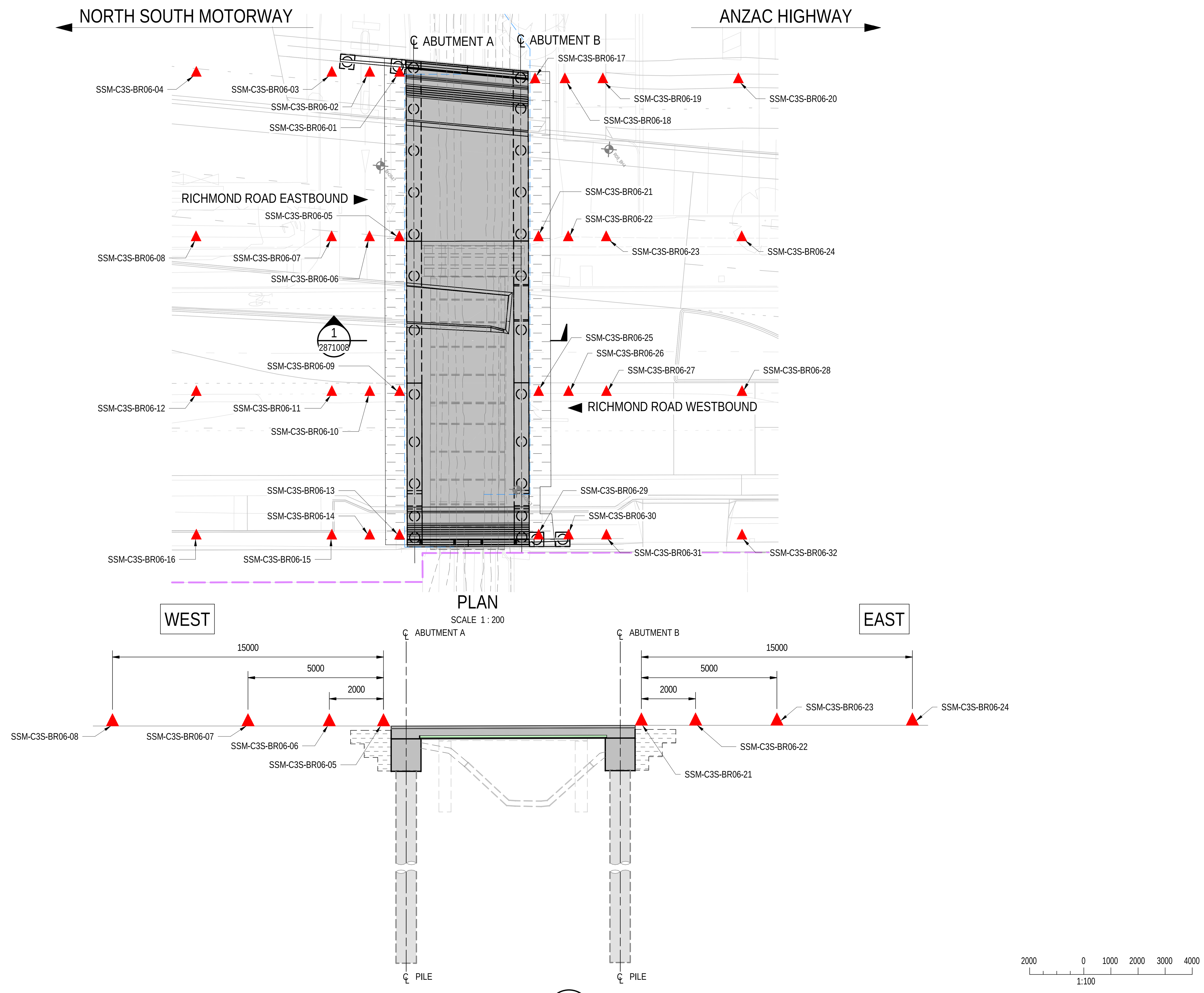
BRIDGE SETTLEMENT TRIGGER LEVELS (mm)				
LOCATION		AFTER COMPLETION OF BRIDGE CONSTRUCTION AND BEFORE LAYING A WEARING COURSE		
ABUTMENT	MARKER	GREEN	AMBER	RED
ABUTMENT B	SSM-C3S-BR06-17	2	4	6
ABUTMENT B	SSM-C3S-BR06-18	2	4	6
ABUTMENT B	SSM-C3S-BR06-19	2	4	6
ABUTMENT B	SSM-C3S-BR06-20	2	4	6
ABUTMENT B	SSM-C3S-BR06-21	2	4	6
ABUTMENT B	SSM-C3S-BR06-22	2	4	6
ABUTMENT B	SSM-C3S-BR06-23	2	4	6
ABUTMENT B	SSM-C3S-BR06-24	2	4	6
ABUTMENT B	SSM-C3S-BR06-25	2	4	6
ABUTMENT B	SSM-C3S-BR06-26	2	4	6
ABUTMENT B	SSM-C3S-BR06-27	2	4	6
ABUTMENT B	SSM-C3S-BR06-28	2	4	6

INSTRUMENTATION AND MONITORING (CONTINUED)

BRIDGE SETTLEMENT TRIGGER LEVELS (mm)				
LOCATION		AFTER COMPLETION OF BRIDGE CONSTRUCTION AND BEFORE LAYING A WEARING COURSE		
ABUTMENT	MARKER	GREEN	AMBER	RED
ABUTMENT A	SSM-C3S-BR06-13	2	4	6
ABUTMENT A	SSM-C3S-BR06-14	2	4	6
ABUTMENT A	SSM-C3S-BR06-15	2	4	6
ABUTMENT A	SSM-C3S-BR06-16	2	4	6

BRIDGE SETTLEMENT TRIGGER LEVELS (mm)				
LOCATION		AFTER COMPLETION OF BRIDGE CONSTRUCTION AND BEFORE LAYING A WEARING COURSE		
ABUTMENT	MARKER	GREEN	AMBER	RED
ABUTMENT B	SSM-C3S-BR06-29	2	4	6
ABUTMENT B	SSM-C3S-BR06-30	2	4	6
ABUTMENT B	SSM-C3S-BR06-31	2	4	6
ABUTMENT B	SSM-C3S-BR06-32	2	4	6

IM5. LONG-TERM MONITORING SHALL BE UNDERTAKEN IN ACCORDANCE WITH THE SPECIFICATIONS.



SECTION 1
SCALE 1 : 100
2871008

PROJECT DOCUMENT REFERENCE
T2DPAA-T2D-C3S-BR-DRG-101008

FOR CONSTRUCTION

No.	AMENDMENT DESCRIPTION	BY	CHECK	ACCEPTANCE	DATE
0	ISSUED FOR CONSTRUCTION (KNet No: 24467656)	T2DA	T2DA	M. SHORT	28.03.26

INDEX SHEET REFERENCE: 8011 SHEET 2871002

T2D → TORRENS TO DARLINGTON

DESIGNER: A.SINGHAL / S.PODDAR
QUALIFICATION: MEng BEng
DATE: 25.03.26
REVIEWER: R.WOZNIAK
QUALIFICATION: FIEAust CPEng NER RPEV RPEQ MICE CEng
DATE: 25.03.26
INDEPENDENT DESIGN CERTIFIER (IF REQUIRED): T. JACKSON
QUALIFICATION: BEng (Hons), MEngSc
DATE: 27.03.26

Government of South Australia
Department for Infrastructure and Transport

PROJECT No.: 31921901
FILE No.: 2020/10577
DESIGN No.: #15551401
SURVEY No.: #15551401
PROJECT START ROAD RUNNING DISTANCE: NOT APPLICABLE
PROJECT END ROAD RUNNING DISTANCE: NOT APPLICABLE
SCALES: 1:200

ROAD No. 6200
RICHMOND ROAD
RAILWAY TERRACE - SOUTH ROAD; MILE END SOUTH
BRIDGE OVER KESWICK CREEK
GENERAL NOTES - SHEET 04

DESIGNED:	DRAFTED:	ACCEPTED FOR USE:	ACCEPTANCE FORM KNET No.:	DRAWING No.:	SHEET No.:	AMEND No.:
T2DA	T2DA	M. SHORT TITLE: DIRECTOR OF ENG. DATE: 28.03.26	24467656 IN ACCORDANCE WITH DP013	8011 SHEET LATITUDE -34.94188	2871008 SHEET LONGITUDE 138.57397	0

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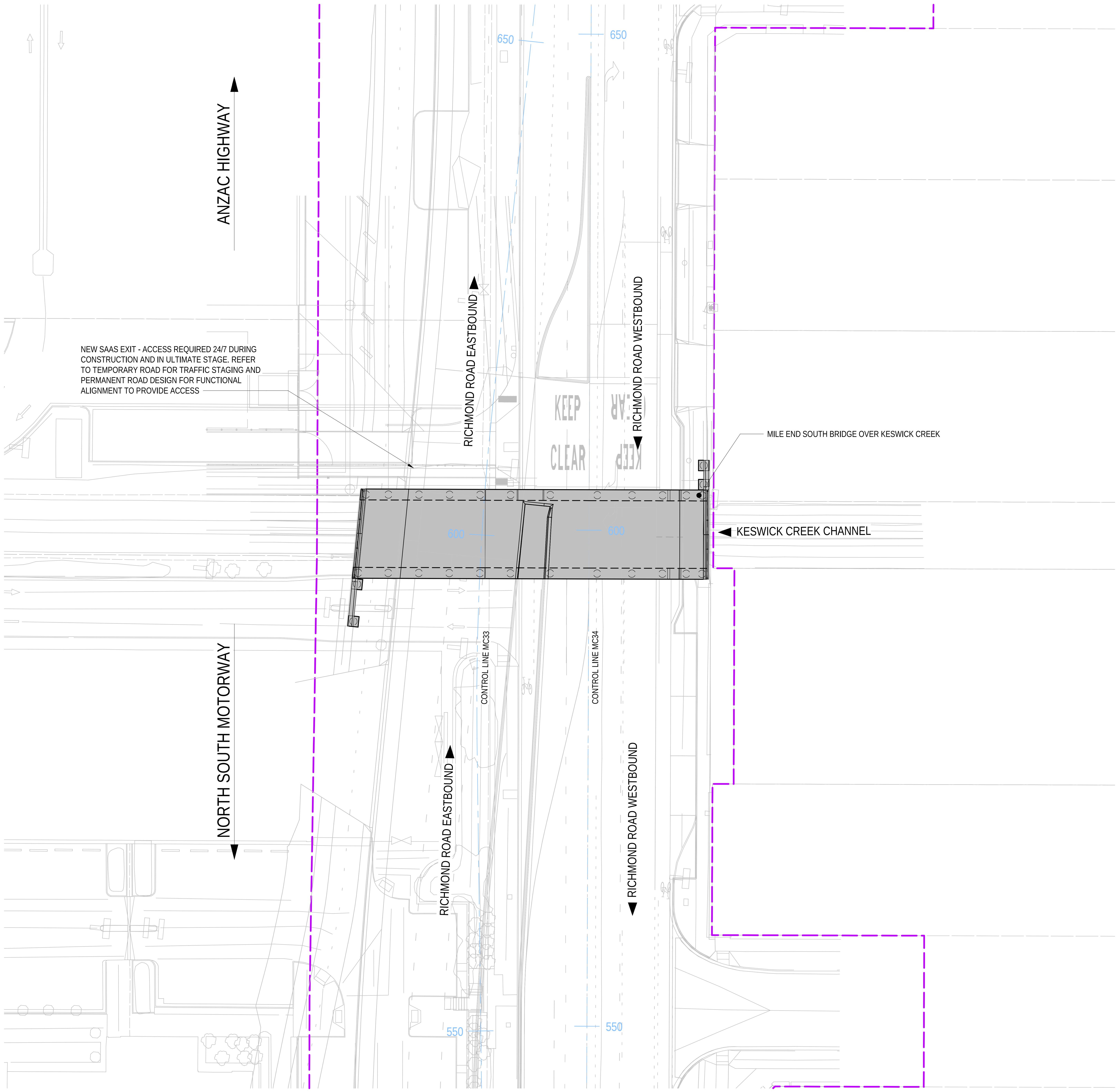
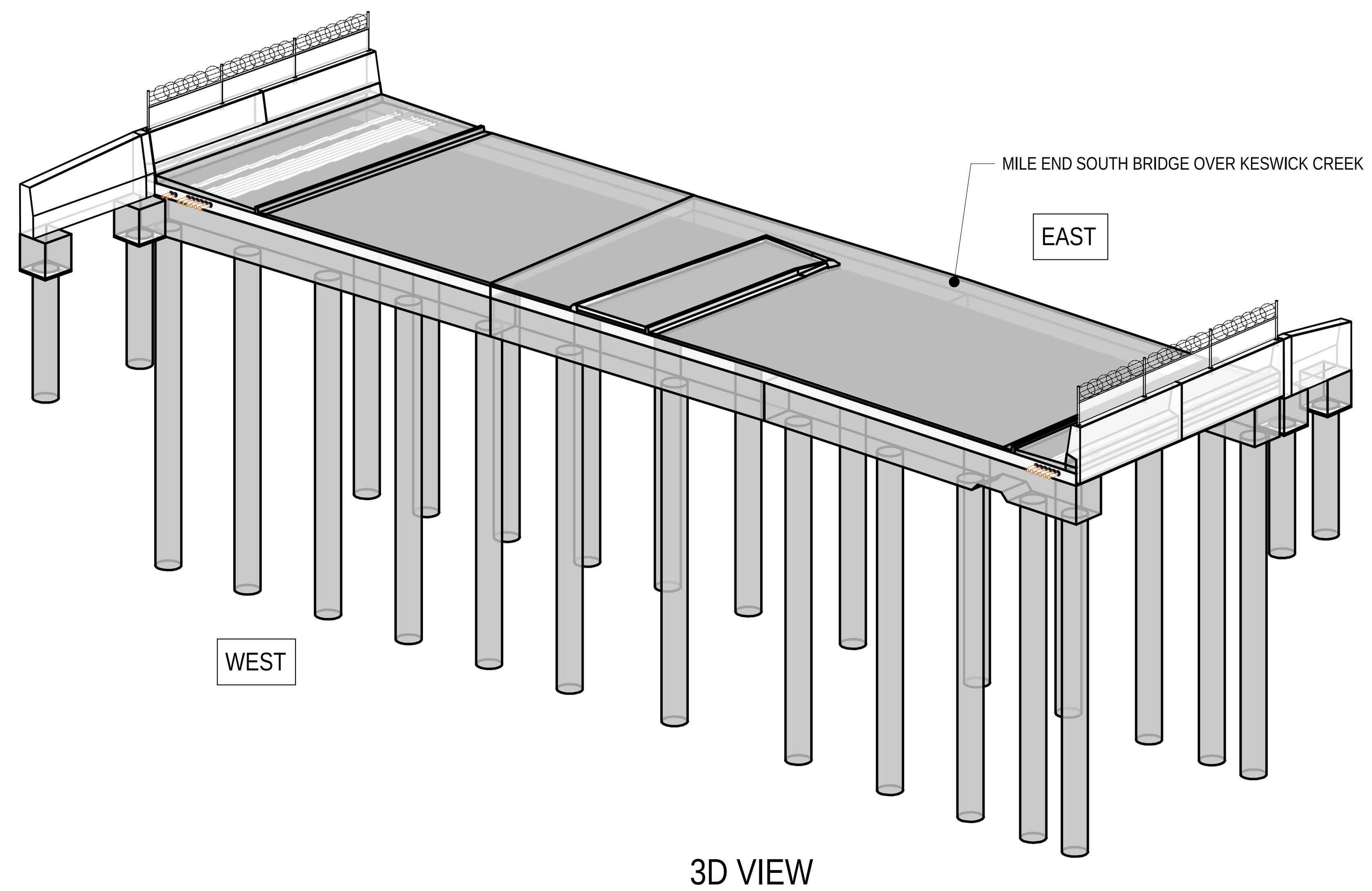


PLAN

SCALE 1 : 250



3D VIEW

LEGEND

GENERAL

PROJECT BOUNDARY

PROJECT DOCUMENT REFERENCE
T2DPAA-T2D-C3S-BR-DRG-101011

FOR CONSTRUCTION

No.	AMENDMENT DESCRIPTION	BY	CHECK	ACCEPTANCE	DATE
0	ISSUED FOR CONSTRUCTION (KNet No: 24467656)	T2DA	T2DA	M. SHORT	28.03.26

INDEX SHEET REFERENCE: 8011 SHEET 2871002

T2D TORRENS TO DARLINGTON

DESIGNER: A.SINGHAL / S.PODDAR
QUALIFICATION: MEng BEng
DATE: 25.03.26
REVIEWER: R.WOZNIAK
QUALIFICATION: FIEAust CPEng NER RPEV RPEQ MICE CEng
DATE: 25.03.26
INDEPENDENT DESIGN CERTIFIER (IF REQUIRED): T. JACKSON
QUALIFICATION: BEng (Hons), MEngSc
DATE: 27.03.26

Government of South Australia
Department for Infrastructure and Transport

PROJECT No.: 31921901
FILE No.: 2020/10577
DESIGN No.: #15551401
SURVEY No.: #15551401
PROJECT START ROAD RUNNING DISTANCE: NOT APPLICABLE
PROJECT END ROAD RUNNING DISTANCE: NOT APPLICABLE
SCALES: 5000 0 2500 5000 7500 10000
1:250

ROAD No. 6200
RICHMOND ROAD
RAILWAY TERRACE - SOUTH ROAD; MILE END SOUTH BRIDGE OVER KESWICK CREEK
SITE PLAN

DESIGNED: T2DA
DRAFTED: T2DA
ACCEPTED FOR USE: M. SHORT
TITLE: DIRECTOR OF ENG.
DATE: 28.03.26
ACCEPTANCE FORM KNET No.: 24467656
IN ACCORDANCE WITH DP013
DRAWING No.: 8011
SHEET No.: 2871011
AMEND No.: 0
SHEET LATITUDE -34.94188 SHEET LONGITUDE 138.57397

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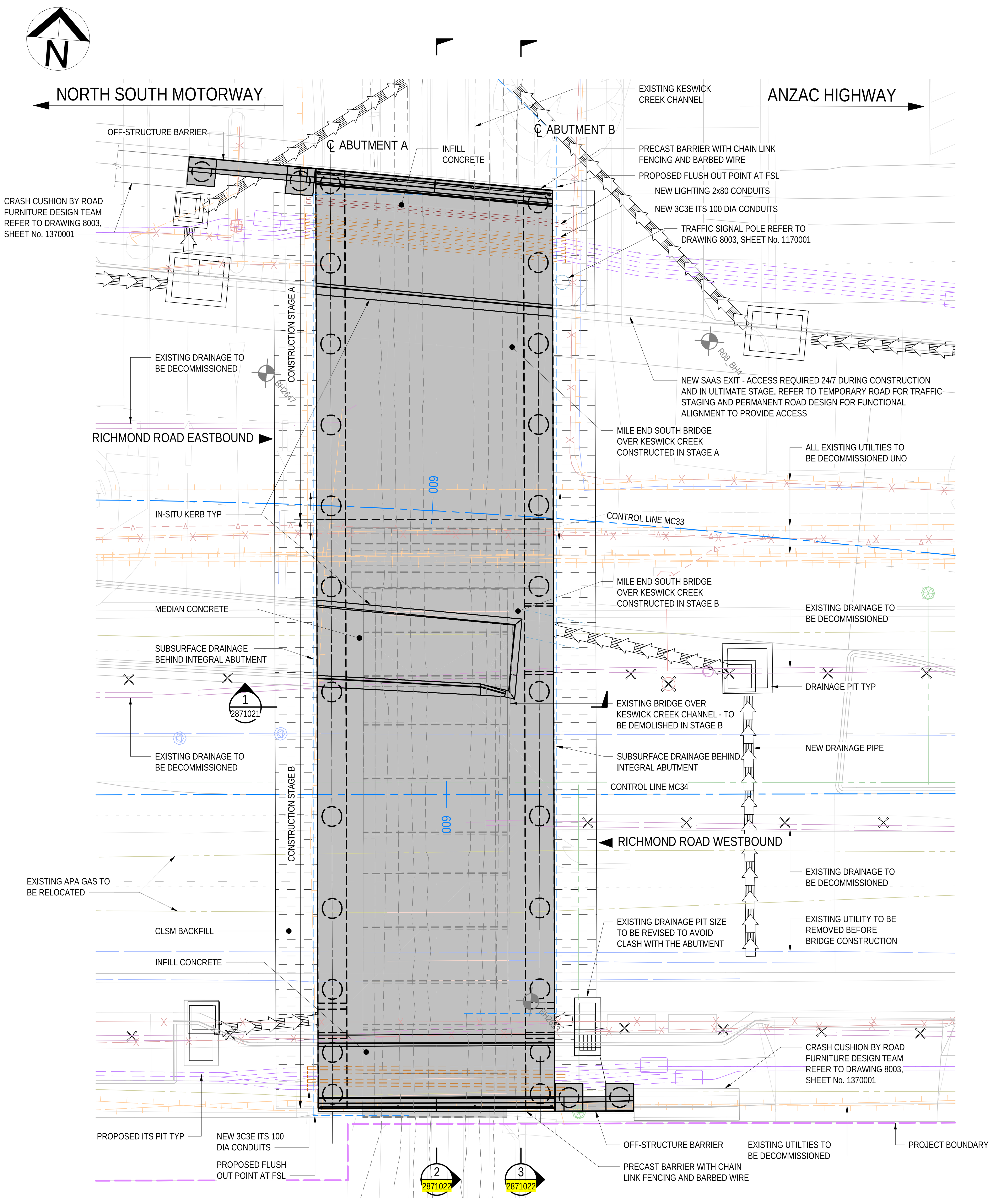
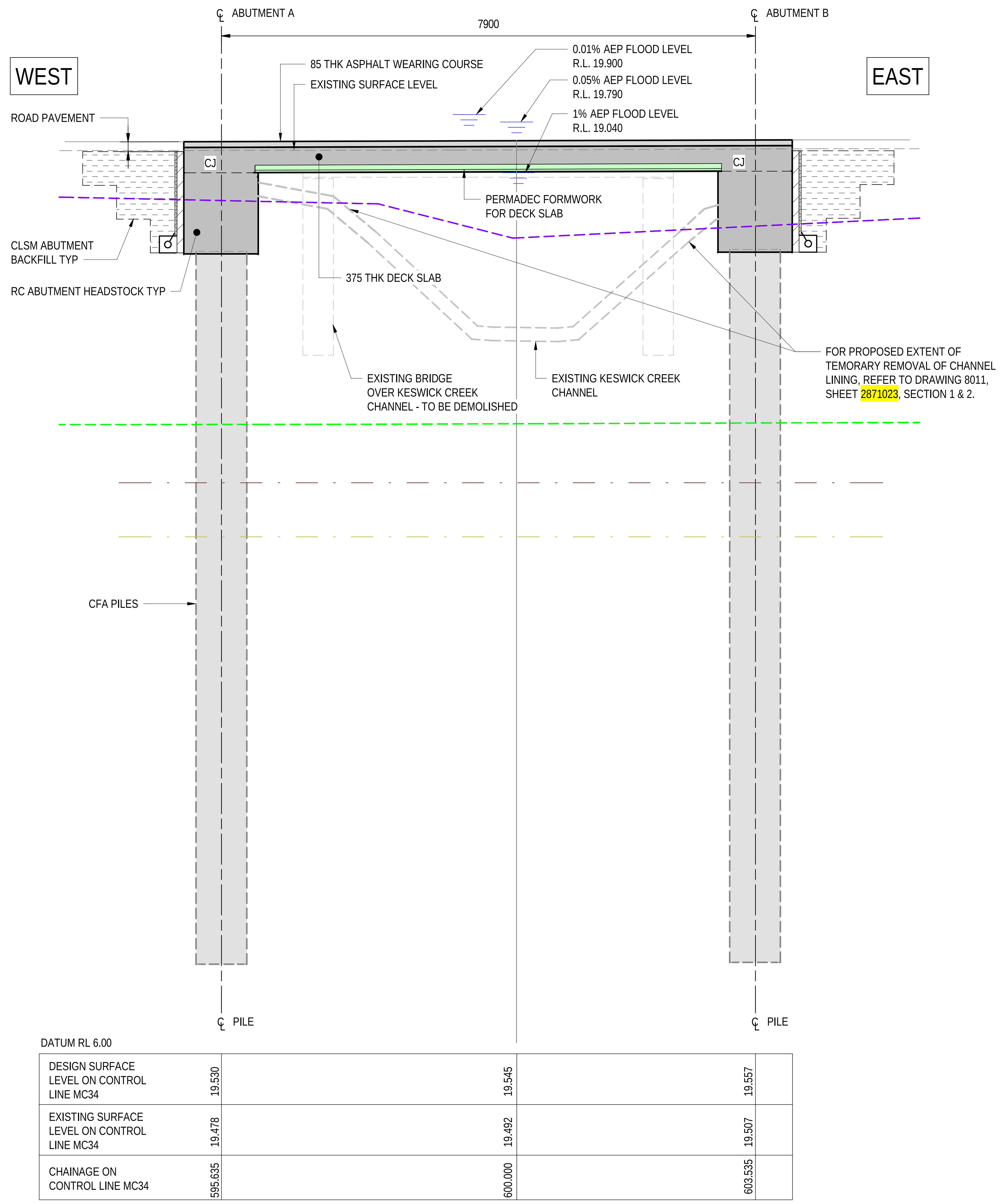
FILE NAME: Autodesk Docs://T2D - Torrens to Darlington /T2DPAA-T2D-C3S-BR-M3D-100302.rvt



PLAN
SCALE 1 : 100



DESIGN SURFACE LEVEL ON CONTROL LINE MC34	19.530	19.545	19.557
EXISTING SURFACE LEVEL ON CONTROL LINE MC34	19.478	19.492	19.507
CHAINAGE ON CONTROL LINE MC34	595.635	600.000	603.635

SECTION 1 (2871021)
SCALE 1 : 50

LEGEND

GEOTECHNICAL

- EXISTING BOREHOLE
- EXISTING PENETRATION TEST
- EXISTING TEST PIT
- PROPOSED BOREHOLE
- PROPOSED PENETRATION TEST
- PROPOSED TEST PIT

UTILITIES - EXISTING

- EXISTING ELECTRICAL - OVERHEAD (SAPN)
- EXISTING ELECTRICAL - UNDERGROUND (SAPN)
- EXISTING ELECTRICAL - UNDERGROUND (DIT)
- EXISTING COMMUNICATIONS
- EXISTING STORMWATER - PIPE / CULVERT
- EXISTING GAS
- EXISTING SEWER
- EXISTING WATER
- UTILITIES TO BE DECOMMISSIONED / REMOVED

UTILITIES - PROPOSED

- PROPOSED ELECTRICAL - OVERHEAD (SAPN)
- PROPOSED ELECTRICAL - UNDERGROUND (SAPN)
- PROPOSED ELECTRICAL - UNDERGROUND (DIT)
- PROPOSED COMMUNICATIONS
- PROPOSED GAS
- PROPOSED SEWER
- PROPOSED WATER

LIGHTING AND ITS - PROPOSED

- PROPOSED ITS - CONDUIT ELEC/COMMS
- PROPOSED ITS - PIT
- PROPOSED LIGHTING - CONDUIT ELEC
- PROPOSED LIGHTING - OUTREACH + LUMINAIRE

DRAINAGE

- PROPOSED DRAINAGE - PIPE
- PROPOSED DRAINAGE - OPEN SPOON CHANNEL
- PROPOSED DRAINAGE - OPEN VEGETATED CHANNEL
- PROPOSED GRATED FIELD PIT
- PROPOSED GRATED INLET PIT
- PROPOSED DOUBLE GRATED INLET PIT
- PROPOSED DOUBLE GRATED SIDE ENTRY PIT
- PROPOSED SIDE ENTRY PIT

GROUND WATER LEVEL LEGEND:

- ULS GROUND WATER LEVEL
- SLS GROUND WATER LEVEL

GEOLOGICAL UNIT LEVEL LEGEND:

GEOLOGICAL UNIT LEVELS ARE INFERRED AND MAY VARY ON SITE

- TOP OF POORAKA FORMATION
- TOP OF ALLUVIUM
- TOP OF HINDMARSH CLAY (UPPER)
- TOP OF HINDMARSH CLAY (LOWER)
- TOP OF TRANSITIONAL
- TOP OF BURNHAM LIMESTONE
- TOP OF HALLET COVE SANDSTONE
- TOP OF PORT WILLUNGA FORMATION
- INFERRED FAULT

NOTES

1. FOR GENERAL NOTES REFER SHEET No. 2871005 TO 2871008.

1000 0 500 1000 1500 2000
1:50

PROJECT DOCUMENT REFERENCE
T2DPAA-T2D-C3S-BR-DRG-101021

FOR CONSTRUCTION

No.	AMENDMENT DESCRIPTION	BY	CHECK	ACCEPTANCE	DATE
0	ISSUED FOR CONSTRUCTION (KNet No: 24467656)	T2DA	T2DA	M. SHORT	28.03.26

INDEX SHEET REFERENCE: 8011 SHEET 2871002

T2D TORRENS TO DARLINGTON

DESIGNER: A.SINGHAL / S.PODDAR
QUALIFICATION: MEng BEng
DATE: 25.03.26
REVIEWER: R.WOZNIAK
QUALIFICATION: FIEAust CPEng NER RPEV RPEQ MICE CEng
DATE: 25.03.26
INDEPENDENT DESIGN CERTIFIER (IF REQUIRED): T. JACKSON
QUALIFICATION: BEng (Hons), MEngSc
DATE: 27.03.26

Government of South Australia
Department for Infrastructure and Transport

PROJECT No.: 31921901
FILE No.: 2020/10577
DESIGN No.: #15551401
SURVEY No.: #15551401
PROJECT START ROAD RUNNING DISTANCE: NOT APPLICABLE
PROJECT END ROAD RUNNING DISTANCE: NOT APPLICABLE
SCALES: 2000 0 1000 2000 3000 4000 1:100

ROAD No. 6200
RICHMOND ROAD
RAILWAY TERRACE - SOUTH ROAD; MILE END SOUTH BRIDGE OVER KESWICK CREEK
GENERAL ARRANGEMENT - SHEET 01

DESIGNED: T2DA
DRAFTED: T2DA
ACCEPTED FOR USE: M. SHORT
TITLE: DIRECTOR OF ENG.
DATE: 28.03.26
ACCEPTANCE FORM KNET No.: 24467656
IN ACCORDANCE WITH DP013
DRAWING No.: 8011
SHEET No.: 2871021
AMEND No.: 0
SHEET LATITUDE -34.94188
SHEET LONGITUDE 138.57397

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FILE NAME: Autodesk Docs://T2D - Torrens to Darlington /T2DPAA-T2D-C3S-BR-M3D-100302.rvt

SECTION 2
SCALE 1 : 50
2871021

SECTION 3
SCALE 1 : 50
2871021

GROUND WATER LEVEL LEGEND:

- ULS GROUND WATER LEVEL
- SLS GROUND WATER LEVEL

GEOLOGICAL UNIT LEVEL LEGEND:

GEOLOGICAL UNIT LEVELS ARE INFERRED AND MAY VARY ON SITE

- TOP OF POORAKA FORMATION
- TOP OF ALLUVIUM
- TOP OF HINDMARSH CLAY (UPPER)
- TOP OF HINDMARSH CLAY (LOWER)
- TOP OF TRANSITIONAL
- TOP OF BURNHAM LIMESTONE
- TOP OF HALLET COVE SANDSTONE
- TOP OF PORT WILLUNGA FORMATION
- INFERRED FAULT

NOTES

1. FOR GENERAL NOTES REFER SHEET No. 2871005 TO 2871008.

PROJECT DOCUMENT REFERENCE
T2DPAA-T2D-C3S-BR-DRG-101022

FOR CONSTRUCTION

No.	AMENDMENT DESCRIPTION	BY	CHECK	ACCEPTANCE	DATE
0	ISSUED FOR CONSTRUCTION (KNet No: 24467656)	T2DA	T2DA	M. SHORT	28.03.26

INDEX SHEET REFERENCE: 8011 SHEET 2871002

T2D TORRENS TO DARLINGTON

DESIGNER: A.SINGHAL / S.PODDAR
QUALIFICATION: MEng BEng
DATE: 25.03.26
REVIEWER: R.WOZNIAK
QUALIFICATION: FIEAust CPEng NER RPEV RPEQ MICE CEng
DATE: 25.03.26
INDEPENDENT DESIGN CERTIFIER (IF REQUIRED): T. JACKSON
QUALIFICATION: BEng (Hons), MEngSc
DATE: 27.03.26

Government of South Australia
Department for Infrastructure and Transport

PROJECT No.: 31921901
FILE No.: 2020/10577
DESIGN No.: #15551401
SURVEY No.: #15551401
PROJECT START ROAD RUNNING DISTANCE: NOT APPLICABLE
PROJECT END ROAD RUNNING DISTANCE: NOT APPLICABLE
SCALES: 1000 0 500 1000 1500 2000
1:50

ROAD No. 6200
RICHMOND ROAD
RAILWAY TERRACE - SOUTH ROAD; MILE END SOUTH
BRIDGE OVER KESWICK CREEK
GENERAL ARRANGEMENT - SHEET 02

DESIGNED: T2DA
DRAFTED: T2DA
ACCEPTED FOR USE: M. SHORT
TITLE: DIRECTOR OF ENG.
DATE: 28.03.26
ACCEPTANCE FORM KNET No.: 24467656
IN ACCORDANCE WITH DP013
DRAWING No.: 8011
SHEET No.: 2871022
AMEND No.: 0
SHEET LATITUDE -34.94188
SHEET LONGITUDE 138.57397

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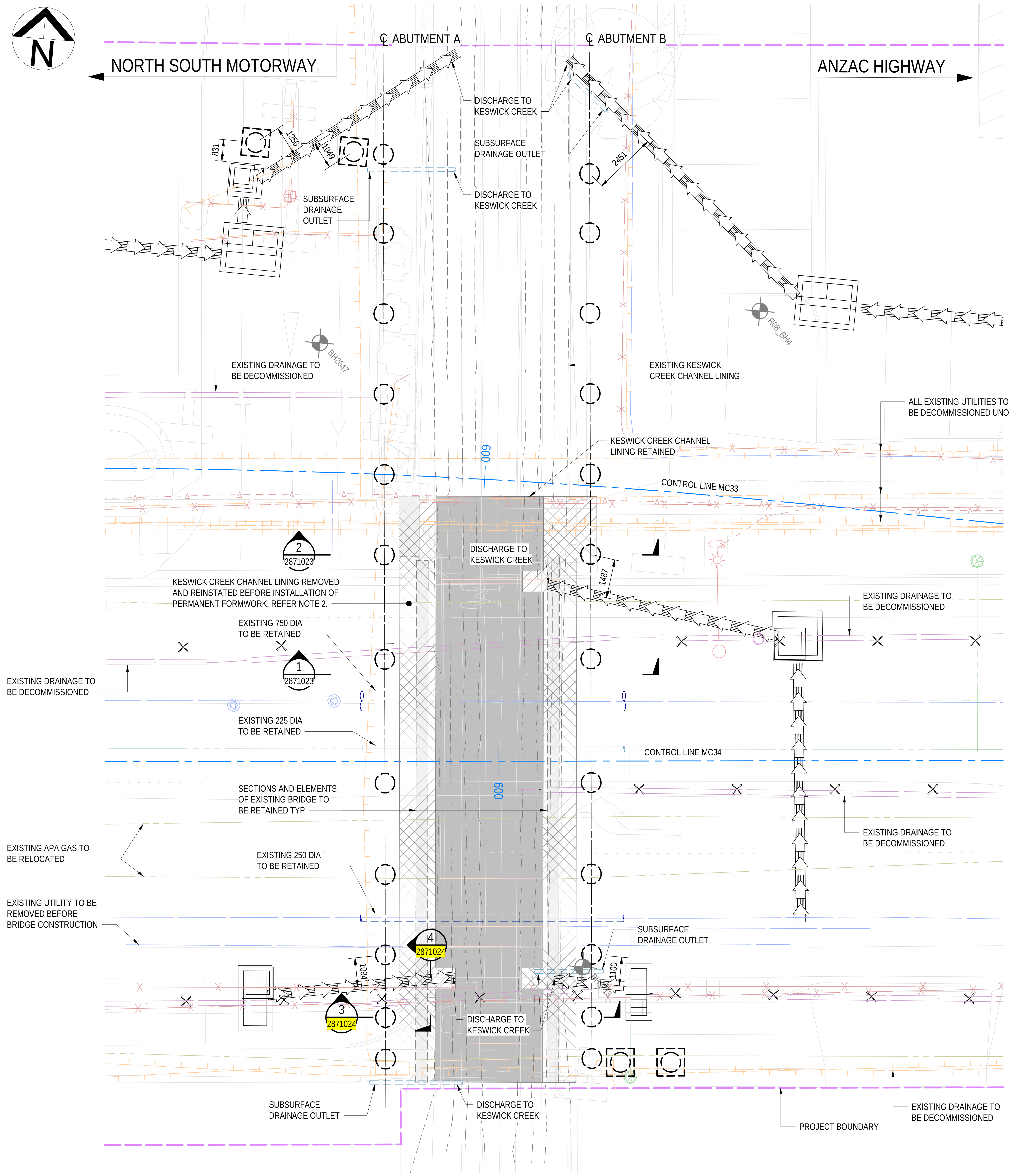
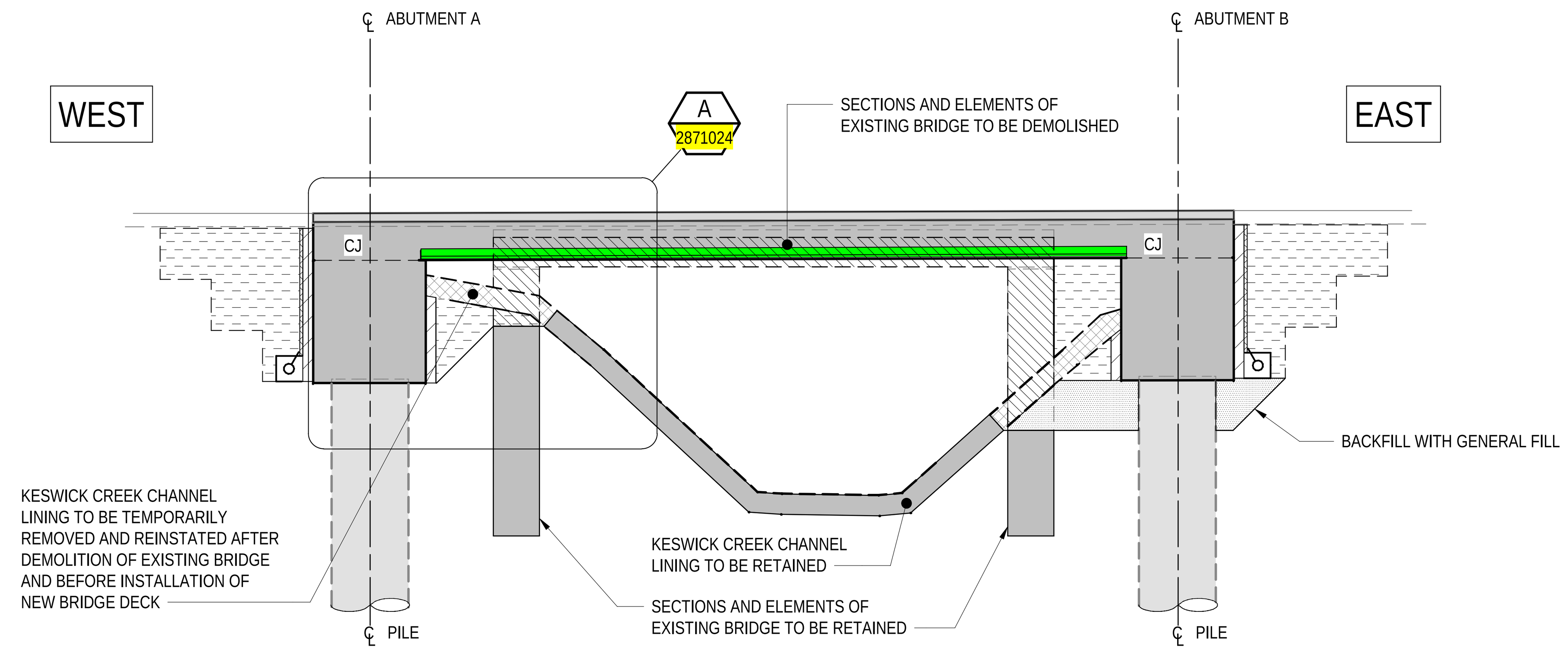
FILE NAME: Autodesk Docs://T2D - Torrens to Darlington /T2DPAA-T2D-C3S-BR-M3D-100302.rvt



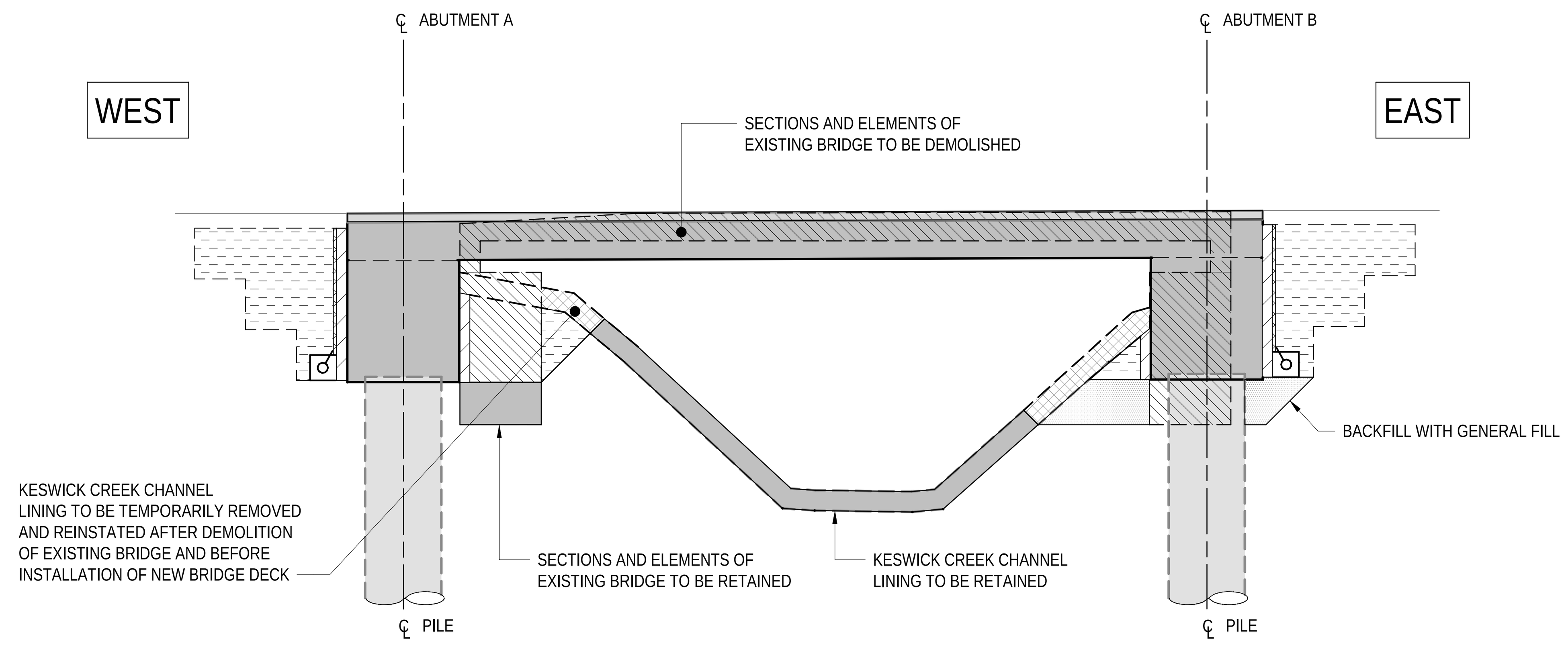
PLAN UNDER THE BRIDGE SOFFIT

SCALE 1 : 100



SECTION 1 (2871023)

SCALE 1 : 50



SECTION 2 (2871023)

SCALE 1 : 50

NOTES

1. FOR GENERAL NOTES REFER TO SHEET No. 2871005 TO 2871008.
2. EXTENT OF TEMPORARILY REMOVAL AND REINSTATEMENT OF CONCRETE LINING ALONG KESWICK CREEK CHANNEL TO MATCH THE FOOTPRINT OF THE EXISTING BRIDGE AND TO BE CONFIRMED BY CONSTRUCTION TEAM PRIOR WORKS.

1000 0 500 1000 1500 2000
1:50

PROJECT DOCUMENT REFERENCE
T2DPAA-T2D-C3S-BR-DRG-101023

FOR CONSTRUCTION

No.	AMENDMENT DESCRIPTION	BY	CHECK	ACCEPTANCE	DATE
0	ISSUED FOR CONSTRUCTION (KNet No: 24467656)	T2DA	T2DA	M. SHORT	28.03.26

INDEX SHEET REFERENCE: 8011 SHEET 2871002

T2D → TORRENS TO DARLINGTON

DESIGNER: A.SINGHAL / S.PODDAR
QUALIFICATION: MEng BEng
DATE: 25.03.26

REVIEWER: R.WOZNIAK
QUALIFICATION: FIEAust CPEng NER RPEV RPEQ MICE CEng
DATE: 25.03.26

INDEPENDENT DESIGN CERTIFIER (IF REQUIRED): T. JACKSON
QUALIFICATION: BEng (Hons), MEngSc
DATE: 27.03.26

Government of South Australia
Department for Infrastructure and Transport

PROJECT No.: 31921901 | FILE No.: 2020/10577
DESIGN No.: #15551401 | SURVEY No.: #15551401
PROJECT START ROAD RUNNING DISTANCE: NOT APPLICABLE
PROJECT END ROAD RUNNING DISTANCE: NOT APPLICABLE
SCALES: 2000 0 1000 2000 3000 4000 1:100

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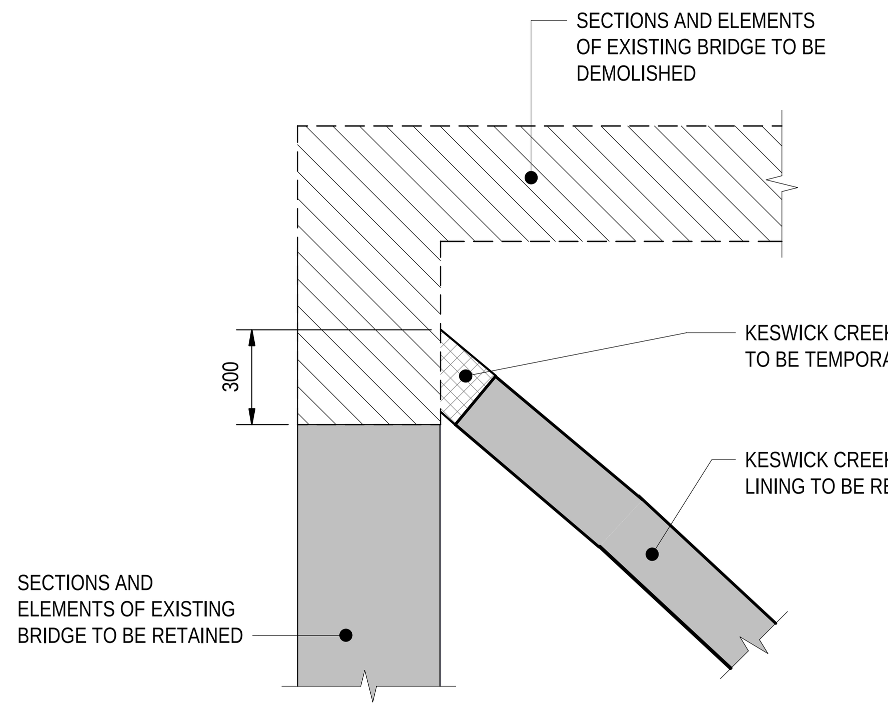
ROAD No. 6200
RICHMOND ROAD
RAILWAY TERRACE - SOUTH ROAD; MILE END SOUTH
BRIDGE OVER KESWICK CREEK
GENERAL ARRANGEMENT - SHEET 03

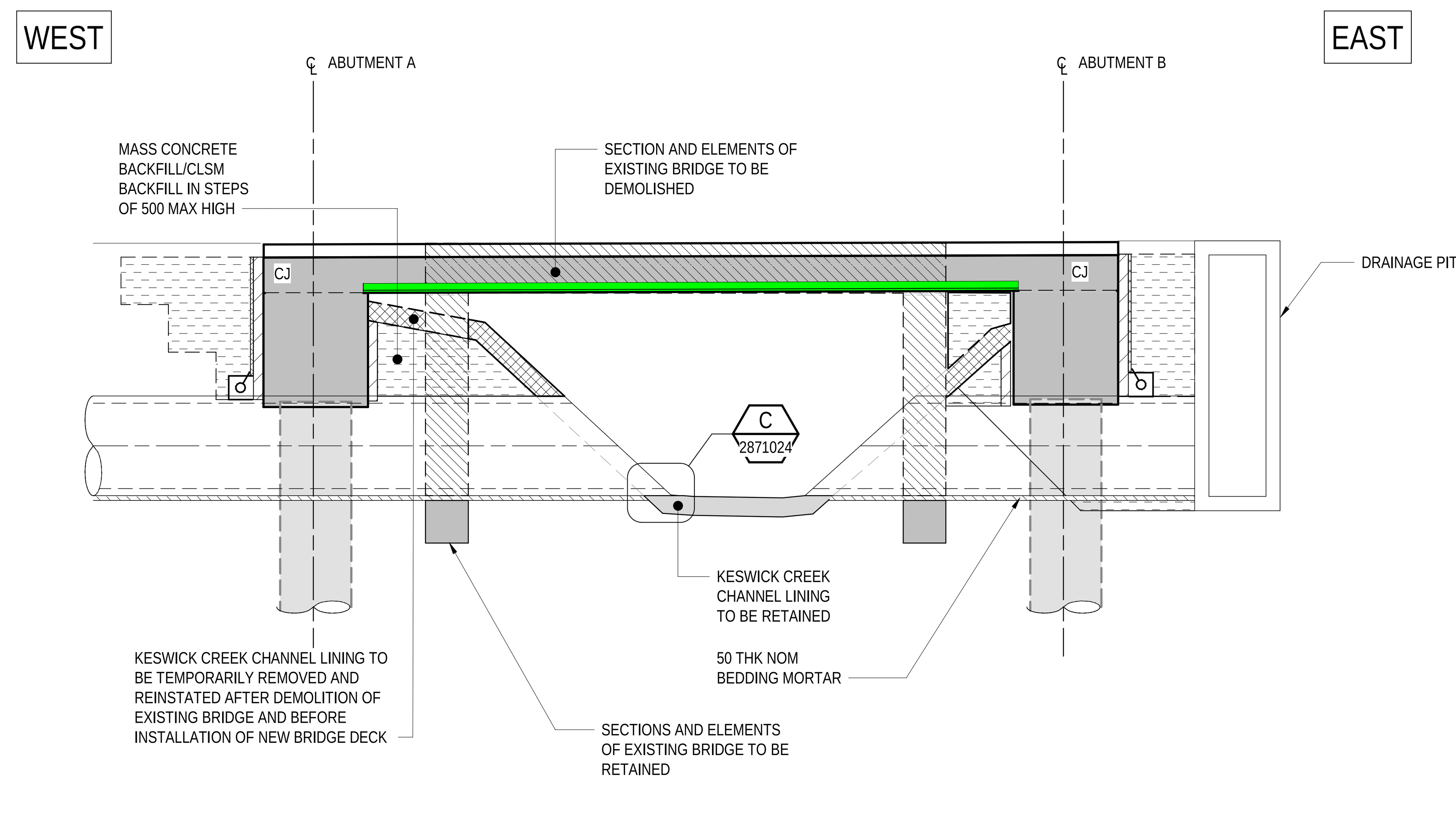
DESIGNED:	DRAFTED:	ACCEPTED FOR USE:	ACCEPTANCE FORM KNET No.:	DRAWING No.:	SHEET No.:	AMEND No.:
T2DA	T2DA	M. SHORT, TITLE: DIRECTOR OF ENG., DATE: 28.03.26	24467656 IN ACCORDANCE WITH DP013	8011	2871023	0

SHEET LATITUDE -34.94188 SHEET LONGITUDE 138.57397

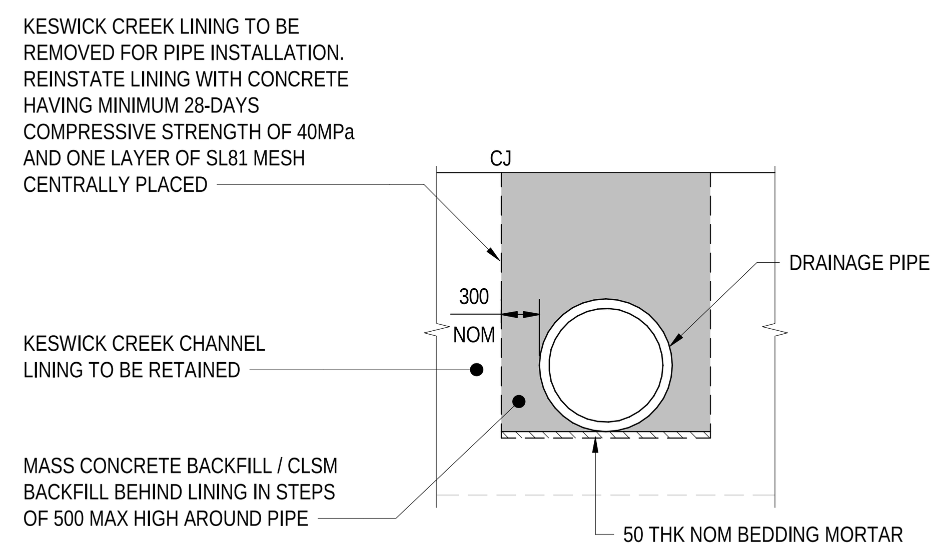
FILE NAME: Autodesk Docs://T2D - Torrens to Darlington /T2DPAA-T2D-C3S-BR-M3D-100302.rvt



STAGE B1 - EXISTING BRIDGE DEMOLITION



SECTION 3
SCALE 1 : 50
2871023



SECTION 4
SCALE 1 : 50
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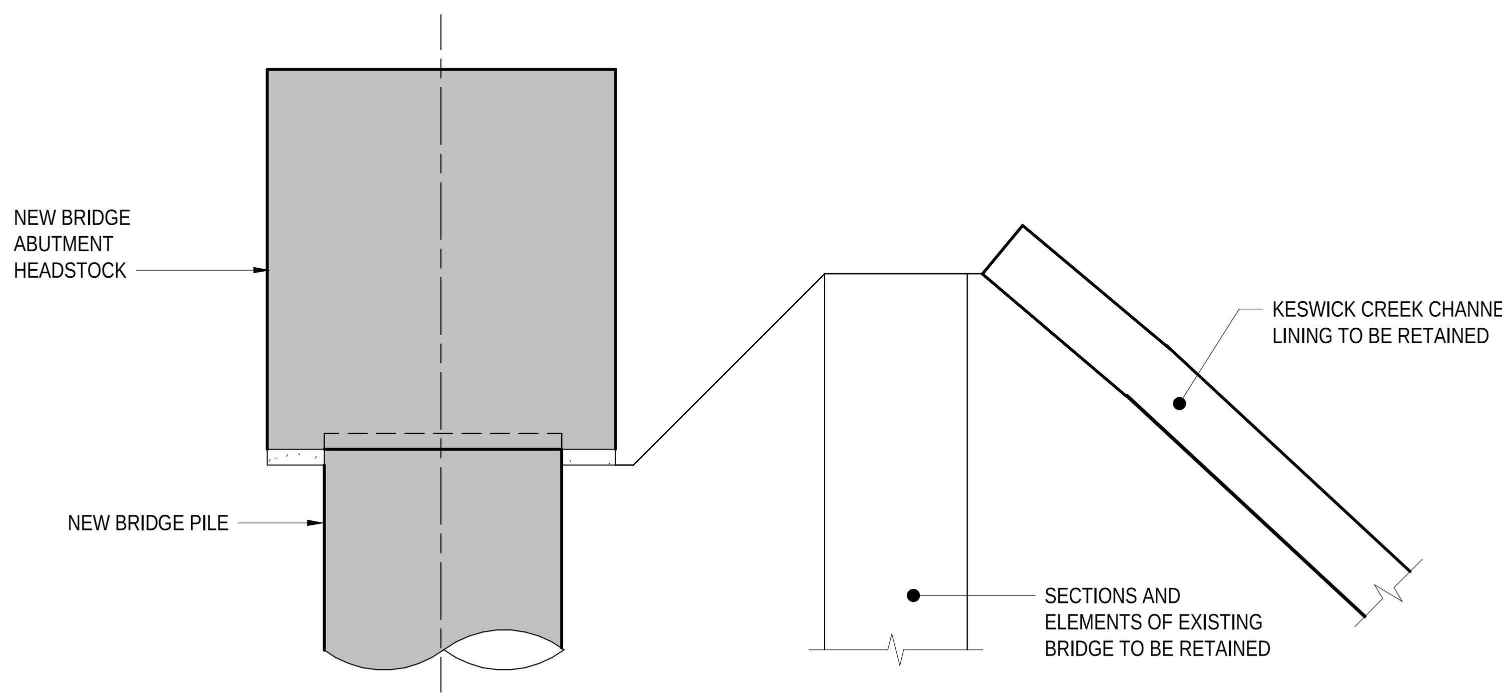
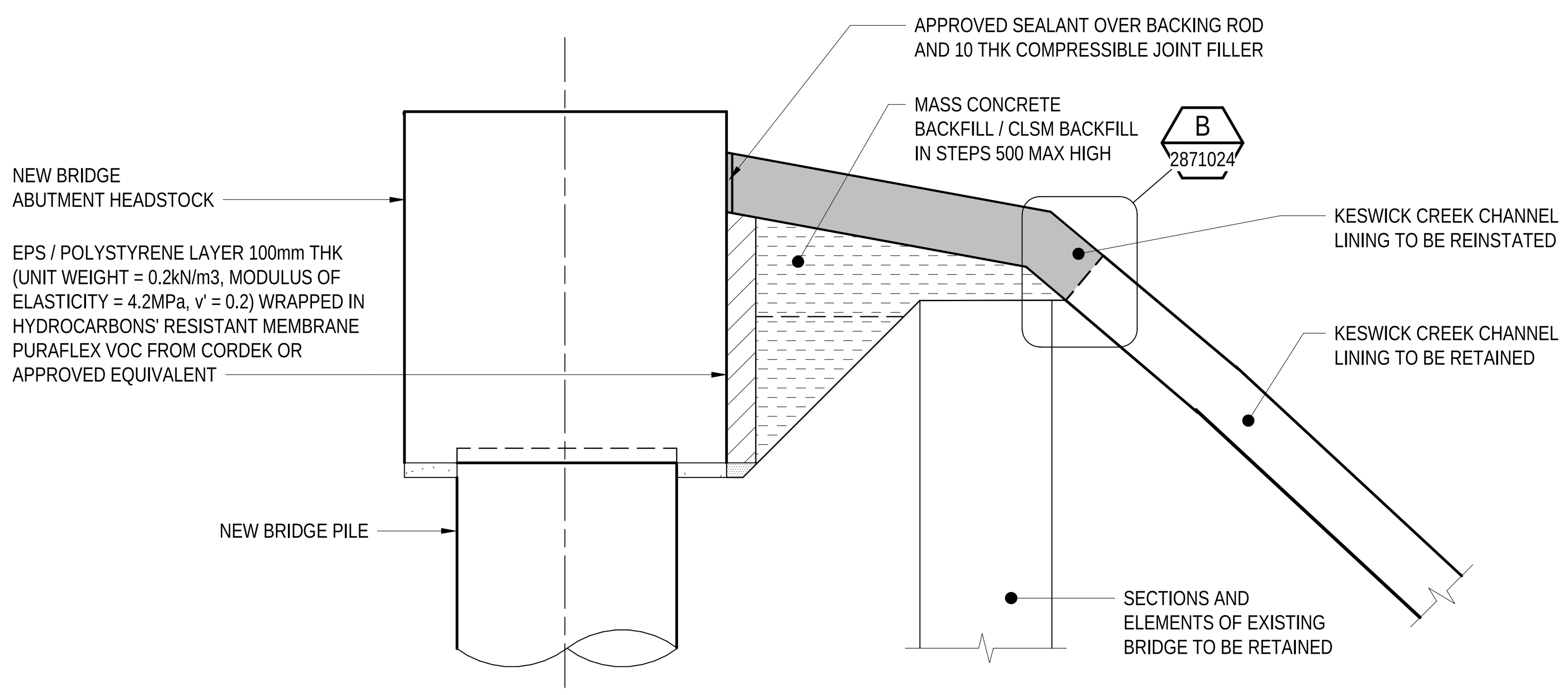
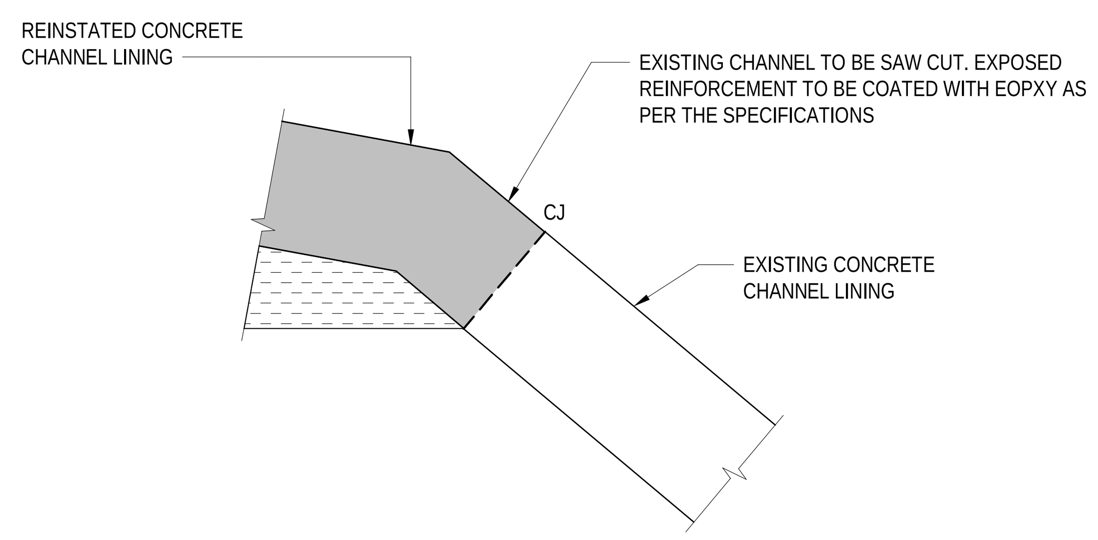


STAGE A2, B2, B3 - CONSTRUCTION OF NEW BRIDGE



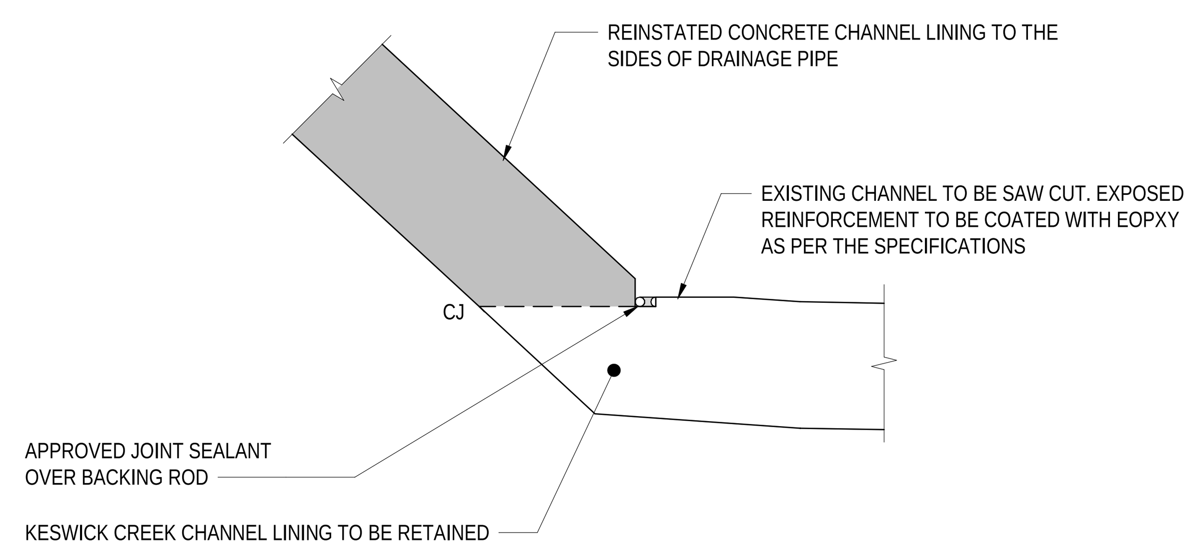
STAGE A3, B4 - REINSTATMENT OF CHANNEL LINING
(TO BE COMPLETED BEFORE CONSTRUCTION OF BRIDGE DECK)

DETAIL A
SCALE 1 : 20
2871023



DIMENSION AND LEVEL TO BE CONFIRMED ON SITE

DETAIL B
SCALE 1 : 10
2871024



DETAIL C
SCALE 1 : 10
2871024

(DRAINAGE PIPE NOT SHOWN FOR CLARITY)

NOTES

1. FOR GENERAL NOTES REFER TO SHEET No. 2871005 TO 2871008.
2. FOR CONSTRUCTION SEQUENCE STAGE DETAILS, REFER TO SHEET No. 2871031 TO 2871033.

200 0 100 200 300 400
1:10
400 0 200 400 600 800
1:20

PROJECT DOCUMENT REFERENCE
T2DPAA-T2D-C3S-BR-DRG-101024

FOR CONSTRUCTION

No.	AMENDMENT DESCRIPTION	BY	CHECK	ACCEPTANCE	DATE
0	ISSUED FOR CONSTRUCTION (KNet No: 24467656)	T2DA	T2DA	M. SHORT	28.03.26

INDEX SHEET REFERENCE: 8011 SHEET 2871002

T2D → TORRENS TO DARLINGTON

DESIGNER: A.SINGHAL / S.PODDAR
QUALIFICATION: MEng BEng
DATE: 25.03.26
REVIEWER: R.WOZNIAK
QUALIFICATION: FIEAust CPEng NER RPEV RPEQ MICE CEng
DATE: 25.03.26
INDEPENDENT DESIGN CERTIFIER (IF REQUIRED): T. JACKSON
QUALIFICATION: BEng (Hons), MEngSc
DATE: 27.03.26

Government of South Australia
Department for Infrastructure and Transport

PROJECT No.: 31921901
FILE No.: 2020/10577
DESIGN No.: #15551401
SURVEY No.: #15551401
PROJECT START ROAD RUNNING DISTANCE: NOT APPLICABLE
PROJECT END ROAD RUNNING DISTANCE: NOT APPLICABLE
SCALES: 1000 0 500 1000 1500 2000 1:50

ROAD No. 6200
RICHMOND ROAD
RAILWAY TERRACE - SOUTH ROAD; MILE END SOUTH
BRIDGE OVER KESWICK CREEK
GENERAL ARRANGEMENT - SHEET 04

DESIGNED:	DRAFTED:	ACCEPTED FOR USE:	ACCEPTANCE FORM KNET No.:	DRAWING No.:	SHEET No.:	AMEND No.:
T2DA	T2DA	M. SHORT TITLE: DIRECTOR OF ENG. DATE: 28.03.26	24467656 IN ACCORDANCE WITH DP013	8011 SHEET LATITUDE -34.94188	2871024 SHEET LONGITUDE 138.57397	0

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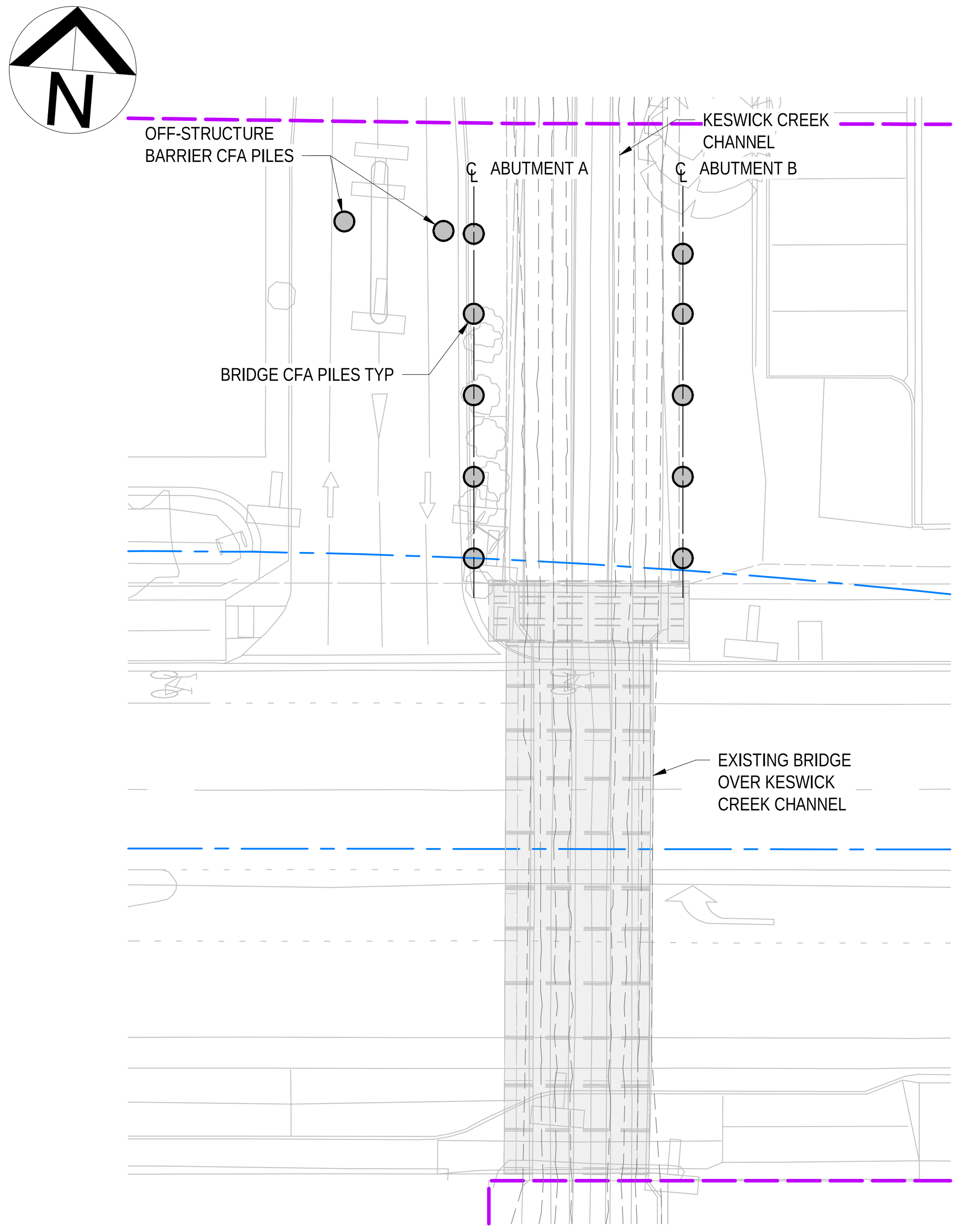
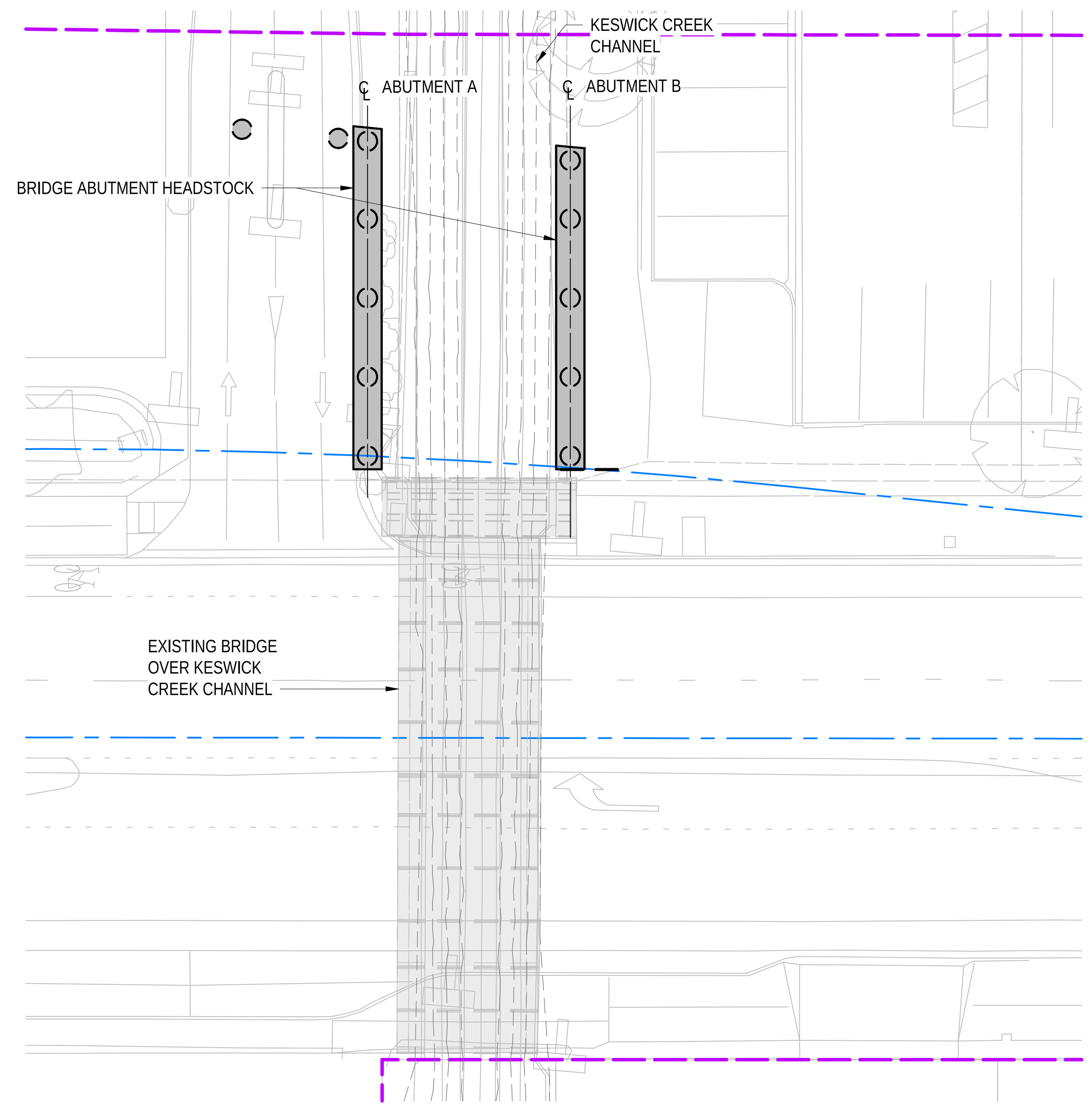
FILE NAME: Autodesk Docs://T2D - Torrens to Darlington /T2DPAA-T2D-C3S-BR-M3D-100302.rvt



CONSTRUCTION STAGE A1

SCALE 1 : 200

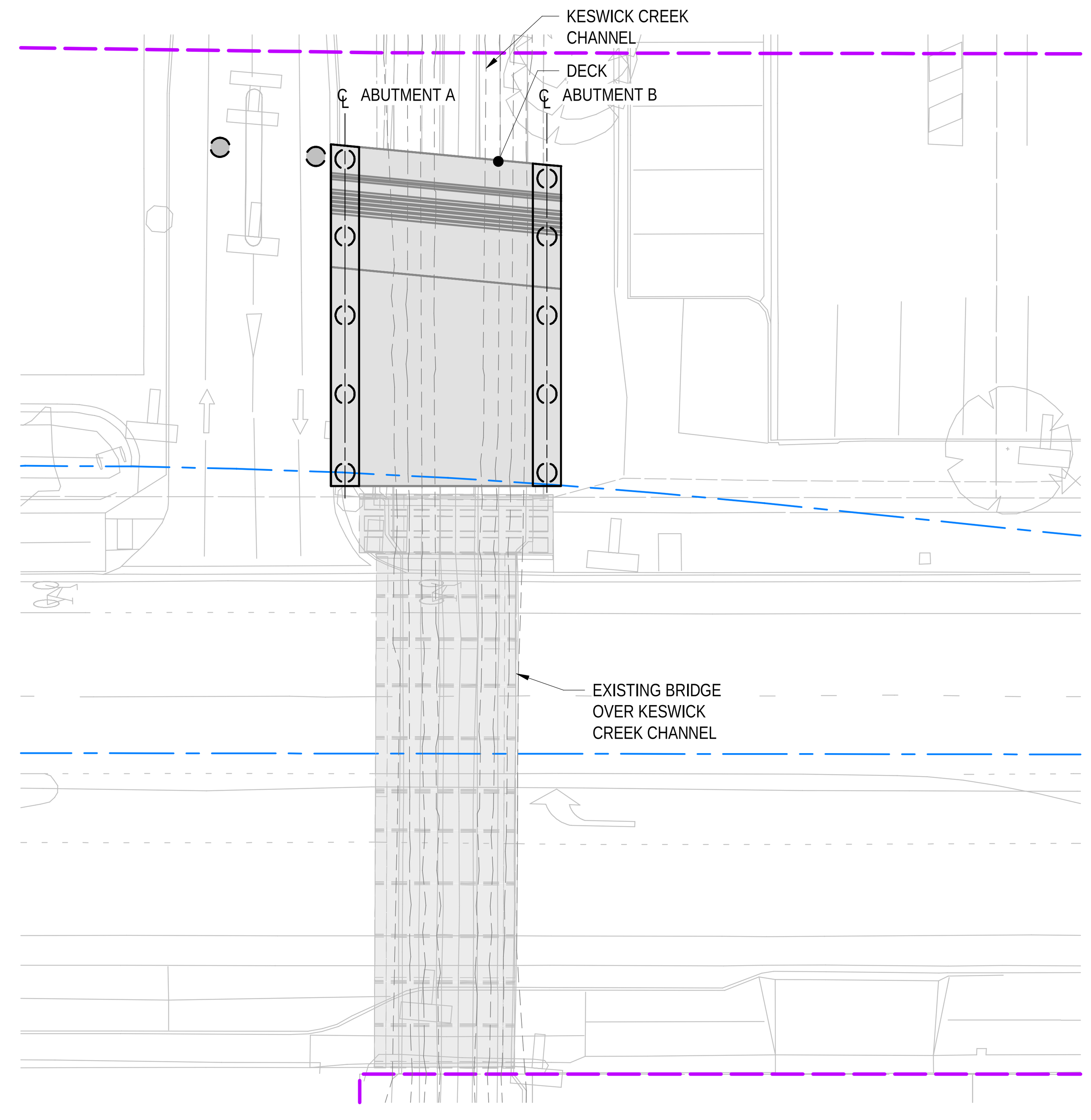
1.1 EXCAVATE EXISTING SOIL AND INSTALL PILING PLATFORMS.
1.2 CONSTRUCT CFA PILES AT ABUTMENT A & B FOR NORTHERN SECTION AND FOR OFF-STRUCTURE BARRIER FOR EASTBOUND CARRIAGEWAY.



CONSTRUCTION STAGE A2

SCALE 1 : 200

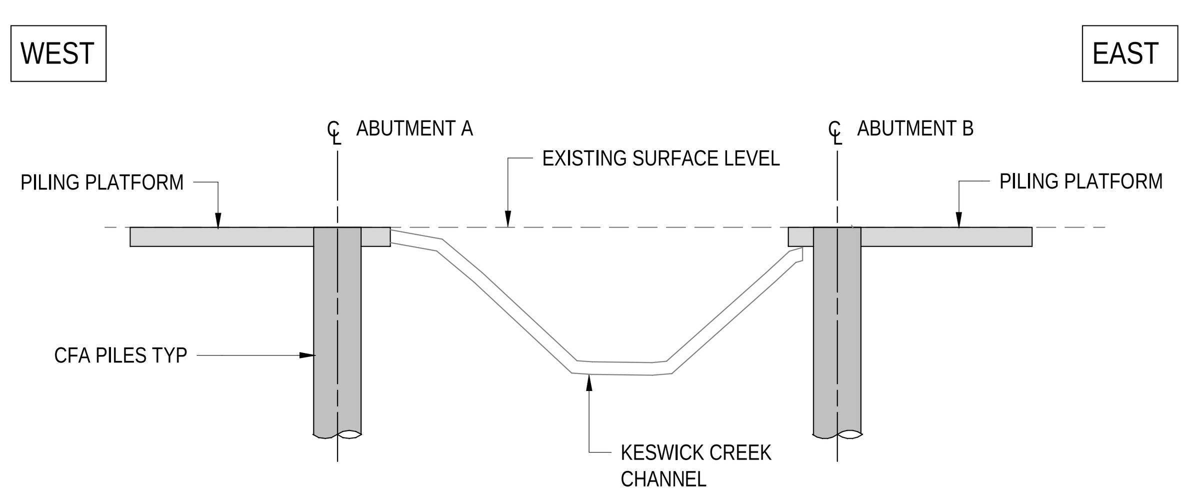
2.1 LOCALLY EXCAVATE SOIL AT BRIDGE ABUTMENTS.
2.2 BREAK BACK CFA PILES TO CUT-OFF LEVELS.
2.3 CAST ABUTMENT HEADSTOCKS. APPLY PROTECTION TO EXPOSED STITCH STARTER BARS.



CONSTRUCTION STAGE A3

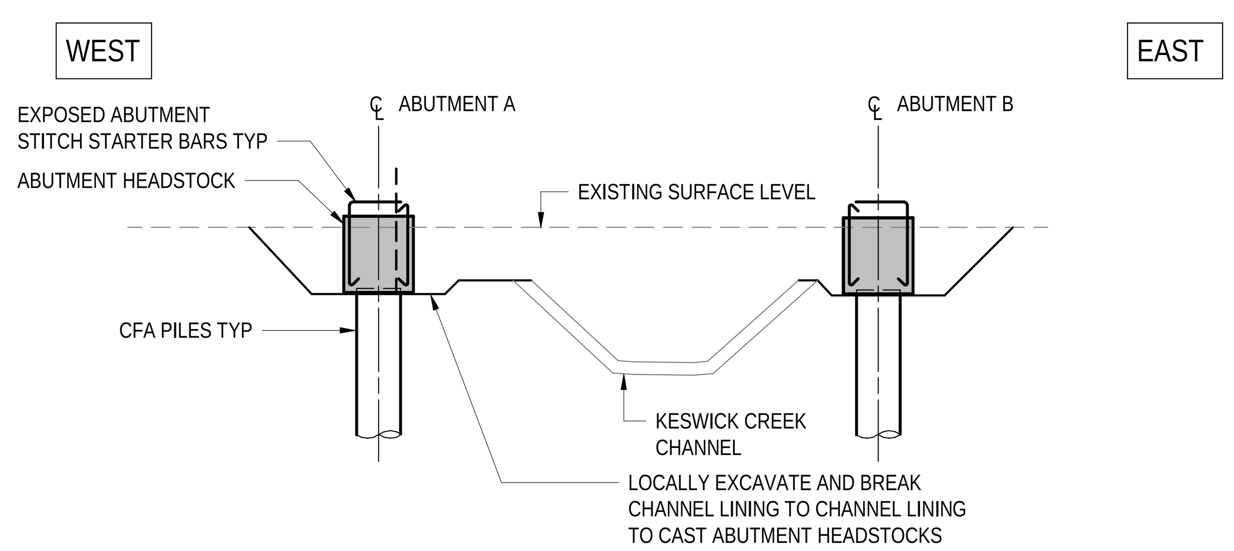
SCALE 1 : 200

3.1 LOCALLY REPAIR THE KESWICK CREEK CHANNEL LINING AS REQUIRED.
3.2 INSTALL TEMPORARY VERTICAL AND LATERAL SUPPORTS FOR DECK SLAB PERMANENT FORMWORK (TO BE DETAILED IN THE RELEVANT TEMPORARY WORKS PACKAGE).
3.3 ONCE HEADSTOCK HAS REACHED 14 DAYS CONCRETE COMPRESSIVE STRENGTH OF MINIMUM 25MPa, PLACE PERMADEC PERMANENT FORMWORK AND REINFORCEMENT FOR RC DECK SLAB AND INTEGRAL ABUTMENT STITCH.
3.4 INSTALL CONDUITS FOR NEW SERVICES.
3.5 CAST RC DECK SLAB AND INTEGRAL ABUTMENT STITCH MONOLITHICALLY IN ONE POUR.
3.6 REMOVE TEMPORARY SUPPORT AFTER DECK CONCRETE HAS REACHED 28 DAYS CONCRETE COMPRESSIVE STRENGTH OF MINIMUM 50MPa.



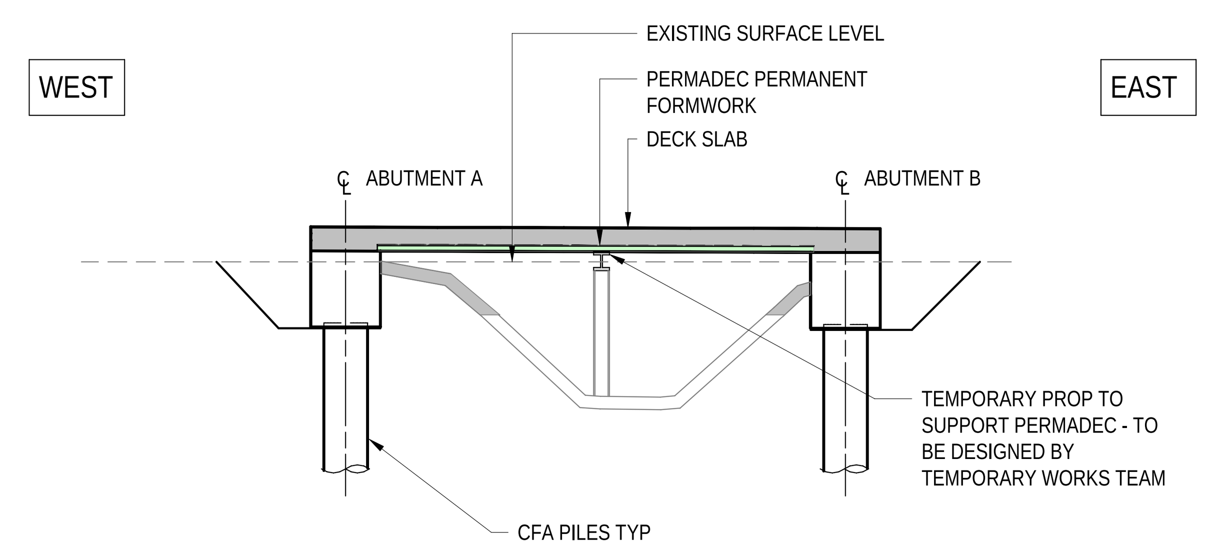
SECTION-STAGE A1

SCALE 1 : 100



SECTION-STAGE A2

SCALE 1 : 100



SECTION-STAGE A3

SCALE 1 : 100

NOTES

1. FOR GENERAL NOTES REFER TO SHEET No. 2871005 TO 2871008.

2000 0 1000 2000 3000 4000
1:100

PROJECT DOCUMENT REFERENCE
T2DPAA-T2D-C3S-BR-DRG-101031

FOR CONSTRUCTION

No.	AMENDMENT DESCRIPTION	BY	CHECK	ACCEPTANCE	DATE
0	ISSUED FOR CONSTRUCTION (KNet No: 24467656)	T2DA	T2DA	M. SHORT	28.03.26

INDEX SHEET REFERENCE: 8011 SHEET 2871002

T2D → TORRENS TO DARLINGTON

DESIGNER: A.SINGHAL / S.PODDAR
QUALIFICATION: MEng BEng
DATE: 25.03.26

REVIEWER: R.WOZNIAK
QUALIFICATION: FIEAust CPEng NER RPEV RPEQ MICE CEng
DATE: 25.03.26

INDEPENDENT DESIGN CERTIFIER (IF REQUIRED): T. JACKSON
QUALIFICATION: BEng (Hons), MEngSc
DATE: 27.03.26

Government of South Australia
Department for Infrastructure and Transport

PROJECT No.: 31921901
FILE No.: 2020/10577
DESIGN No.: #15551401
SURVEY No.: #15551401
PROJECT START ROAD RUNNING DISTANCE: NOT APPLICABLE
PROJECT END ROAD RUNNING DISTANCE: NOT APPLICABLE
SCALES: 4000 0 2000 4000 6000 8000 1:200

ROAD No. 6200
RICHMOND ROAD
RAILWAY TERRACE - SOUTH ROAD; MILE END SOUTH
BRIDGE OVER KESWICK CREEK
CONSTRUCTION SEQUENCE - SHEET 01

DESIGNED: T2DA
DRAFTED: T2DA
ACCEPTED FOR USE: M. SHORT
TITLE: DIRECTOR OF ENG.
DATE: 28.03.26
ACCEPTANCE FORM KNET No.: 24467656
IN ACCORDANCE WITH DP013
DRAWING No.: 8011
SHEET No.: 2871031
AMEND No.: 0
SHEET LATITUDE -34.94188
SHEET LONGITUDE 138.57397

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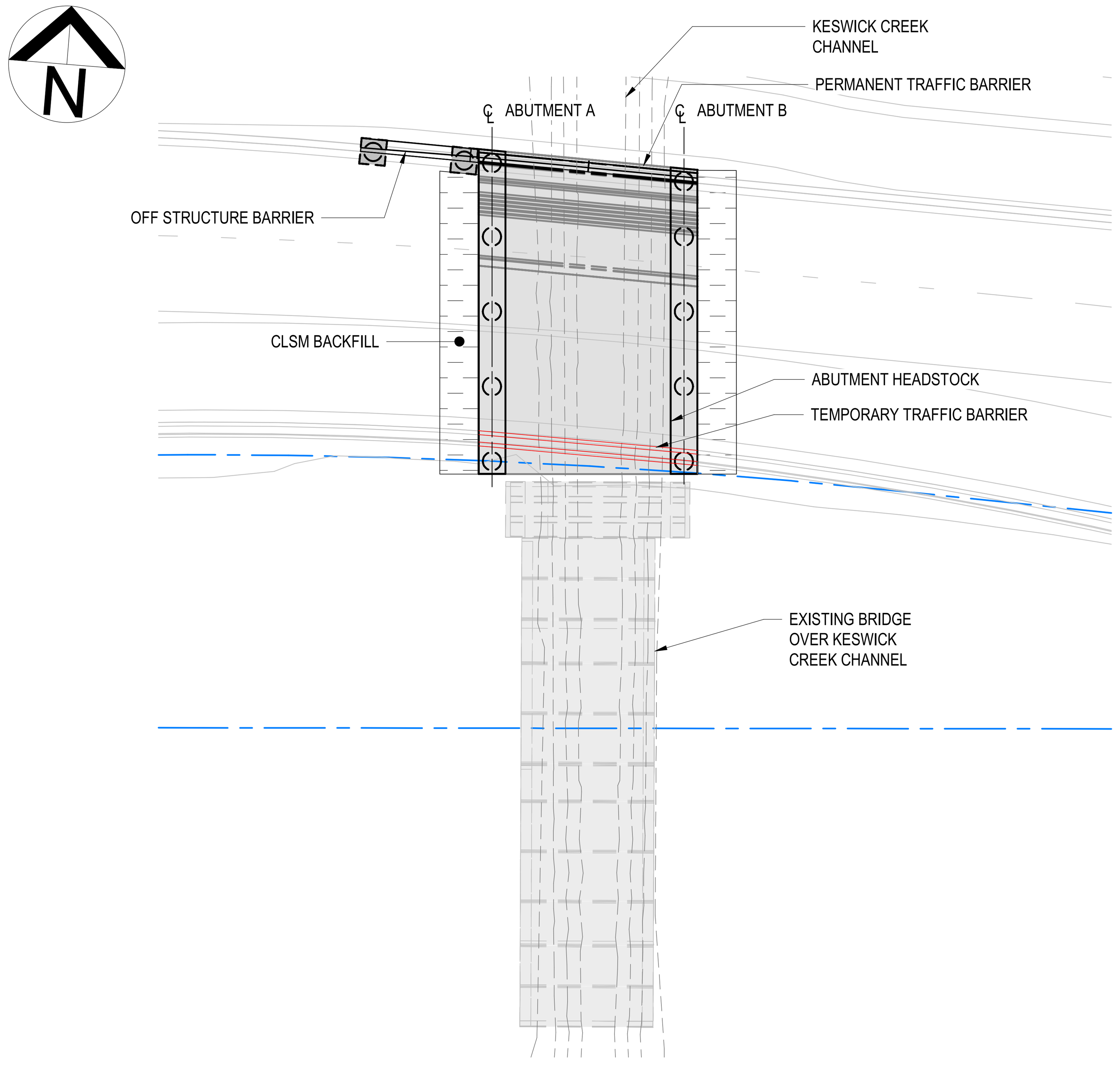
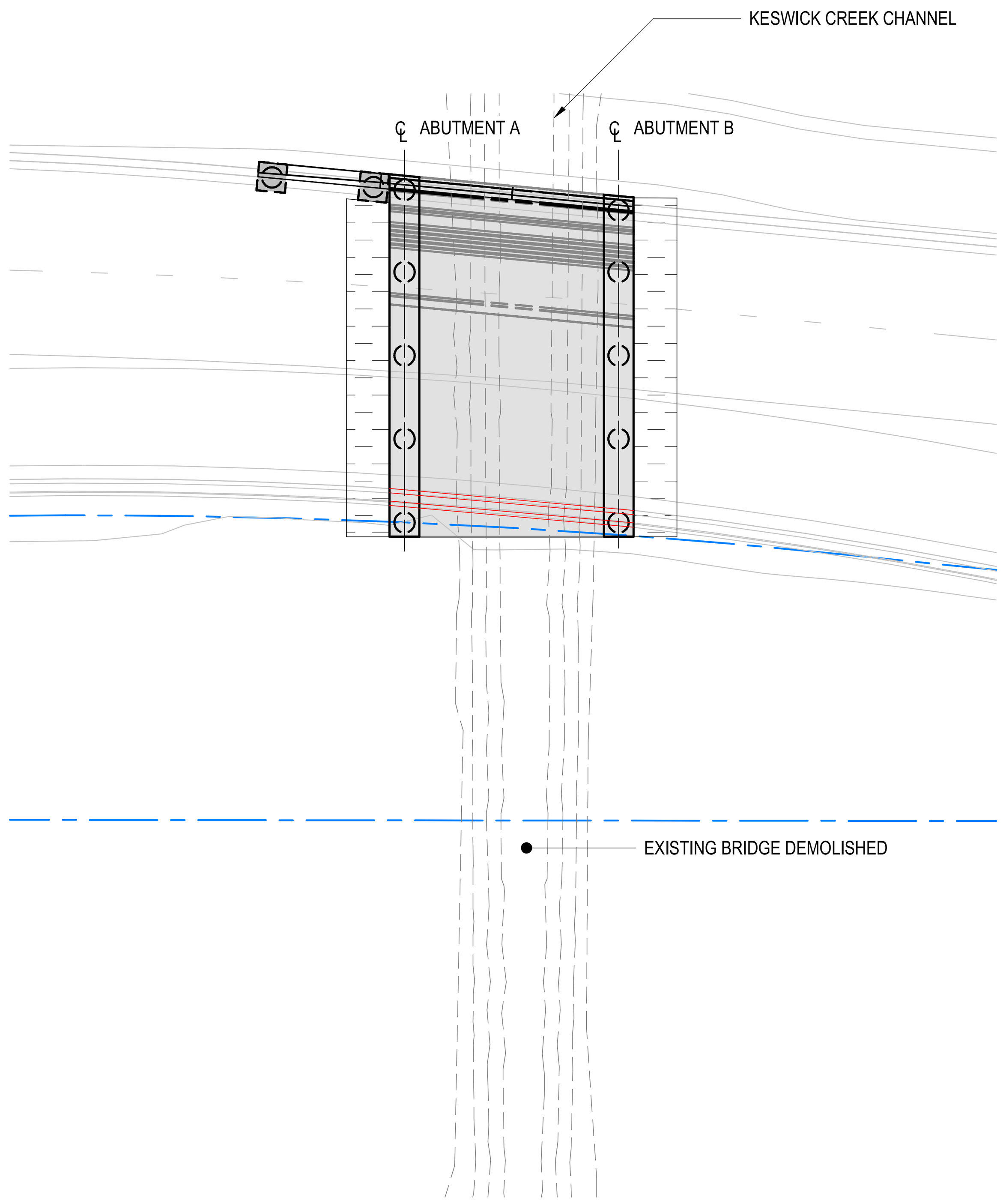
FILE NAME: Autodesk Docs://T2D - Torrens to Darlington /T2DPAA-T2D-C3S-BR-M3D-100302.rvt



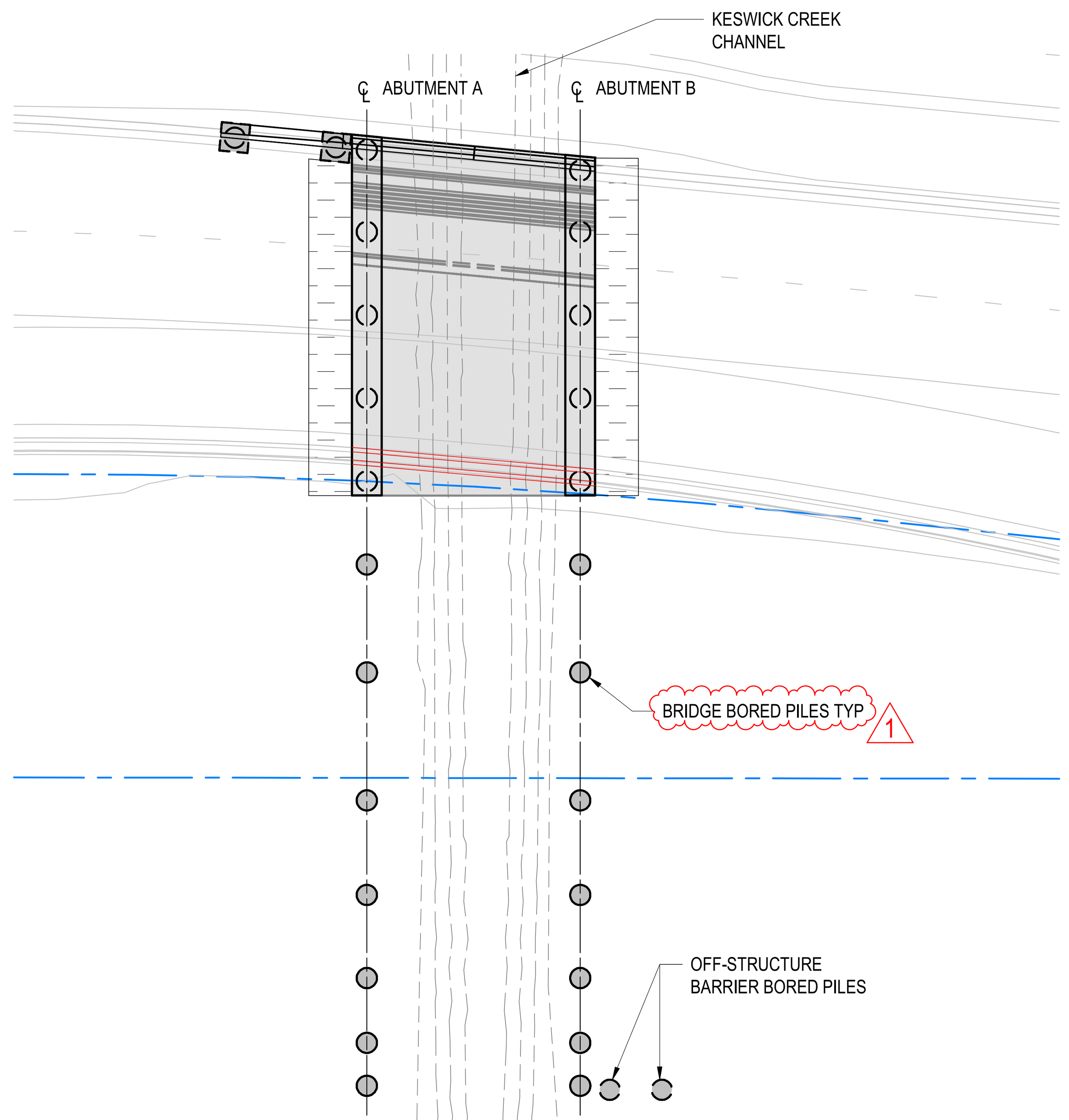
CONSTRUCTION STAGE A4
SCALE 1 : 200

4.1 RELOCATE AND / OR DIVERT EXISTING UNDERBORED SERVICES AND INSTALL ITS AND LIGHTING CABLES IN THE BRIDGE DECK.
4.2 LOCALLY EXCAVATE FURTHER BEHIND EACH ABUTMENT AND INSTALL WATERPROOFING, COMPRESSIBLE POLYSTERENE LAYER AND SUBSURFACE DRAINAGE AT BACK OF ABUTMENTS.
4.3 INSTALL NEW UNDERBORED SERVICES.
4.4 INSTALL OFF-STRUCTURE BARRIER AND CONSTRUCT PILE CAP.
4.5 BACKFILL BEHIND EACH ABUTMENTS WITH CONTROLLED LOW-STRENGTH MATERIAL (CLSM) FILL.
4.6 PREPARE WORKS FOR TRAFFIC DIVERSION.
4.7 INSTALL TEMPORARY PAVEMENT, PERMANENT AND TEMPORARY BARRIER AND MARK TRAFFIC LANES.
4.8 DIVERT RICHMOND ROAD TRAFFIC TO THE NEW SECTION OF THE BRIDGE TO THE NORTH.



CONSTRUCTION STAGE B1
SCALE 1 : 200

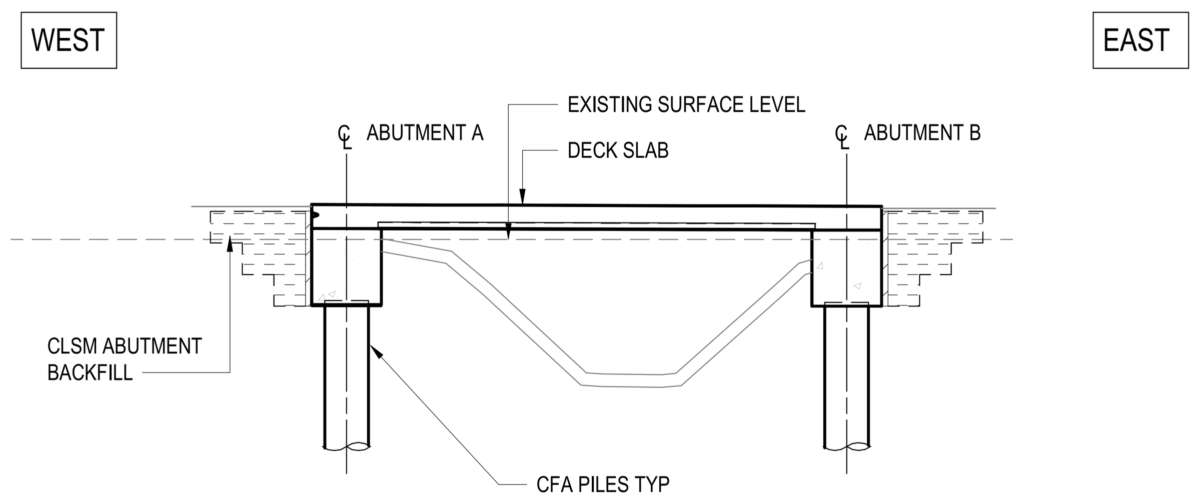
1.1 RELOCATE, DIVERT AND PROTECT EXISTING SERVICES.
1.2 DEMOLISH EXISTING BRIDGE SUPERSTRUCTURE.
1.3 LOCALLY REMOVE SECTION OF CHANNEL CONCRETE LINING TO ALLOW DEMOLITION OF EXISTING SUBSTRUCTURE.
1.4 DEMOLISH EXISTING BRIDGE SUBSTRUCTURE AND BACKFILL WITH CLSM.



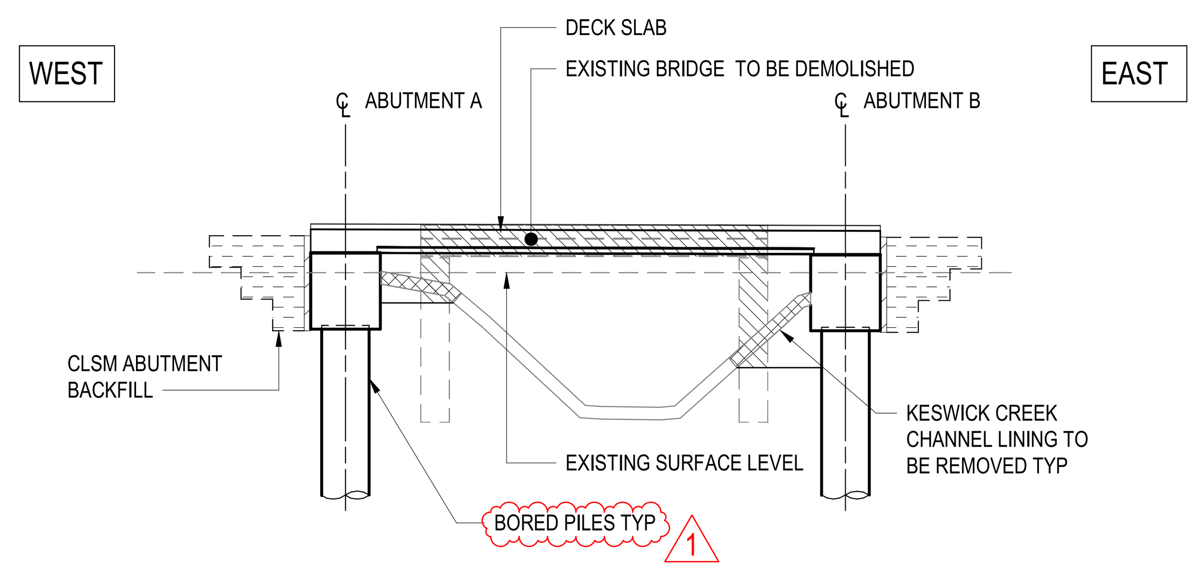
CONSTRUCTION STAGE B2
SCALE 1 : 200

2.1 EXCAVATE EXISTING SOIL AND INSTALL PILING PLATFORMS.
2.2 CONSTRUCT BORED PILES AT ABUTMENT A & B FOR SOUTHERN SECTION AND FOR OFF-STRUCTURE BARRIER FOR WESTBOUND CARRIAGEWAY.

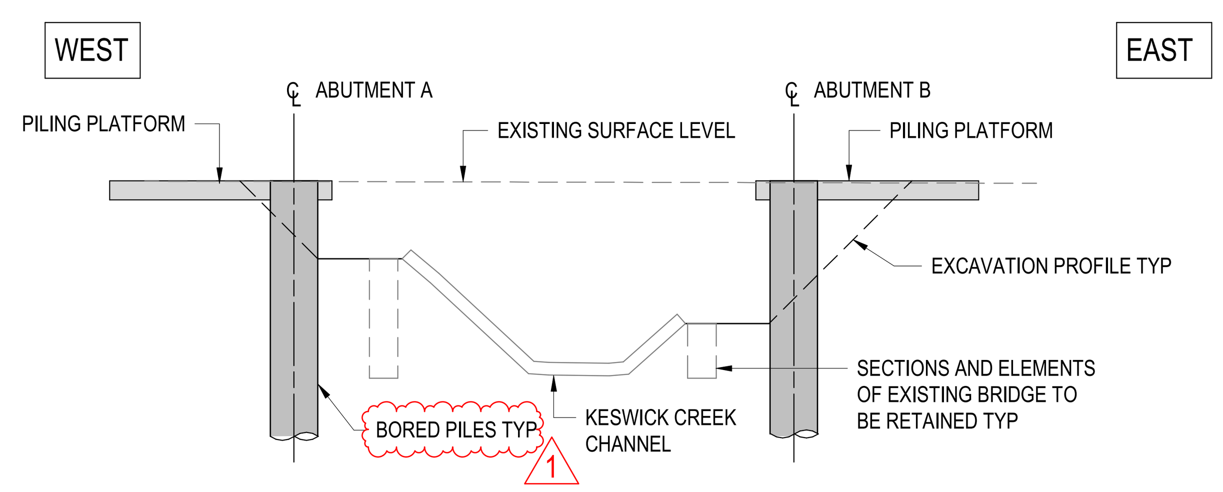
1



SECTION-STAGE A4
SCALE 1 : 100



SECTION-STAGE B1
SCALE 1 : 100



SECTION-STAGE B2
SCALE 1 : 100
NOTE:- STAGE A NOT SHOWN FOR CLARITY

NOTES

1. FOR GENERAL NOTES REFER TO SHEET No. 2871005 TO 2871008.

2000 0 1000 2000 3000 4000
1:100

PROJECT DOCUMENT REFERENCE
T2DPAA-T2D-C3S-BR-DRG-101032

FOR CONSTRUCTION

No.	AMENDMENT DESCRIPTION	BY	CHECK	ACCEPTANCE	DATE
1	ISSUED FOR CONSTRUCTION (KNet No. 24941076)	T2DA	T2DA	M. SHORT	17.07.26
0	ISSUED FOR CONSTRUCTION (KNet No. 24467656)	T2DA	T2DA	M. SHORT	28.03.26

INDEX SHEET REFERENCE: 8011 SHEET 2871002

T2D TORRENS TO DARLINGTON

DISCLAIMER.
BACKGROUND MODEL INFORMATION SHOWN ON THIS DRAWING IS PROVIDED FOR REFERENCE ONLY. THE BACKGROUND DESIGN MAY BE SUBJECT TO ONGOING DESIGN DEVELOPMENT AND REFINEMENT.

DESIGNER: A.SINGHAL / S.PODDAR
QUALIFICATION: MEng BEng
DATE: 25.03.26
REVIEWER: R.WOZNIAK
QUALIFICATION: FIEAust CPEng NER RPEV RPEQ MICE CEng
DATE: 25.03.26
INDEPENDENT DESIGN CERTIFIER (IF REQUIRED): T. JACKSON
QUALIFICATION: BEng (Hons), MEngSc
DATE: 27.03.26

Government of South Australia
Department for Infrastructure and Transport

PROJECT No.: 31921901
FILE No.: 2020/10577
DESIGN No.: #15551401
SURVEY No.: #15551401
PROJECT START ROAD RUNNING DISTANCE: NOT APPLICABLE
PROJECT END ROAD RUNNING DISTANCE: NOT APPLICABLE
SCALES: 4000 0 2000 4000 6000 8000 1:200

ROAD No. 6200
RICHMOND ROAD
RAILWAY TERRACE - SOUTH ROAD; MILE END SOUTH
BRIDGE OVER KESWICK CREEK
CONSTRUCTION SEQUENCE - SHEET 02

DESIGNED:	DRAFTED:	ACCEPTED FOR USE:	ACCEPTANCE FORM KNET No.:	DRAWING No.:	SHEET No.:	AMEND No.:
T2DA	T2DA	M. SHORT TITLE: DIRECTOR ENG. DATE: 28.03.26	24467656 IN ACCORDANCE WITH DP013	8011 SHEET LATITUDE -34.94188	2871032 SHEET LONGITUDE 138.57397	1

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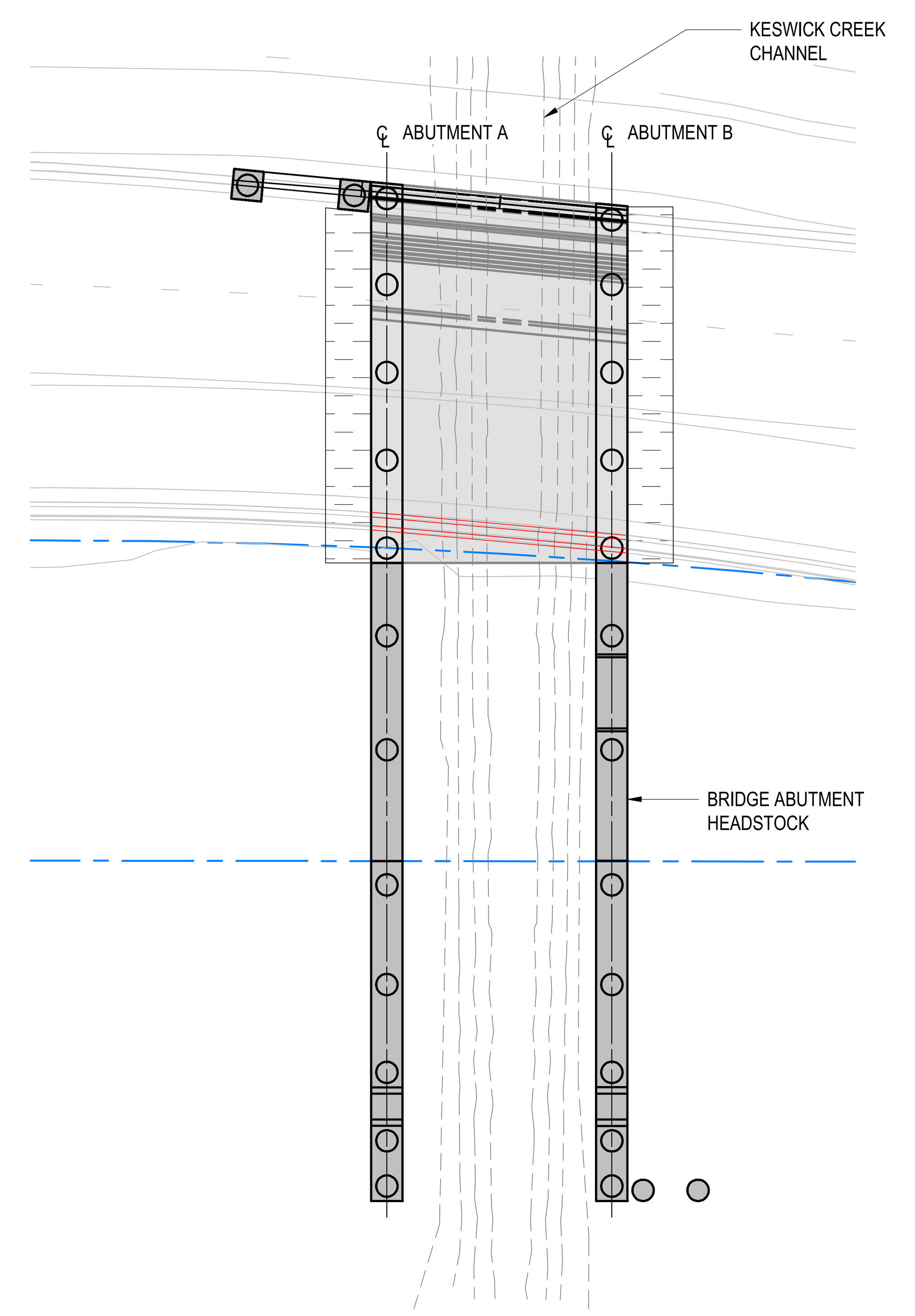
ALL DIMENSIONS ARE IN MILLIMETRES UNLESS SHOWN OTHERWISE

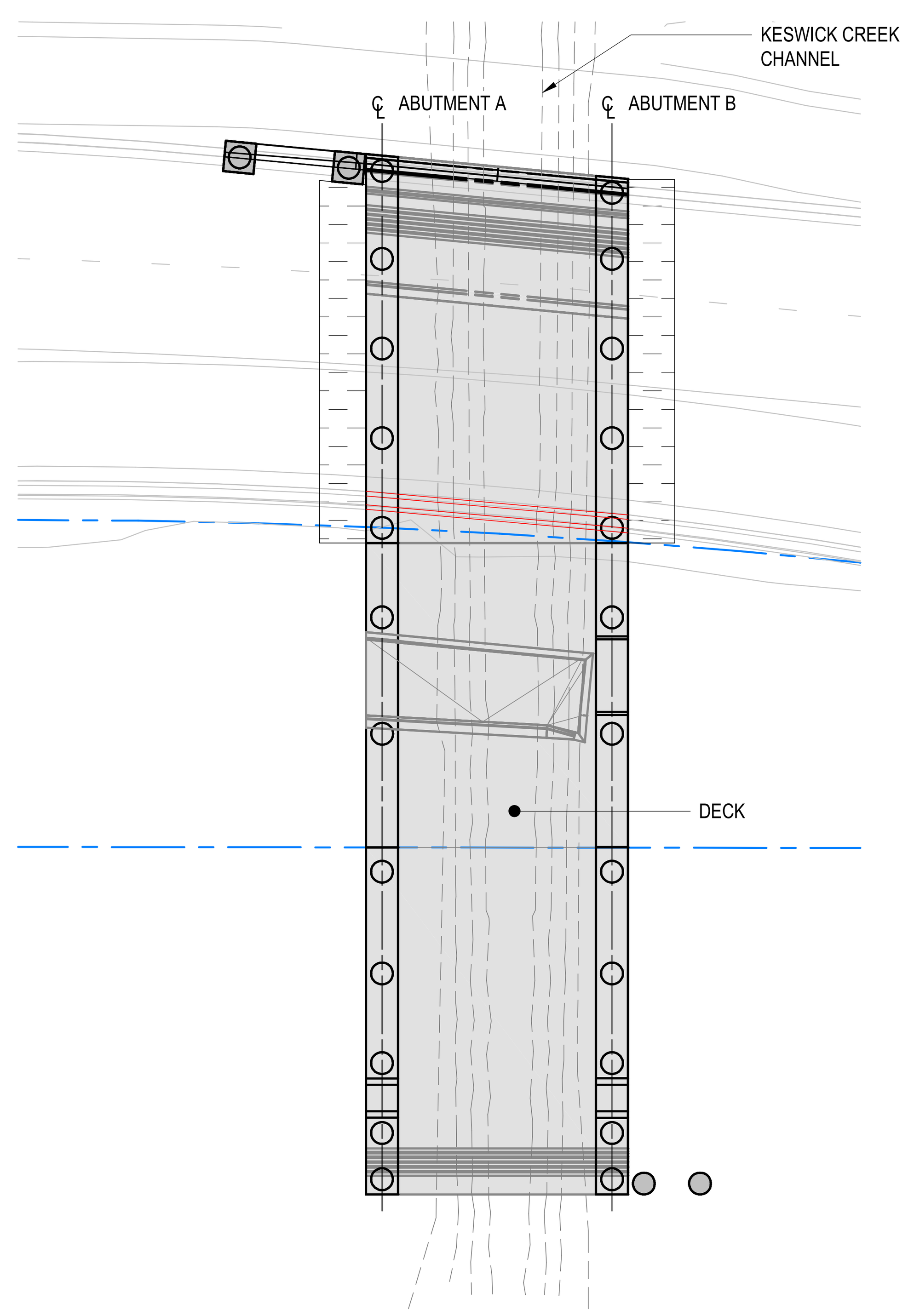
FILE NAME: Autodesk Docs://T2D - Torrens to Darlington /T2DPAA-T2D-C3S-BR-M3D-100302.rvt



CONSTRUCTION STAGE B3

SCALE 1 : 200

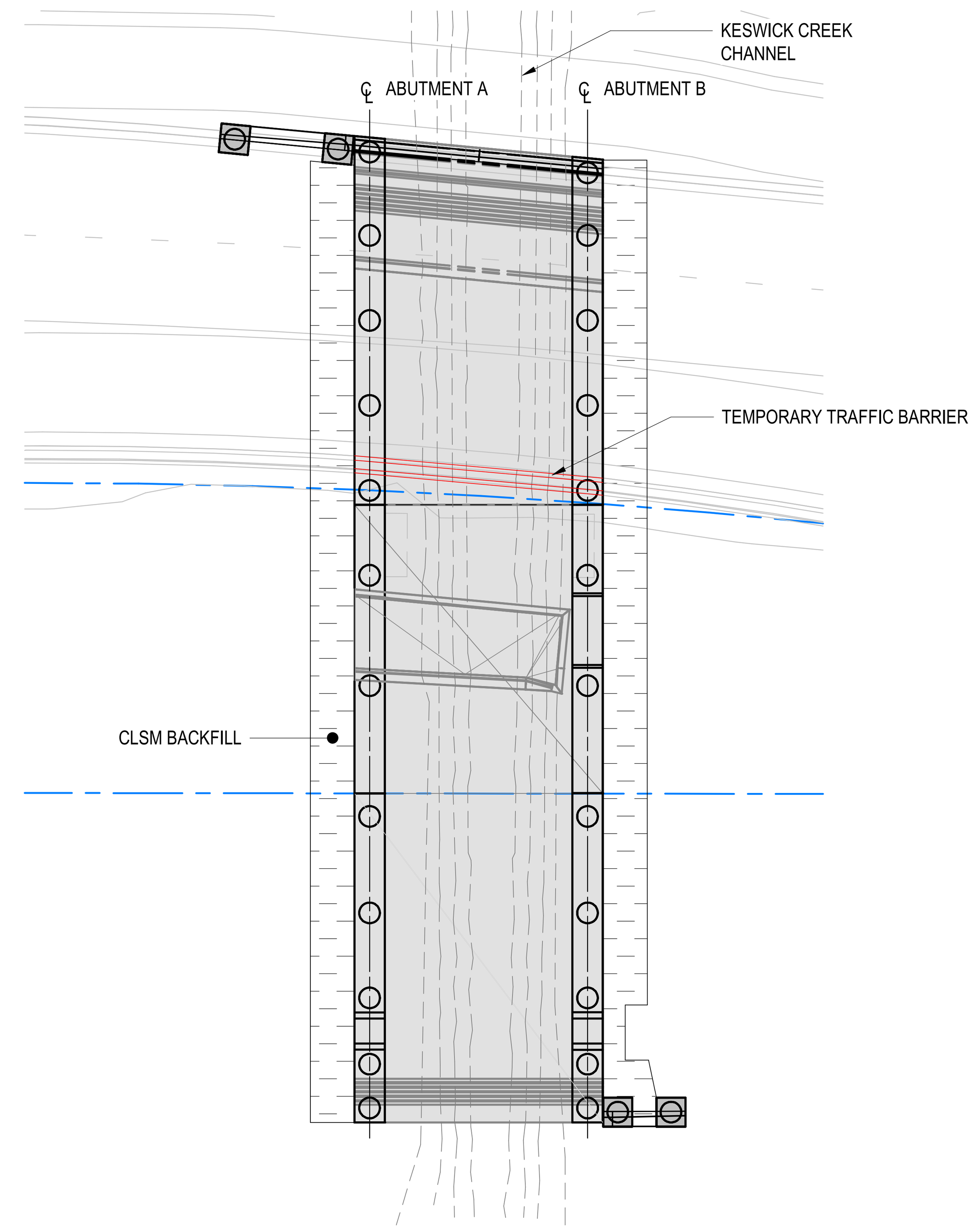
3.1 LOCALLY EXCAVATE SOIL AT BRIDGE ABUTMENTS.
3.2 BREAK BACK BORED PILES TO CUT-OFF LEVELS.
3.3 CAST ABUTMENT HEADSTOCKS. APPLY PROTECTION TO EXPOSED STITCH STARTER BARS.



CONSTRUCTION STAGE B4

SCALE 1 : 200

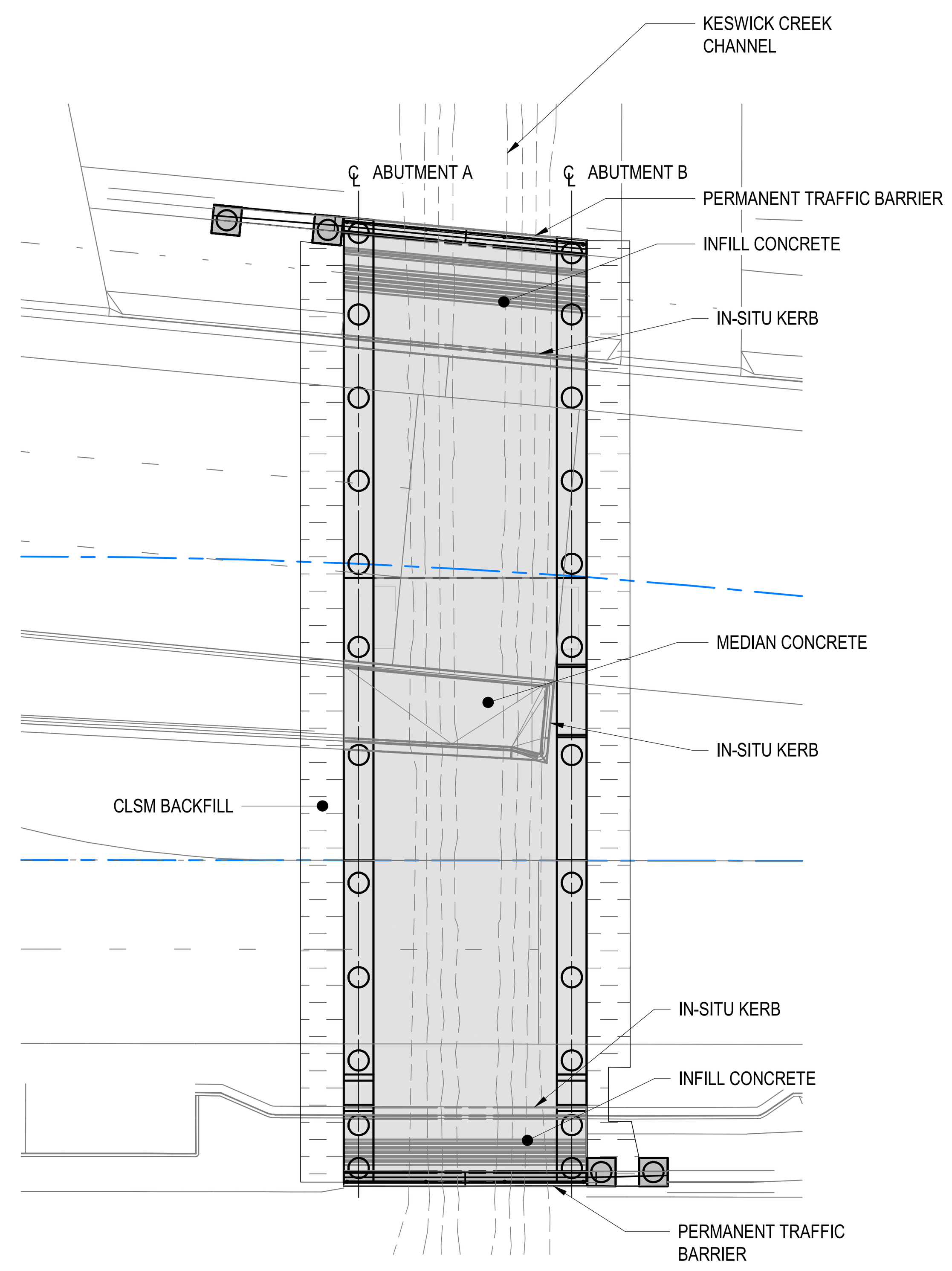
4.1 BACKFILL IN FRONT OF ABUTMENT AND REINSTATE KESWICK CREEK CHANNEL LINING.
4.2 INSTALL TEMPORARY VERTICAL AND LATERAL SUPPORTS FOR DECK SLAB PERMANENT FORMWORK (TO BE DETAILED IN THE RELEVANT TEMPORARY WORKS PACKAGE).
4.3 AFTER 14 DAYS OR ONCE HEADSTOCK HAS REACHED CONCRETE COMPRESSIVE STRENGTH OF MINIMUM 25MPa, PLACE PERMADEC PERMANENT FORMWORK AND REINFORCEMENT FOR RC DECK SLAB AND INTEGRAL ABUTMENT STITCH.
4.4 INSTALL CONDUITS FOR NEW SERVICES.
4.5 CAST RC DECK SLAB AND INTEGRAL ABUTMENT STITCH MONOLITHICALLY IN ONE POUR.
4.6 REMOVE TEMPORARY SUPPORT AFTER DECK CONCRETE 28 DAYS OR ONCE CONCRETE HAS REACHED COMPRESSIVE STRENGTH OF MINIMUM 50MPa.



CONSTRUCTION STAGE B5

SCALE 1 : 200

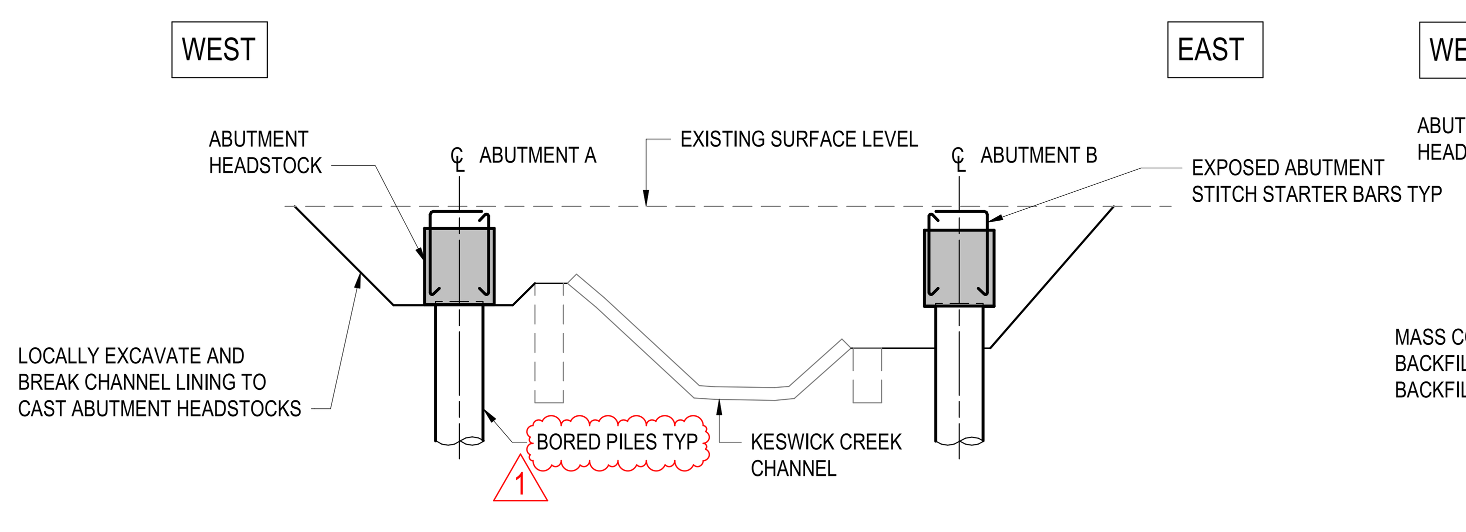
5.1 RELOCATE AND / OR DIVERT EXISTING UNDERBORED SERVICES AND INSTALL ITS AND LIGHTING CABLES IN THE BRIDGE DECK.
5.2 LOCALLY EXCAVATE FURTHER BEHIND EACH ABUTMENT AND INSTALL WATERPROOFING, COMPRESSIBLE POLYSTYRENE LAYER AND SUBSURFACE DRAINAGE AT BACK OF ABUTMENTS.
5.3 INSTALLATION OF PITS AND OTHER NEW UNDERBORED SERVICES.
5.4 INSTALL OFF-STRUCTURE BARRIER AND CONSTRUCT PILE CAP.
5.5 BACKFILL BEHIND EACH ABUTMENTS WITH CONTROLLED LOW-STRENGTH MATERIAL (CLSM) FILL.



CONSTRUCTION STAGE B6

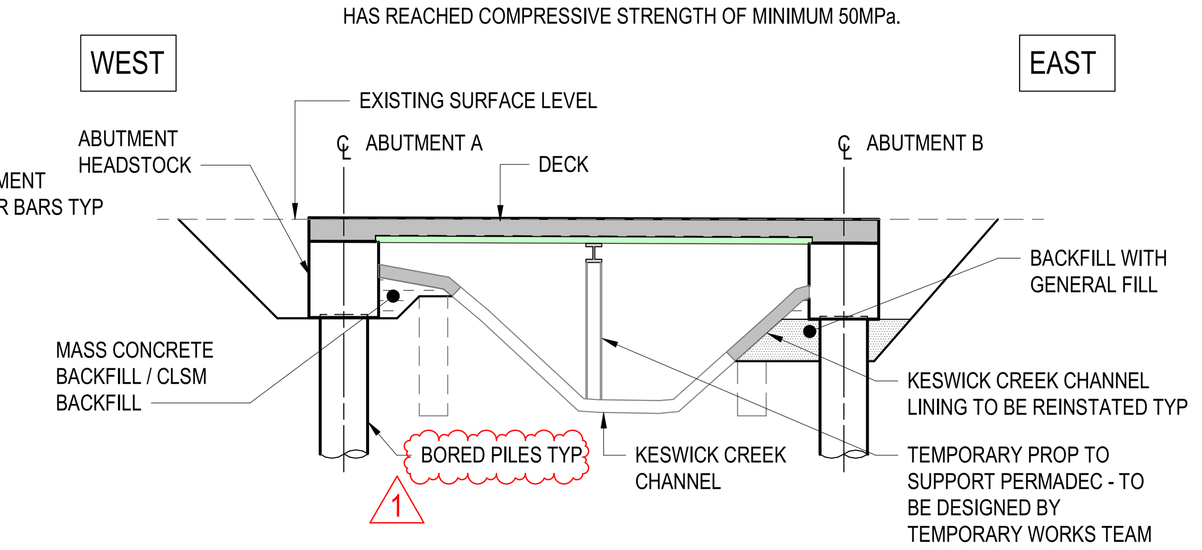
SCALE 1 : 200

6.1 REMOVE TEMPORARY PAVEMENT AND TEMPORARY BARRIER.
6.2 INSTALL ON-STRUCTURE TRAFFIC BARRIERS, FENCING, FOOTPATHS AND RAISED KERBS FOR THE ENTIRE STRUCTURE.
6.3 CONSTRUCT ON-STRUCTURE ROAD FORMATION AND ASPHALT WEARING SURFACE FOR THE ENTIRE STRUCTURE.
6.4 MARK TRAFFIC LANES AND OPEN THE BRIDGE FOR TRAFFIC.



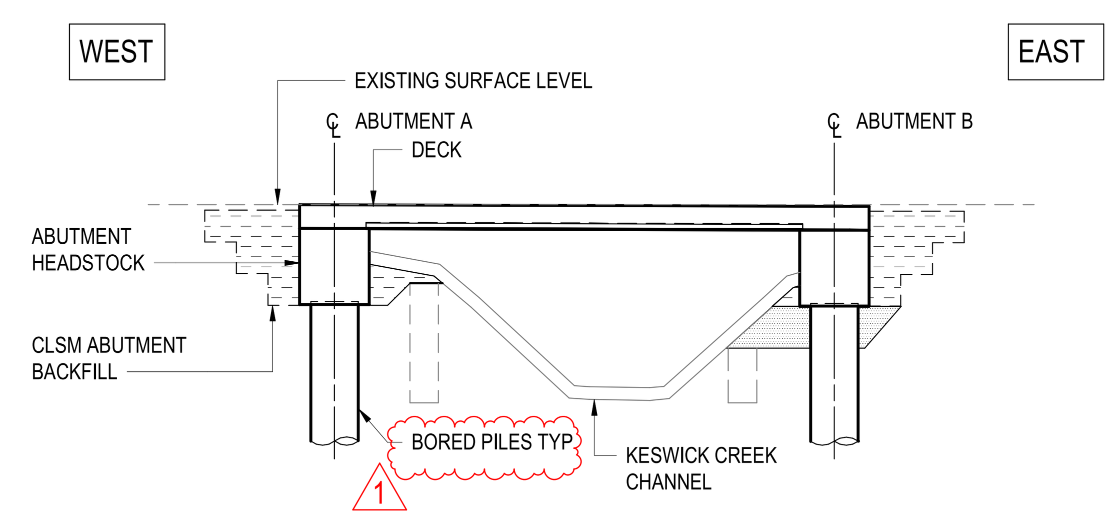
SECTION-STAGE B3

SCALE 1 : 100



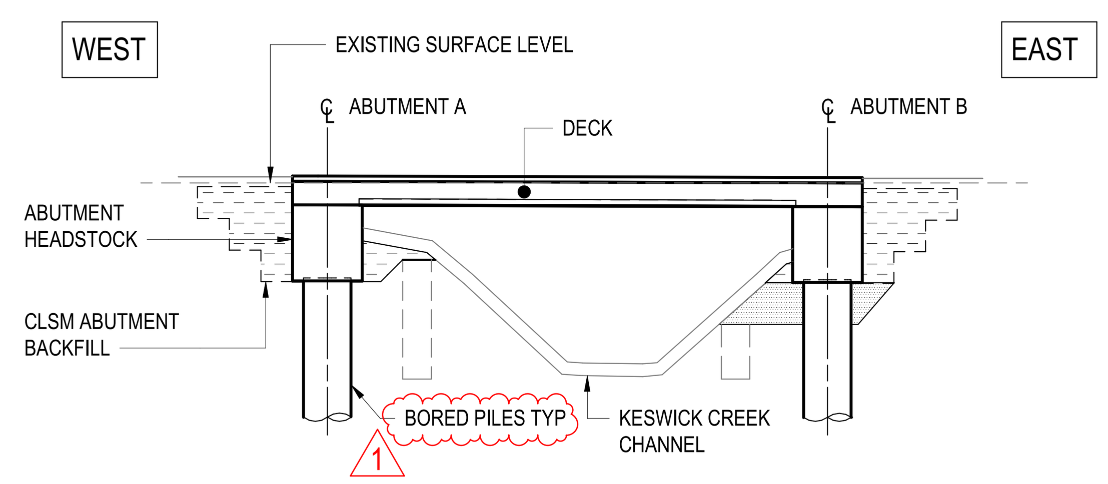
SECTION-STAGE B4

SCALE 1 : 100



SECTION-STAGE B5

SCALE 1 : 100



SECTION-STAGE B6

SCALE 1 : 100

NOTES

1. FOR GENERAL NOTES REFER TO SHEET No. 2871005 TO 2871008.

2000 0 1000 2000 3000 4000
1:100

PROJECT DOCUMENT REFERENCE
T2DPAA-T2D-C3S-BR-DRG-101033

FOR CONSTRUCTION

No.	AMENDMENT DESCRIPTION	BY	CHECK	ACCEPTANCE	DATE
1	ISSUED FOR CONSTRUCTION (KNet No. 24941076)	T2DA	T2DA	M. SHORT	17.07.26
0	ISSUED FOR CONSTRUCTION (KNet No. 24467656)	T2DA	T2DA	M. SHORT	28.03.26

INDEX SHEET REFERENCE: 8011 SHEET 2871002

T2D → TORRENS TO DARLINGTON

DISCLAIMER.
BACKGROUND MODEL INFORMATION SHOWN ON THIS DRAWING IS PROVIDED FOR REFERENCE ONLY. THE BACKGROUND DESIGN MAY BE SUBJECT TO ONGOING DESIGN DEVELOPMENT AND REFINEMENT.

DESIGNER: A.SINGHAL / S.PODDAR
QUALIFICATION: MEng BEng
DATE: 25.03.26

REVIEWER: R.WOZNIAK
QUALIFICATION: FIEAust CPEng NER RPEV RPEQ MICE CEng
DATE: 25.03.26

INDEPENDENT DESIGN CERTIFIER (IF REQUIRED): T. JACKSON
QUALIFICATION: BEng (Hons), MEngSc
DATE: 27.03.26

Government of South Australia
Department for Infrastructure and Transport

PROJECT No.: 31921901
FILE No.: 2020/10577
DESIGN No.: #15551401
SURVEY No.: #15551401
PROJECT START ROAD RUNNING DISTANCE: NOT APPLICABLE
PROJECT END ROAD RUNNING DISTANCE: NOT APPLICABLE
SCALES: 4000 0 2000 4000 6000 8000 1:200

ROAD No. 6200
RICHMOND ROAD
RAILWAY TERRACE - SOUTH ROAD; MILE END SOUTH
BRIDGE OVER KESWICK CREEK
CONSTRUCTION SEQUENCE - SHEET 03

DESIGNED:	DRAFTED:	ACCEPTED FOR USE:	ACCEPTANCE FORM KNET No.:	DRAWING No.:	SHEET No.:	AMEND No.:
T2DA	T2DA	M. SHORT TITLE: DIRECTOR ENG. DATE: 28.03.26	24467656 IN ACCORDANCE WITH DP013	8011	2871033	1

SHEET LATITUDE -34.94188 SHEET LONGITUDE 138.57397

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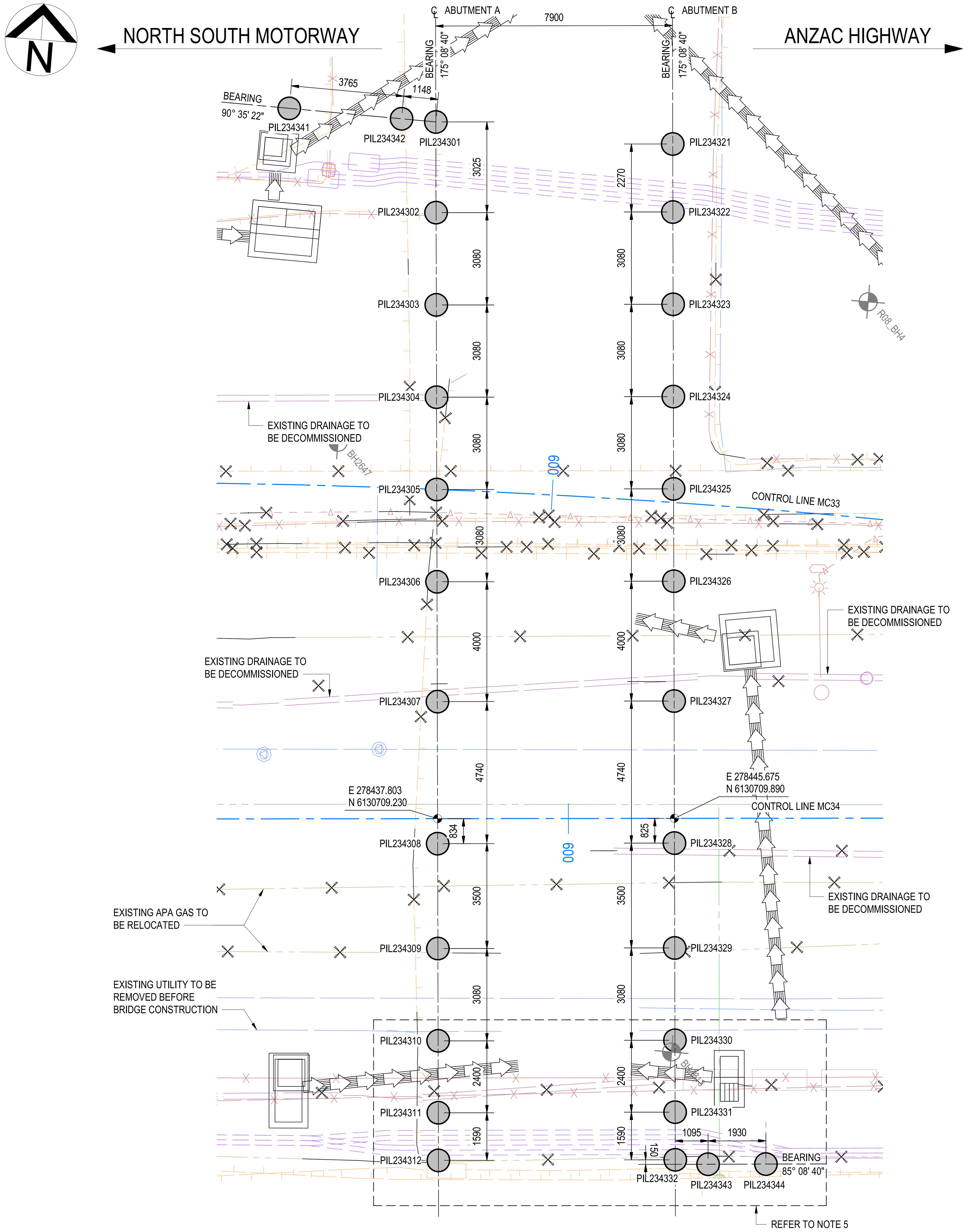
FILE NAME: Autodesk Docs://T2D - Torrens to Darlington /T2DPAA-T2D-C3S-BR-M3D-100302.rvt



PLAN

SCALE 1 : 100

ABUTMENT A PILE SCHEDULE

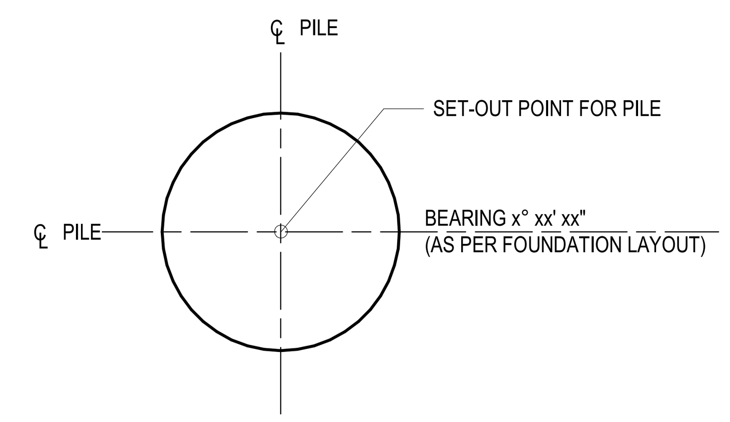
SITE ID	LOCATION	PILE CONSTRUCTION TYPE	PILE DIAMETER (mm)	PILE TOE LEVEL RL 'A' (m)	PILE CUT OFF LEVEL RL 'B' (m)	PILE LENGTH (mm)	EASTING (m)	NORTHING (m)	FOUNDING MATERIAL	PILE STRUCTURAL DESIGN LOAD		MINIMUM REQUIRED DESIGN ULTIMATE GEOTECHNICAL STRENGTH Rd, ug (kN)	PILE TESTING (CAPWAP OR EQUIVALENT FOR REPRESENTATIVE PILE)	TEST LOAD Pg (COMPRESSION) (kN)
										SLS N* (kN)	ULS Nu* (kN)			
PIL234301	ABUTMENT A	CFA	750	7.143	17.693	10550	278435.835	6130732.397	ALLUVIUM (CLAY)	1050	1486	2010	-	-
PIL234302	ABUTMENT A	CFA	750	7.173	17.723	10550	278436.090	6130729.383	ALLUVIUM (CLAY)	1050	1486	2010	-	-
PIL234303	ABUTMENT A	CFA	750	7.203	17.753	10550	278436.352	6130726.314	ALLUVIUM (CLAY)	1050	1486	2010	YES	2010
PIL234304	ABUTMENT A	CFA	750	7.234	17.784	10550	278436.612	6130723.245	ALLUVIUM (CLAY)	1050	1486	2010	-	-
PIL234305	ABUTMENT A	CFA	750	7.264	17.814	10550	278436.873	6130720.176	ALLUVIUM (CLAY)	1050	1486	2010	-	-
PIL234306	ABUTMENT A	BORED	750	6.345	17.845	11500	278437.134	6130717.107	ALLUVIUM (SAND / GRAVEL)	1050	1486	2700	-	-
PIL234307	ABUTMENT A	BORED	750	6.376	17.876	11500	278437.479	6130713.122	ALLUVIUM (SAND / GRAVEL)	1050	1486	2700	-	-
PIL234308	ABUTMENT A	BORED	750	6.411	17.911	11500	278437.873	6130708.399	ALLUVIUM (SAND / GRAVEL)	1050	1486	2700	-	-
PIL234309	ABUTMENT A	BORED	750	6.380	17.880	11500	278438.170	6130704.911	ALLUVIUM (SAND / GRAVEL)	1050	1486	2700	-	-
PIL234310	ABUTMENT A	BORED	750	7.300	17.850	10550	278438.430	6130701.843	ALLUVIUM (SAND / GRAVEL)	768	1165	2080	-	-
PIL234311	ABUTMENT A	BORED	750	7.278	17.828	10550	278438.634	6130699.451	ALLUVIUM (SAND / GRAVEL)	768	1165	2080	-	-
PIL234312	ABUTMENT A	BORED	750	7.255	17.805	10550	278438.768	6130697.867	ALLUVIUM (SAND / GRAVEL)	768	1165	2080	-	-

ABUTMENT B PILE SCHEDULE

SITE ID	LOCATION	PILE CONSTRUCTION TYPE	PILE DIAMETER (mm)	PILE TOE LEVEL RL 'A' (m)	PILE CUT OFF LEVEL RL 'B' (m)	PILE LENGTH (mm)	EASTING (m)	NORTHING (m)	FOUNDING MATERIAL	PILE STRUCTURAL DESIGN LOAD		MINIMUM REQUIRED DESIGN ULTIMATE GEOTECHNICAL STRENGTH Rd, ug (kN)	PILE TESTING (CAPWAP OR EQUIVALENT FOR REPRESENTATIVE PILE)	TEST LOAD Pg (COMPRESSION) (kN)
										SLS N* (kN)	ULS Nu* (kN)			
PIL234321	ABUTMENT B	CFA	750	7.175	17.725	10550	278443.770	6130732.314	ALLUVIUM (CLAY)	1050	1486	2010	-	-
PIL234322	ABUTMENT B	CFA	750	7.199	17.749	10550	278443.962	6130730.052	ALLUVIUM (CLAY)	1050	1486	2010	-	-
PIL234323	ABUTMENT B	CFA	750	7.230	17.780	10550	278444.223	6130726.983	ALLUVIUM (CLAY)	1050	1486	2010	-	-
PIL234324	ABUTMENT B	CFA	750	7.260	17.810	10550	278444.484	6130723.914	ALLUVIUM (CLAY)	1050	1486	2010	-	-
PIL234325	ABUTMENT B	CFA	750	7.291	17.841	10550	278444.745	6130720.845	ALLUVIUM (CLAY)	1050	1486	2010	-	-
PIL234326	ABUTMENT B	BORED	750	6.372	17.872	11500	278445.005	6130717.776	ALLUVIUM (SAND / GRAVEL)	1050	1486	2700	-	-
PIL234327	ABUTMENT B	BORED	750	6.403	17.903	11500	278445.344	6130713.791	ALLUVIUM (SAND / GRAVEL)	1050	1486	2700	YES	2700
PIL234328	ABUTMENT B	BORED	750	6.438	17.938	11500	278445.745	6130709.068	ALLUVIUM (SAND / GRAVEL)	1050	1486	2700	-	-
PIL234329	ABUTMENT B	BORED	750	6.407	17.907	11500	278446.041	6130705.580	ALLUVIUM (SAND / GRAVEL)	1050	1486	2700	-	-
PIL234330	ABUTMENT B	BORED	750	7.327	17.877	10550	278446.302	6130702.511	ALLUVIUM (SAND / GRAVEL)	768	1165	2080	-	-
PIL234331	ABUTMENT B	BORED	750	7.305	17.855	10550	278446.505	6130700.120	ALLUVIUM (SAND / GRAVEL)	768	1165	2080	-	-
PIL234332	ABUTMENT B	BORED	750	7.287	17.837	10550	278446.641	6130698.535	ALLUVIUM (SAND / GRAVEL)	768	1165	2080	-	-

OFF STRUCTURE BARRIER PILE SCHEDULE

SITE ID	LOCATION	PILE CONSTRUCTION TYPE	PILE DIAMETER (mm)	PILE TOE LEVEL RL 'A' (m)	PILE CUT OFF LEVEL RL 'B' (m)	PILE LENGTH (mm)	EASTING (m)	NORTHING (m)	FOUNDING MATERIAL	PILE STRUCTURAL DESIGN LOAD		MINIMUM REQUIRED DESIGN ULTIMATE GEOTECHNICAL STRENGTH Rd, ug (kN)
										SLS N* (kN)	ULS Nu* (kN)	
PIL234341	OFF STRUCTURE BARRIER	CFA	750	13.711	17.761	4050	278430.923	6130732.448	ALLUVIUM (CLAY)	85	253	342
PIL234342	OFF STRUCTURE BARRIER	CFA	750	13.727	17.777	4050	278434.687	6130732.409	ALLUVIUM (CLAY)	85	253	342
PIL234343	OFF STRUCTURE BARRIER	BORED	750	13.880	17.930	4050	278447.744	6130698.479	ALLUVIUM (CLAY)	85	253	452
PIL234344	OFF STRUCTURE BARRIER	BORED	750	13.881	17.931	4050	278449.667	6130698.642	ALLUVIUM (CLAY)	85	253	452



PILE SET-OUT DETAILS

NOTES

1. FOR GENERAL NOTES, REFER TO SHEET No. 2871005 TO 2871008.
2. FOR PILING NOTES, REFER TO SHEET No. 2871056.
3. FOR DRAWING LEGEND, REFER TO SHEET No. 2871021.
4. PILE DESIGN AND LENGTHS ARE BASED ON
   - DYNAMIC LOAD TESTING SHALL BE UNDERTAKEN FOR THE GREATER OF 10% OF THE PILES OR 1 PILE PER ABUTMENT GROUP.
   - ALL PILES SHALL BE INTEGRITY TESTED USING ONE OF THE TEST METHODS SPECIFIED IN AS 2159:2009. THE PROPOSED TESTING METHODOLOGY SHALL BE SUBMITTED TO THE DESIGN ENGINEER FOR APPROVAL.
5. PILES IN PROXIMITY OF 66kV OVERHEAD LINE SHALL BE CAST-IN-SITU, BORED USING LOW HEADROOM PILING RIG WITH TEMPORARY CASING.

PROJECT DOCUMENT REFERENCE
T2DPAA-T2D-C3S-BR-DRG-101051

FOR CONSTRUCTION

No.	AMENDMENT DESCRIPTION	BY	CHECK	ACCEPTANCE	DATE
1	ISSUED FOR CONSTRUCTION (KNet No. 24941076)	T2DA	T2DA	M. SHORT	17.07.26
0	ISSUED FOR CONSTRUCTION (KNet No. 24467656)	T2DA	T2DA	M. SHORT	28.03.26

INDEX SHEET REFERENCE: 8011 SHEET 2871002

T2D → TORRENS TO DARLINGTON

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DESIGNER: A.SINGHAL / S.PODDAR
QUALIFICATION: MEng BEng
DATE: 25.03.26

REVIEWER: R.WOZNIAK
QUALIFICATION: FIEAust CPEng NER RPEV RPEQ MICE CEng
DATE: 25.03.26

INDEPENDENT DESIGN CERTIFIER (IF REQUIRED): T. JACKSON
QUALIFICATION: BEng (Hons), MEngSc
DATE: 27.03.26

Government of South Australia
Department for Infrastructure and Transport

PROJECT No.: 31921901
FILE No.: 2020/10577
DESIGN No.: #15551401
SURVEY No.: #15551401
PROJECT START ROAD RUNNING DISTANCE: NOT APPLICABLE
PROJECT END ROAD RUNNING DISTANCE: NOT APPLICABLE
SCALES: 2000 0 1000 2000 3000 4000
1:100

ROAD No. 6200
RICHMOND ROAD
RAILWAY TERRACE - SOUTH ROAD; MILE END SOUTH
BRIDGE OVER KESWICK CREEK
FOUNDATION LAYOUT

DESIGNED: T2DA
DRAFTED: T2DA
ACCEPTED FOR USE: M. SHORT
TITLE: DIRECTOR ENG.
DATE: 28.03.26
ACCEPTANCE FORM KNET No.: 24467656
IN ACCORDANCE WITH DP013
DRAWING No.: 8011
SHEET No.: 2871051
AMEND No.: 1
SHEET LATITUDE -34.94188
SHEET LONGITUDE 138.57397

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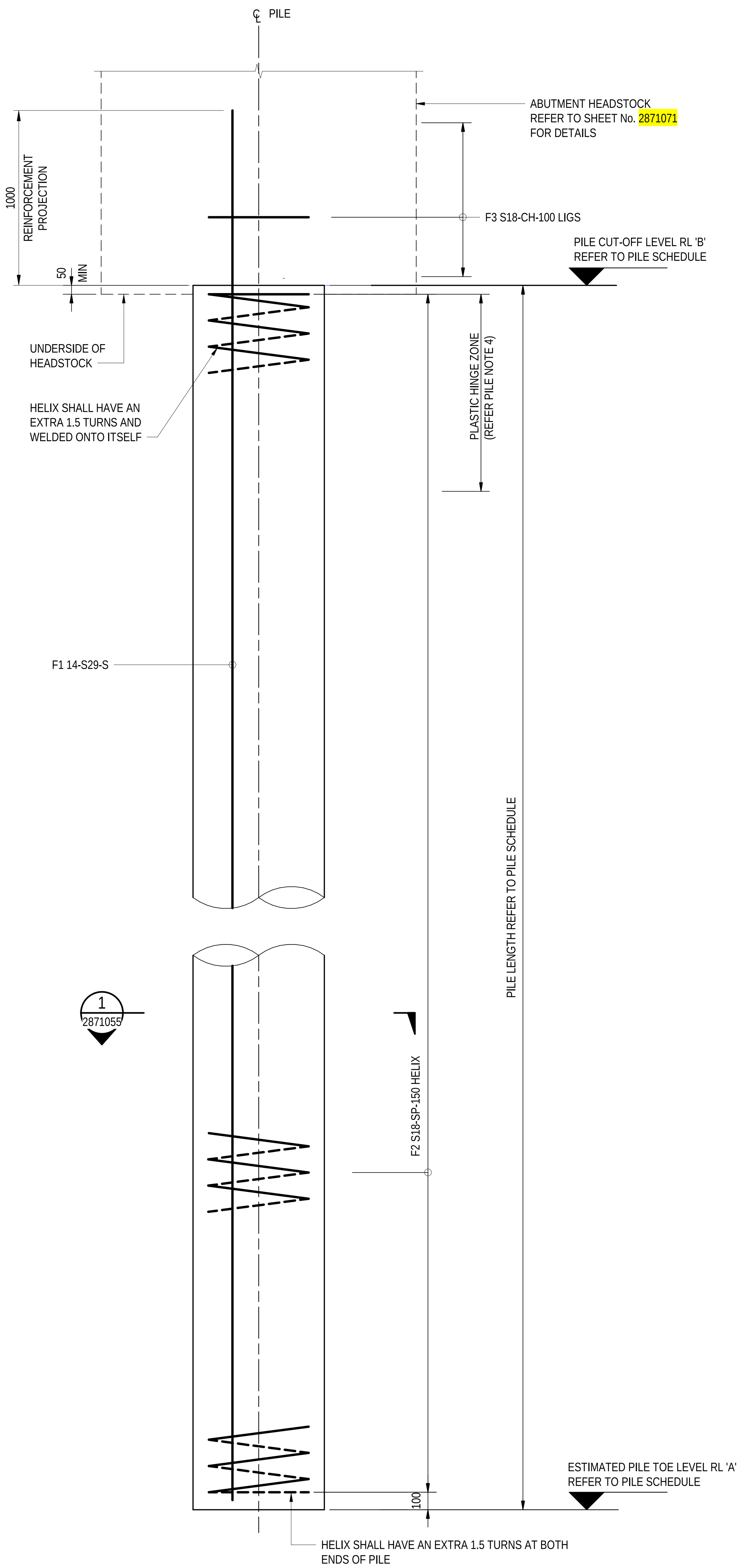
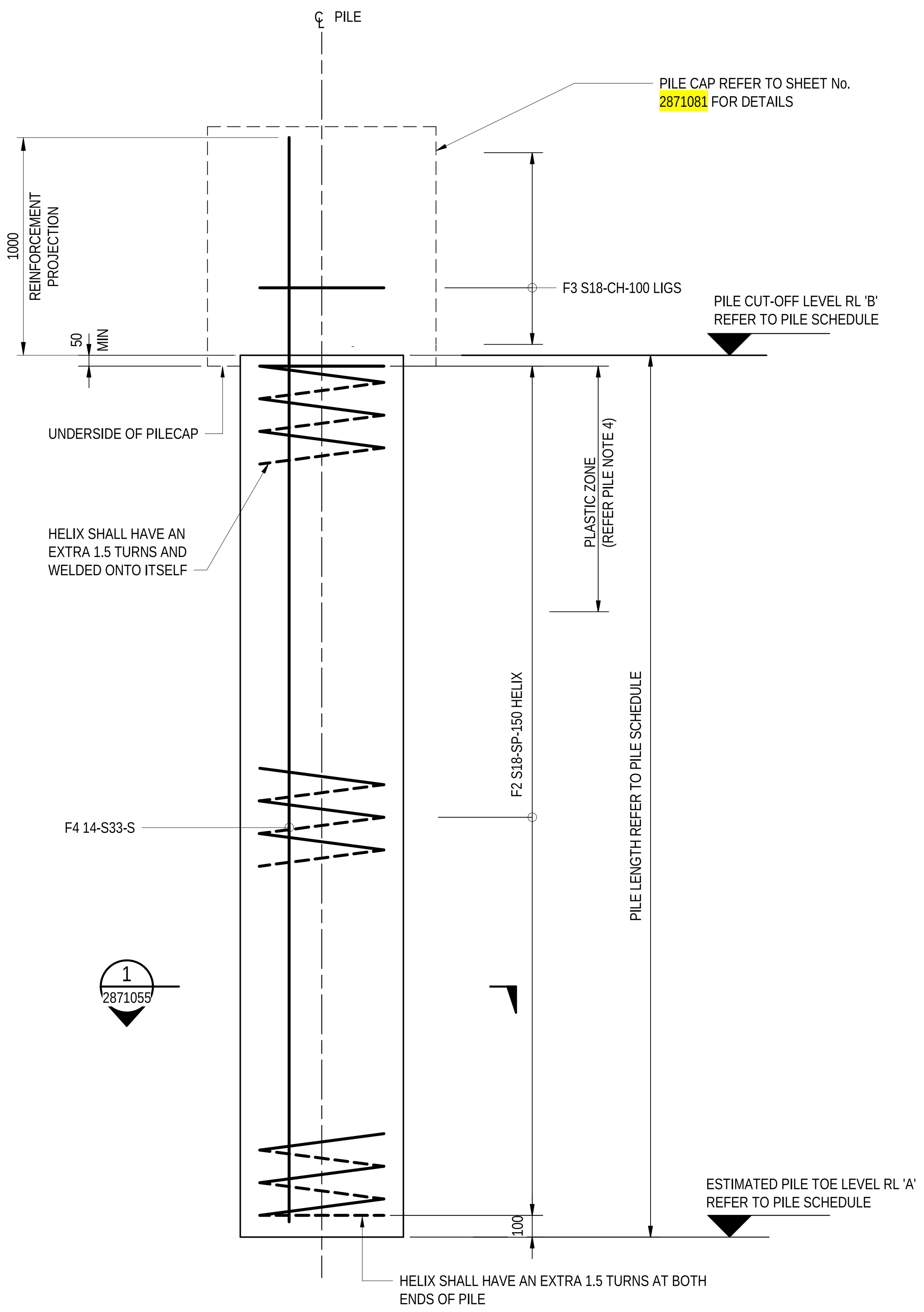


ELEVATION

SCALE 1 : 20

FOR PILES PIL234101 TO TO PIL234112 & PIL234121 TO PIL234132

(MILE END SOUTH BRIDGE PILES)

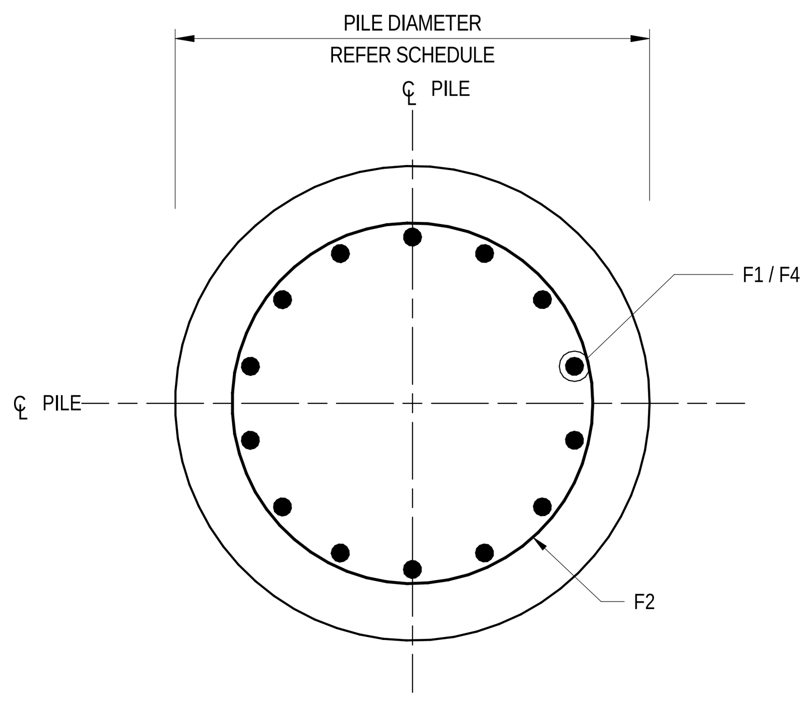


ELEVATION

SCALE 1 : 20

FOR PILES PIL234141 TO PIL234144

(OFF STRUCTURE BARRIER PILES)



SECTION 1

SCALE 1 : 10

2871055

NOTES

1. FOR GENERAL NOTES, REFER TO SHEET No. 2871005 TO 2871008.
2. FOR PILE SCHEDULE, DESIGN LOADS AND PILE TESTING REFER TO SHEET No. 2871051.
3. LENGTH OF PLASTIC HINGE ZONE EQUALS TO A MINIMUM OF PILE DIAMETER, D=750mm.
4. SPLICING OF LONGITUDINAL PILE REINFORCEMENT MUST BE OUTSIDE PLASTIC HINGE ZONES.
5. SPLICING OF LONGITUDINAL PILE REINFORCEMENT IN ADJACENT PILES MUST BE STAGGERED.

200 0 100 200 300 400
1:10

PROJECT DOCUMENT REFERENCE
T2DPAA-T2D-C3S-BR-DRG-101055

FOR CONSTRUCTION

No.	AMENDMENT DESCRIPTION	BY	CHECK	ACCEPTANCE	DATE
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INDEX SHEET REFERENCE: 8011 SHEET 2871002

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SCALES: 400 0 200 400 600 800 1:20

ROAD No. 6200
RICHMOND ROAD
RAILWAY TERRACE - SOUTH ROAD; MILE END SOUTH
BRIDGE OVER KESWICK CREEK
PILE DETAILS - SHEET 01

DESIGNED:	DRAFTED:	ACCEPTED FOR USE:	ACCEPTANCE FORM KNET No.:	DRAWING No.:	SHEET No.:	AMEND No.:
T2DA	T2DA	M. SHORT TITLE: DIRECTOR OF ENG. DATE: 28.03.26	24467656 IN ACCORDANCE WITH DP013	8011 SHEET LATITUDE -34.94188	2871055 SHEET LONGITUDE 138.57397	0

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PILE INSTALLATION METHOD

PI1. PRIOR TO THE INSTALLATION OF ANY PILES, THE PILING CONTRACTOR SHALL SUPPLY TO THE ALLIANCE FOR APPROVAL, DETAILS OF THE INSTALLATION PLANS. THIS SHALL INCLUDE AS A MINIMUM, BUT NOT LIMITED TO, THE FOLLOWING ITEMS:
- PROPOSED DRILLING METHODOLOGY
- DRILLING RIG

PI2. A PILE CONSTRUCTION METHOD STATEMENT SHALL BE SUBMITTED FOR APPROVAL PRIOR TO ANY PILE CONSTRUCTION. THE METHOD STATEMENT SHALL INCLUDE AS A MINIMUM, BUT NOT BE LIMITED TO THE ITEMS OUTLINED IN THE SPECIFICATIONS AND AS 2159.

PI3. WHEN REQUIREMENTS DIFFER BETWEEN SPECIFICATIONS, CONTRACT DOCUMENTS SHALL TAKE PRECEDENCE.

PI4. PLACING OF CONCRETE SHALL BE IN ACCORDANCE WITH THE NOTES ON THIS DRAWING, THE SPECIFICATIONS AND AS 2159 – SECTION 7 – MATERIALS AND CONTRUCTION REQUIREMENTS.

PI5. THE REINFORCEMENT CAGE SHALL BE FREE FROM OIL, EXCESSIVE RUST AND DEBRIS AND SHALL BE STRAIGHT. THE CAGE SHALL NOT BE PLACED UNTIL ACCEPTED BY THE CONSTRUCTION VERIFIER.

PI6. ALL HOOKS, COGS, LAPS AND BENDS AND CLOSED TIES TO BE IN ACCORDANCE WITH AS5100.5.

PI7. BUNDLED BARS SHALL BE TIED TOGETHER IN CONTACT.

PI8. EACH PILE LOCATION, AND LEVEL SHALL BE SURVEYED AS SOON AS PRACTICAL AFTER PILE CONSTRUCTION.

PI9. PILES SHALL BE CONSTRUCTED WITH A TOLERANCE IN PLAN POSITION OF +/- 75mm AT THE CUT-OFF LEVEL AND A MAXIMUM ALLOWABLE VERTICALITY OF 1 IN 50.

PI10. ALL PILES TO BE INTEGRITY TESTED. INTEGRITY TESTING SHALL BE CARRIED OUT ON THE PILES IN ACCORDANCE WITH INTEGRITY TEST METHODS SPECIFIED IN AS 2159. INTEGRITY TESTING EQUIPMENT MUST BE CAPABLE OF CHECKING CROSS-SECTIONAL IRREGULARITIES IN PILES AND IDENTIFYING THE LOCATION AND CHARACTERISTICS OF ANY SIGNIFICANT ANOMALIES SUCH AS VOIDS OR CONTAMINANTS. ACCEPTANCE CRITERIA, SUPERVISION AND REPORTING OF INTEGRITY TESTING SHALL BE IN ACCORDANCE WITH THE REQUIREMENTS OF AS 2159.

PI11. THE PILE SCHEDULE SHOWN ON SHEET No. 2871051 HAS BEEN DEVELOPED BASED ON THE ANTICIPATED GROUND CONDITIONS AND GEOTECHNICAL DESIGN PARAMETERS PRESENTED IN THE GEOTECHNICAL TECHNICAL NOTE. THE PILE TOE LEVEL CORRESPONDS TO THE LENGTH OF PILE WHERE THE DESIGN GEOTECHNICAL STRENGTH (Rd,g) IS GREATER THAN THE ULS STRUCTURAL DESIGN LOAD (N*u).

PI12. THE DESIGN GEOTECHNICAL STRENGTH (Rd,g) SHALL BE CALCULATED BY MULTIPLYING THE DESIGN ULTIMATE GEOTECHNICAL STRENGTH (Rd,ug) BY THE GEOTECHNICAL STRENGTH REDUCTION FACTOR (Φg) IN ACCORDANCE WITH AS 2159.

PI13. PILES SHALL BE INSTALLED TO THE MINIMUM PILE LENGTHS SPECIFIED ON THE PILE SCHEDULE ON SHEET 2871051 OR DEEPER. THE PILE TOE SHALL BE FOUNDED INTO THE ALLUVIUM CLAY OR ALLUVIUM SAND / GRAVEL. THE SITE GEOTECHNICAL REPRESENTATIVE SHALL INSPECT AND RECORD THE MATERIAL ENCOUNTERED AT THE TOE OF THE AUGUR.

PI14. ALL EXISTING SERVICES LOCATIONS SHALL BE VERIFIED ON SITE BY THE CONTRACTOR AND PROTECTED BEFORE COMMENCING WORK.

PI15. ALL LOOSE AND CONTAMINATED CONCRETE AT THE TOP OF THE PILE SHALL BE BROKEN BACK TO SOUND CONCRETE.

PI16. THE CONTRACTOR SHALL BE RESPONSIBLE FOR MAINTAINING ADJACENT STRUCTURES AND SERVICES IN STABLE CONDITION AND ENSURING NO PART IS OVERSTRESSED DURING CONSTRUCTION.

PI17. THE CONTRACTOR SHALL MAINTAIN ADEQUATE RECORDS OF ALL WORKS INCLUDING DRILLING AND PILE INSTALLATION IN ACCORDANCE WITH REQUIREMENTS OF AS 2159.

PI18. FOR THE ESTIMATED GROUNDWATER LEVEL, REFER TO THE GEOTECHNICAL TECHNICAL NOTE T2DPAA-T2D-C3S-GE-TNT-100201.

PI19. PILE SET-OUT COORDINATES HAVE BEEN PROVIDED AT THE REQUEST OF THE CONTRACTOR TO ASSIST IN SETTING-OUT. THE SET-OUT COORDINATES SHALL BE CONFIRMED AND VERIFIED BY THE CONTRACTOR. ANY DISCREPANCIES SHALL BE REPORTED TO THE DESIGN ENGINEER.

CFA PILES

CF1. PILES SHALL BE CONSTRUCTED AS CONTINUOUS FLIGHT AUGER (CFA) PILES OR BORED PILES AS SPECIFIED ON THE DRAWINGS. 1

CF2. PILE DESIGN AND CONSTRUCTION SHALL BE IN ACCORDANCE WITH AS 2159, AS5100.3 AND THE SPECIFICATIONS.

CF3. HIGH STRAIN DYNAMIC LOAD TESTING TO BE CARRIED OUT ON THE NOMINATED PILE IN ACCORDANCE WITH AS 2159-2009 AND THE SPECIFICATIONS. 1

CF4. A GEOTECHNICAL STRENGTH REDUCTION FACTOR OF 0.74 FOR AXIAL COMPRESSION, AND 0.64 FOR LATERAL PILE CAPACITY HAS BEEN DETERMINED IN ACCORDANCE WITH AS 2159.

CF5. DRILLING AND PILE INSTALLATION TO BE ASSESSED BY THE SITE GEOTECHNICAL REPRESENTATIVE TO VERIFY THAT GROUND CONDITIONS ENCOUNTERED ARE CONSISTENT WITH DESIGN ASSUMPTIONS. ADDITIONAL PILE TESTING MAY BE SPECIFIED OR REQUESTED ON THE BASIS OF REVIEW OF PILE CONSTRUCTION INSTALLATION RECORDS, PILE INTEGRITY TESTING AND DYNAMIC LOAD TESTING RESULTS.

CF6. DURING CONSTRUCTION OF ALL CFA PILES, THE DRILLING PARAMETERS, DEPTH, AUGER EXTRACTION RATE. CONCRETE / GROUT PRESSURE AND CONCRETE / GROUT VOLUME ARE TO BE MONITORED AND RECORDED IN REAL TIME. THE MONITORING / REAL TIME DATA WILL BE INCLUDED IN THE RELATED WORK LOT MANAGED UNDER PROJECT PACK WEB (PPW) SYSTEM TO WHICH ACCESS WILL BE GIVEN TO THE CONSTRUCTION VERIFIER AND DIT PRINCIPAL'S REPRESENTATIVE. UNEXPECTED GROUND CONDITIONS, IF FOUND, SHALL BE REFERRED TO THE DESIGNER THROUGH THE PROJECT RFI SYSTEM.

CF7. PILES REQUIRED TO PROJECT 1000mm MIN ABOVE CUT-OFF LEVEL FOR PDA TESTING. AFTER TESTING, BREAK DOWN PILE TO EXPOSE REINFORCEMENT.

CF8. TEST LOADS TO BE IN ACCORDANCE WITH TABLE 8.3.3.2 AND 8.3.3.3 OF AS 2159. ACCEPTANCE CRITERIA ARE TO BE IN ACCORDANCE WITH TABLE 8.4.3.1 AND 8.5.2 OF AS 2159. REFER TO PILE SCHEDULE FOR TEST LOADS. THE PILING CONTRACTOR SHALL EXCLUDE ANY SOIL RESISTANCE FROM THE PILE TEST LEVEL TO THE TO TOP 1 METRE LENGTH OF THE PILE BELOW PILE CUT-OFF LEVEL.

CF9. PILE LOAD TEST RESULTS SHALL BE USED TO ASSESS THE PILE TOE LEVEL ADEQUACY OF TESTED PILES AND REMAINING NON-TESTED PILES IN EACH PILE GROUP. PILE ARRANGEMENT AS SHOWN ON THE DRAWINGS MAY BE ADJUSTED ON THE BASIS OF PILE TESTING RESULTS. PILING CONTRACTOR SHALL DEMONSTRATE AND ADOPT A CONSTRUCTION METHODOLOGY THAT WILL ENSURE THAT THE PILE CAPACITIES ON THE DRAWINGS HAVE BEEN ACHIEVED.

CF10. RESULTS OF ALL PILE TESTING SHALL BE SUBMITTED TO THE SITE GEOTECHNICAL REPRESENTATIVE FOR REVIEW.

CF11. THE MINIMUM PERIOD BETWEEN INSTALLATION OF THE PILE AND COMMENCEMENT OF THE PILE TEST SHALL BE 7 DAYS, OR UNTIL A MINIMUM 35MPa COMPRESSIVE STRENGTH IS ACHIEVED, WHICHEVER IS GREATER.

CF12. WHEN UNDERTAKING LOAD TESTING, THE TEST HAMMER SHALL BE LARGE ENOUGH TO FULLY MOBILISE PILE CAPACITY BUT NOT OVERSTRESS THE PILE IN ACCORDANCE WITH AS 2159. THE PILE REINFORCEMENT IN THE PILES NOMINATED FOR TESTING SHALL BE EXTENDED TO THE PILE TOE TO REDUCE THE RISK OF PILE CRACKING DURING TESTING.

CAST-IN-PLACE PILES

CI1. BORED PILES IDENTIFIED IN SCHEDULE ON SHEET NO. 2871051 SHALL BE CONSTRUCTED AS CAST-IN-PLACE CONCRETE PILES.

CI2. PILE DESIGN AND CONSTRUCTION SHALL BE IN ACCORDANCE WITH AS 2159, AS5100.3 AND THE SPECIFICATIONS.

CI3. A GEOTECHNICAL STRENGTH REDUCTION FACTOR OF 0.70 FOR AXIAL COMPRESSION, AND 0.64 FOR LATERAL AND TORSIONAL PILE CAPACITY HAS BEEN DETERMINED IN ACCORDANCE WITH AS 2159 FOR THE BELOW PILES. AN ADDITIONAL REDUCTION FACTOR OF 0.8 HAS BEEN APPLIED IN CALCULATION OF THE AXIAL CAPACITY TO ACCOUNT FOR THE EFFECT OF SUPPORT FLUIDS. THE PILE DESIGN ASSUMES THAT NO LOAD TESTING IS CARRIED OUT ON THE BELOW CAST-IN-PLACE PILES. PIL234310, PIL234311, PIL234312, PIL234330, PIL234331, PIL234332.

CI4. A GEOTECHNICAL STRENGTH REDUCTION FACTOR OF 0.74 FOR AXIAL COMPRESSION, AND 0.64 FOR LATERAL AND TORSIONAL PILE CAPACITY HAS BEEN DETERMINED IN ACCORDANCE WITH AS 2159 FOR THE BELOW PILES. AN ADDITIONAL REDUCTION FACTOR OF 0.8 HAS BEEN APPLIED IN CALCULATION OF THE AXIAL CAPACITY TO ACCOUNT FOR THE EFFECT OF SUPPORT FLUIDS. HIGH STRAIN DYNAMIC LOAD TESTING TO BE CARRIED OUT ON THE NOMINATED PILE IN ACCORDANCE WITH AS 2159-2009 AND THE SPECIFICATION.PIL234306, PIL234307, PIL234308, PIL234309, PIL234326, PIL234327, PIL234328, PIL234329. 1

CI5. DRILLING AND PILE INSTALLATION TO BE ASSESSED BY THE SITE GEOTECHNICAL REPRESENTATIVE TO VERIFY THAT GROUND CONDITIONS ENCOUNTERED ARE CONSISTENT WITH DESIGN ASSUMPTIONS. ANY UNUSUAL OR UNANTICIPATED CONDITIONS OBSERVED DURING DRILLING INCLUDING BUT NOT LIMITED TO: THE PRESENCE OF BOULDERS OR OTHER OBSTRUCTIONS; EVIDENCE OF CAVING, BOTTOM HEAVE OR OTHER SIGNS OF HOLE INSTABILITY; SOIL OR ROCK CONDITIONS SIGNIFICANTLY DIFFERENT THAN THOSE INDICATED IN NEARBY INVESTIGATION BORINGS; SLOW OR DIFFICULT DRILLING PROGRESS; LOSS OF DRILLING FLUID; SURFACE SETTLEMENT OR OTHER SIGNS OF GROUND LOSS AROUND THE SHAFT EXCAVATION; OR ANY OTHER CONDITIONS THAT MAY POTENTIALLY IMPACT SHAFT FRICTION OR END BEARING RESISTANCE OF THE BORED PILE, SHALL BE DOCUMENTED AS NOTED BY THE SITE GEOTECHNICAL REPRESENTATIVE AND THE DESIGNER NOTIFIED.

CI6. THE CONTRACTOR SHALL PROVIDE A FIELD LOG OF THE MATERIAL ENCOUNTERED OVER THE FULL DEPTH OF THE PILE EXCAVATION. THE MATERIAL RETRIEVED AT 1m INTERVALS DURING THE COURSE OF DRILLING SHALL BE RECORDED AND INSPECTED BY THE SITE GEOTECHNICAL REPRESENTATIVE.

CI7. PILE CONSTRUCTION RECORDS ARE TO BE ASSESSED BY THE SITE GEOTECHNICAL REPRESENTATIVE TO CONFIRM THAT GROUND CONDITIONS ENCOUNTERED ARE CONSISTENT WITH DESIGN ASSUMPTIONS.

CI8. THE CONTRACTOR SHALL SUBMIT CONSTRUCTION AND INSPECTION PLANS DETAILING FINAL BASE CLEANING OPERATIONS, INCLUDING THE TYPE(S) OF EQUIPMENT USED AND METHOD (E.G. USE OF CLEANING BUCKET, WEIGHTED TAPE, SHAFT INSPECTION DEVICE (SID), AND/OR CHECKING OF SAND CONTENT OF FLUID JUST PRIOR TO CASTING OF CONCRETE), AND THE ACCEPTANCE CRITERIA FOR EACH METHOD PROPOSED TO VERIFY PILE EXCAVATION BASE CONDITION FOR APPROVAL PRIOR TO PILE CONSTRUCTION.

CI9. THE PILING CONTRACTOR SHALL CONSIDER THE IMPACT OF PILING THROUGH CLAYEY SAND/GRAVEL LAYERS TO ENSURE MINIMUM HOLE COLLAPSE DURING THE TEMPORARY WORKS. ALL BORED PILING WORKS SHALL BE CARRIED OUT IN ACCORDANCE WITH THE SPECIFICATIONS.

GEOTECHNICAL UNCERTAINTY NOTES

1. THE SUBSURFACE PROFILES THAT ARE SHOWN ON THE DESIGN DRAWINGS ARE INDICATIVE INTERPOLATIONS BETWEEN BOREHOLES. THERE IS NO WARRANTY OR CERTAINTY ON THEIR CORRECTNESS AT ANY PARTICULAR POINT.
2. CONSTRUCTION WORK IS TO BE CARRIED OUT WITH CAREFUL RECOGNITION OF THE UNCERTAINTIES THAT EXIST OVER THE GROUND CONDITIONS.
3. DESIGNS HAVE BEEN PREPARED BASED ON THE AVAILABLE GEOTECHNICAL INFORMATION AND REASONABLE ASSUMPTIONS OVER GROUND CONDITIONS.
4. THE PRINCIPAL UNCERTAINTIES IN THE SUBSURFACE CONDITIONS ARE THE VARIOUS THICKNESS AND EXTENT OF THE EXISTING FILL, AND ALLUVIAL GRANULAR DEPOSITS, AND THE GROUNDWATER. GEOTECHNICAL UNCERTAINTY SHALL BE ADDRESSED BY REGULAR ASSESSMENT OF THE CONDITIONS AS THEY ARE REVEALED BY CONSTRUCTION WORKS. THIS INFORMATION WILL BE ASSESSED AS IT BECOMES AVAILABLE BY SITE GEOTECHNICAL REPRESENTATIVE.
5. NO PROVISION FOR FUTURE UTILITY EXCAVATION HAS BEEN CONSIDERED FROM THE DESIGN SURFACE LEVEL WITHIN THE RICHMOND ROAD IN THE VICINITY OF THE BRIDGE.

PROJECT DOCUMENT REFERENCE
T2DPAA-T2D-C3S-BR-DRG-101056

FOR CONSTRUCTION

No.	AMENDMENT DESCRIPTION	BY	CHECK	ACCEPTANCE	DATE
1	ISSUED FOR CONSTRUCTION (KNet No. 24941076)	T2DA	T2DA	M. SHORT	17.07.26
0	ISSUED FOR CONSTRUCTION (KNet No. 24467656)	T2DA	T2DA	M. SHORT	28.03.26

INDEX SHEET REFERENCE: 8011 SHEET 2871002

T2D TORRENS TO DARLINGTON

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QUALIFICATION: MEng BEng
DATE: 25.03.26

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QUALIFICATION: FIEAust CPEng NER RPEV RPEQ MICE CEng
DATE: 25.03.26

INDEPENDENT DESIGN CERTIFIER (IF REQUIRED): T. JACKSON
QUALIFICATION: BEng (Hons), MEngSc
DATE: 27.03.26

Government of South Australia
Department for Infrastructure and Transport

PROJECT No.: 31921901	FILE No.: 2020/10577
DESIGN No.: #15551401	SURVEY No.: #15551401
PROJECT START ROAD RUNNING DISTANCE: NOT APPLICABLE	
PROJECT END ROAD RUNNING DISTANCE: NOT APPLICABLE	
SCALES: NOT TO SCALE	

ROAD No. 6200
RICHMOND ROAD
RAILWAY TERRACE - SOUTH ROAD; MILE END SOUTH
BRIDGE OVER KESWICK CREEK
PILE DETAILS - SHEET 02

DESIGNED:	DRAFTED:	ACCEPTED FOR USE:	ACCEPTANCE FORM KNET No.:	DRAWING No.:	SHEET No.:	AMEND No.:
T2DA	T2DA	M. SHORT TITLE: DIRECTOR ENG. DATE: 28.03.26	24467656 IN ACCORDANCE WITH DP013	8011 SHEET LATITUDE -34.94188	2871056 SHEET LONGITUDE 138.57397	1

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ABUTMENT A - PLAN

SCALE 1 : 50

ABUTMENT A - ELEVATION

SCALE 1 : 50

NOTES

1. FOR GENERAL NOTES REFER TO SHEET No. 2871005 TO 2871008.

PROJECT DOCUMENT REFERENCE
T2DPAA-T2D-C3S-BR-DRG-101071

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SCALES: 1000 0 500 1000 1500 2000 — 1:50

ROAD No. 6200
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RAILWAY TERRACE - SOUTH ROAD; MILE END SOUTH
BRIDGE OVER KESWICK CREEK
ABUTMENT CONCRETE - SHEET 01

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T2DA	T2DA	M. SHORT TITLE: DIRECTOR OF ENG. DATE: 28.03.26	24467656 IN ACCORDANCE WITH DP013	8011 SHEET LATITUDE -34.94188	2871071 SHEET LONGITUDE 138.57397	0

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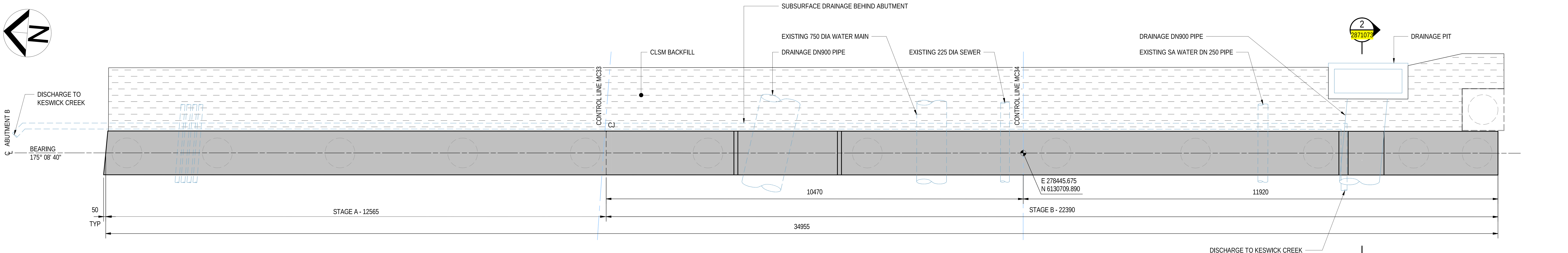
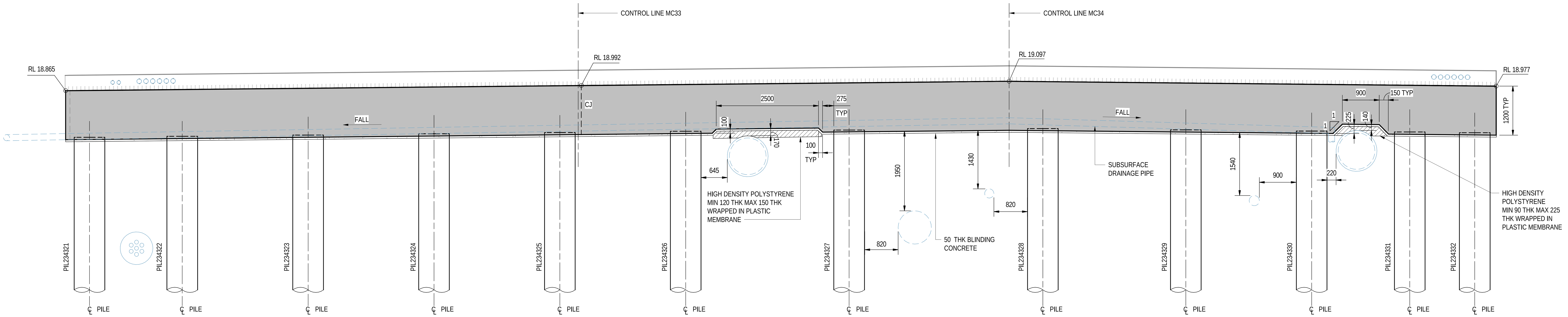


ABUTMENT B - PLAN

SCALE 1 : 50



ABUTMENT B - ELEVATION

SCALE 1 : 50

NOTES

1. FOR GENERAL NOTES REFER TO SHEET No. 2871005 TO 2871008.

PROJECT DOCUMENT REFERENCE
T2DPAA-T2D-C3S-BR-DRG-101072

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1:50

ROAD No. 6200
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RAILWAY TERRACE - SOUTH ROAD; MILE END SOUTH
BRIDGE OVER KESWICK CREEK
ABUTMENT CONCRETE - SHEET 02

DESIGNED: T2DA
DRAFTED: T2DA
ACCEPTED FOR USE: M. SHORT
TITLE: DIRECTOR OF ENG.
DATE: 28.03.26
ACCEPTANCE FORM KNET No.: 24467656
IN ACCORDANCE WITH DP013
DRAWING No.: 8011
SHEET No.: 2871072
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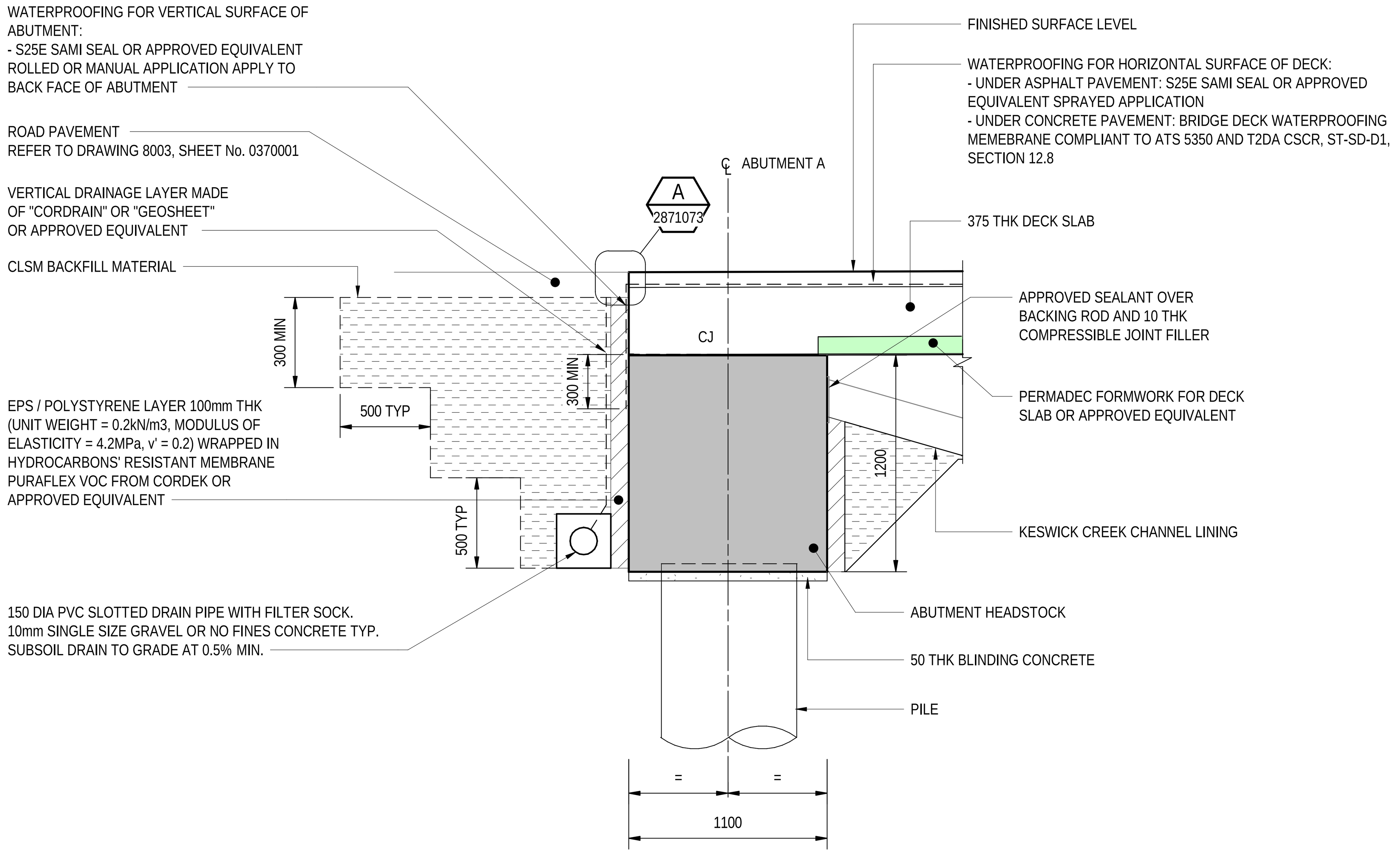
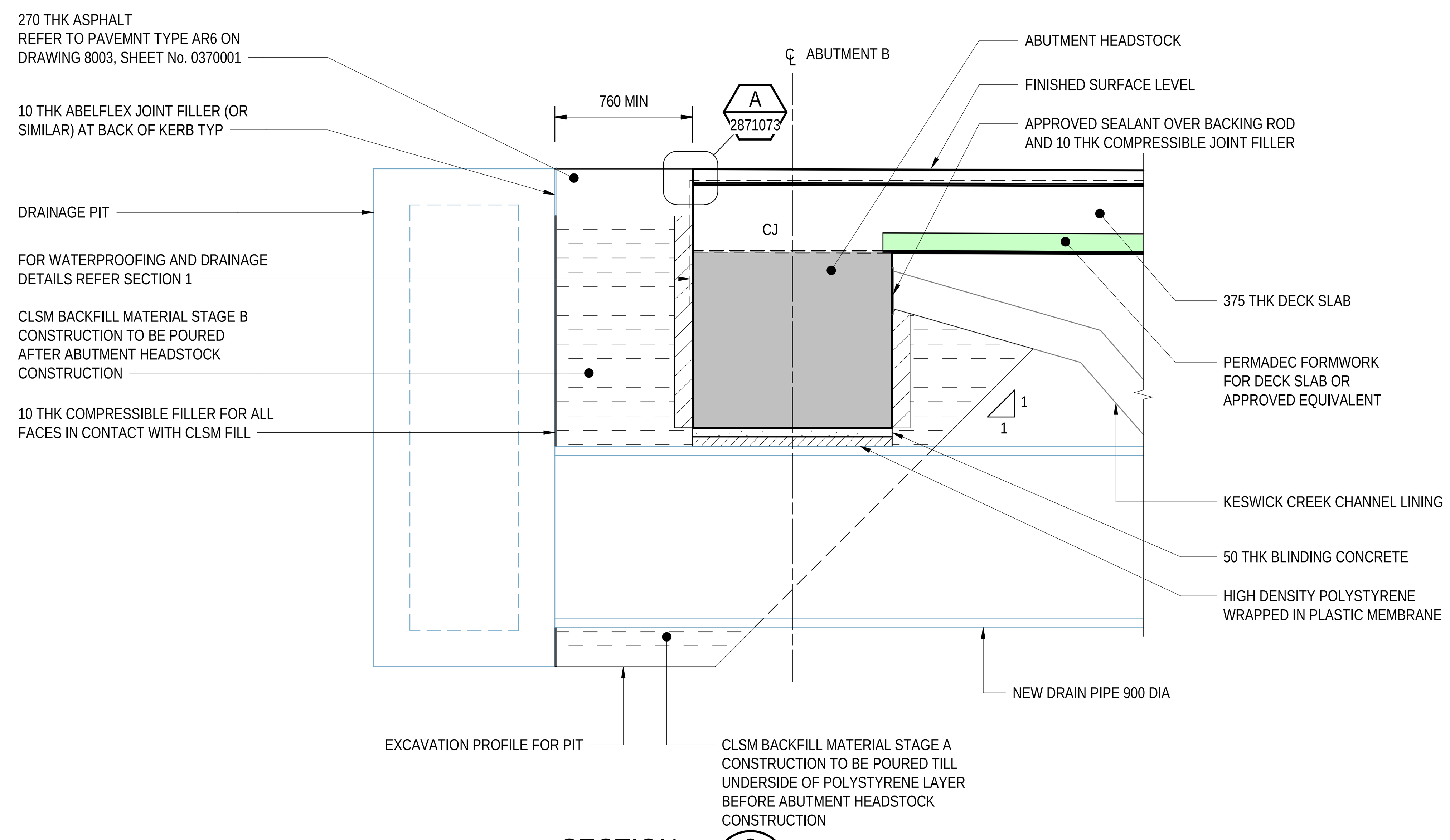
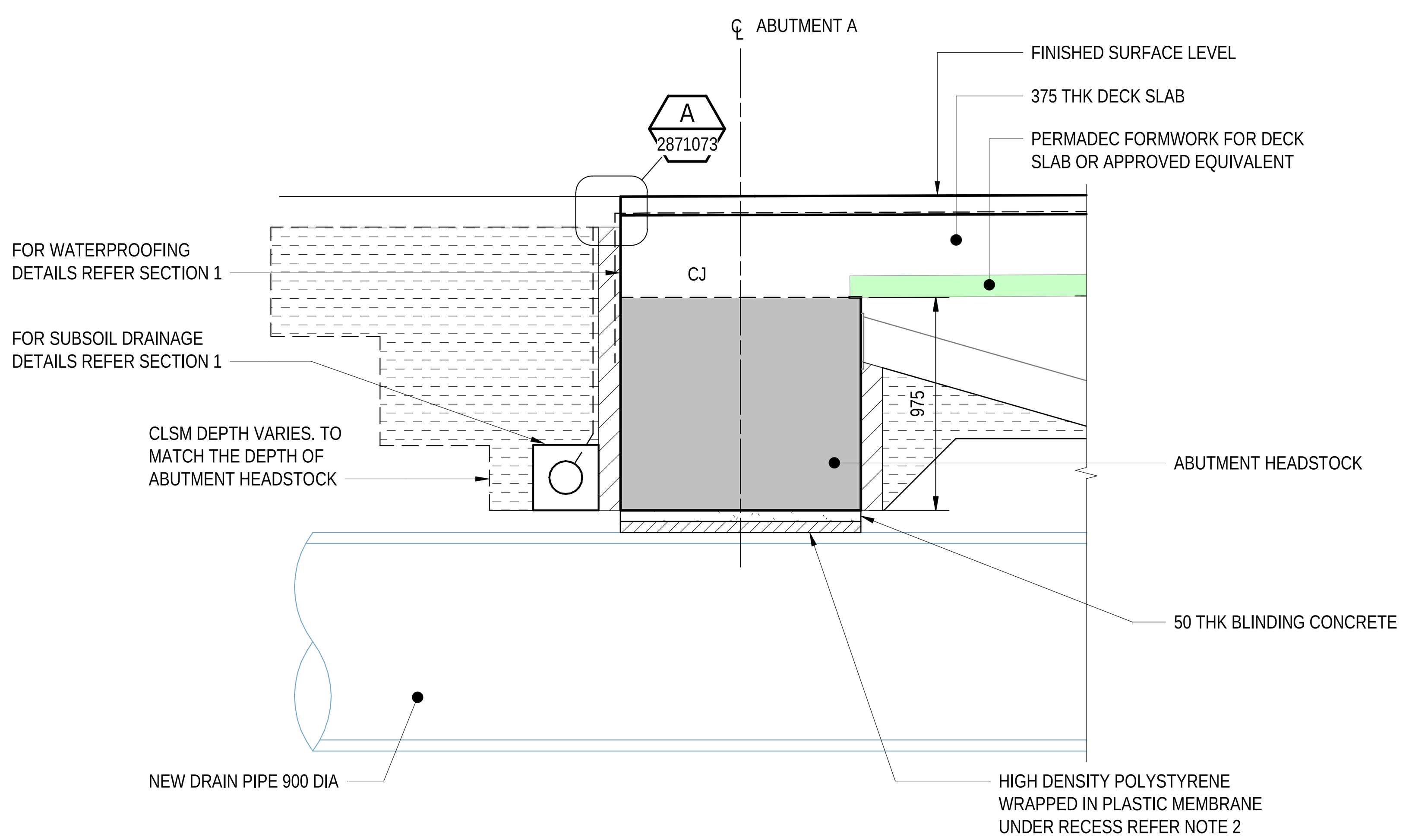


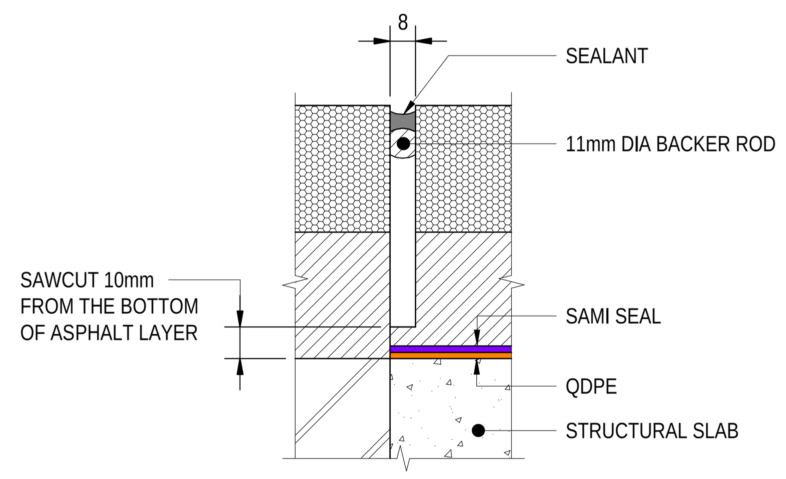
SECTION 1
SCALE 1 : 25
2871071



SECTION 2
SCALE 1 : 25
2871072



SECTION 3
SCALE 1 : 25
2871071



TYPICAL SEALED SAWCUT DETAIL

DETAIL A
SCALE 1 : 2
2871073

NOTES

1. FOR GENERAL NOTES REFER TO SHEET No. 2871005 TO 2871008.
2. FOR THICKNESS OF HIGH DENSITY POLYSTYRENE LAYER AT DRAINAGE PIPE LOCATIONS REFER TO SHEET No. 2871071 AND 2871072.

40 0 20 40 60 80
1:2

PROJECT DOCUMENT REFERENCE
T2DPAA-T2D-C3S-BR-DRG-101073

FOR CONSTRUCTION

No.	AMENDMENT DESCRIPTION	BY	CHECK	ACCEPTANCE	DATE
0	ISSUED FOR CONSTRUCTION (KNet No: 24467656)	T2DA	T2DA	M. SHORT	28.03.26

INDEX SHEET REFERENCE: 8011 SHEET 2871002

T2D TORRENS TO DARLINGTON

UNCONTROLLED COPY WHEN PRINTED | 100 MILLIMETRES ON ORIGINAL DRAWING | ALL DIMENSIONS ARE IN MILLIMETRES UNLESS SHOWN OTHERWISE

DESIGNER: A.SINGHAL / S.PODDAR
QUALIFICATION: MEng BEng
DATE: 25.03.26
REVIEWER: R.WOZNIAK
QUALIFICATION: FIEAust CPEng NER RPEV RPEQ MICE CEng
DATE: 25.03.26
INDEPENDENT DESIGN CERTIFIER (IF REQUIRED): T. JACKSON
QUALIFICATION: BEng (Hons), MEngSc
DATE: 27.03.26

Government of South Australia
Department for Infrastructure and Transport

PROJECT No.: 31921901
FILE No.: 2020/10577
DESIGN No.: #15551401
SURVEY No.: #15551401
PROJECT START ROAD RUNNING DISTANCE: NOT APPLICABLE
PROJECT END ROAD RUNNING DISTANCE: NOT APPLICABLE
SCALES: 500 0 250 500 750 1000
1:25

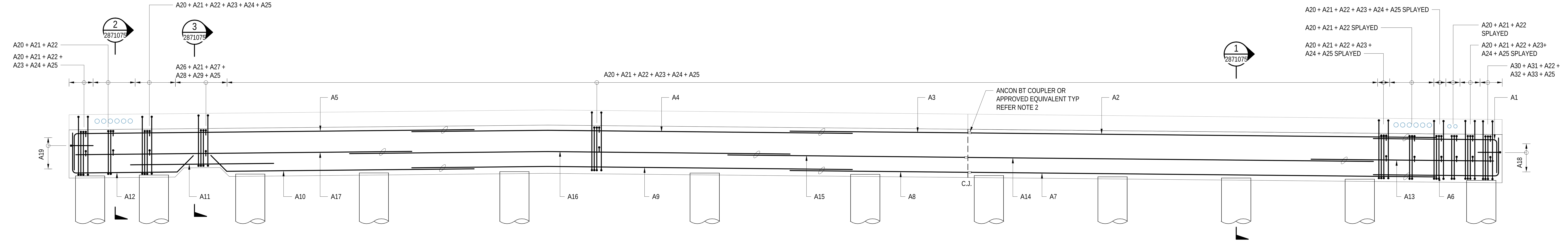
ROAD No. 6200
RICHMOND ROAD
RAILWAY TERRACE - SOUTH ROAD; MILE END SOUTH
BRIDGE OVER KESWICK CREEK
ABUTMENT CONCRETE - SHEET 03

DESIGNED:	DRAFTED:	ACCEPTED FOR USE:	ACCEPTANCE FORM KNET No.:	DRAWING No.:	SHEET No.:	AMEND No.:
T2DA	T2DA	M. SHORT TITLE: DIRECTOR OF ENG. DATE: 28.03.26	24467656 IN ACCORDANCE WITH DP013	8011	2871073	0

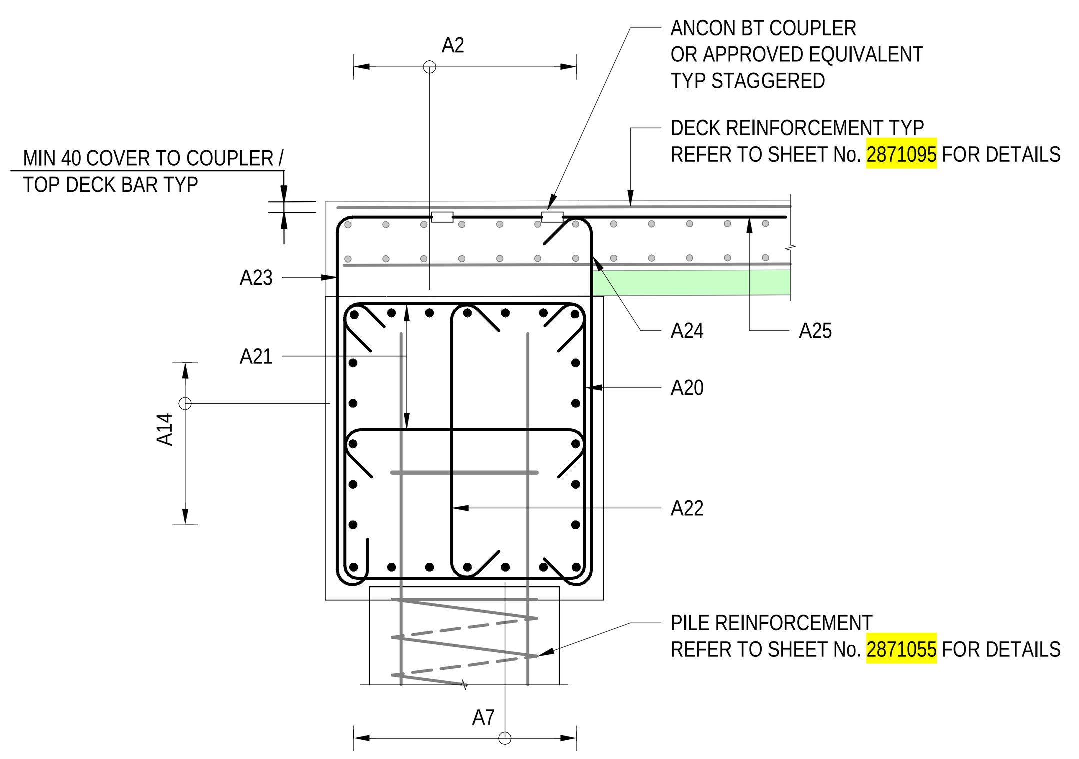
SHEET LATITUDE -34.94188 SHEET LONGITUDE 138.57397

FILE NAME: Autodesk Docs://T2D - Torrens to Darlington /T2DPAA-T2D-C3S-BR-M3D-100302.rvt

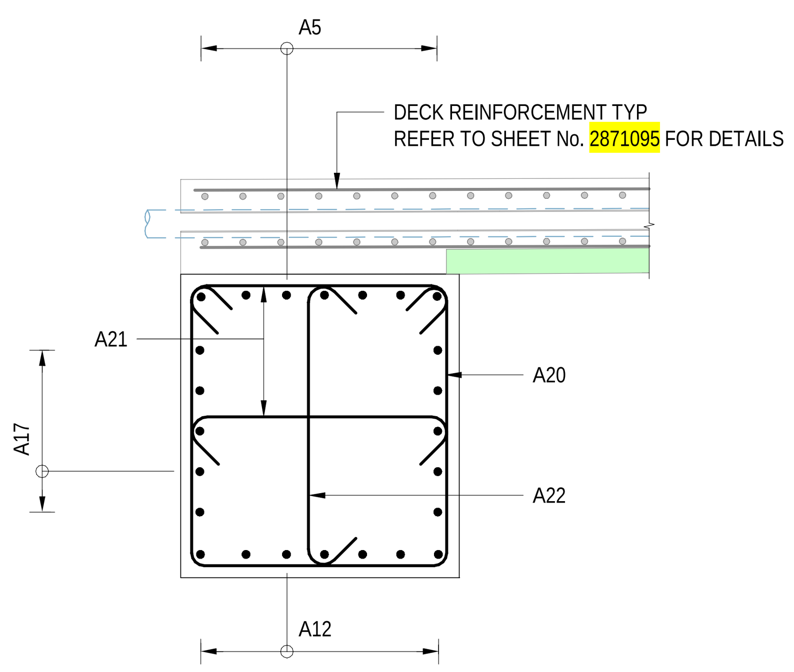
ABUTMENT A - PLAN
SCALE 1 : 50



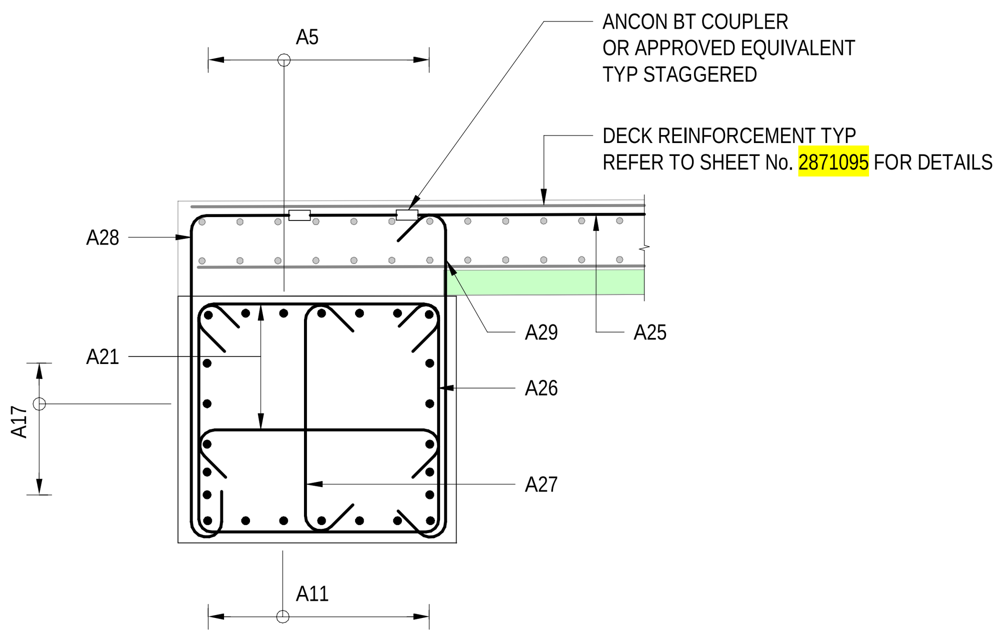
ELEVATION
SCALE 1 : 50



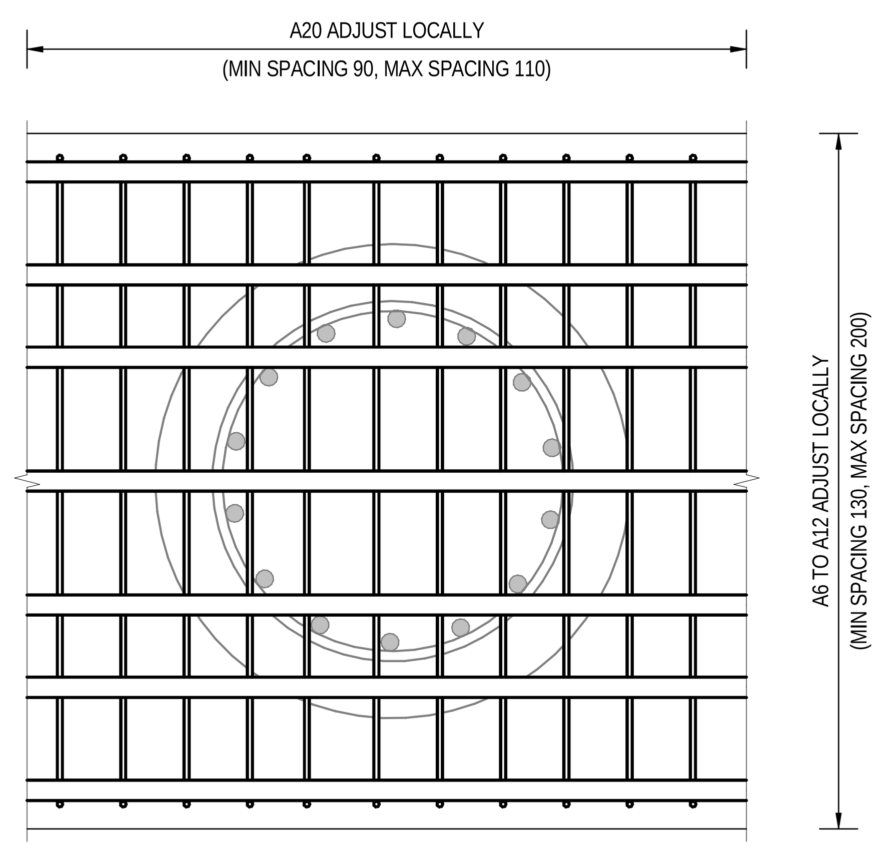
SECTION 1
SCALE 1 : 25
2871075



SECTION 2
SCALE 1 : 25
2871075



SECTION 3
SCALE 1 : 25
2871075



TYPICAL PLAN ABUTMENT A HEADSTOCK BOTTOM LEVEL
SCALE 1 : 10

NOTES

1. FOR GENERAL NOTES REFER TO SHEET No. 2871005 TO 2871008.
2. STAGGER EVERY SECOND COUPLER BY 150mm NOMINAL.

1:10
1:25

PROJECT DOCUMENT REFERENCE
T2DPAA-T2D-C3S-BR-DRG-101075

FOR CONSTRUCTION

No.	AMENDMENT DESCRIPTION	BY	CHECK	ACCEPTANCE	DATE
0	ISSUED FOR CONSTRUCTION (KNet No: 24467656)	T2DA	T2DA	M. SHORT	28.03.26

INDEX SHEET REFERENCE: 8011 SHEET 2871002

T2D → TORRENS TO DARLINGTON

DESIGNER: A.SINGHAL / S.PODDAR
QUALIFICATION: MEng BEng
DATE: 25.03.26

REVIEWER: R.WOZNIAK
QUALIFICATION: FIEAust CPEng NER RPEV RPEQ MICE CEng
DATE: 25.03.26

INDEPENDENT DESIGN CERTIFIER (IF REQUIRED): T. JACKSON
QUALIFICATION: BEng (Hons), MEngSc
DATE: 27.03.26

Government of South Australia
Department for Infrastructure and Transport

PROJECT No.: 31921901
FILE No.: 2020/10577
DESIGN No.: #15551401
SURVEY No.: #15551401
PROJECT START ROAD RUNNING DISTANCE: NOT APPLICABLE
PROJECT END ROAD RUNNING DISTANCE: NOT APPLICABLE
SCALES: 1:50

ROAD No. 6200
RICHMOND ROAD
RAILWAY TERRACE - SOUTH ROAD; MILE END SOUTH
BRIDGE OVER KESWICK CREEK
ABUTMENT REINFORCEMENT - SHEET 01

DESIGNED: T2DA
DRAFTED: T2DA
ACCEPTED FOR USE: M. SHORT
TITLE: DIRECTOR OF ENG.
DATE: 28.03.26
ACCEPTANCE FORM KNET No.: 24467656
IN ACCORDANCE WITH DP013
DRAWING No.: 8011
SHEET No.: 2871075
AMEND No.: 0
SHEET LATITUDE -34.94188
SHEET LONGITUDE 138.57397

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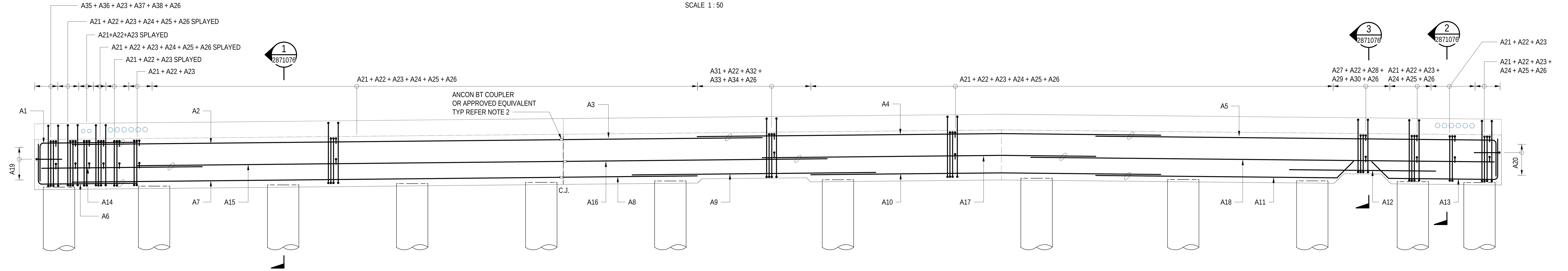
FILE NAME: Autodesk Docs://T2D - Torrens to Darlington /T2DPAA-T2D-C3S-BR-M3D-100302.rvt

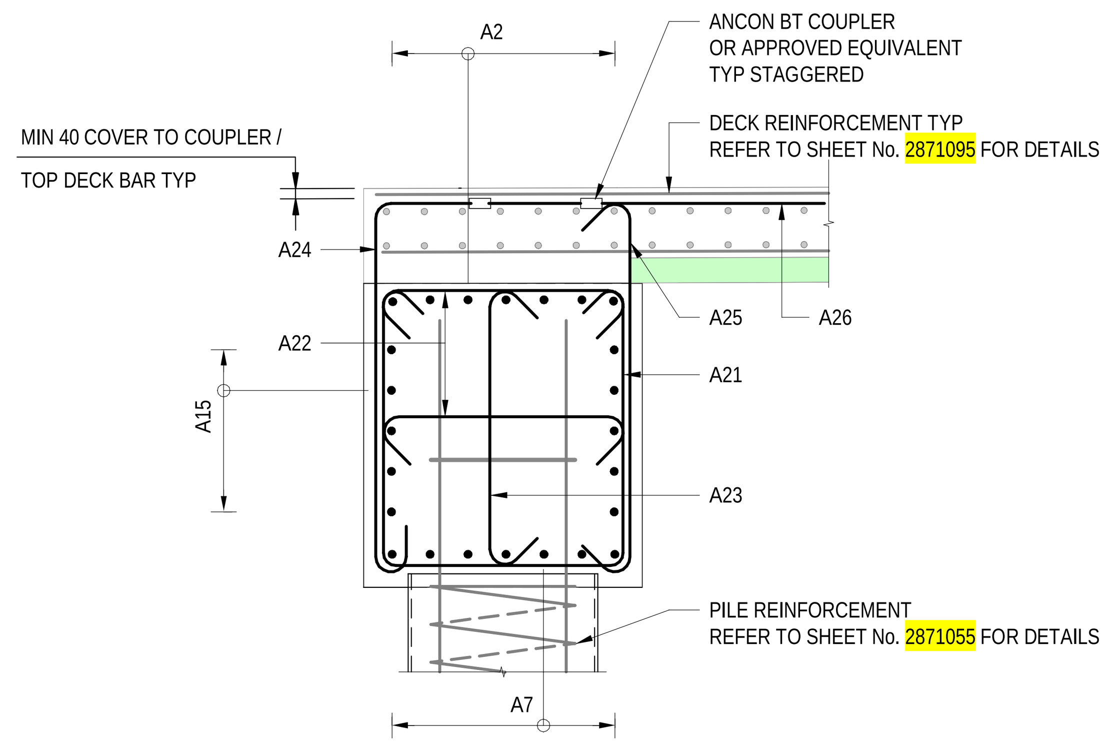
ABUTMENT B - PLAN

SCALE 1 : 50

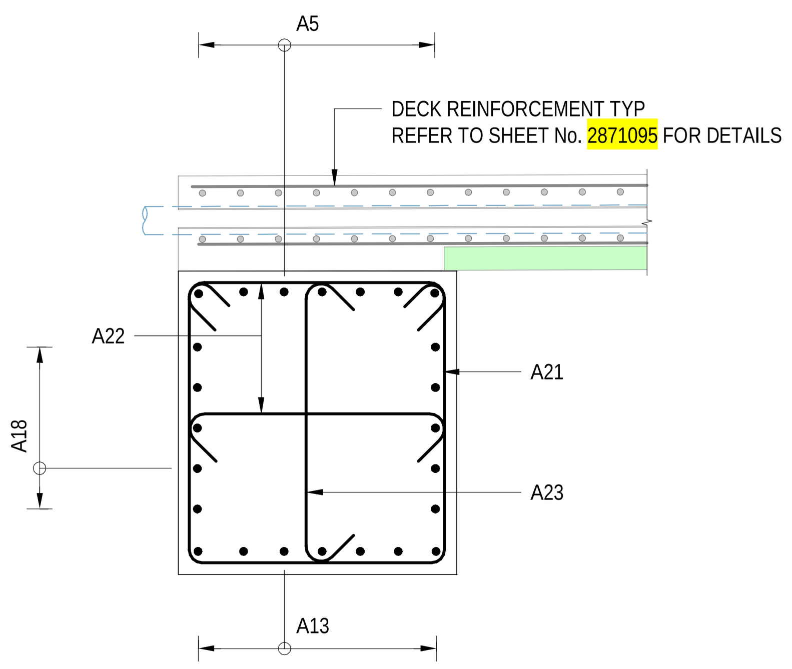


ELEVATION

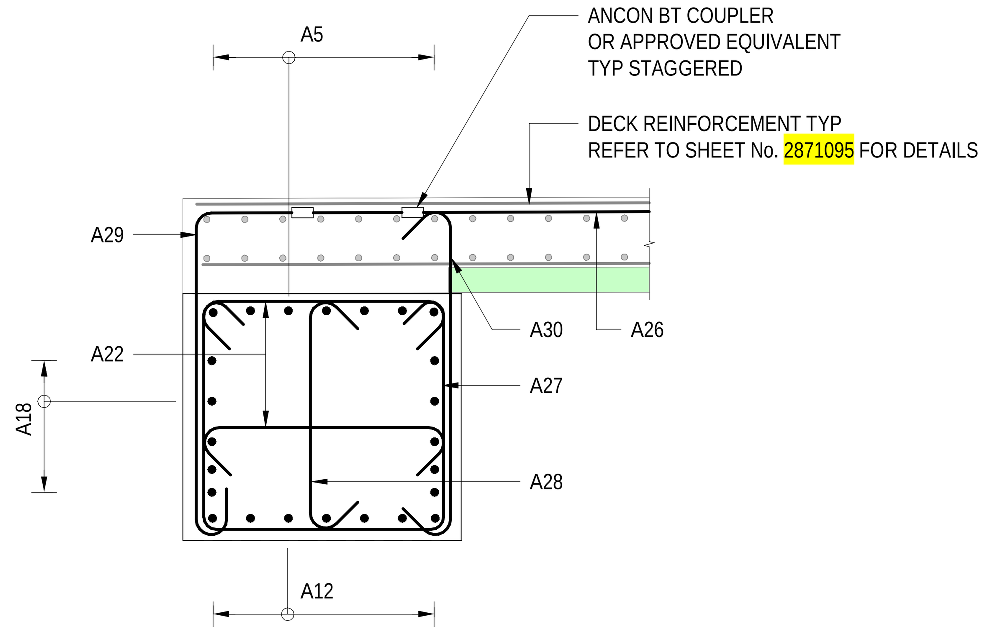
SCALE 1 : 50



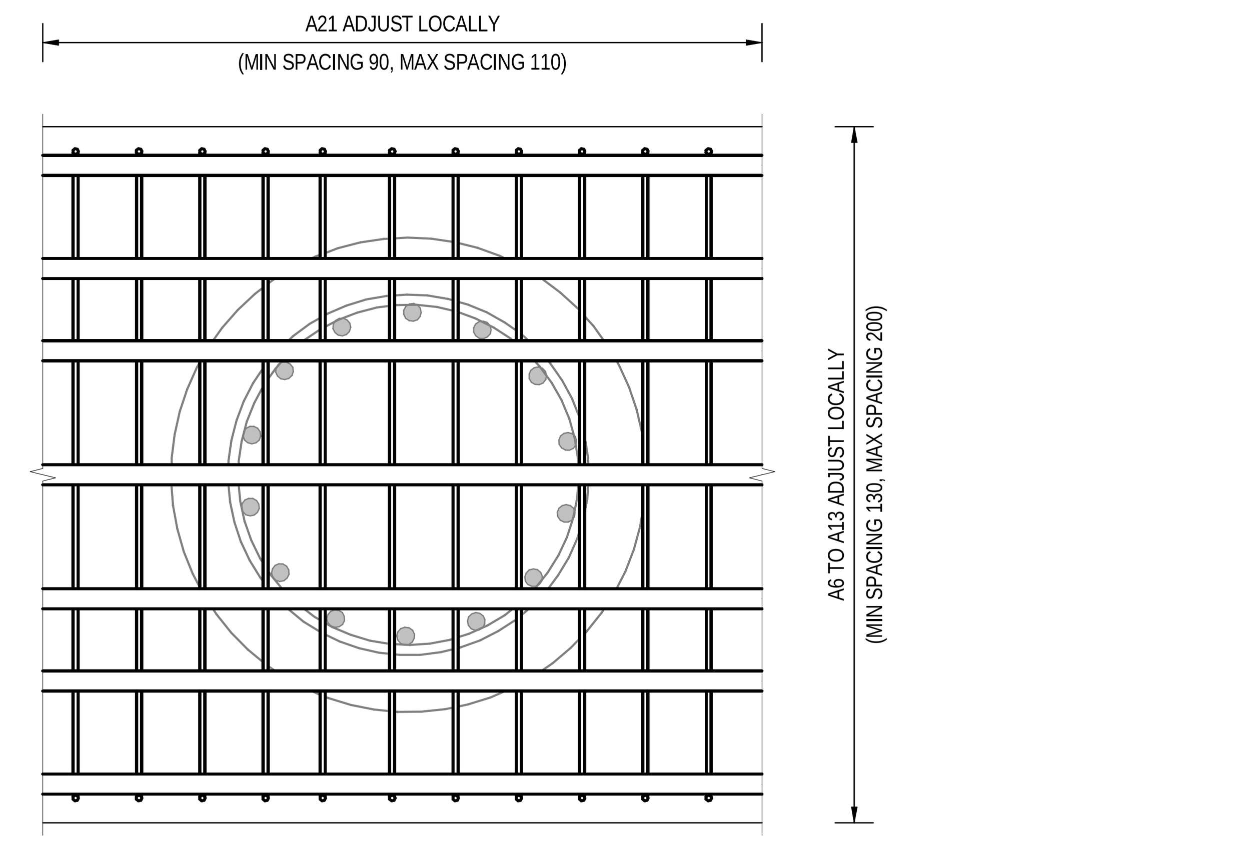
SECTION 1
SCALE 1 : 25 2871076



SECTION 2
SCALE 1 : 25 2871076



SECTION 3
SCALE 1 : 25 2871076



TYPICAL PLAN ABUTMENT B HEADSTOCK BOTTOM LEVEL

SCALE 1 : 10

200 0 100 200 300 400
1:10
500 0 250 500 750 1000
1:25

NOTES

1. FOR GENERAL NOTES REFER TO SHEET No. 2871005 TO 2871008.
2. STAGGER EVERY SECOND COUPLER BY 150mm NOMINAL.

PROJECT DOCUMENT REFERENCE
T2DPAA-T2D-C3S-BR-DRG-101076

FOR CONSTRUCTION

No.	AMENDMENT DESCRIPTION	BY	CHECK	ACCEPTANCE	DATE
0	ISSUED FOR CONSTRUCTION (KNet No: 24467656)	T2DA	T2DA	M. SHORT	28.03.26

INDEX SHEET REFERENCE: 8011 SHEET 2871002

T2D → TORRENS TO DARLINGTON

DESIGNER: A.SINGHAL / S.PODDAR
QUALIFICATION: MEng BEng
DATE: 25.03.26

REVIEWER: R.WOZNIAK
QUALIFICATION: FIEAust CPEng NER RPEV RPEQ MICE CEng
DATE: 25.03.26

INDEPENDENT DESIGN CERTIFIER (IF REQUIRED): T. JACKSON
QUALIFICATION: BEng (Hons), MEngSc
DATE: 27.03.26

Government of South Australia
Department for Infrastructure and Transport

PROJECT No.: 31921901
FILE No.: 2020/10577
DESIGN No.: #15551401
SURVEY No.: #15551401
PROJECT START ROAD RUNNING DISTANCE: NOT APPLICABLE
PROJECT END ROAD RUNNING DISTANCE: NOT APPLICABLE
SCALES: 1000 0 500 1000 1500 2000 1:50

ROAD No. 6200
RICHMOND ROAD
RAILWAY TERRACE - SOUTH ROAD; MILE END SOUTH
BRIDGE OVER KESWICK CREEK
ABUTMENT REINFORCEMENT - SHEET 02

DESIGNED: T2DA
DRAFTED: T2DA
ACCEPTED FOR USE: M. SHORT
TITLE: DIRECTOR OF ENG.
DATE: 28.03.26
ACCEPTANCE FORM KNET No.: 24467656
IN ACCORDANCE WITH DP013
DRAWING No.: 8011
SHEET No.: 2871076
AMEND No.: 0
SHEET LATITUDE -34.94188
SHEET LONGITUDE 138.57397

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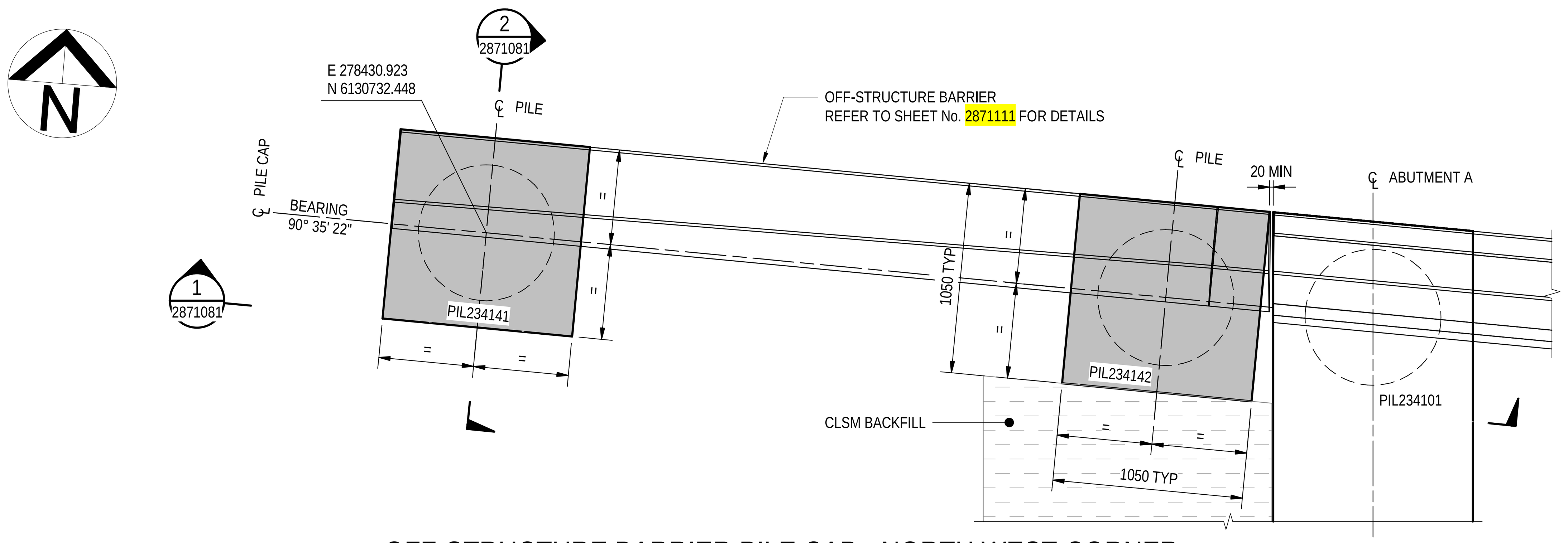
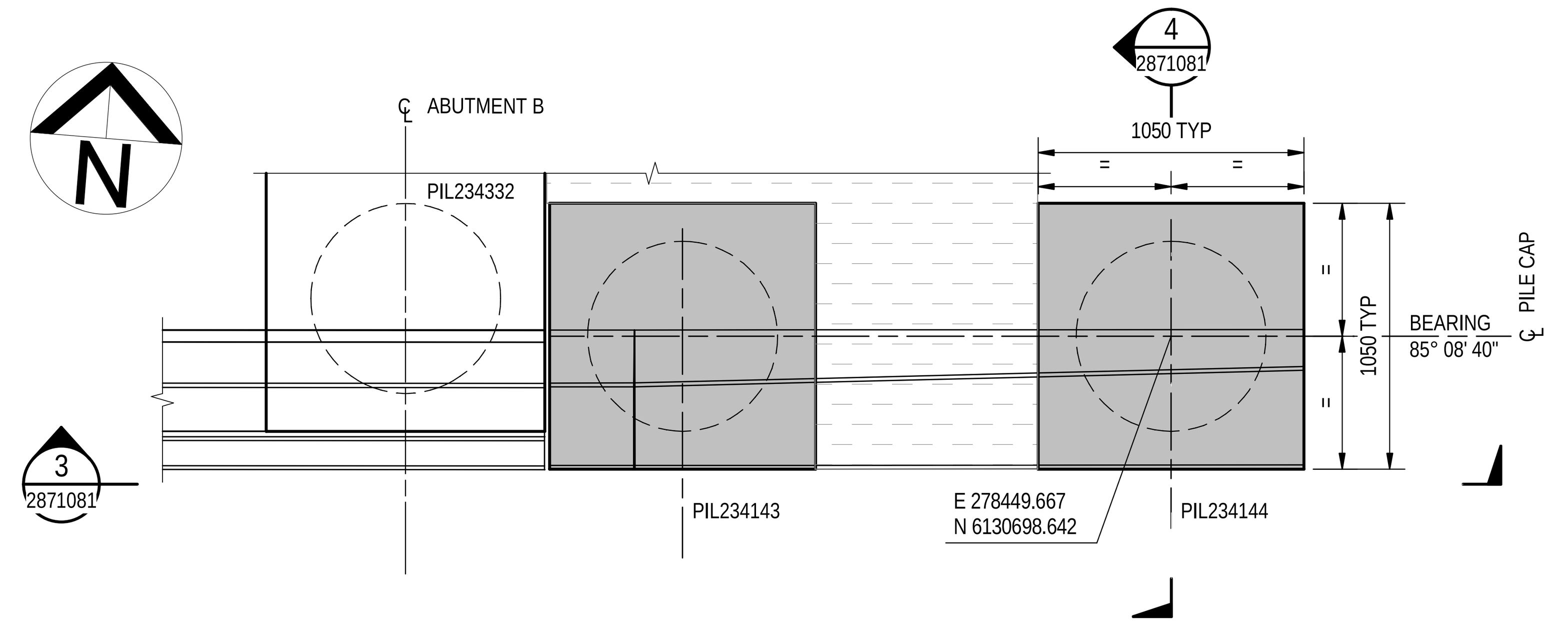
FILE NAME: Autodesk Docs://T2D - Torrens to Darlington /T2DPAA-T2D-C3S-BR-M3D-100302.rvt



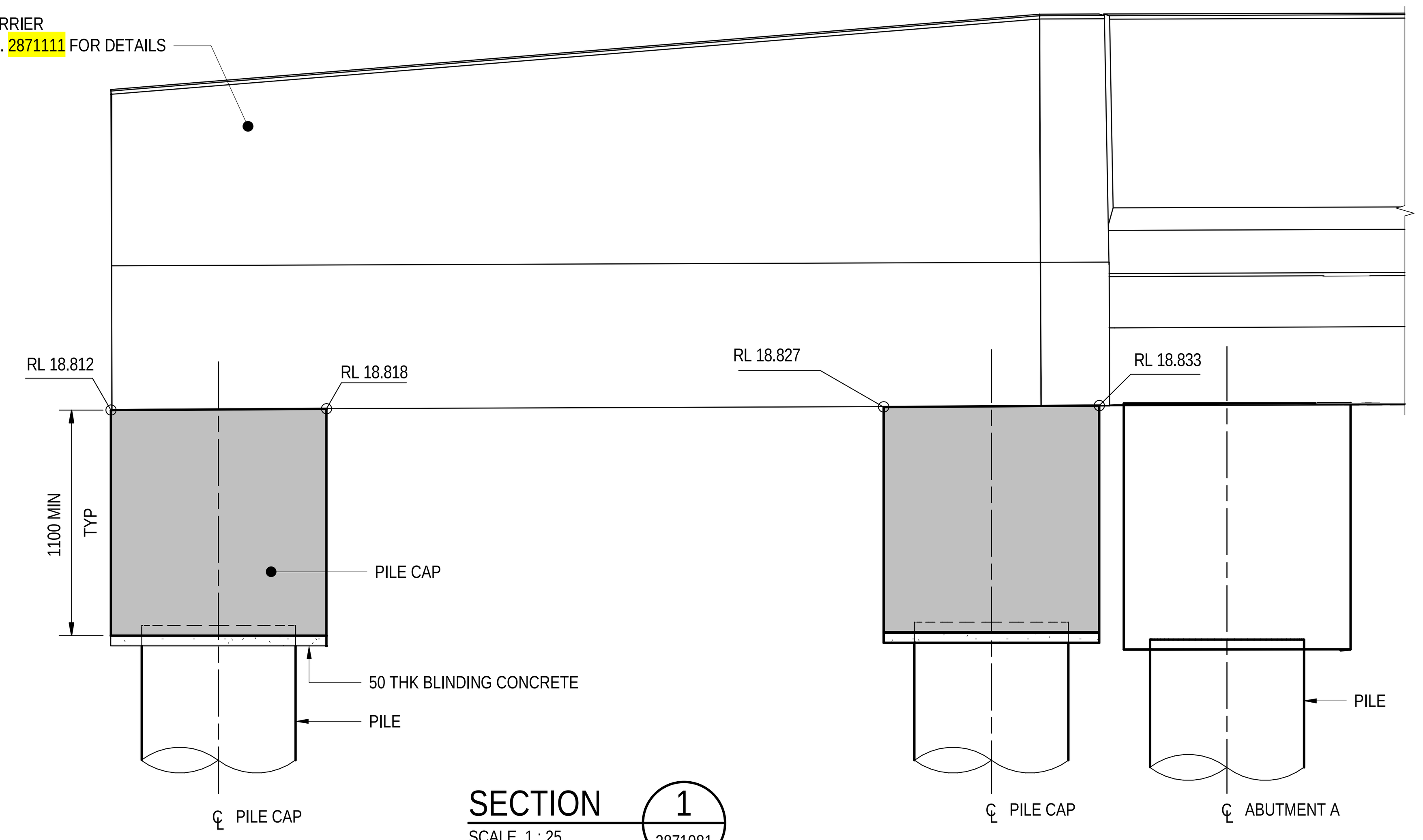
OFF-STRUCTURE BARRIER PILE CAP - NORTH WEST CORNER

SCALE 1 : 25

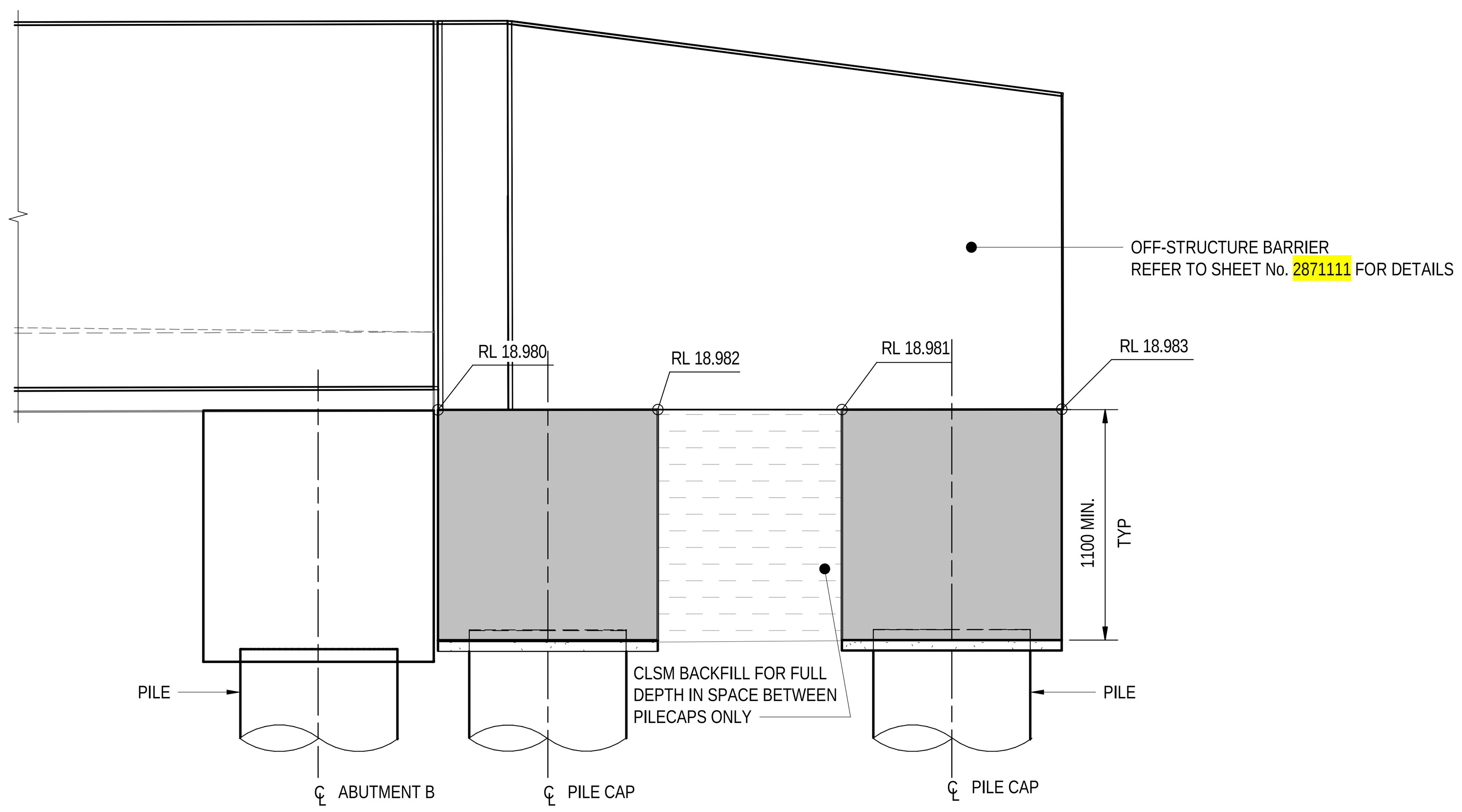


OFF-STRUCTURE BARRIER PILE CAP - SOUTH EAST CORNER

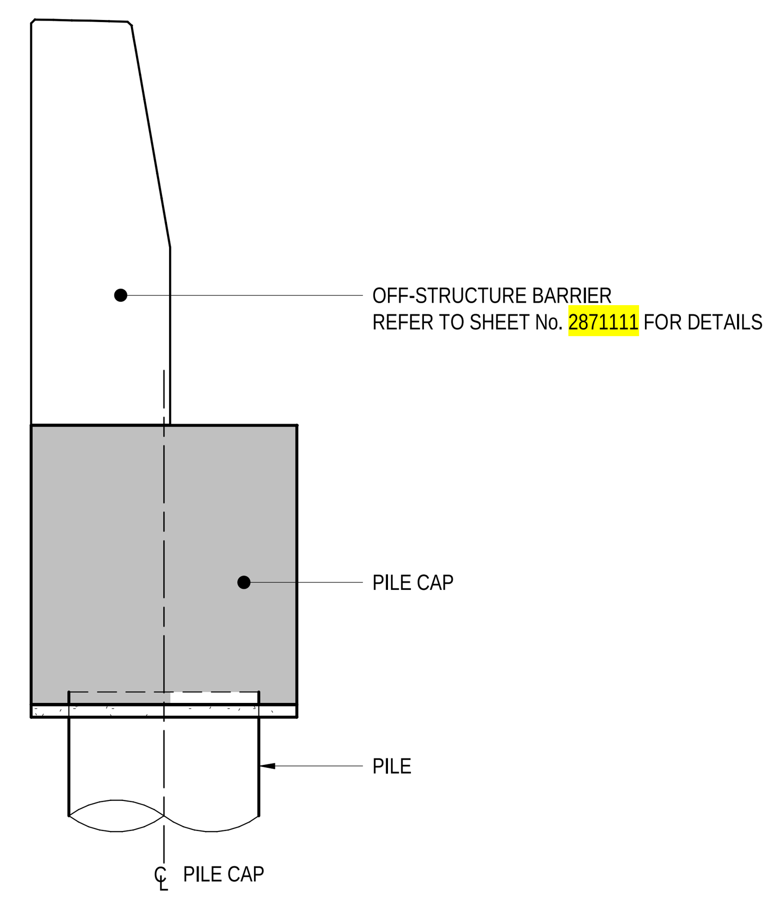
SCALE 1 : 25



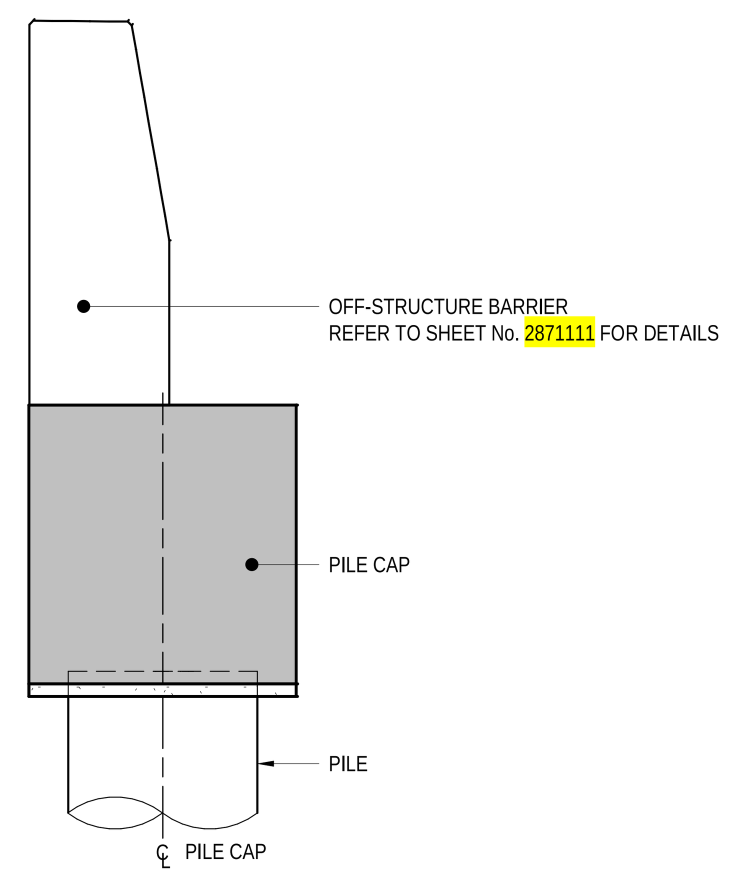
SECTION 1
SCALE 1 : 25
2871081



SECTION 3
SCALE 1 : 25
2871081



SECTION 2
SCALE 1 : 25
2871081



SECTION 4
SCALE 1 : 25
2871081

NOTES

1. FOR GENERAL NOTES REFER TO SHEET No. 2871005 TO 2871008.

PROJECT DOCUMENT REFERENCE
T2DPAA-T2D-C3S-BR-DRG-101081

FOR CONSTRUCTION

No.	AMENDMENT DESCRIPTION	BY	CHECK	ACCEPTANCE	DATE
0	ISSUED FOR CONSTRUCTION (KNet No: 24467656)	T2DA	T2DA	M. SHORT	28.03.26

INDEX SHEET REFERENCE: 8011 SHEET 2871002

T2D → TORRENS TO DARLINGTON

DESIGNER: A.SINGHAL / S.PODDAR
QUALIFICATION: MEng BEng
DATE: 25.03.26

REVIEWER: R.WOZNIAK
QUALIFICATION: FIEAust CPEng NER RPEV RPEQ MICE CEng
DATE: 25.03.26

INDEPENDENT DESIGN CERTIFIER (IF REQUIRED): T. JACKSON
QUALIFICATION: BEng (Hons), MEngSc
DATE: 27.03.26

Government of South Australia
Department for Infrastructure and Transport

PROJECT No.: 31921901
FILE No.: 2020/10577
DESIGN No.: #15551401
SURVEY No.: #15551401
PROJECT START ROAD RUNNING DISTANCE: NOT APPLICABLE
PROJECT END ROAD RUNNING DISTANCE: NOT APPLICABLE
SCALES: 500 0 250 500 750 1000 1:25

ROAD No. 6200
RICHMOND ROAD
RAILWAY TERRACE - SOUTH ROAD; MILE END SOUTH
BRIDGE OVER KESWICK CREEK
PILE CAP CONCRETE

DESIGNED: T2DA
DRAFTED: T2DA
ACCEPTED FOR USE: M. SHORT
TITLE: DIRECTOR OF ENG.
DATE: 28.03.26
ACCEPTANCE FORM KNET No.: 24467656
IN ACCORDANCE WITH DP013
DRAWING No.: 8011
SHEET No.: 2871081
AMEND No.: 0
SHEET LATITUDE -34.94188
SHEET LONGITUDE 138.57397

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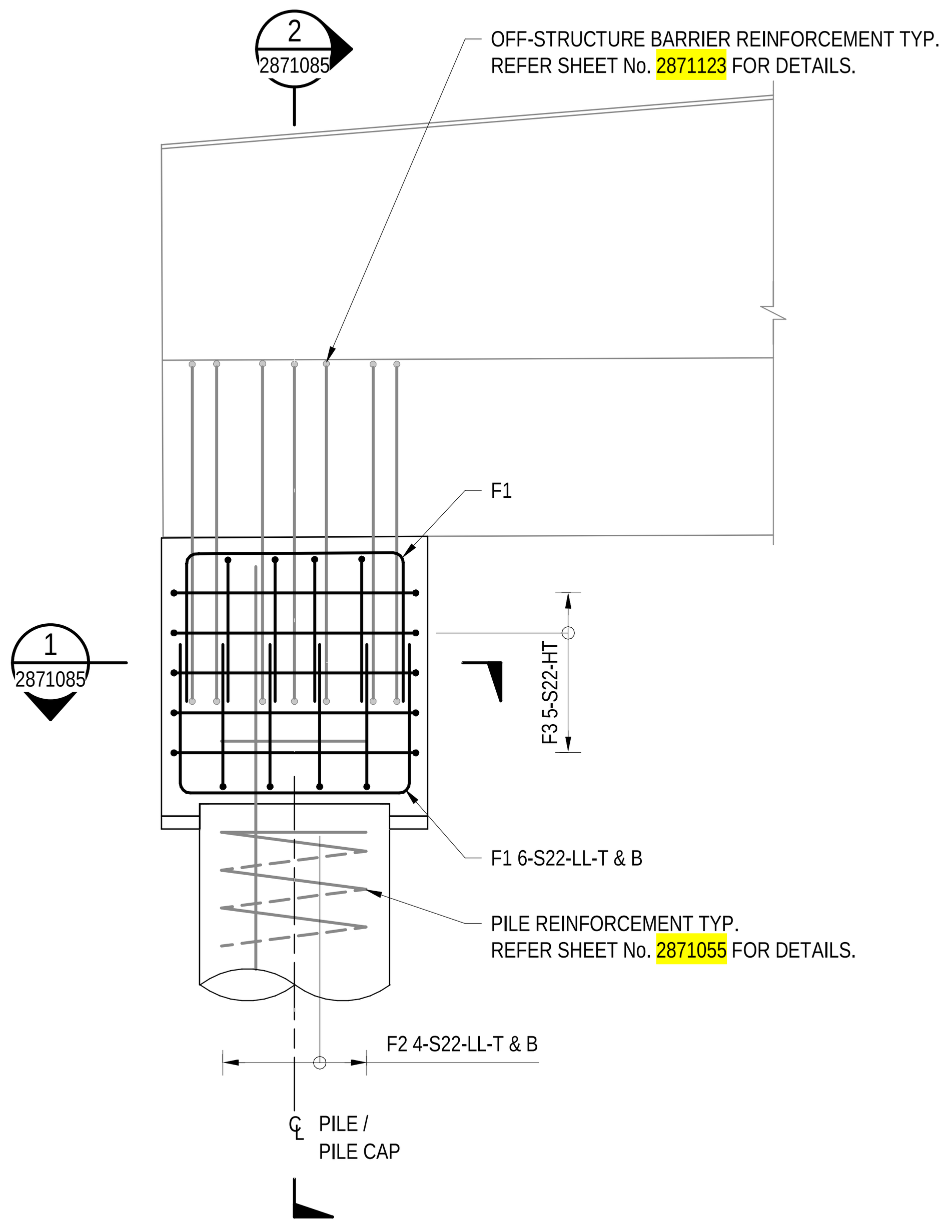
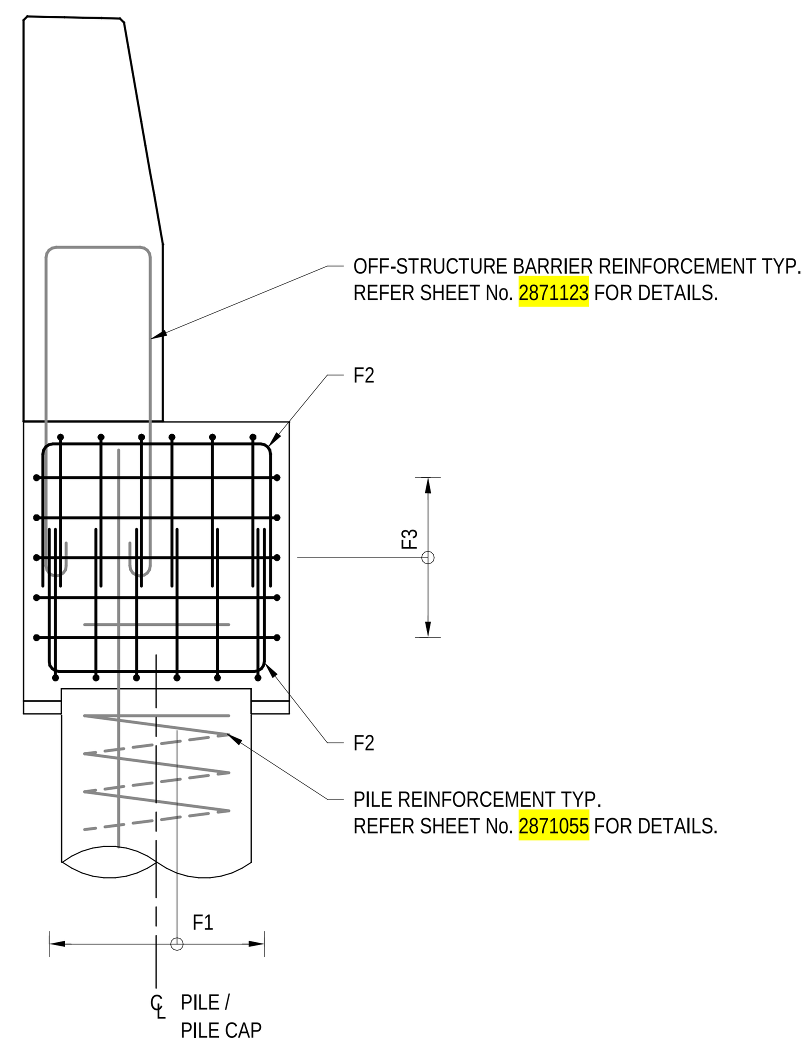
100 MILLIMETRES ON ORIGINAL DRAWING
ALL DIMENSIONS ARE IN MILLIMETRES UNLESS SHOWN OTHERWISE

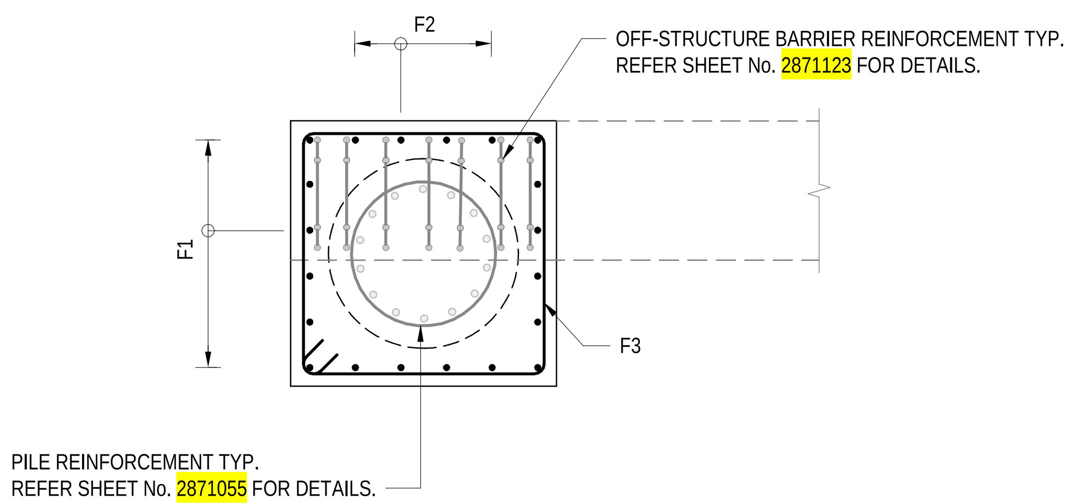
FILE NAME: Autodesk Docs://T2D - Torrens to Darlington /T2DPAA-T2D-C3S-BR-M3D-100302.rvt



ELEVATION - TYPICAL PILE CAP REINFORCEMENT DETAILS
SCALE 1 : 25



SECTION 2
SCALE 1 : 25
2871085



SECTION 1
SCALE 1 : 25
2871085

NOTES

1. FOR GENERAL NOTES, REFER TO SHEET No. 2871005 TO 2871008.
2. LOCALLY ADJUST PILECAP BARS TO AVOID CLASH WITH BARRIER STITCH BARS.

PROJECT DOCUMENT REFERENCE
T2DPAA-T2D-C3S-BR-DRG-101085

FOR CONSTRUCTION

No.	AMENDMENT DESCRIPTION	BY	CHECK	ACCEPTANCE	DATE
0	ISSUED FOR CONSTRUCTION (KNet No: 24467656)	T2DA	T2DA	M. SHORT	28.03.26

INDEX SHEET REFERENCE: 8011 SHEET 2871002

T2D TORRENS TO DARLINGTON

UNCONTROLLED COPY WHEN PRINTED | 100 MILLIMETRES ON ORIGINAL DRAWING | ALL DIMENSIONS ARE IN MILLIMETRES UNLESS SHOWN OTHERWISE

DESIGNER: A.SINGHAL / S.PODDAR
QUALIFICATION: MEng BEng
DATE: 25.03.26
REVIEWER: R.WOZNIAK
QUALIFICATION: FIEAust CPEng NER RPEV RPEQ MICE CEng
DATE: 25.03.26
INDEPENDENT DESIGN CERTIFIER (IF REQUIRED): T. JACKSON
QUALIFICATION: BEng (Hons), MEngSc
DATE: 27.03.26

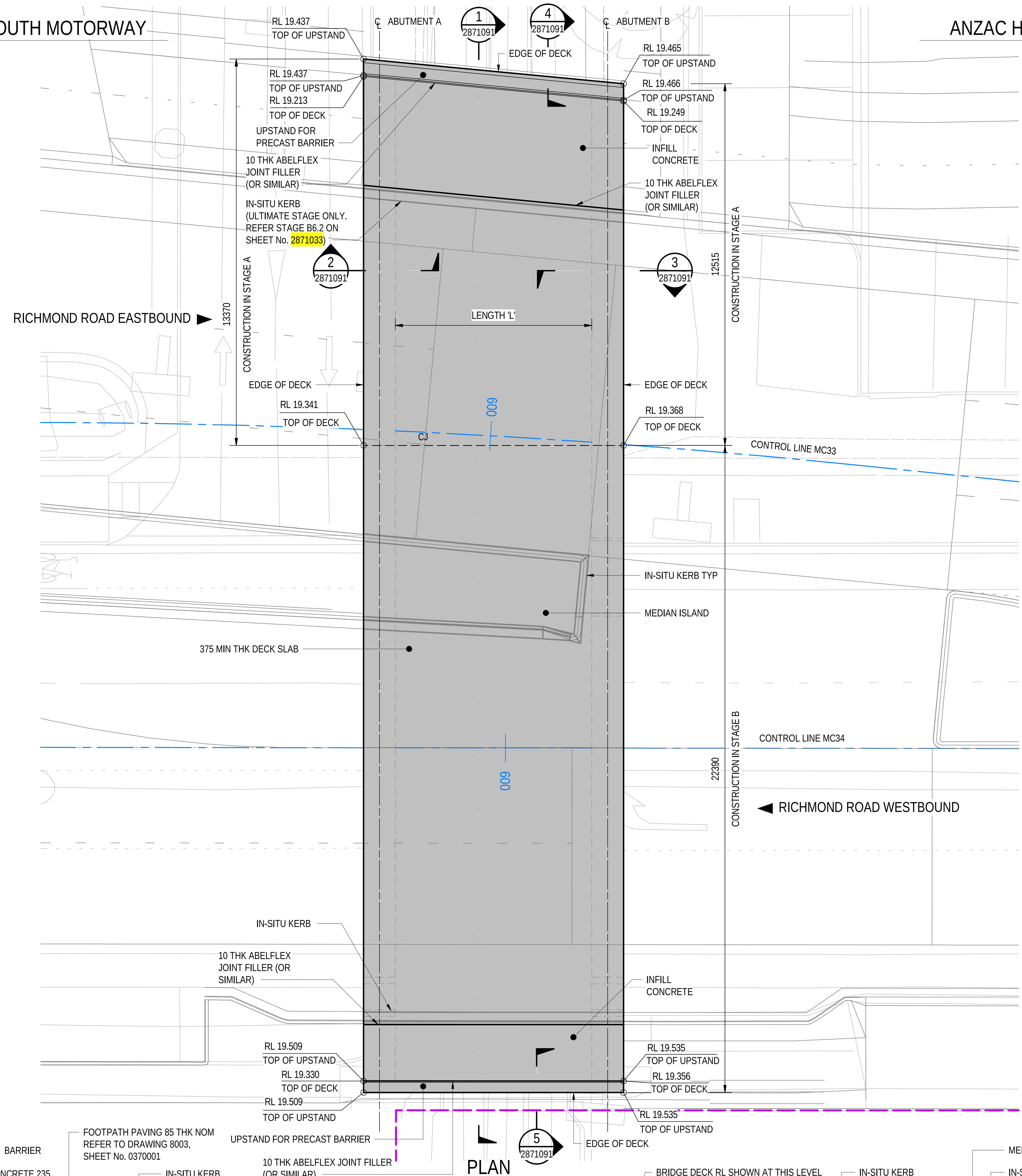
Government of South Australia
Department for Infrastructure and Transport

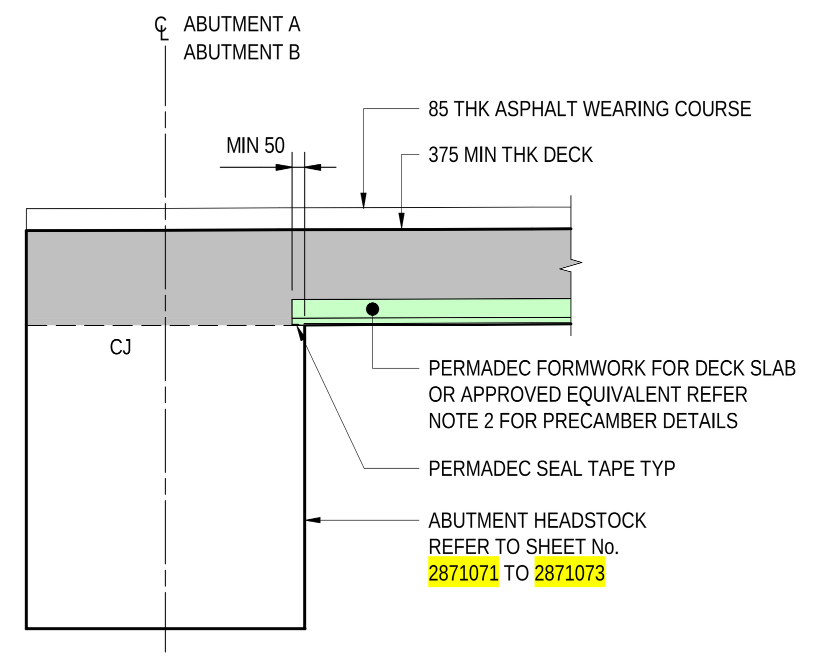
PROJECT No.: 31921901	FILE No.: 2020/10577
DESIGN No.: #15551401	SURVEY No.: #15551401
PROJECT START ROAD RUNNING DISTANCE: NOT APPLICABLE	
PROJECT END ROAD RUNNING DISTANCE: NOT APPLICABLE	
SCALES: 500 0 250 500 750 1000 1:25	

ROAD No. 6200
RICHMOND ROAD
RAILWAY TERRACE - SOUTH ROAD; MILE END SOUTH
BRIDGE OVER KESWICK CREEK
PILE CAP REINFORCEMENT

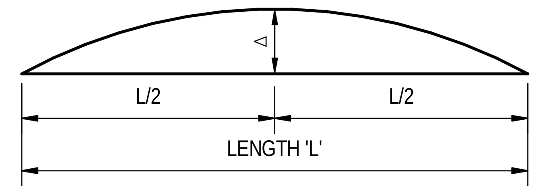
DESIGNED:	DRAFTED:	ACCEPTED FOR USE:	ACCEPTANCE FORM KNET No.:	DRAWING No.:	SHEET No.:	AMEND No.:
T2DA	T2DA	M. SHORT TITLE: DIRECTOR OF ENG. DATE: 28.03.26	24467656 IN ACCORDANCE WITH DP013	8011 SHEET LATITUDE -34.94188	2871085 SHEET LONGITUDE 138.57397	0

FILE NAME: Autodesk Docs://T2D - Torrens to Darlington /T2DPAA-T2D-C3S-BR-M3D-100302.rvt





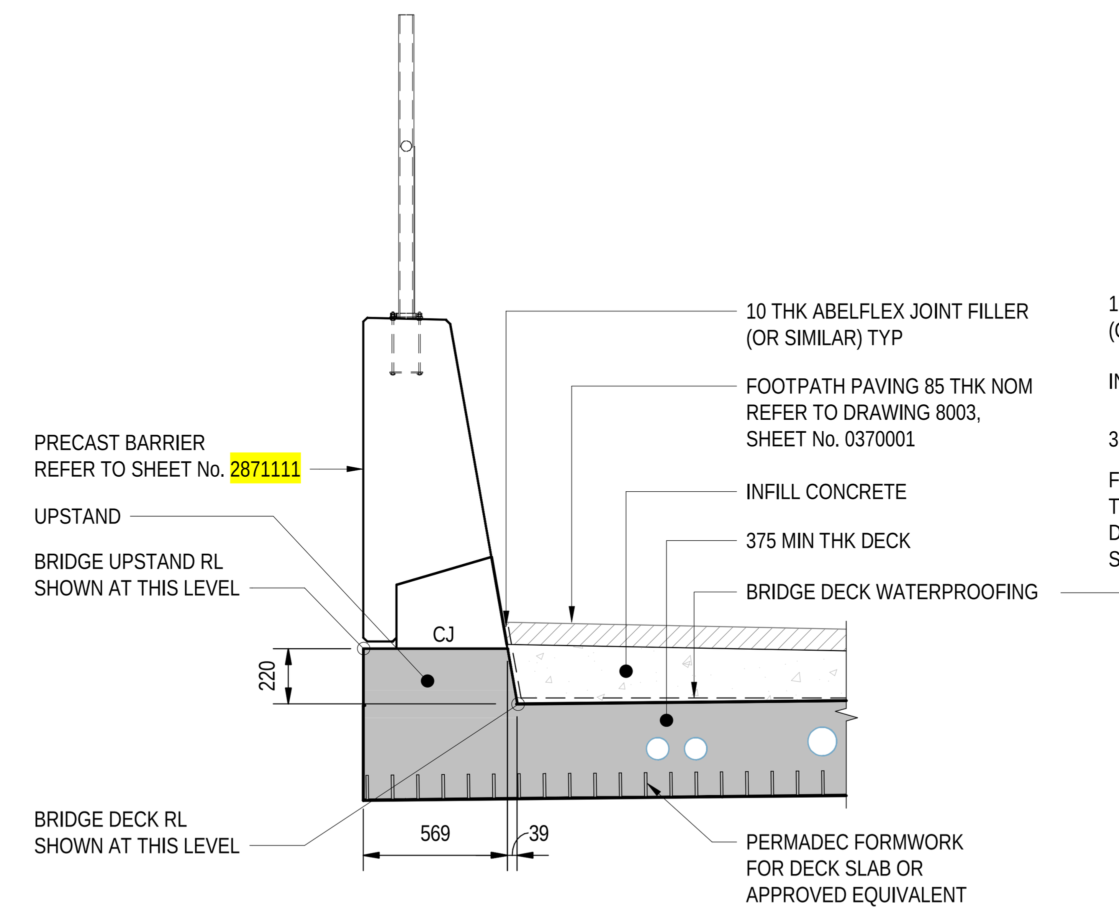
SECTION 2 / 3 (2871091) SCALE 1 : 25 SIMILAR AND MIRRORED



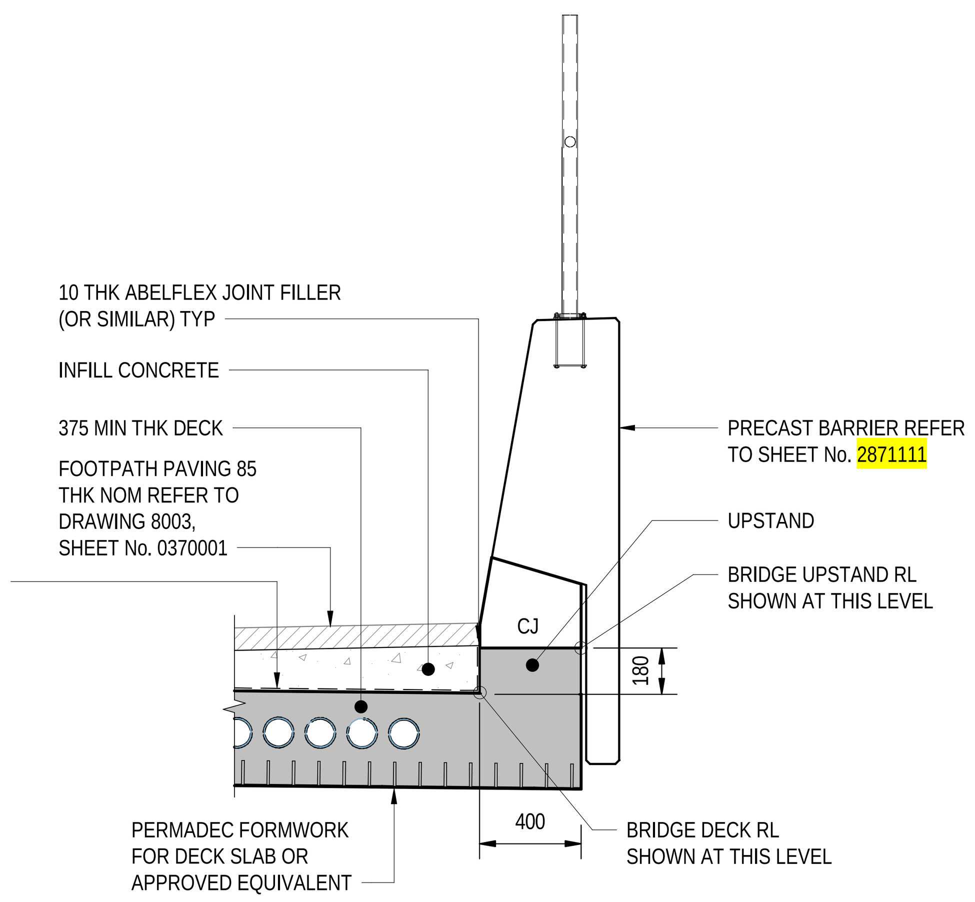
DECK PRECAMBER VALUE

REFER NOTE 2.

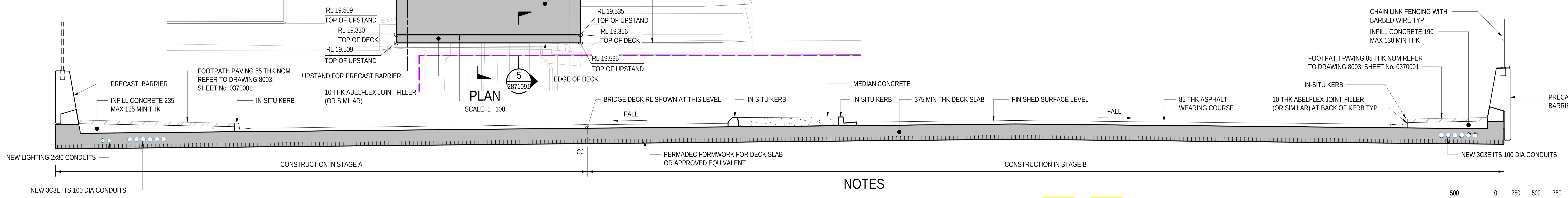
DECK DEFLECTION VALUE	
DECK	△ = 12mm



SECTION 4 (2871091) SCALE 1 : 25



SECTION 5 (2871091) SCALE 1 : 25



SECTION 1 (2871091) SCALE 1 : 50

NOTES

1. FOR GENERAL NOTES REFER TO SHEET No. 2871005 TO 2871008.
2. PERMANENT FORMWORK AND TEMPORARY WORKS TO IMPLEMENT PRECAMBER FOR ESTIMATED DOWNWARD DEFLECTION DUE TO COMBINED DEAD LOAD AND SUPERIMPOSED DEAD LOADS OF SUPERSTRUCTURE AS PER VALUE PROVIDED ON THIS SHEET.
3. CONSTRUCTION TOLERANCES ARE NOT ACCOUNTED FOR IN ESTIMATION OF THE DECK DEFLECTION VALUE.

SCALES 1:25, 1:50

PROJECT DOCUMENT REFERENCE T2DPAA-T2D-C3S-BR-DRG-101091

FOR CONSTRUCTION

No.	AMENDMENT DESCRIPTION	BY	CHECK	ACCEPTANCE	DATE
0	ISSUED FOR CONSTRUCTION (KNet No: 24467656)	T2DA	T2DA	M. SHORT	28.03.26

INDEX SHEET REFERENCE: 8011 SHEET 2871002

T2D → TORRENS TO DARLINGTON

DESIGNER: A.SINGHAL / S.PODDAR QUALIFICATION: MEng BEng DATE: 25.03.26

REVIEWER: R.WOZNIAK QUALIFICATION: FIEAust CPEng NER RPEV RPEQ MICE CEng DATE: 25.03.26

INDEPENDENT DESIGN CERTIFIER (IF REQUIRED): T. JACKSON QUALIFICATION: BEng (Hons), MEngSc DATE: 27.03.26

Government of South Australia
Department for Infrastructure and Transport

PROJECT No.: 31921901 | FILE No.: 2020/10577
DESIGN No.: #15551401 | SURVEY No.: #15551401
PROJECT START ROAD RUNNING DISTANCE: NOT APPLICABLE
PROJECT END ROAD RUNNING DISTANCE: NOT APPLICABLE
SCALES: 1:100

ROAD No. 6200
RICHMOND ROAD
RAILWAY TERRACE - SOUTH ROAD; MILE END SOUTH
BRIDGE OVER KESWICK CREEK
DECK CONCRETE

DESIGNED:	DRAFTED:	ACCEPTED FOR USE:	ACCEPTANCE FORM KNET No.:	DRAWING No.:	SHEET No.:	AMEND No.:
T2DA	T2DA	M. SHORT TITLE: DIRECTOR OF ENG. DATE: 28.03.26	24467656 IN ACCORDANCE WITH DP013	8011 SHEET LATITUDE -34.94188	2871091 SHEET LONGITUDE 138.57397	0

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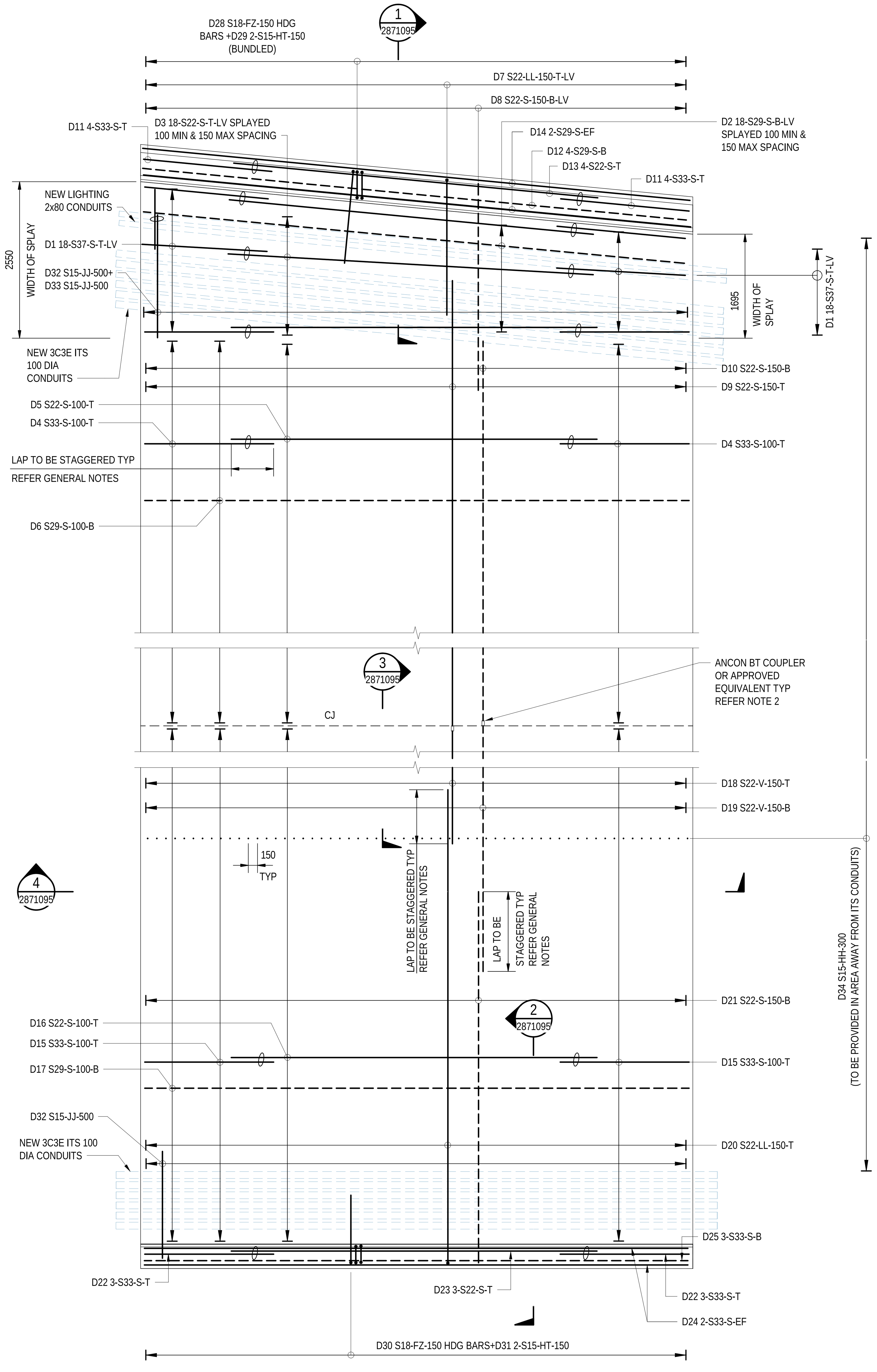
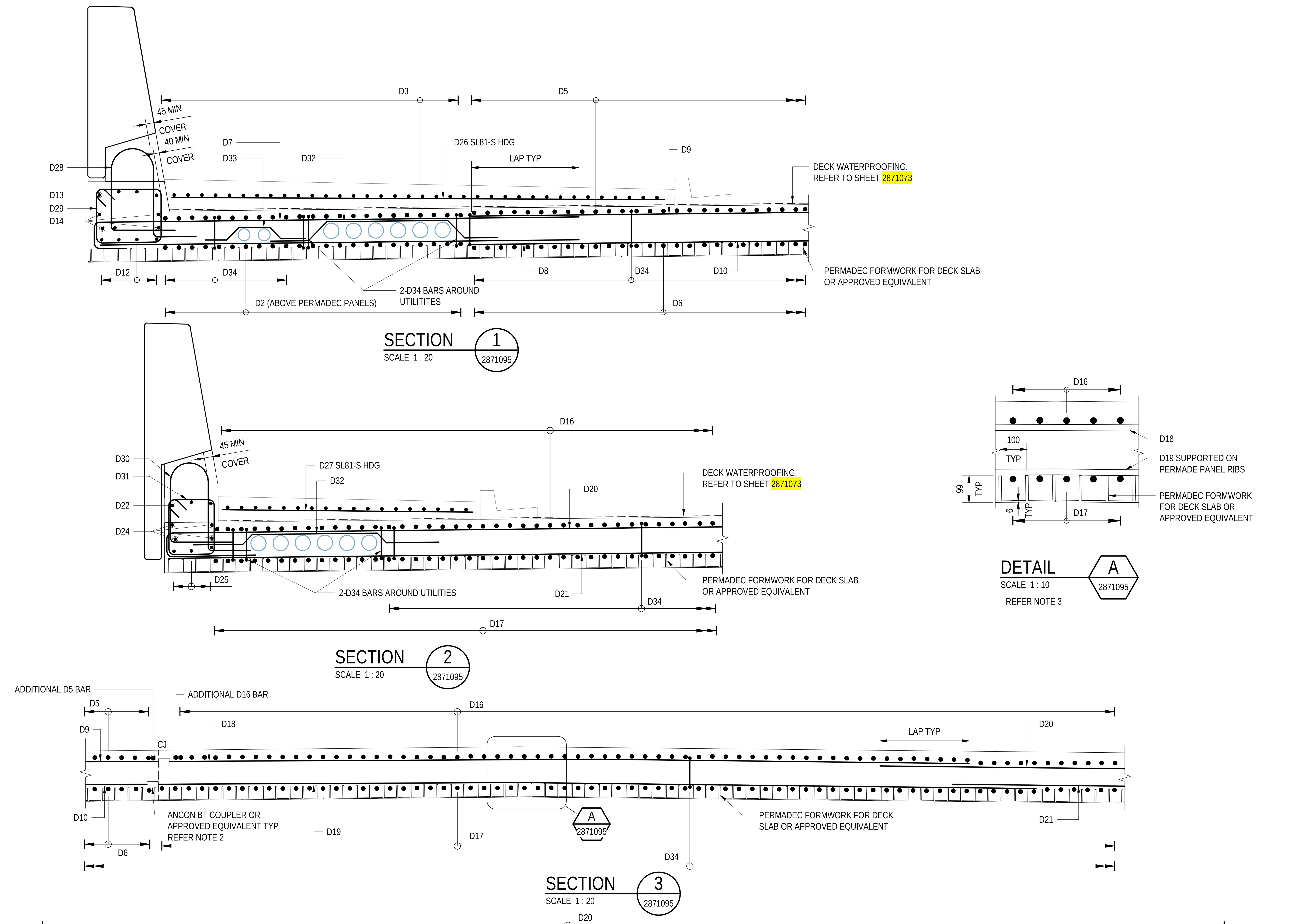
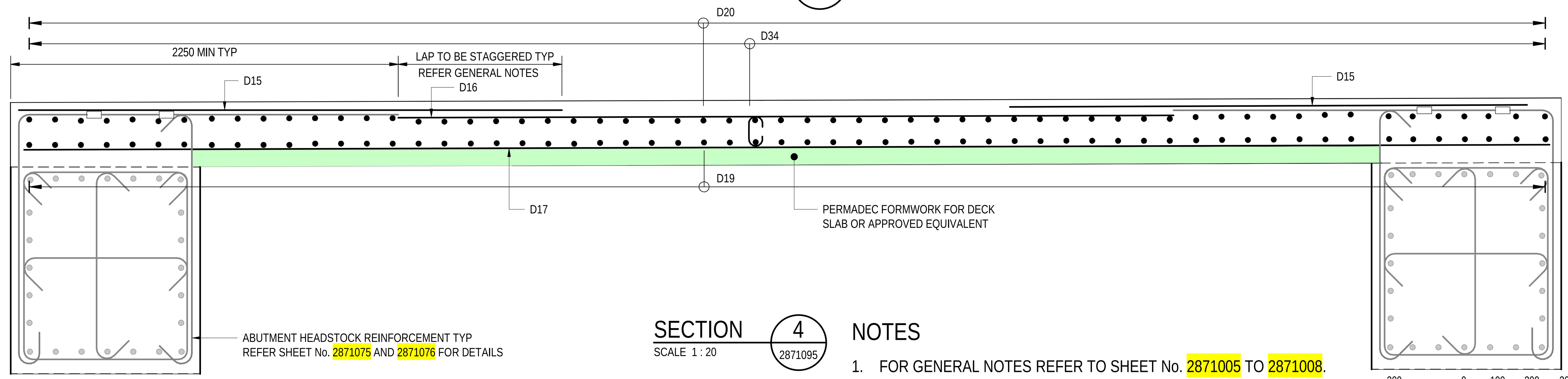
FILE NAME: Autodesk Docs://T2D - Torrens to Darlington /T2DPAA-T2D-C3S-BR-MSD-100302.rvt



PLAN
SCALE 1 : 50





SECTION 4
SCALE 1 : 20

NOTES

1. FOR GENERAL NOTES REFER TO SHEET No. 2871005 TO 2871008.
2. STAGGER EVERY SECOND COUPLER BY 150mm NOMINAL.
3. FOR DETAILS OF PERMADEC PERMANENT FORMWORK CONSIDERED IN DESIGN, REFER TO DESIGN REPORT T2DPAA-T2D-C3S-BR-DRP-100201 APPENDIX K. ANY DEVIATION IN THE PANEL GEOMETRY MUST BE APPROVED BY THE DESIGN ENGINEER BEFORE PROCEEDING FOR CONSTRUCTION.

1:10 — 200 0 100 200 300 400
1:20 — 400 0 200 400 600 800

PROJECT DOCUMENT REFERENCE
T2DPAA-T2D-C3S-BR-DRG-101095

FOR CONSTRUCTION

No.	AMENDMENT DESCRIPTION	BY	CHECK	ACCEPTANCE	DATE
0	ISSUED FOR CONSTRUCTION (KNet No: 24467656)	T2DA	T2DA	M. SHORT	28.03.26

INDEX SHEET REFERENCE: 8011 SHEET 2871002

T2D TORRENS TO DARLINGTON

UNCONTROLLED COPY WHEN PRINTED — 100 MILLIMETRES ON ORIGINAL DRAWING — ALL DIMENSIONS ARE IN MILLIMETRES UNLESS SHOWN OTHERWISE

DESIGNER: A.SINGHAL / S.PODDAR
QUALIFICATION: MEng BEng
DATE: 25.03.26
REVIEWER: R.WOZNIAK
QUALIFICATION: FIEAust CPEng NER RPEV RPEQ MICE CEng
DATE: 25.03.26
INDEPENDENT DESIGN CERTIFIER (IF REQUIRED): T. JACKSON
QUALIFICATION: BEng (Hons), MEngSc
DATE: 27.03.26

Government of South Australia
Department for Infrastructure and Transport

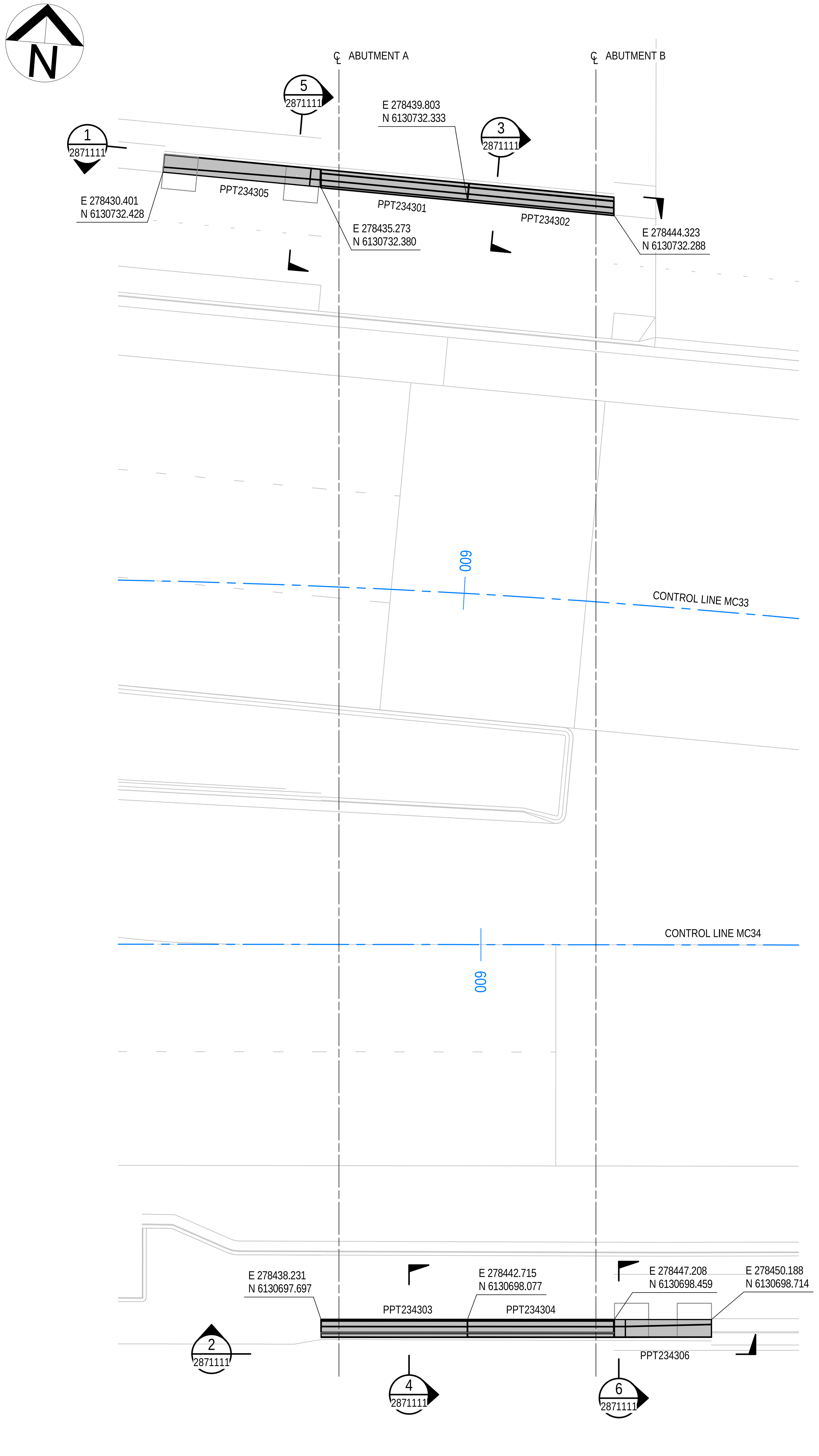
PROJECT No.: 31921901 | FILE No.: 2020/10577
DESIGN No.: #15551401 | SURVEY No.: #15551401
PROJECT START ROAD RUNNING DISTANCE: NOT APPLICABLE
PROJECT END ROAD RUNNING DISTANCE: NOT APPLICABLE
SCALES: 1000 0 500 1000 1500 2000 — 1:50

ROAD No. 6200
RICHMOND ROAD
RAILWAY TERRACE - SOUTH ROAD; MILE END SOUTH
BRIDGE OVER KESWICK CREEK
DECK REINFORCEMENT

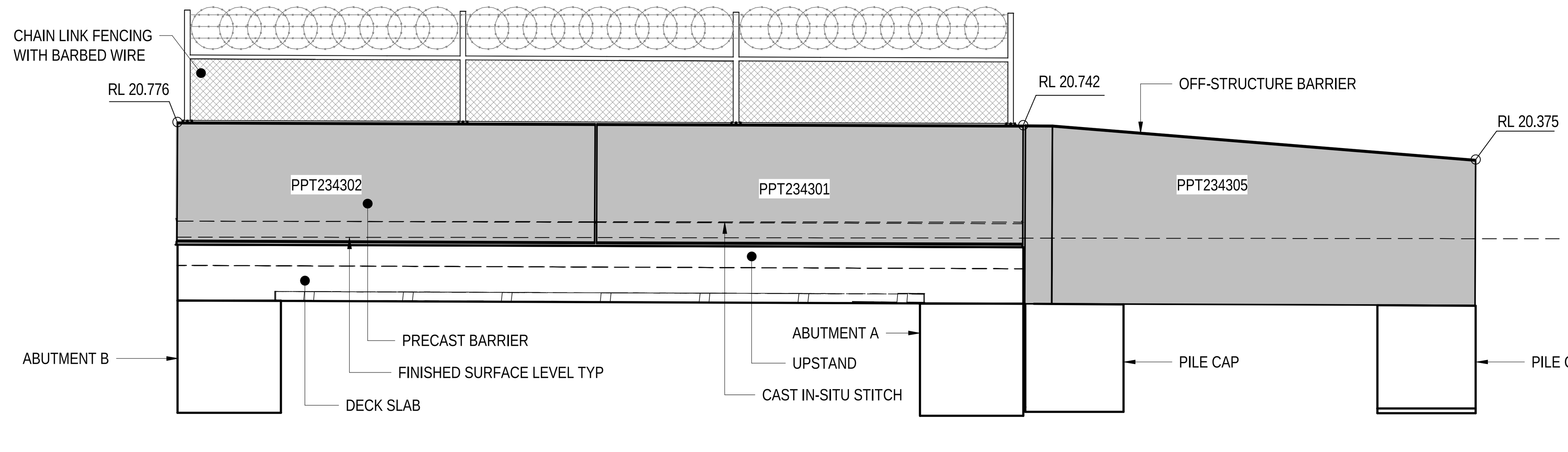
DESIGNED:	DRAFTED:	ACCEPTED FOR USE:	ACCEPTANCE FORM KNET No.:	DRAWING No.:	SHEET No.:	AMEND No.:
T2DA	T2DA	M. SHORT, TITLE: DIRECTOR OF ENG. DATE: 28.03.26	24467656 IN ACCORDANCE WITH DP013	8011	2871095	0

SHEET LATITUDE -34.94188 SHEET LONGITUDE 138.57397

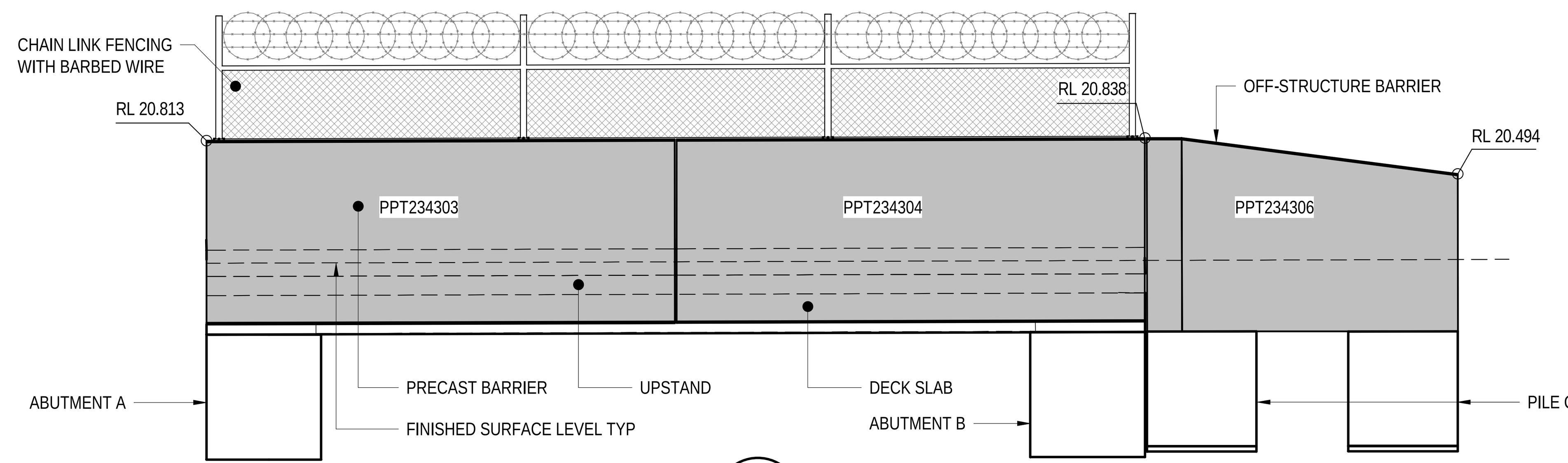
FILE NAME: Autodesk Docs://T2D - Torrens to Darlington /T2DPAA-T2D-C3S-BR-M3D-100302.rvt



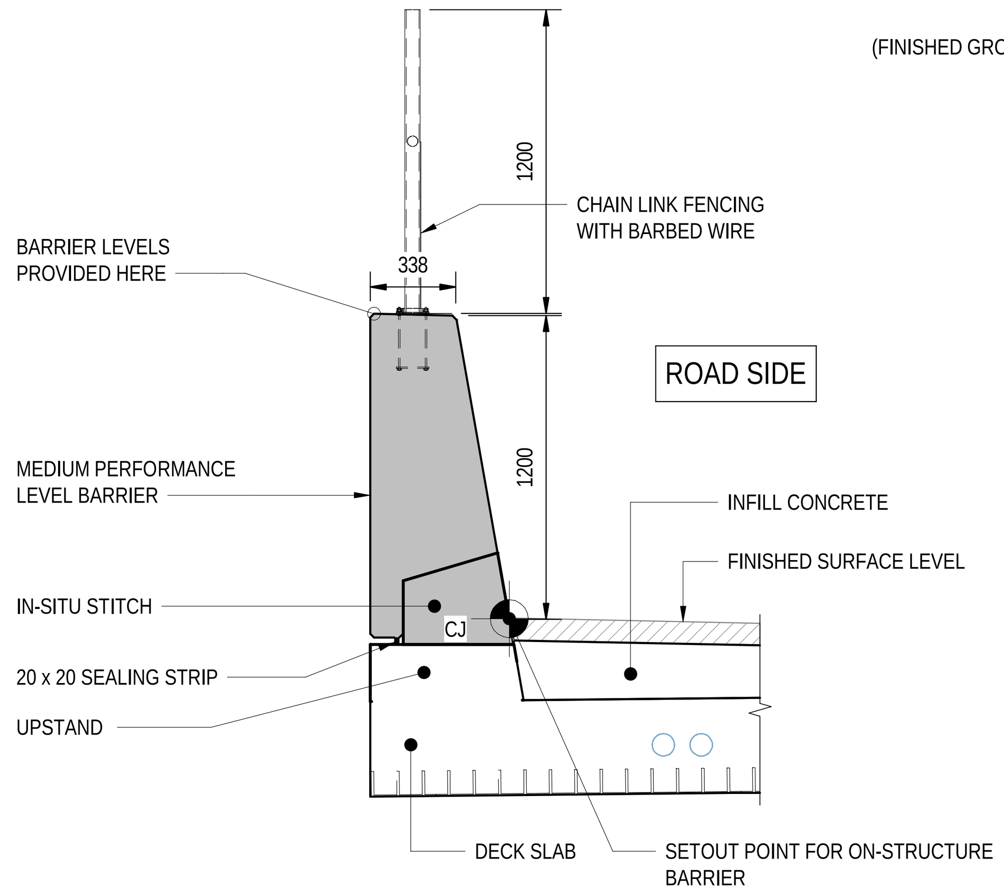
PLAN
SCALE 1 : 100



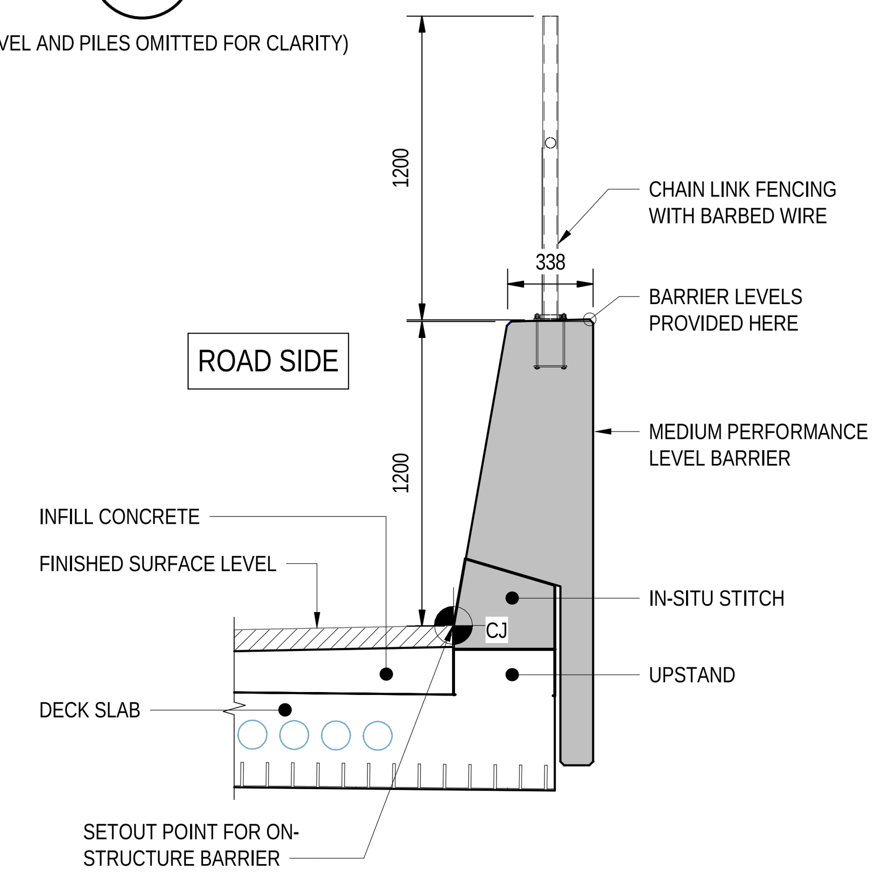
SECTION 1 (2871111)
SCALE 1 : 50
(FINISHED GROUND LEVEL AND PILES OMITTED FOR CLARITY)



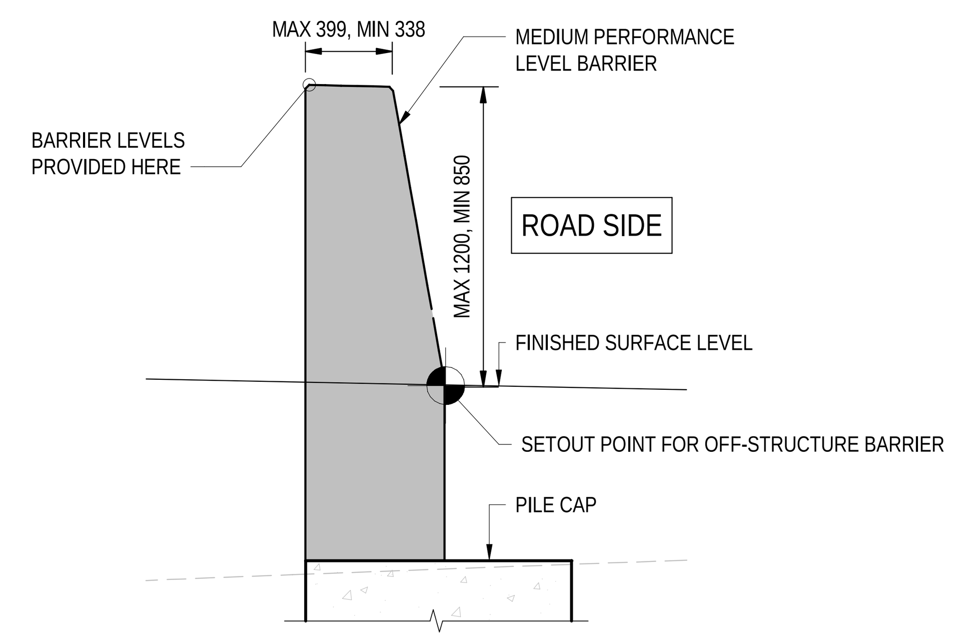
SECTION 2 (2871111)
SCALE 1 : 50
(FINISHED GROUND LEVEL AND PILES OMITTED FOR CLARITY)



SECTION 3 (2871111)
SCALE 1 : 25



SECTION 4 (2871111)
SCALE 1 : 25



SECTION 5 6 (2871111)
SCALE 1 : 25
SIMILAR AND
MIRRORED

NOTES

1. FOR GENERAL NOTES REFER TO SHEET No. 2871005 TO 2871008.

BARRIER SCHEDULE

SITE ID	BARRIER TYPE	BARRIER PERFORMANCE LEVEL	LENGTH "L" (mm)	TAPERED END SKEW ANGLES		ARCHITECTURAL RELIEF PATTERN	COVER PLATE TYPE		CAST IN ELEMENTS				ATTACHMENT TYPE	POST ANCHOR SPACING		REMARKS
				"A" LEFT	"B RIGHT"		LEFT HAND SIDE	RIGHT HND SIDE	HORIZONTAL CONDUIT CONFIGURATION	VERTICAL CONDUIT FROM ROAD	JUNCTION BOX TYPE	OFFSET FROM LEFT "Y"		L1	L2	
PPT234301	MPB-02-B	MEDIUM	4510	85° 08' 40"	90° 00' 00"	NO	CP-02	N/A	N/A	N/A	N/A	N/A	FENCING	120	1455	
PPT234302	MPB-02-B	MEDIUM	4510	90° 00' 00"	94° 51' 20"	NO	N/A	N/A	N/A	N/A	N/A	N/A	FENCING	1455	120	
PPT234303	MPB-02-A	MEDIUM	4490	90° 00' 00"	90° 00' 00"	NO	N/A	N/A	N/A	N/A	N/A	N/A	FENCING	1450	120	TAIL LENGTH DIFFERS FROM STANDARD BARRIER DETAILS
PPT234304	MPB-02-A	MEDIUM	4490	90° 00' 00"	90° 00' 00"	NO	CP-02	N/A	N/A	N/A	N/A	N/A	FENCING	120	1450	TAIL LENGTH DIFFERS FROM STANDARD BARRIER DETAILS
PPT234305	NON STANDARD	MEDIUM	4860	90° 00' 00"	94° 51' 20"	NO	N/A	CP-02	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
PPT234306	NON STANDARD	MEDIUM	2980	90° 00' 00"	90° 00' 00"	NO	N/A	CP-02	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

500 0 250 500 750 1000
1:25
1000 0 500 1000 1500 2000
1:50

PROJECT DOCUMENT REFERENCE
T2DPAA-T2D-C3S-BR-DRG-101111

FOR CONSTRUCTION

No.	AMENDMENT DESCRIPTION	BY	CHECK	ACCEPTANCE	DATE
0	ISSUED FOR CONSTRUCTION (KNet No: 24467656)	T2DA	T2DA	M. SHORT	28.03.26

INDEX SHEET REFERENCE: 8011 SHEET 2871002

T2D TORRENS TO DARLINGTON

UNCONTROLLED COPY WHEN PRINTED | 100 MILLIMETRES ON ORIGINAL DRAWING | ALL DIMENSIONS ARE IN MILLIMETRES UNLESS SHOWN OTHERWISE

DESIGNER: A.SINGHAL / S.PODDAR
QUALIFICATION: MEng BEng
DATE: 25.03.26
REVIEWER: R.WOZNIAK
QUALIFICATION: FIEAust CPEng NER RPEV RPEQ MICE CEng
DATE: 25.03.26
INDEPENDENT DESIGN CERTIFIER (IF REQUIRED): T. JACKSON
QUALIFICATION: BEng (Hons), MEngSc
DATE: 27.03.26

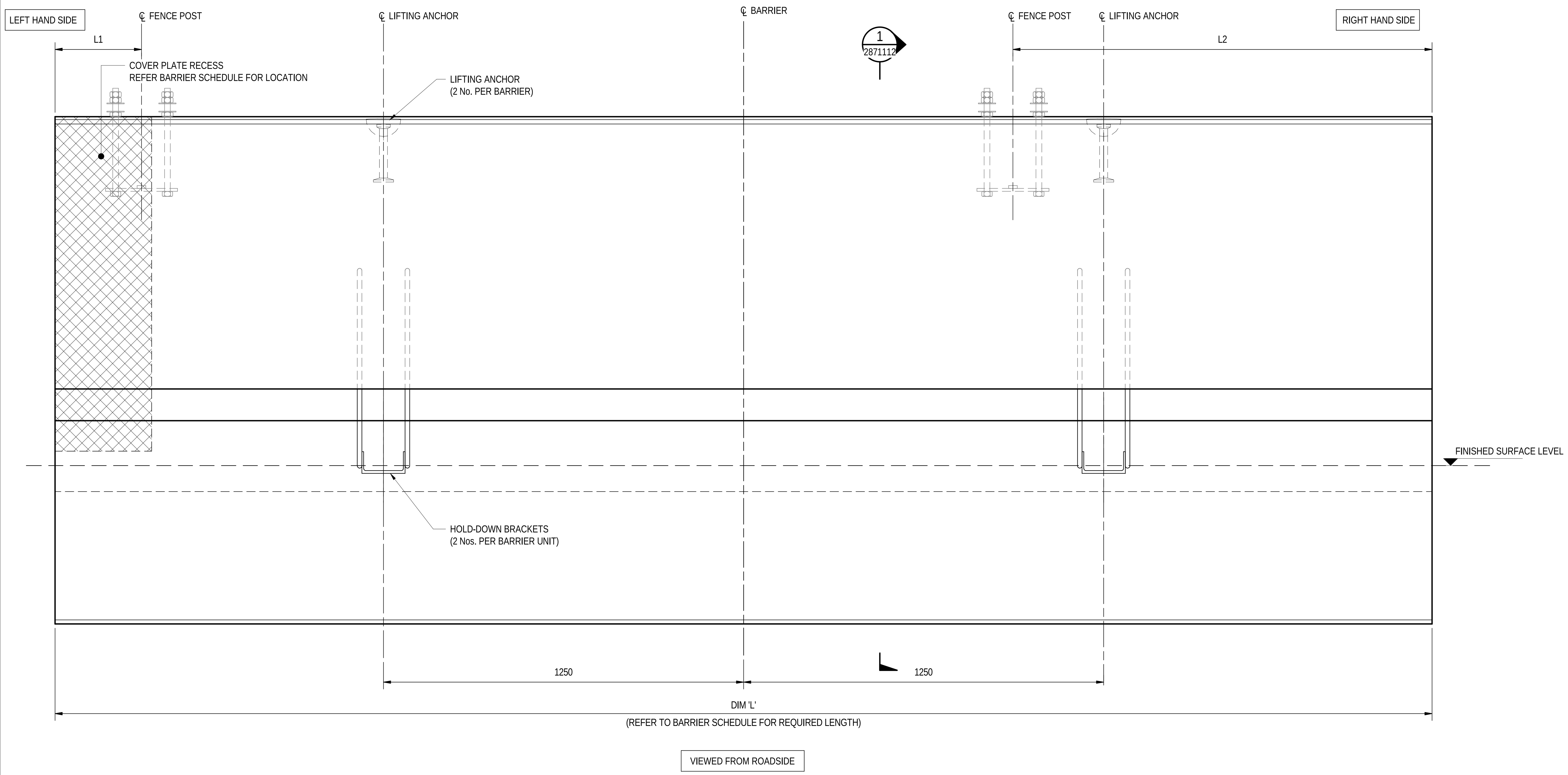
Government of South Australia
Department for Infrastructure and Transport

PROJECT No.: 31921901
FILE No.: 2020/10577
DESIGN No.: #15551401
SURVEY No.: #15551401
PROJECT START ROAD RUNNING DISTANCE: NOT APPLICABLE
PROJECT END ROAD RUNNING DISTANCE: NOT APPLICABLE
SCALES: 2000 0 1000 2000 3000 4000 1:100

ROAD No. 6200
RICHMOND ROAD
RAILWAY TERRACE - SOUTH ROAD; MILE END SOUTH BRIDGE OVER KESWICK CREEK BARRIER LAYOUT

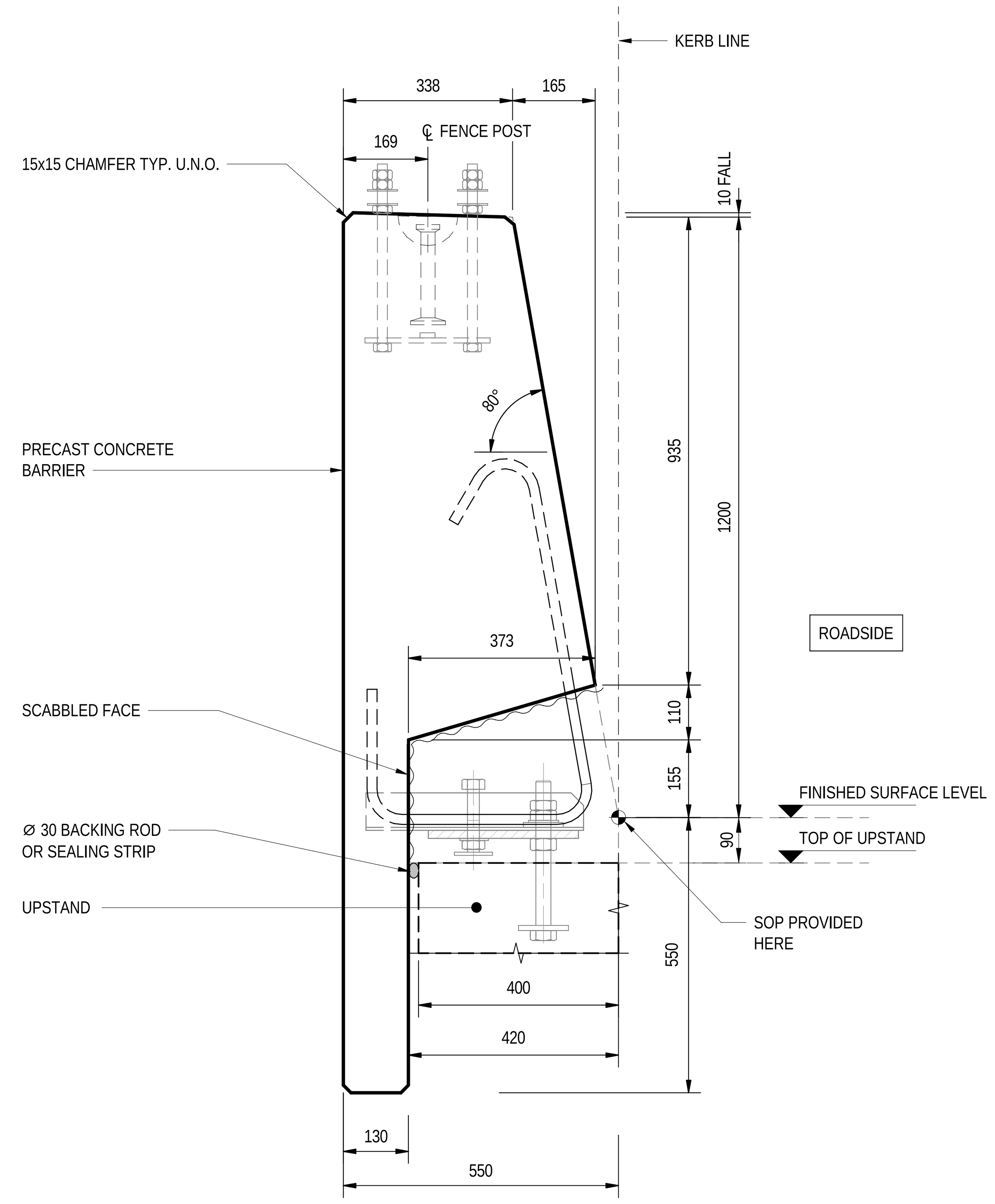
DESIGNED: T2DA
DRAFTED: T2DA
ACCEPTED FOR USE: M. SHORT
TITLE: DIRECTOR OF ENG.
DATE: 28.03.26
ACCEPTANCE FORM KNET No.: 24467656
IN ACCORDANCE WITH DP013
DRAWING No.: 8011
SHEET No.: 2871111
AMEND No.: 0
SHEET LATITUDE -34.94188 SHEET LONGITUDE 138.57397

FILE NAME: Autodesk Docs://T2D - Torrens to Darlington /T2DPAA-T2D-C3S-BR-M3D-100302.rvt



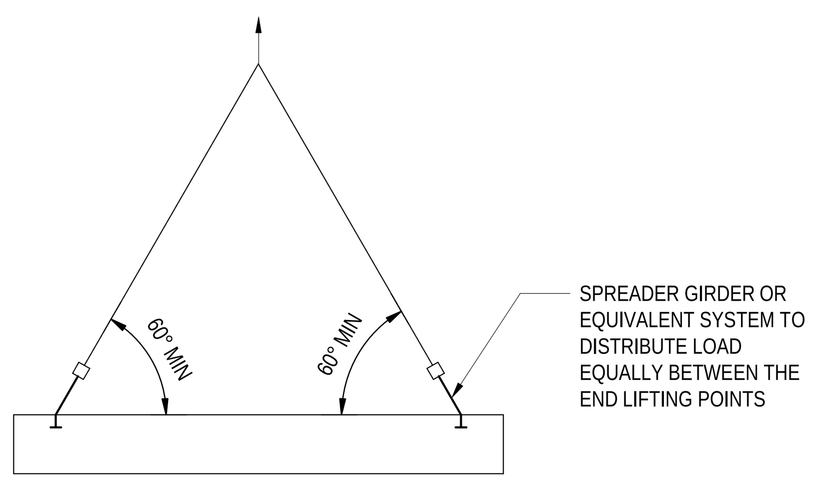
TYPICAL ELEVATION - PRECAST MEDIUM PERFORMANCE LEVEL BARRIER TYPE - MPB-02-A

SCALE 1 : 10



SECTION 1 (2871112)

SCALE 1 : 10



LIFTING DIAGRAM

LIFTING DEVICE DESIGNED AND
CERTIFIED BY PRECAST SUPPLIER
NTS

NOTES

1. FOR GENERAL NOTES REFER TO SHEET No. 2871005 TO 2871008.

PROJECT DOCUMENT REFERENCE
T2DPAA-T2D-C3S-BR-DRG-101112

FOR CONSTRUCTION

No.	AMENDMENT DESCRIPTION	BY	CHECK	ACCEPTANCE	DATE
0	ISSUED FOR CONSTRUCTION (KNet No: 24467656)	T2DA	T2DA	M. SHORT	28.03.26

INDEX SHEET REFERENCE: 8011 SHEET 2871002

T2D → TORRENS TO DARLINGTON

UNCONTROLLED COPY WHEN PRINTED | 100 MILLIMETRES ON ORIGINAL DRAWING | ALL DIMENSIONS ARE IN MILLIMETRES UNLESS SHOWN OTHERWISE

DESIGNER: A.SINGHAL / S.PODDAR
QUALIFICATION: MEng BEng
DATE: 25.03.26

REVIEWER: R.WOZNIAK
QUALIFICATION: FIEAust CPEng NER RPEV RPEQ MICE CEng
DATE: 25.03.26

INDEPENDENT DESIGN CERTIFIER (IF REQUIRED): T. JACKSON
QUALIFICATION: BEng (Hons), MEngSc
DATE: 27.03.26

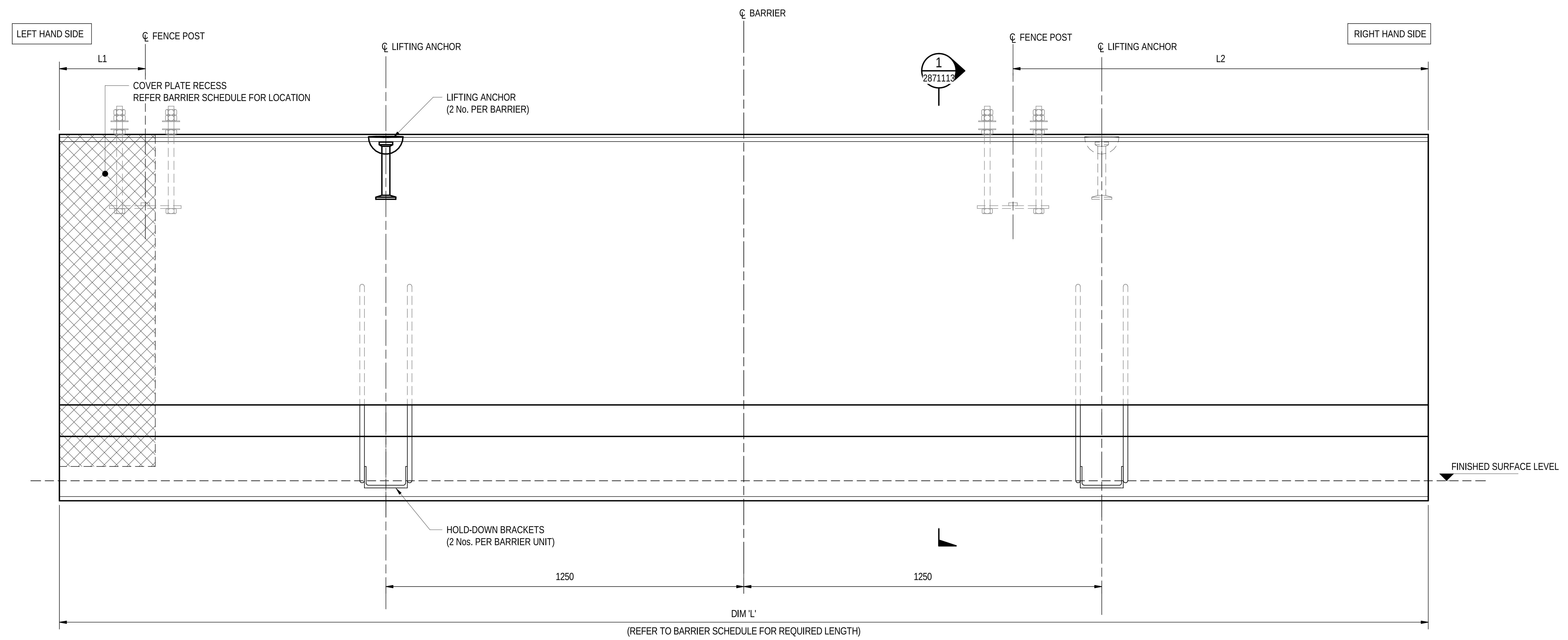
Government of South Australia
Department for Infrastructure and Transport

PROJECT No.: 31921901
FILE No.: 2020/10577
DESIGN No.: #15551401
SURVEY No.: #15551401
PROJECT START ROAD RUNNING DISTANCE: NOT APPLICABLE
PROJECT END ROAD RUNNING DISTANCE: NOT APPLICABLE
SCALES: 200 0 100 200 300 400 1:10

ROAD No. 6200
RICHMOND ROAD
RAILWAY TERRACE - SOUTH ROAD; MILE END SOUTH
BRIDGE OVER KESWICK CREEK
BARRIER DETAILS - SHEET 01

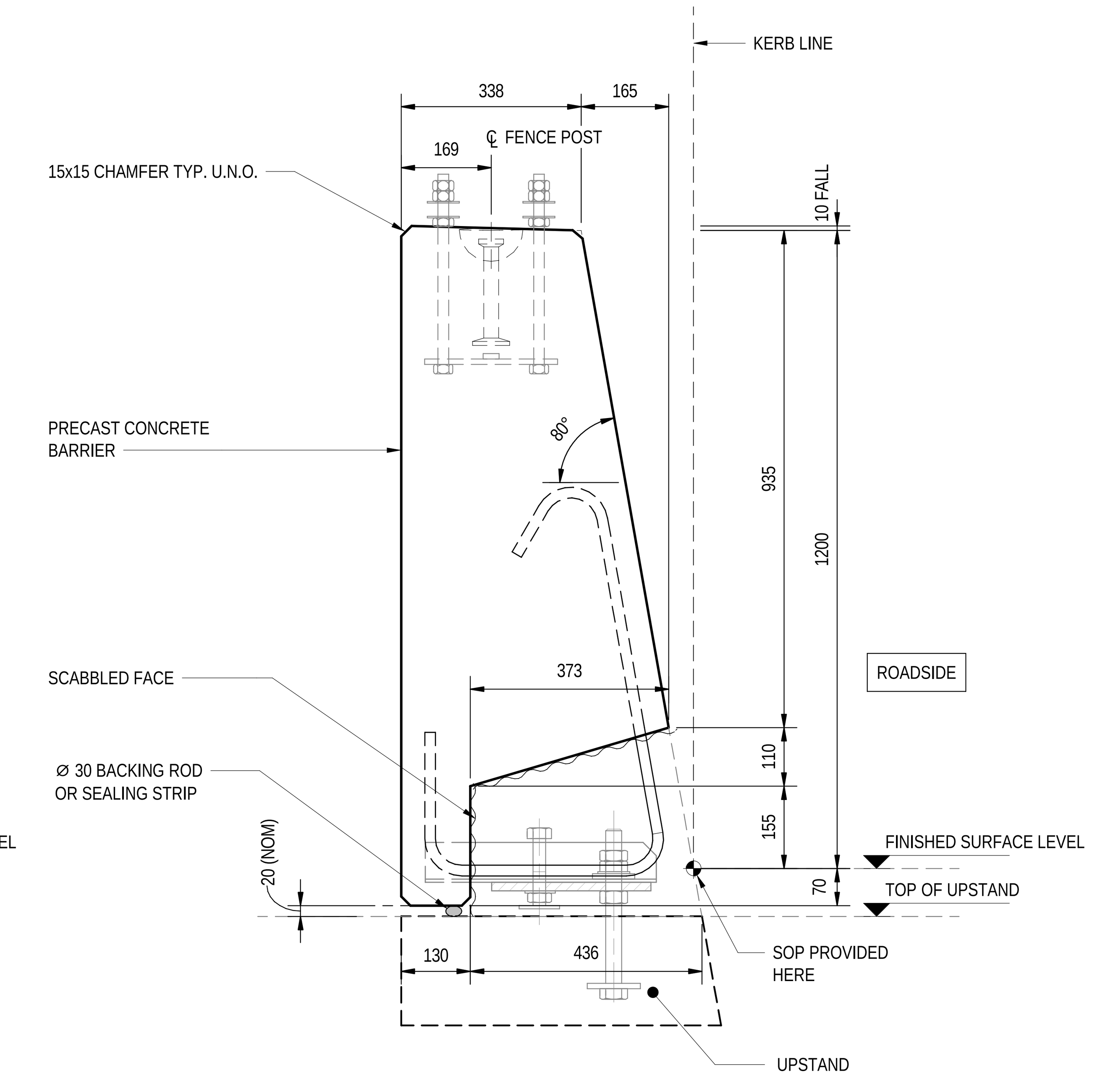
DESIGNED: T2DA
DRAFTED: T2DA
ACCEPTED FOR USE: M. SHORT
TITLE: DIRECTOR OF ENG.
DATE: 28.03.26
ACCEPTANCE FORM KNET No.: 24467656
IN ACCORDANCE WITH DP013
DRAWING No.: 8011
SHEET No.: 2871112
AMEND No.: 0
SHEET LATITUDE -34.94188 SHEET LONGITUDE 138.57397

FILE NAME: Autodesk Docs://T2D - Torrens to Darlington /T2DPAA-T2D-C3S-BR-M3D-100302.rvt



TYPICAL ELEVATION - PRECAST MEDIUM PERFORMANCE LEVEL BARRIER TYPE - MPB-02-B

SCALE 1 : 10



SECTION 1 / 2871113

SCALE 1 : 10

NOTES

1. FOR GENERAL NOTES REFER TO SHEET No. 2871005 TO 2871008.

PROJECT DOCUMENT REFERENCE
T2DPAA-T2D-C3S-BR-DRG-101113

FOR CONSTRUCTION

No.	AMENDMENT DESCRIPTION	BY	CHECK	ACCEPTANCE	DATE
0	ISSUED FOR CONSTRUCTION (KNet No: 24467656)	T2DA	T2DA	M. SHORT	28.03.26

INDEX SHEET REFERENCE: 8011 SHEET 2871002

T2D → TORRENS TO DARLINGTON

UNCONTROLLED COPY WHEN PRINTED ◄ 100 MILLIMETRES ON ORIGINAL DRAWING ► ALL DIMENSIONS ARE IN MILLIMETRES UNLESS SHOWN OTHERWISE

DESIGNER: A.SINGHAL / S.PODDAR
QUALIFICATION: MEng BEng
DATE: 25.03.26

REVIEWER: R.WOZNIAK
QUALIFICATION: FIEAust CPEng NER RPEV RPEQ MICE CEng
DATE: 25.03.26

INDEPENDENT DESIGN CERTIFIER (IF REQUIRED): T. JACKSON
QUALIFICATION: BEng (Hons), MEngSc
DATE: 27.03.26

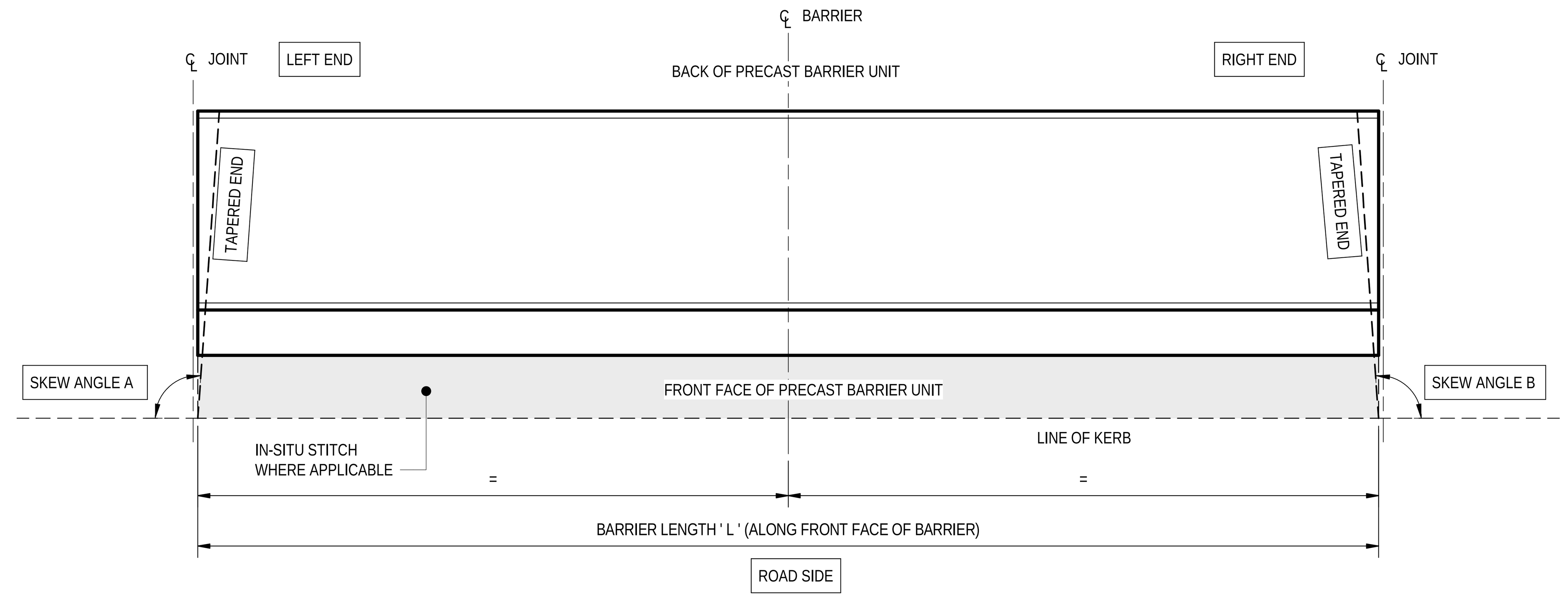
Government of South Australia
Department for Infrastructure and Transport

PROJECT No.: 31921901
FILE No.: 2020/10577
DESIGN No.: #15551401
SURVEY No.: #15551401
PROJECT START ROAD RUNNING DISTANCE: NOT APPLICABLE
PROJECT END ROAD RUNNING DISTANCE: NOT APPLICABLE
SCALES: 200 0 100 200 300 400 1:10

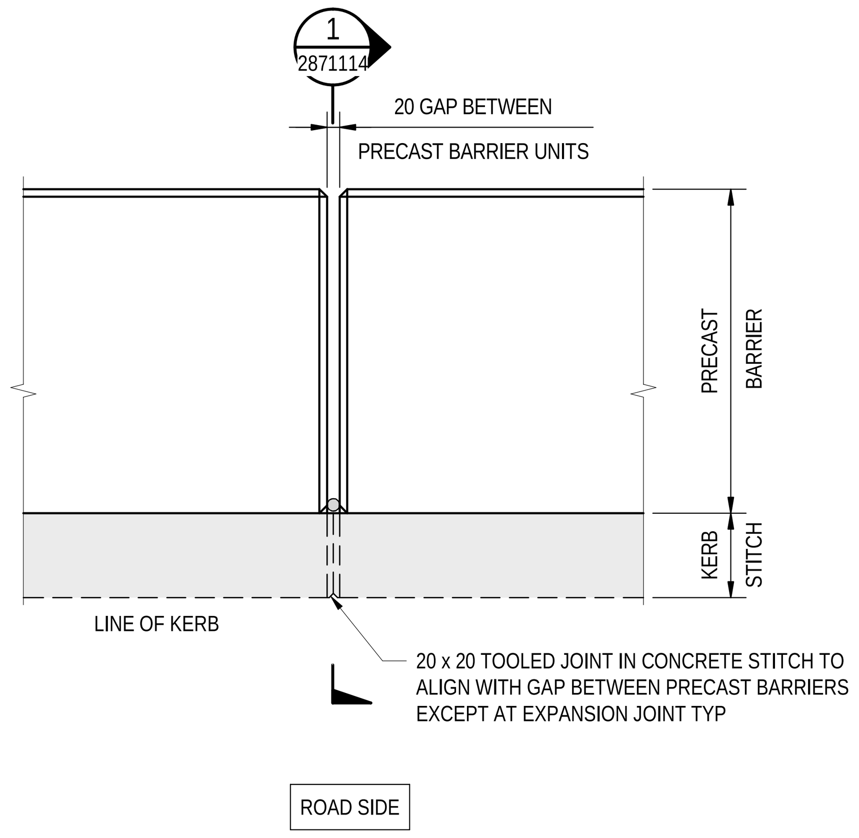
ROAD No. 6200
RICHMOND ROAD
RAILWAY TERRACE - SOUTH ROAD; MILE END SOUTH
BRIDGE OVER KESWICK CREEK
BARRIER DETAILS - SHEET 02

DESIGNED: T2DA
DRAFTED: T2DA
ACCEPTED FOR USE: M. SHORT
TITLE: DIRECTOR OF ENG.
DATE: 28.03.26
ACCEPTANCE FORM KNET No.: 24467656
IN ACCORDANCE WITH DP013
DRAWING No.: 8011
SHEET No.: 2871113
AMEND No.: 0
SHEET LATITUDE -34.94188
SHEET LONGITUDE 138.57397

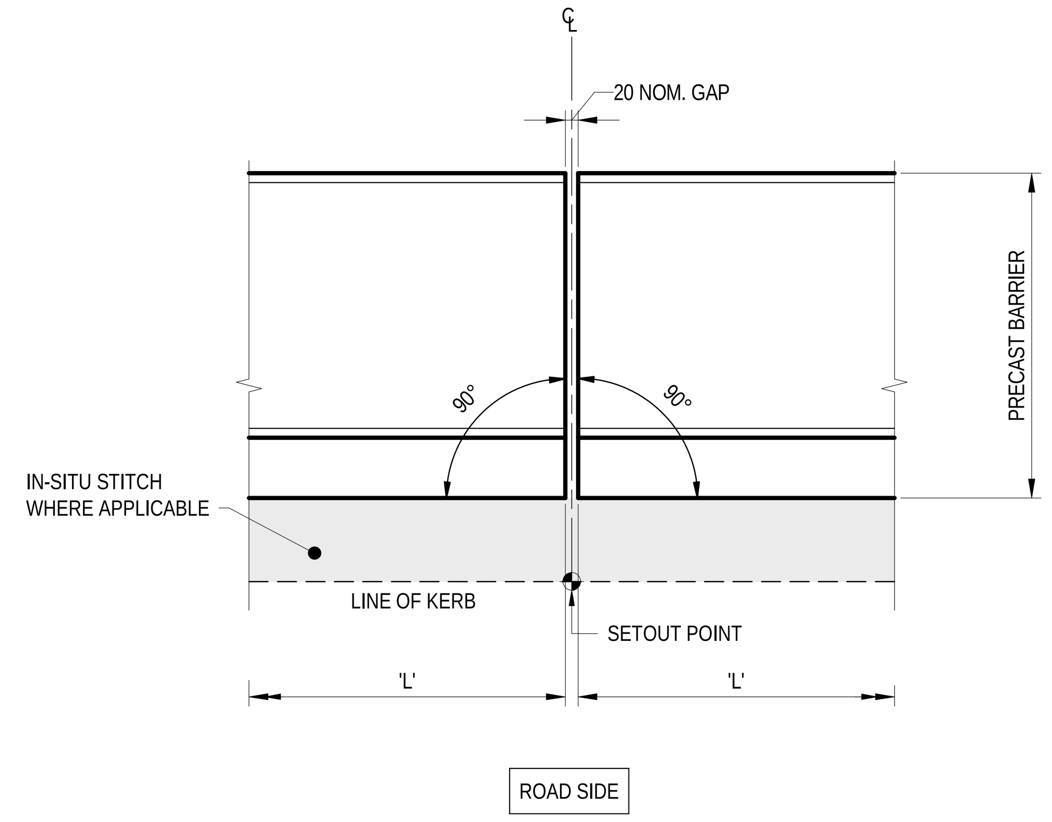
FILE NAME: Autodesk Docs://T2D - Torrens to Darlington /T2DPAA-T2D-C3S-BR-M3D-100302.rvt



PLAN



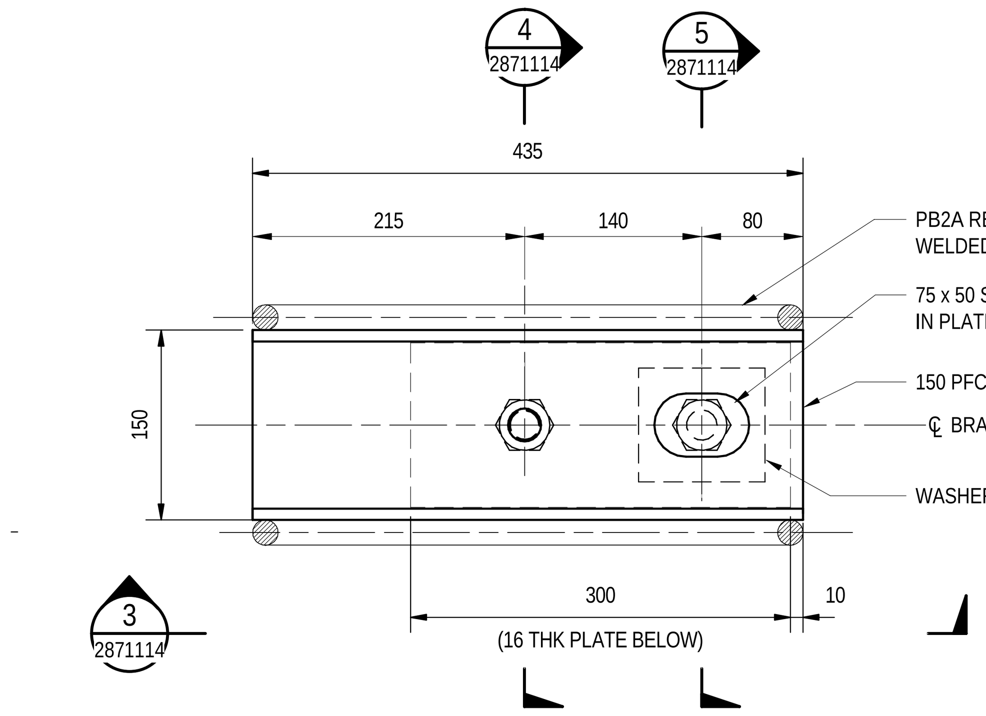
BARRIER JOINT DETAIL - PLAN



STRAIGHT LINE GEOMETRY

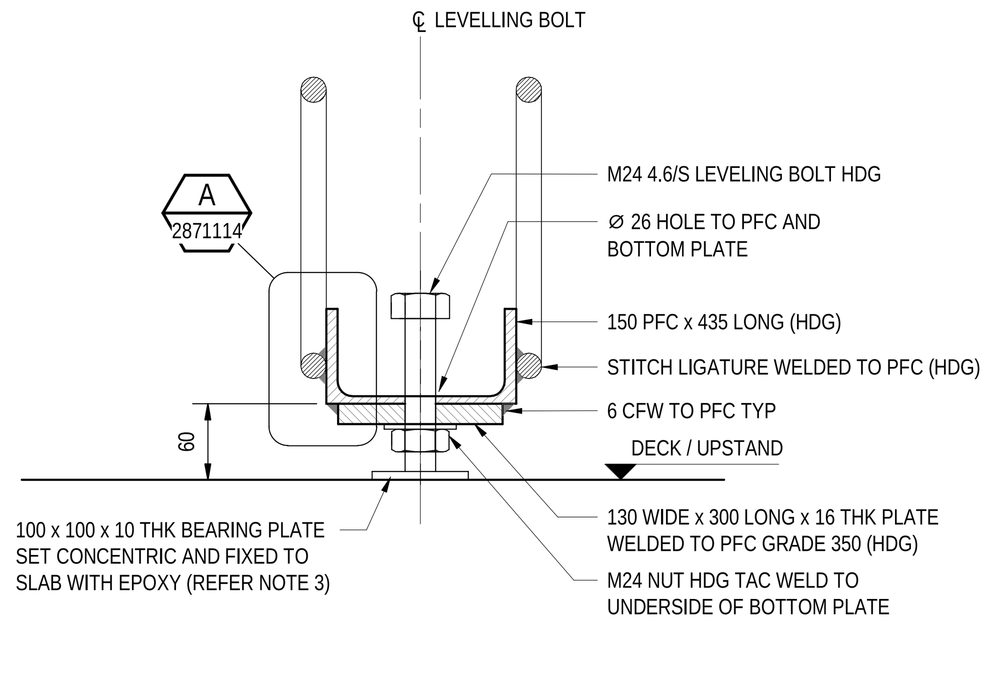
TYPICAL BARRIER SET OUT PLANS

N.T.S



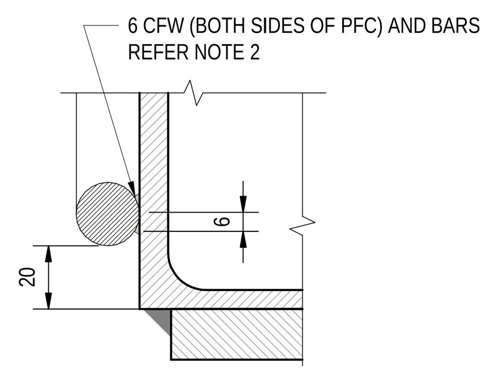
LEVELLING BRACKET - PLAN

SCALE 1 : 5



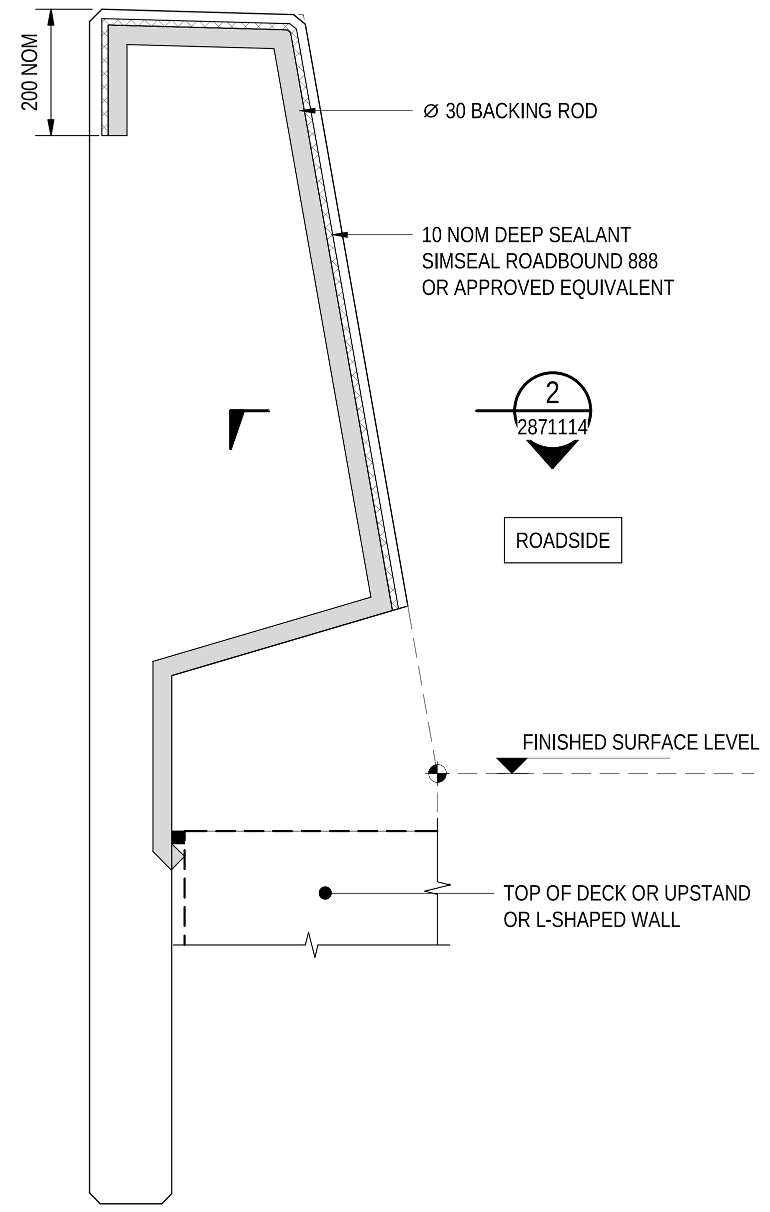
SECTION 4 (2871114)

SCALE 1 : 5



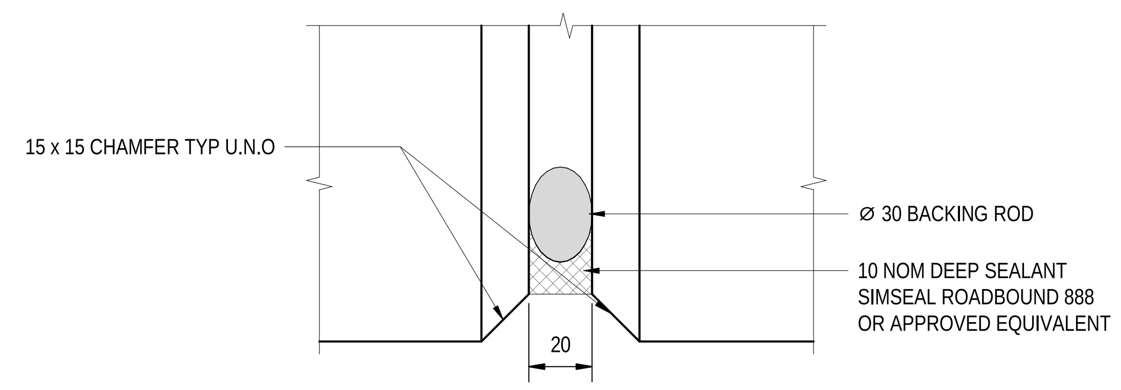
DETAIL A (2871114)

SCALE 1 : 2



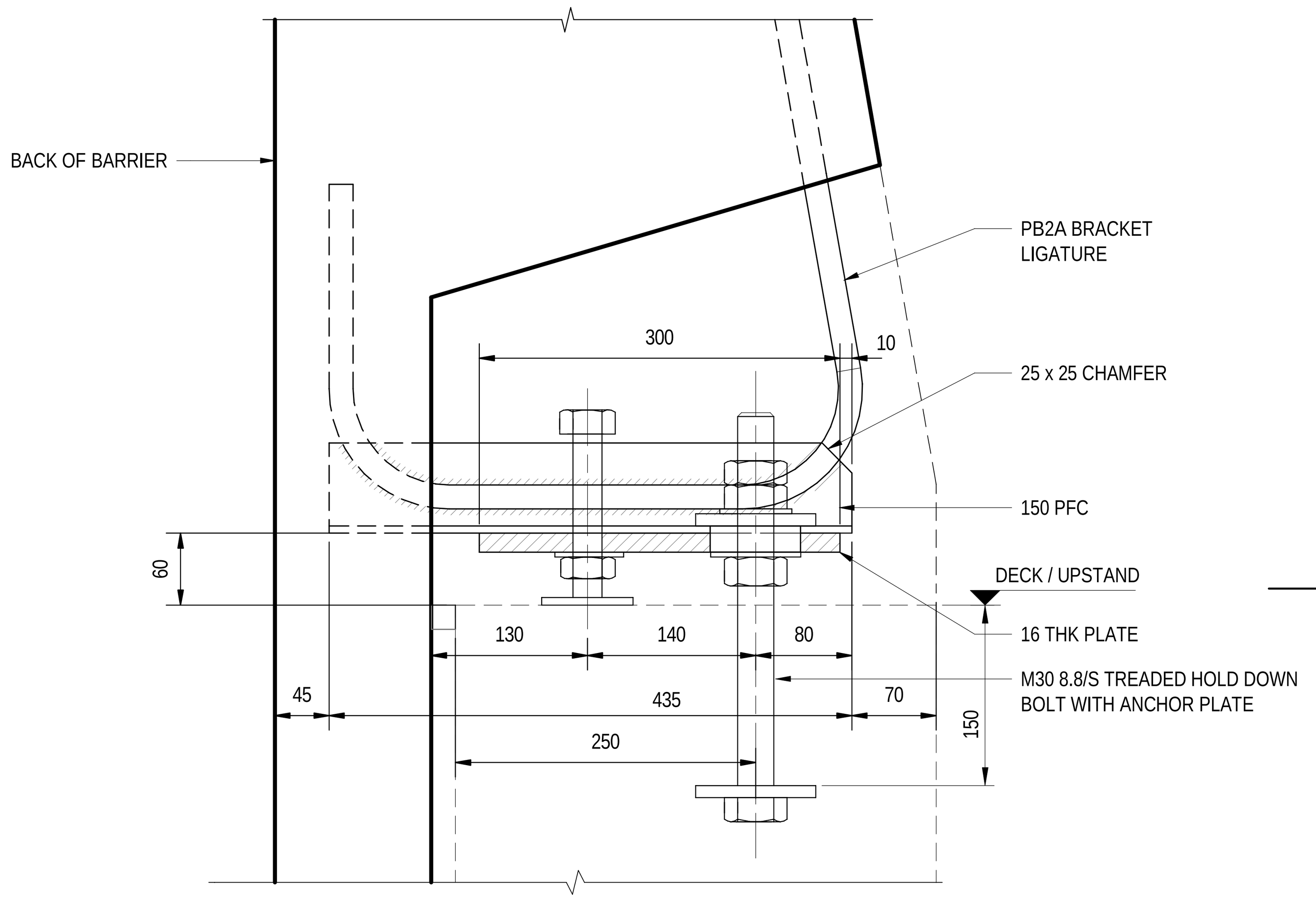
SECTION 1 (2871114)

SCALE 1 : 10



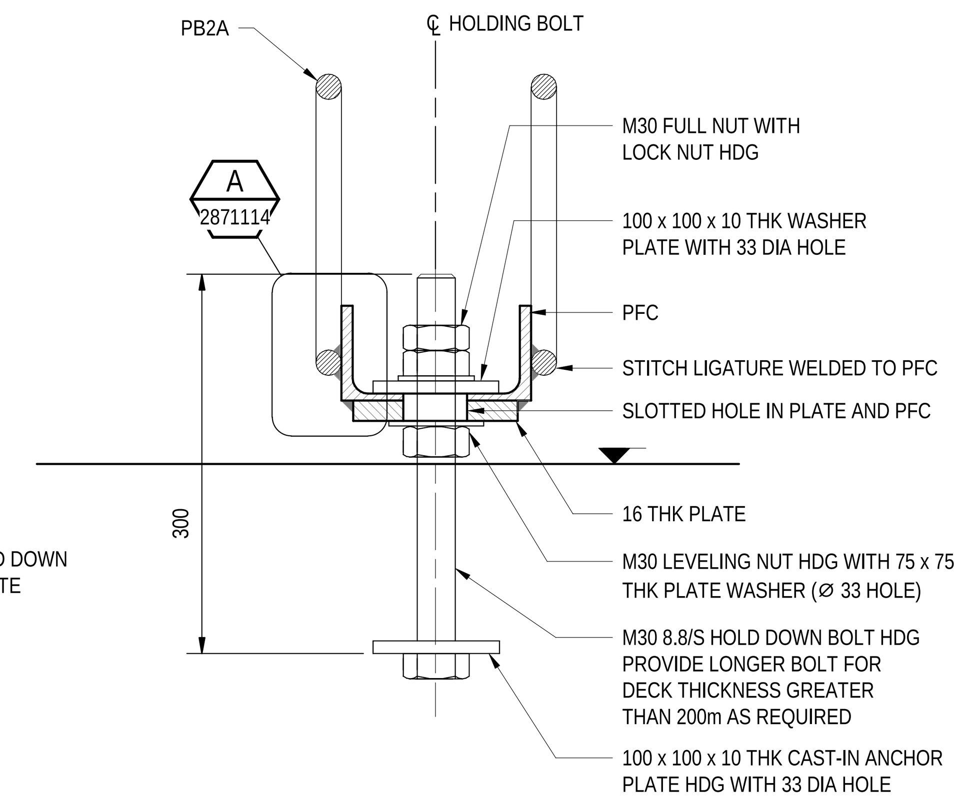
SECTION 2 (2871114)

SCALE 1 : 2



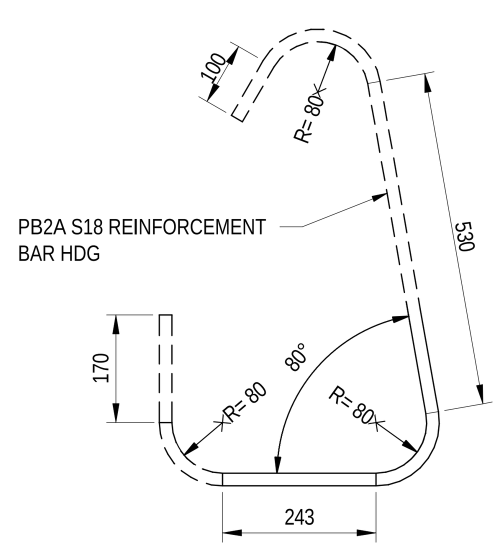
SECTION 3 (2871114)

SCALE 1 : 5



SECTION 5 (2871114)

SCALE 1 : 5



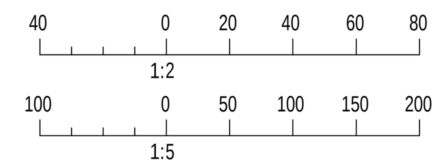
MPB-02-A AND MPB-02-B

HOLD DOWN BRACKET LIGATURE DETAIL

SCALE 1 : 10

NOTES

1. FOR GENERAL NOTES REFER TO SHEET No. 2871005 TO 2871008.
2. S18 LIG WELD SHALL BE CONSIDERED AS JOINT TYPE B1-1A AS PER TABLE F2 OF AS 1554.3.
3. LEVELLING BOLT SHALL NOT PROTRUDE MORE THAN 60mm BELOW THE BRACKET. IF THE SOFFIT OF THE BRACKET IS MORE THAN 60mm ABOVE THE CONCRETE SURFACE, ADD 75 x 75 x 10 PACKERS TO LIMIT THE BOLT PROTRUSION.
4. TEMPORARY WORK DESIGN FOR MAXIMUM WIND SPEEDS OF 25m/s DURING CONSTRUCTION.
5. NO ATTACHMENT SHALL BE INSTALLED WITH THE BARRIER DURING PLACEMENT AND UNTIL STITCH POUR ACHIEVED ITS DESIGN STRENGTH.



PROJECT DOCUMENT REFERENCE
T2DPAA-T2D-C3S-BR-DRG-101114

FILE NAME: Autodesk Docs://T2D - Torrens to Darlington /T2DPAA-T2D-C3S-BR-M3D-100302.rvt

FOR CONSTRUCTION

No.	AMENDMENT DESCRIPTION	BY	CHECK	ACCEPTANCE	DATE
0	ISSUED FOR CONSTRUCTION (KNet No: 24467656)	T2DA	T2DA	M. SHORT	28.03.26

INDEX SHEET REFERENCE: 8011 SHEET 2871002

T2D → TORRENS TO DARLINGTON

UNCONTROLLED COPY WHEN PRINTED | 100 MILLIMETRES ON ORIGINAL DRAWING | ALL DIMENSIONS ARE IN MILLIMETRES UNLESS SHOWN OTHERWISE

DESIGNER: A.SINGHAL / S.PODDAR
QUALIFICATION: MEng BEng
DATE: 25.03.26

REVIEWER: R.WOZNIAK
QUALIFICATION: FIEAust CPEng NER RPEV RPEQ MICE CEng
DATE: 25.03.26

INDEPENDENT DESIGN CERTIFIER (IF REQUIRED): T. JACKSON
QUALIFICATION: BEng (Hons), MEngSc
DATE: 27.03.26

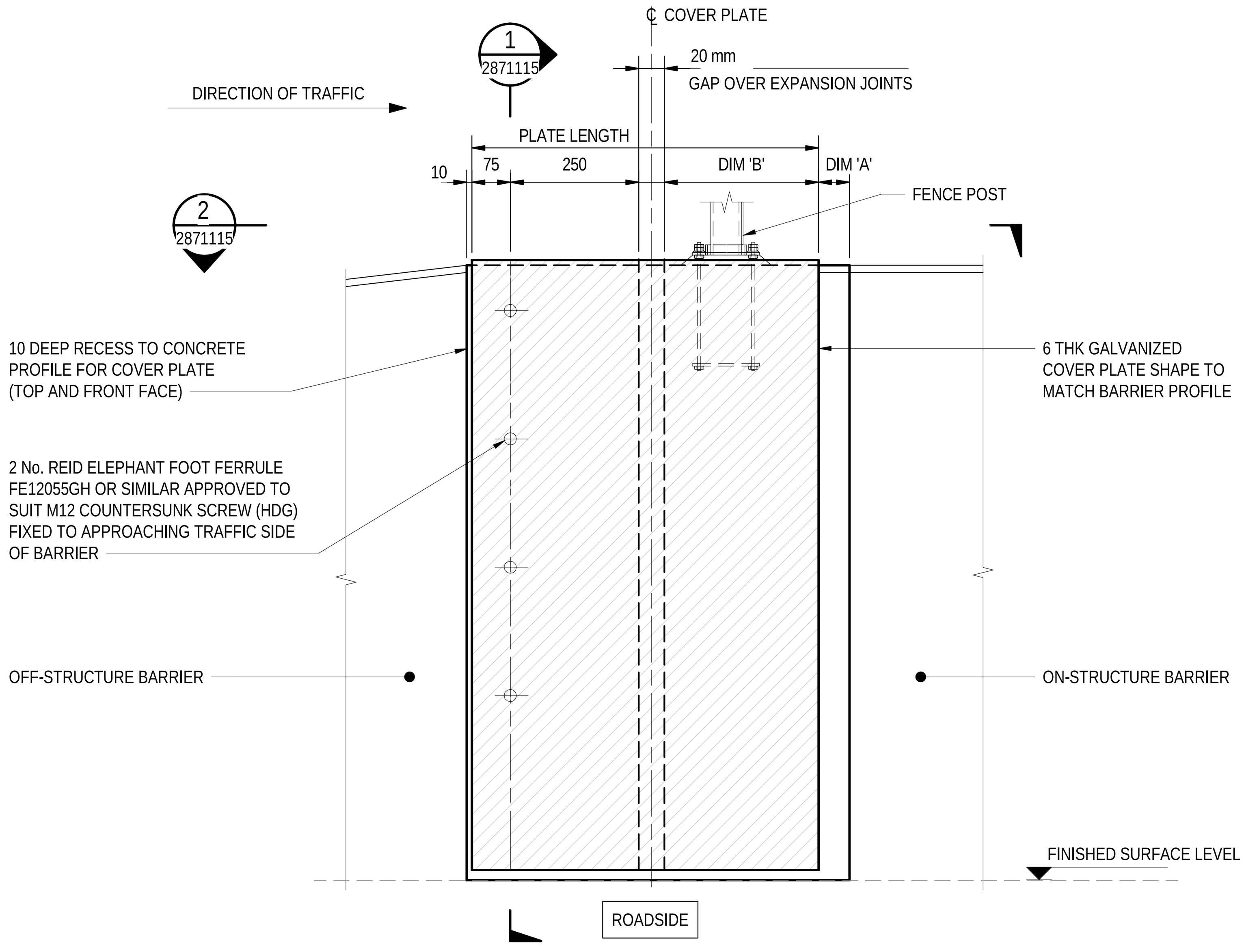
Government of South Australia
Department for Infrastructure and Transport

PROJECT No.: 31921901 | FILE No.: 2020/10577
DESIGN No.: #15551401 | SURVEY No.: #15551401
PROJECT START ROAD RUNNING DISTANCE: NOT APPLICABLE
PROJECT END ROAD RUNNING DISTANCE: NOT APPLICABLE
SCALES: 200 0 100 200 300 400 1:10

ROAD No. 6200
RICHMOND ROAD
RAILWAY TERRACE - SOUTH ROAD; MILE END SOUTH
BRIDGE OVER KESWICK CREEK
BARRIER DETAILS - SHEET 03

DESIGNED:	DRAFTED:	ACCEPTED FOR USE:	ACCEPTANCE FORM KNET No.:	DRAWING No.:	SHEET No.:	AMEND No.:
T2DA	T2DA	M. SHORT TITLE: DIRECTOR OF ENG. DATE: 28.03.26	24467656 IN ACCORDANCE WITH DP013	8011	2871114	0

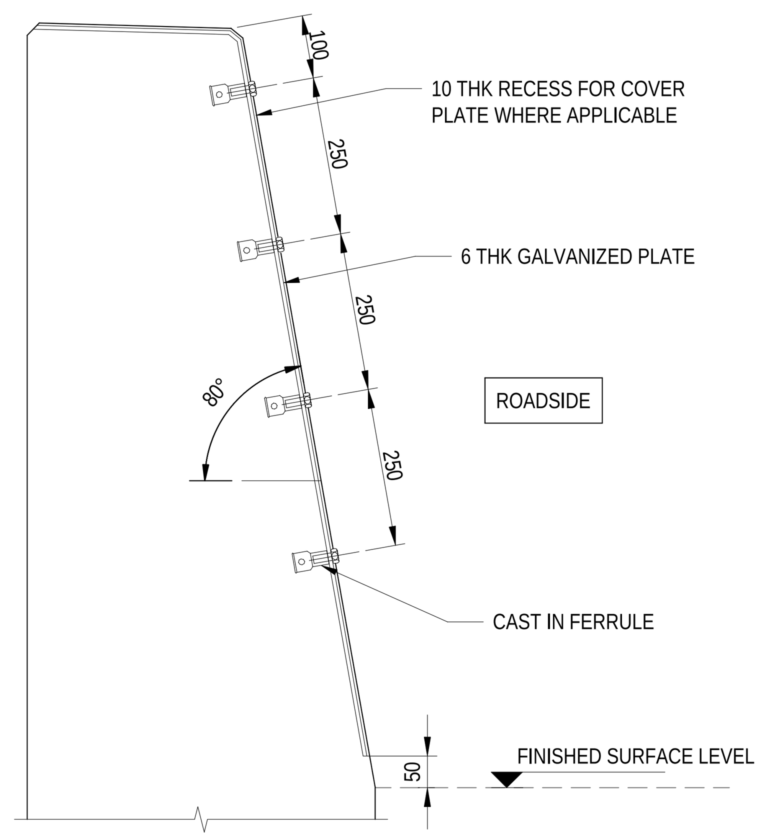
SHEET LATITUDE -34.94188 SHEET LONGITUDE 138.57397



COVER PLATE ELEVATION - TYPE CP-02

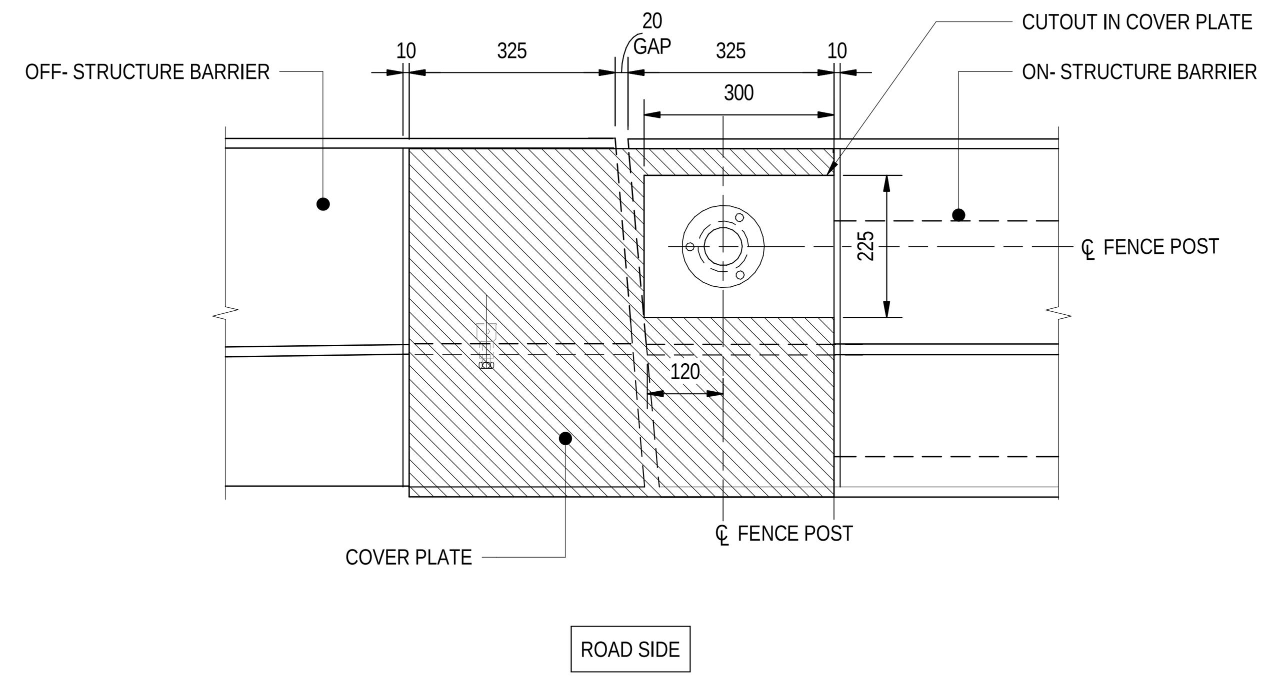
SCALE 1 : 10

COVER PLATE SCHEDULE			
COVER PLATE TYPE	PLATE LENGTH 'L' (mm)	DIM 'A' (mm)	DIM 'B' (mm)
CP-02	670	10	325



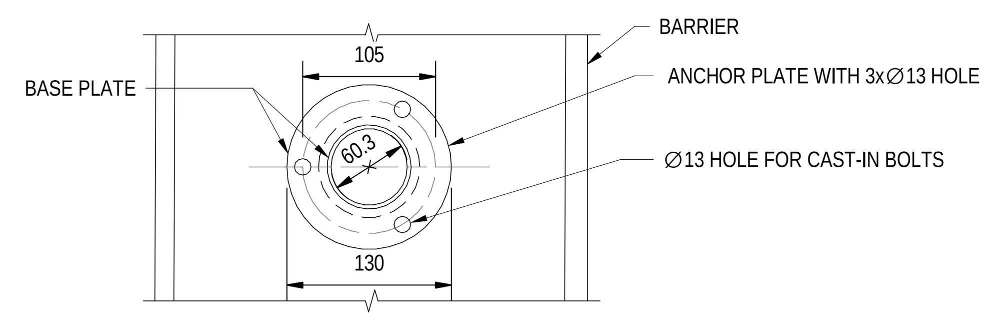
SECTION 1

SCALE 1 : 10 2871115



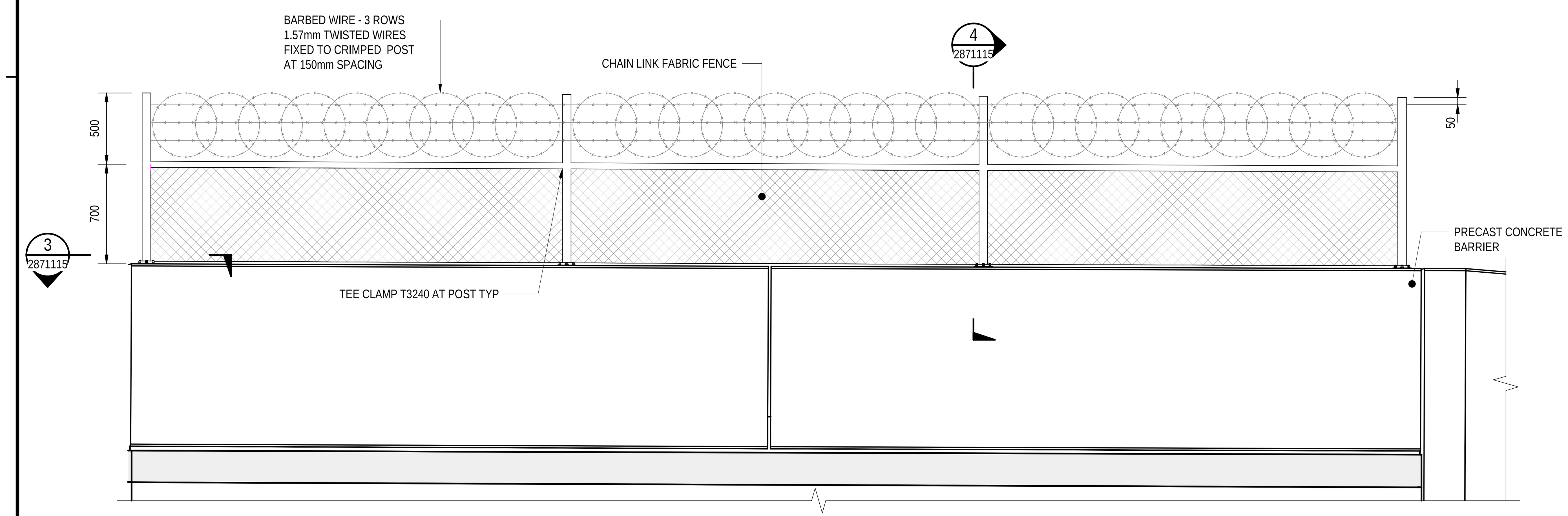
SECTION 2

SCALE 1 : 10 2871115



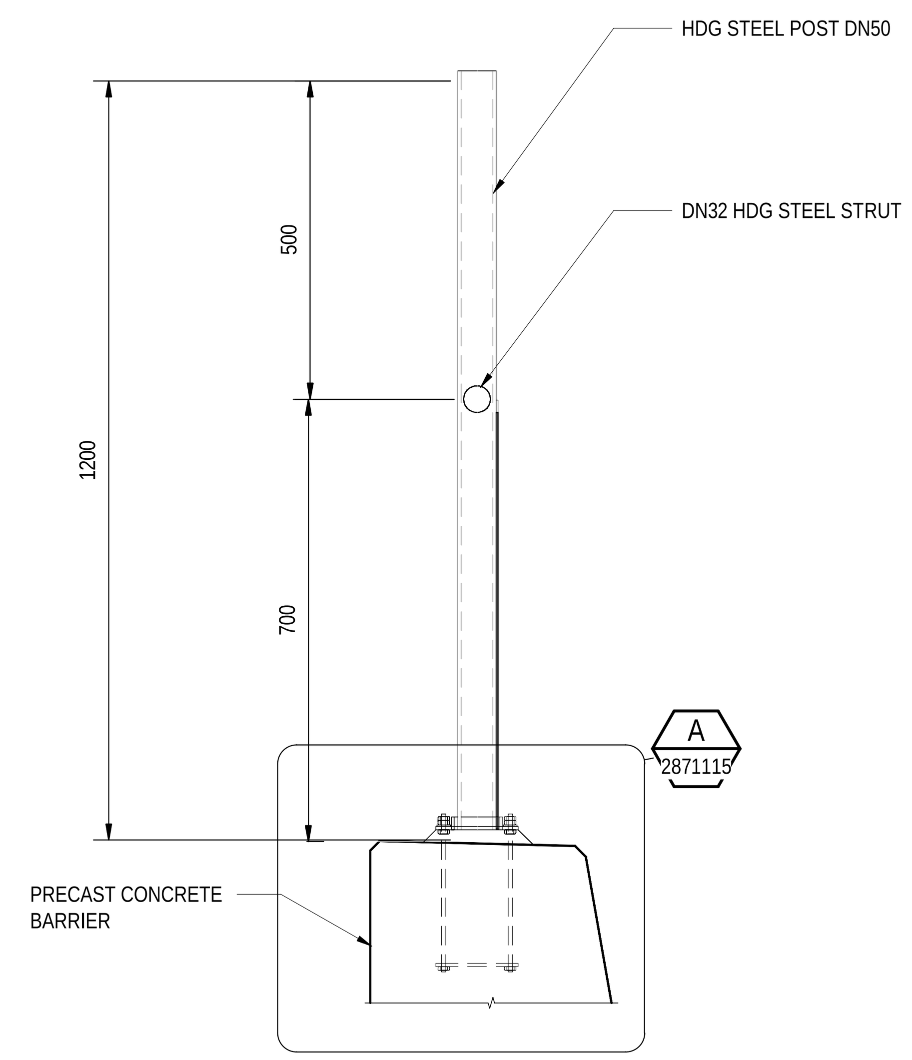
SECTION 3

SCALE 1 : 5 2871115



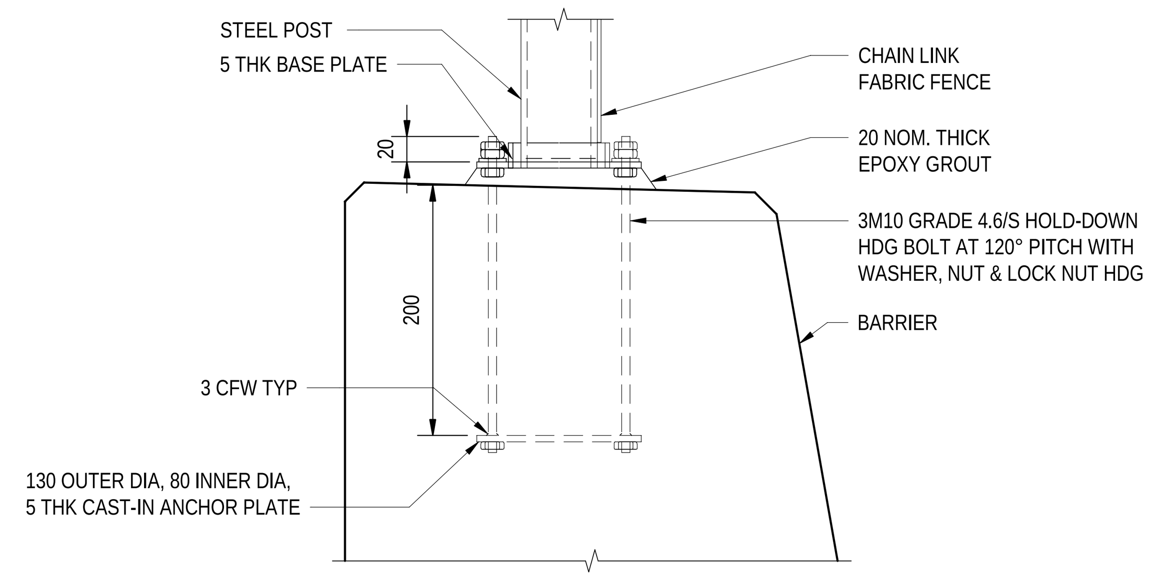
TYPICAL FENCING ELEVATION

SCALE 1 : 25



SECTION 4

SCALE 1 : 10 2871115



DETAIL A

SCALE 1 : 5 2871115

NOTES

1. FOR GENERAL NOTES REFER TO SHEET No. 2871005 TO 2871008.

100 0 50 100 150 200
1:5
200 0 100 200 300 400
1:10

PROJECT DOCUMENT REFERENCE
T2DPAA-T2D-C3S-BR-DRG-101115

FOR CONSTRUCTION

No.	AMENDMENT DESCRIPTION	BY	CHECK	ACCEPTANCE	DATE
0	ISSUED FOR CONSTRUCTION (KNet No: 24467656)	T2DA	T2DA	M. SHORT	28.03.26

INDEX SHEET REFERENCE: 8011 SHEET 2871002

T2D TORRENS TO DARLINGTON

DESIGNER: A.SINGHAL / S.PODDAR
QUALIFICATION: MEng BEng
DATE: 25.03.26
REVIEWER: R.WOZNIAK
QUALIFICATION: FIEAust CPEng NER RPEV RPEQ MICE CEng
DATE: 25.03.26
INDEPENDENT DESIGN CERTIFIER (IF REQUIRED): T. JACKSON
QUALIFICATION: BEng (Hons), MEngSc
DATE: 27.03.26

Government of South Australia
Department for Infrastructure and Transport

PROJECT No.: 31921901
FILE No.: 2020/10577
DESIGN No.: #15551401
SURVEY No.: #15551401
PROJECT START ROAD RUNNING DISTANCE: NOT APPLICABLE
PROJECT END ROAD RUNNING DISTANCE: NOT APPLICABLE
SCALES: 500 0 250 500 750 1000 1:25

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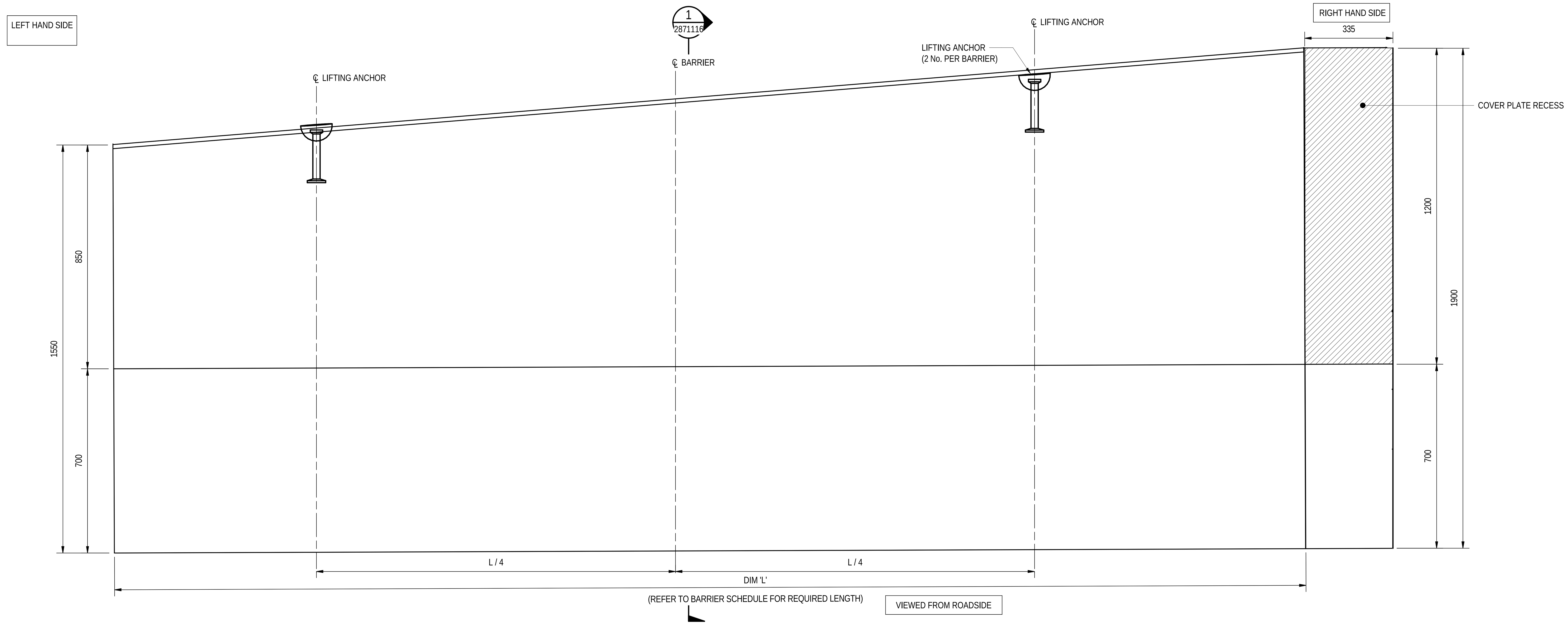
ROAD No. 6200
RICHMOND ROAD
RAILWAY TERRACE - SOUTH ROAD; MILE END SOUTH
BRIDGE OVER KESWICK CREEK
BARRIER DETAILS - SHEET 04

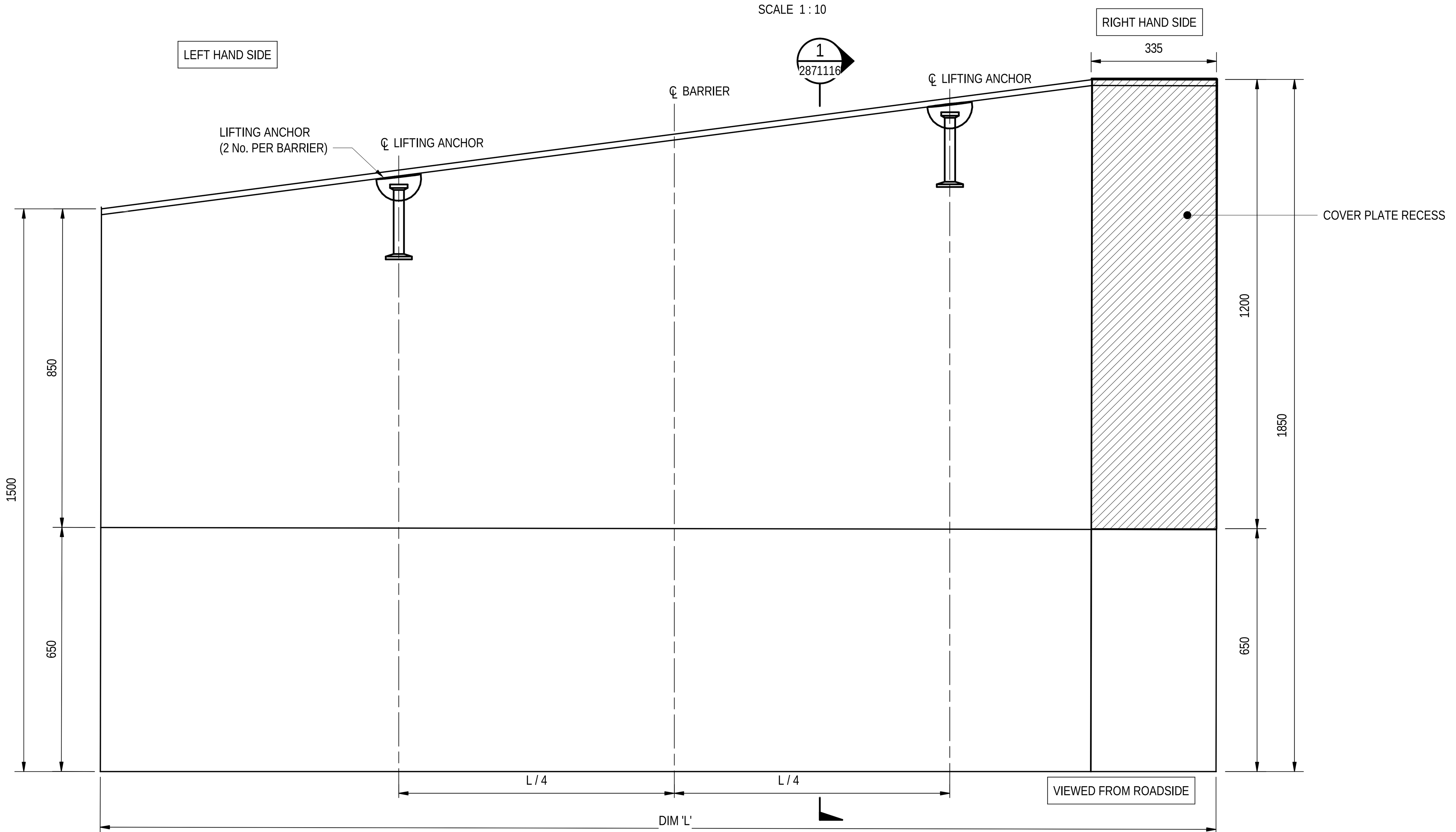
DESIGNED: T2DA
DRAFTED: T2DA
ACCEPTED FOR USE: M. SHORT
TITLE: DIRECTOR OF ENG.
DATE: 28.03.26
ACCEPTANCE FORM KNET No.: 24467656
IN ACCORDANCE WITH DP013
DRAWING No.: 8011
SHEET No.: 2871115
AMEND No.: 0
SHEET LATITUDE -34.94188 SHEET LONGITUDE 138.57397

FILE NAME: Autodesk Docs://T2D - Torrens to Darlington /T2DPAA-T2D-C3S-BR-M3D-100302.rvt



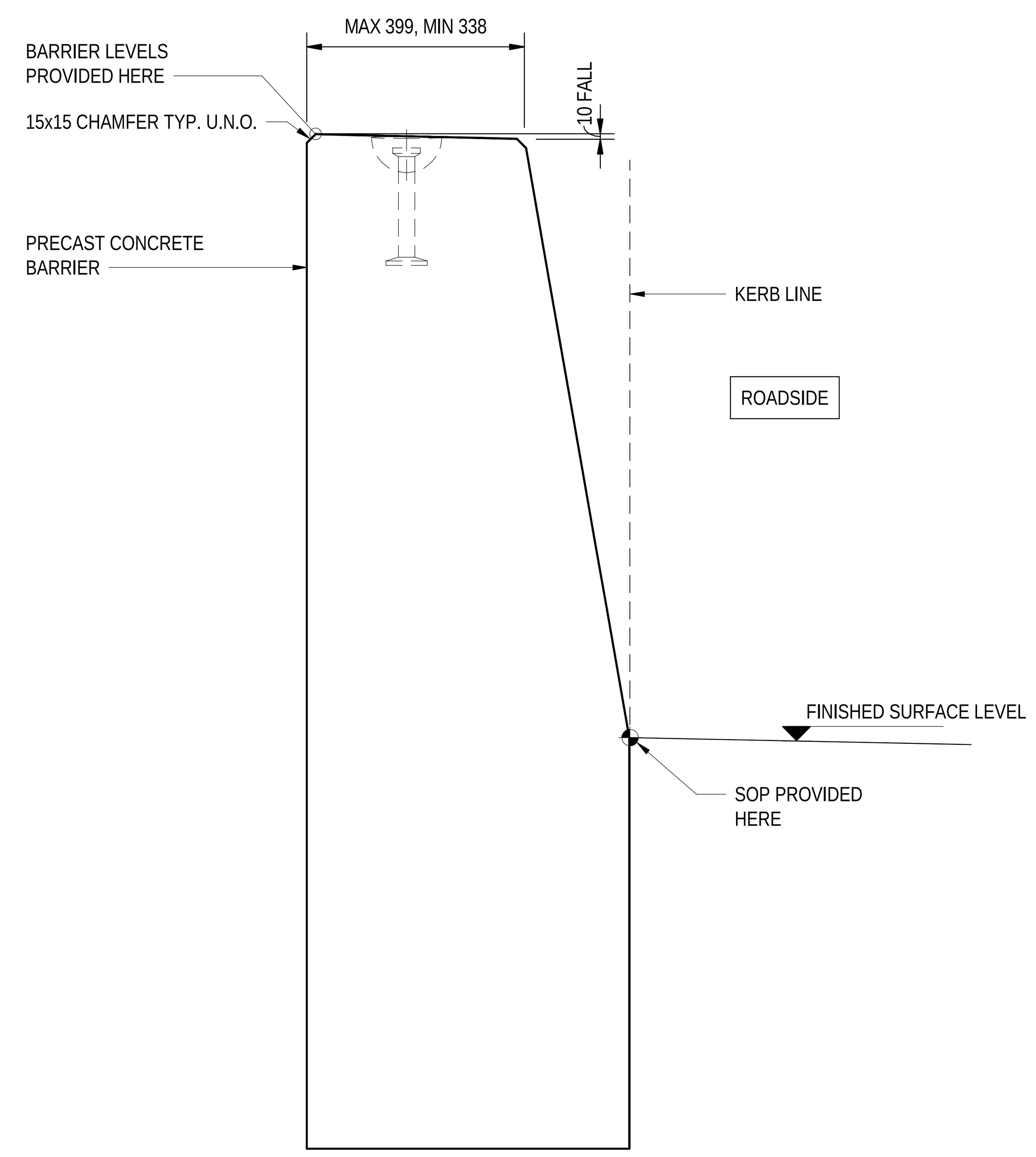
ELEVATION - BARRIER PPT234105

SCALE 1 : 10



ELEVATION - BARRIER PPT234106

SCALE 1 : 10



SECTION 1

SCALE 1 : 10

2871116

NOTES

1. FOR GENERAL NOTES REFER TO SHEET No. 2871005 TO 2871008.

PROJECT DOCUMENT REFERENCE T2DPAA-T2D-C3S-BR-DRG-101116

FOR CONSTRUCTION

No.	AMENDMENT DESCRIPTION	BY	CHECK	ACCEPTANCE	DATE
0	ISSUED FOR CONSTRUCTION (KNet No: 24467656)	T2DA	T2DA	M. SHORT	28.03.26

INDEX SHEET REFERENCE: 8011 SHEET 2871002

T2D → TORRENS TO DARLINGTON

DESIGNER: A.SINGHAL / S.PODDAR
QUALIFICATION: MEng BEng
DATE: 25.03.26

REVIEWER: R.WOZNIAK
QUALIFICATION: FIEAust CPEng NER RPEV RPEQ MICE CEng
DATE: 25.03.26

INDEPENDENT DESIGN CERTIFIER (IF REQUIRED): T. JACKSON
QUALIFICATION: BEng (Hons), MEngSc
DATE: 27.03.26

Government of South Australia
Department for Infrastructure and Transport

PROJECT No.: 31921901
FILE No.: 2020/10577
DESIGN No.: #15551401
SURVEY No.: #15551401
PROJECT START ROAD RUNNING DISTANCE: NOT APPLICABLE
PROJECT END ROAD RUNNING DISTANCE: NOT APPLICABLE
SCALES: 200 0 100 200 300 400 1:10

ROAD No. 6200
RICHMOND ROAD
RAILWAY TERRACE - SOUTH ROAD; MILE END SOUTH
BRIDGE OVER KESWICK CREEK
BARRIER DETAILS - SHEET 05

DESIGNED:	DRAFTED:	ACCEPTED FOR USE:	ACCEPTANCE FORM KNET No.:	DRAWING No.:	SHEET No.:	AMEND No.:
T2DA	T2DA	M. SHORT TITLE: DIRECTOR OF ENG. DATE: 28.03.26	24467656 IN ACCORDANCE WITH DP013	8011	2871116	0

SHEET LATITUDE -34.94188 SHEET LONGITUDE 138.57397

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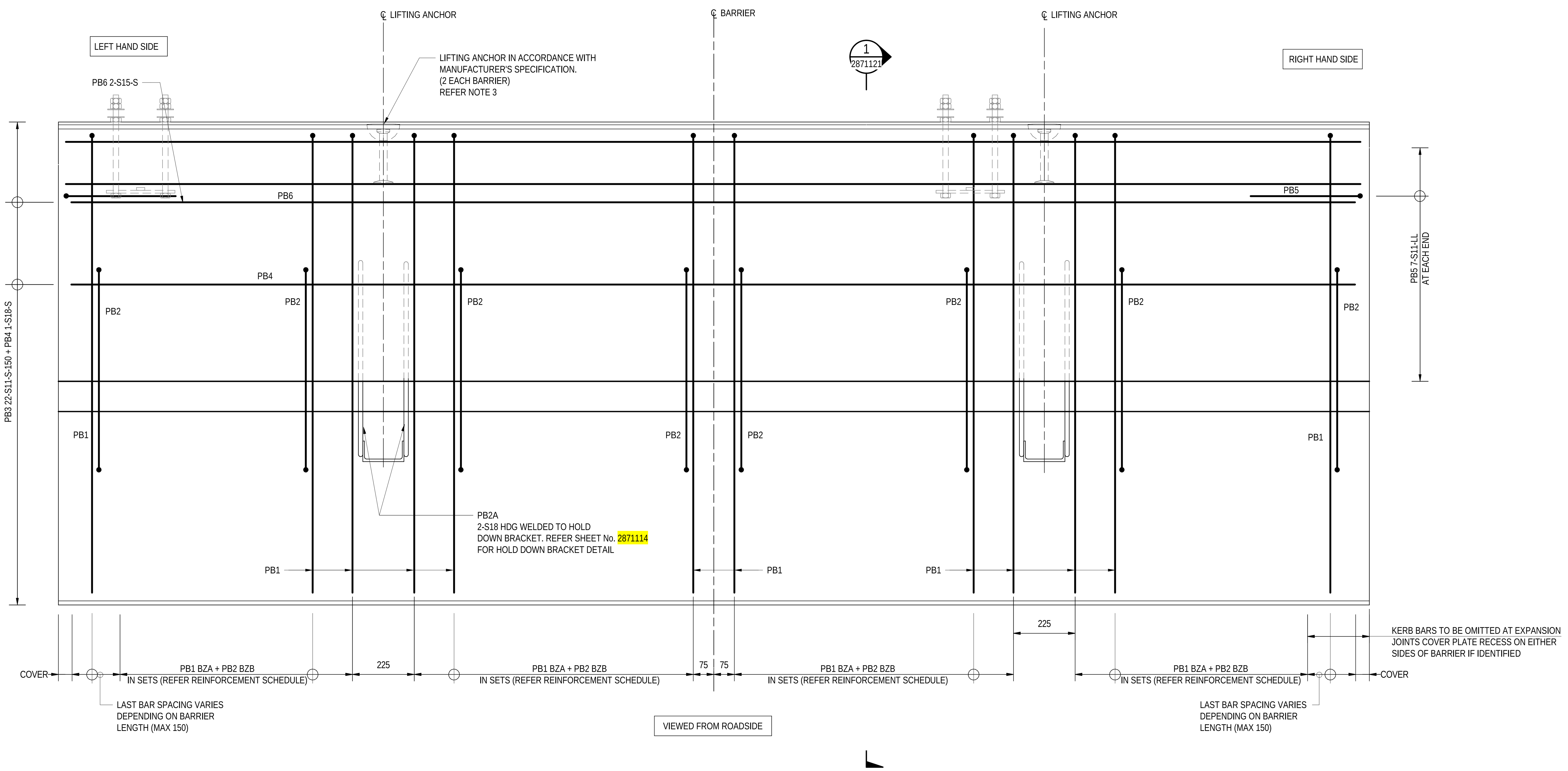
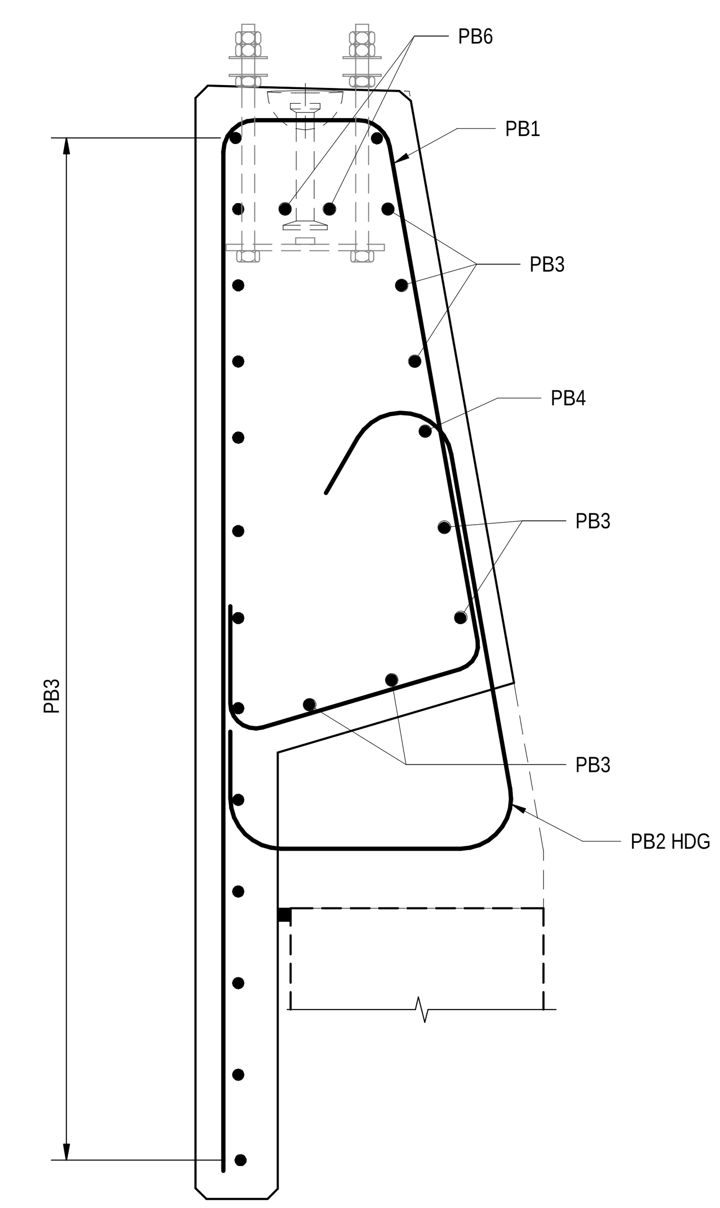


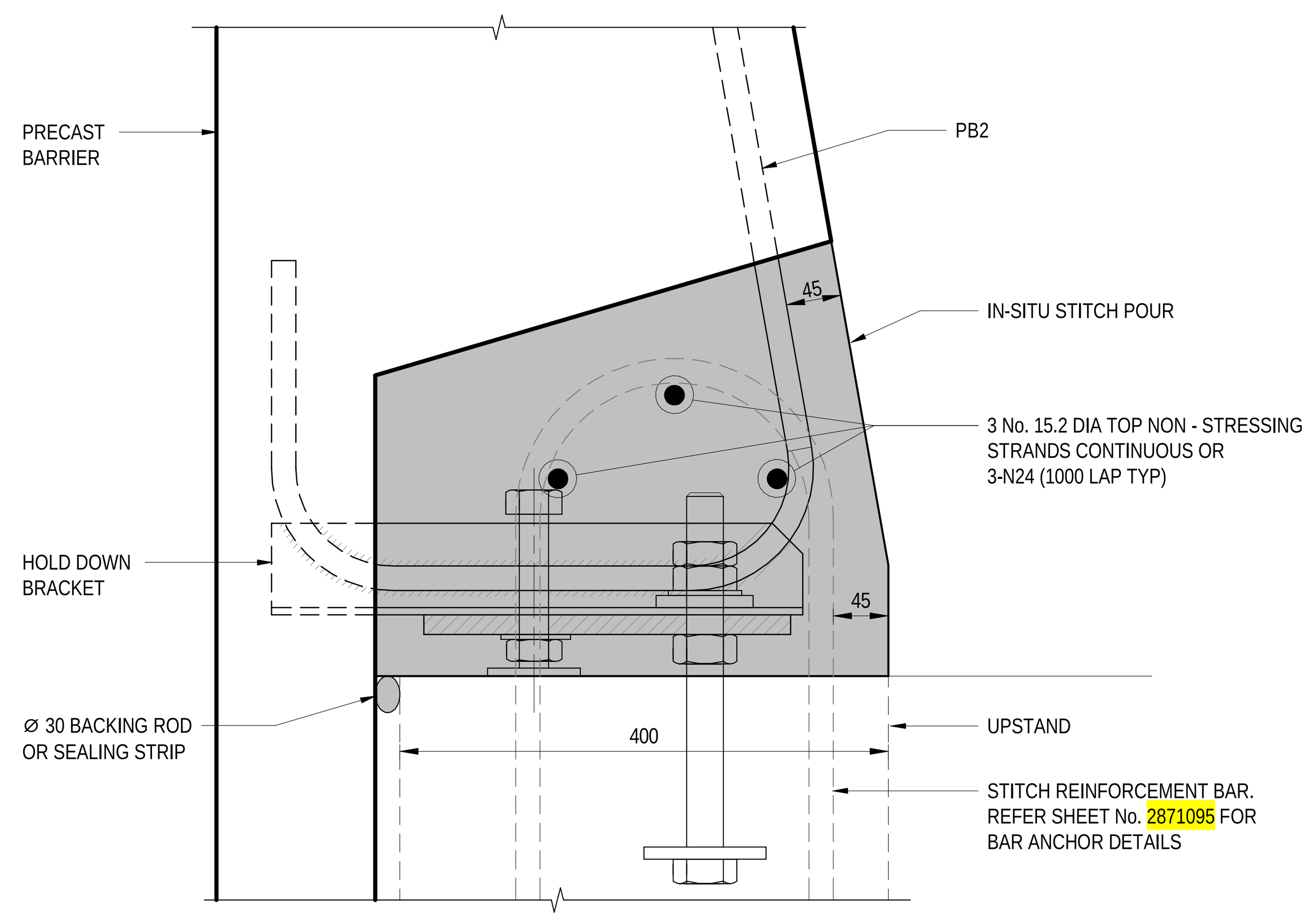
ELEVATION - PRECAST MEDIUM PERFORMANCE LEVEL BARRIER TYPE - MPB-02-A

SCALE 1 : 10



SECTION 1 2871121

SCALE 1 : 10



TYPICAL IN SITU STITCH DETAIL

SCALE 1 : 5

REINFORCEMENT SCHEDULE	
PB1	PB2
S15-150	S15-150

NOTES

1. FOR GENERAL NOTES REFER TO SHEET No. 2871005 TO 2871008.
2. REFER SHEET No. 2871151 AND 2871152 FOR BAR SHAPE SCHEDULE.
3. LIFTING LUGS SHOWN INDICATIVELY. TEMPORARY WORKS DESIGNER TO CONFIRM LOCATION AND DESIGN LIFTING LUGS AS PER MANUFACTURER SPECIFICATION AND WORKSAFE GUIDELINES. LIFTING ANCHOR POCKETS TO BE GROUT INFILLED AFTER BARRIER INSTALLATION.

100 0 50 100 150 200
1:5

PROJECT DOCUMENT REFERENCE
T2DPAA-T2D-C3S-BR-DRG-101121

FOR CONSTRUCTION

No.	AMENDMENT DESCRIPTION	BY	CHECK	ACCEPTANCE	DATE
0	ISSUED FOR CONSTRUCTION (KNet No: 24467656)	T2DA	T2DA	M. SHORT	28.03.26

INDEX SHEET REFERENCE: 8011 SHEET 2871002

T2D → TORRENS TO DARLINGTON

DESIGNER: A.SINGHAL / S.PODDAR
QUALIFICATION: MEng BEng
DATE: 25.03.26
REVIEWER: R.WOZNIAK
QUALIFICATION: FIEAust CPEng NER RPEV RPEQ MICE CEng
DATE: 25.03.26
INDEPENDENT DESIGN CERTIFIER (IF REQUIRED): T. JACKSON
QUALIFICATION: BEng (Hons), MEngSc
DATE: 27.03.26

SOUTH AUSTRALIA
Government of South Australia
Department for Infrastructure and Transport

PROJECT No.: 31921901
FILE No.: 2020/10577
DESIGN No.: #15551401
SURVEY No.: #15551401
PROJECT START ROAD RUNNING DISTANCE: NOT APPLICABLE
PROJECT END ROAD RUNNING DISTANCE: NOT APPLICABLE
SCALES: 200 0 100 200 300 400 1:10

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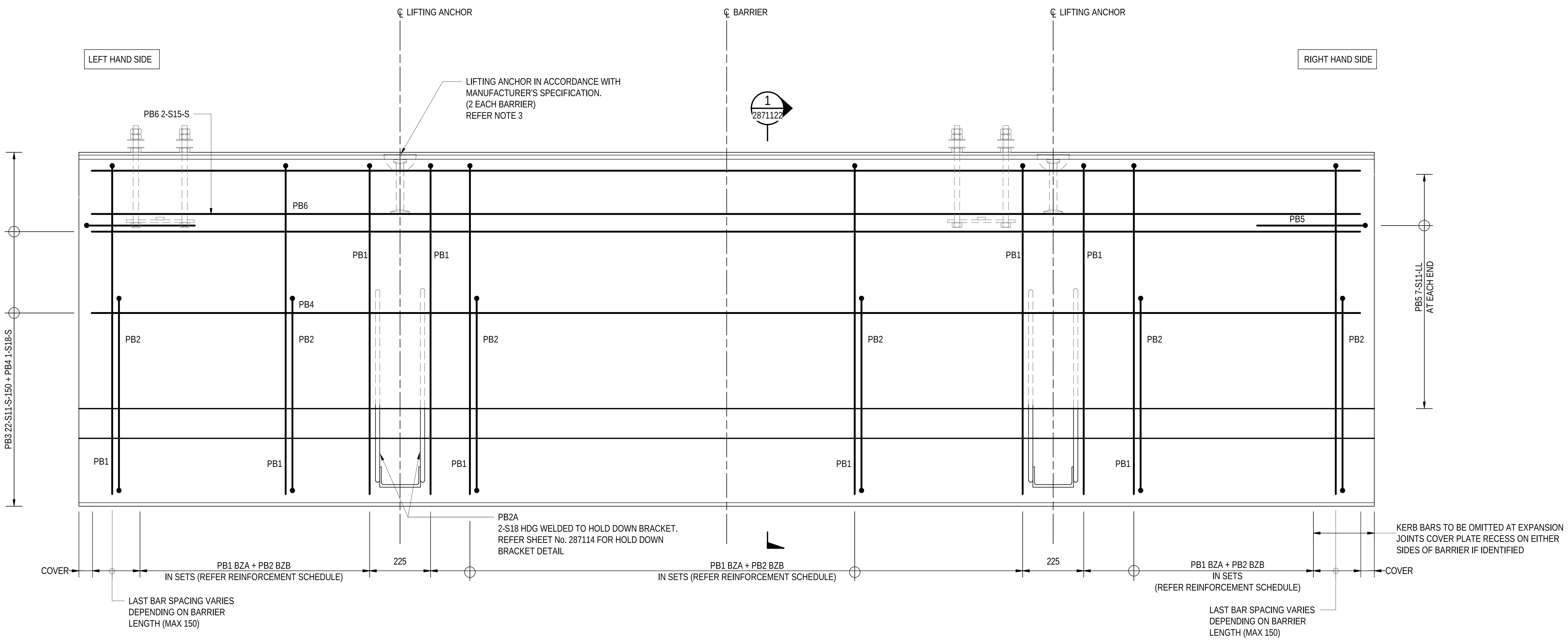
ROAD No. 6200
RICHMOND ROAD
RAILWAY TERRACE - SOUTH ROAD; MILE END SOUTH
BRIDGE OVER KESWICK CREEK
BARRIER REINFORCEMENT - SHEET 01

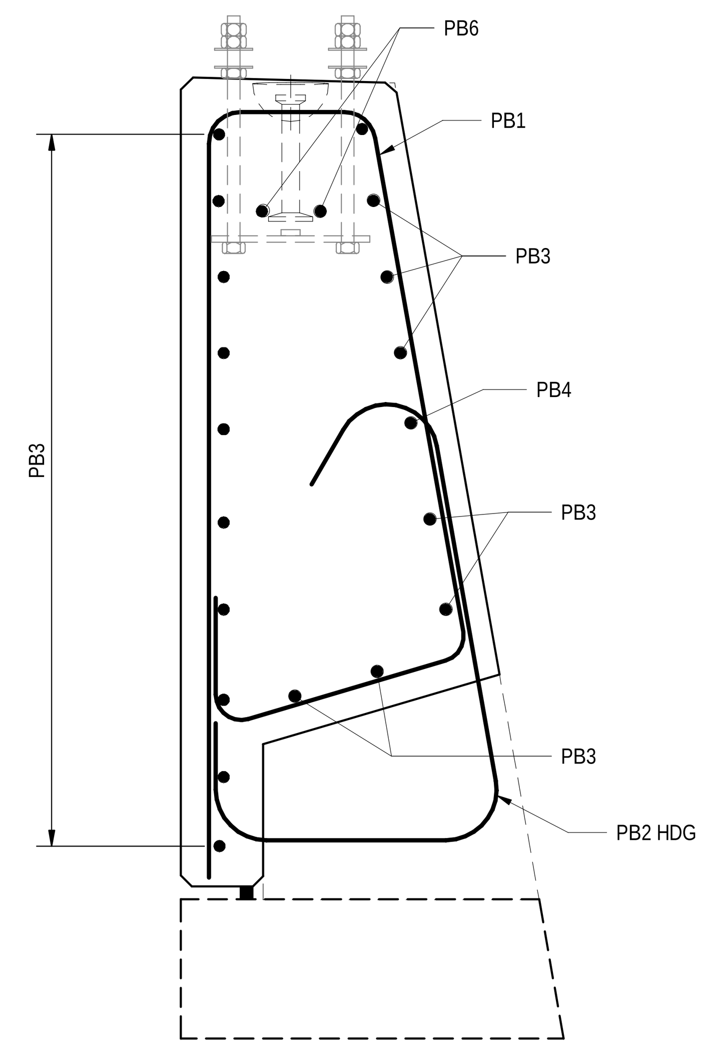
DESIGNED: T2DA
DRAFTED: T2DA
ACCEPTED FOR USE: M. SHORT
TITLE: DIRECTOR OF ENG.
DATE: 28.03.26
ACCEPTANCE FORM KNET No.: 24467656
IN ACCORDANCE WITH DP013
DRAWING No.: 8011
SHEET No.: 2871121
AMEND No.: 0
SHEET LATITUDE -34.94188
SHEET LONGITUDE 138.57397

FILE NAME: Autodesk Docs://T2D - Torrens to Darlington /T2DPAA-T2D-C3S-BR-M3D-100302.rvt



ELEVATION - PRECAST MEDIUM PERFORMANCE LEVEL BARRIER TYPE - MPB-02-B

SCALE 1 : 10



SECTION 1 (2871122)

SCALE 1 : 10

REINFORCEMENT SCHEDULE	
PB1	PB2
N16-150	N16-150

NOTES

1. FOR GENERAL NOTES REFER TO SHEET No. 2871005 TO 2871008.
2. REFER SHEET No. 2871151 AND 2871152 FOR BAR SHAPE SCHEDULE.
3. LIFTING LUGS SHOWN INDICATIVELY. TEMPORARY WORKS DESIGNER TO CONFIRM LOCATION AND DESIGN LIFTING LUGS AS PER MANUFACTURER SPECIFICATION AND WORKSAFE GUIDELINES. LIFTING ANCHOR POCKETS TO BE GROUT INFILLED AFTER BARRIER INSTALLATION.

PROJECT DOCUMENT REFERENCE T2DPAA-T2D-C3S-BR-DRG-101122

FOR CONSTRUCTION

No.	AMENDMENT DESCRIPTION	BY	CHECK	ACCEPTANCE	DATE
0	ISSUED FOR CONSTRUCTION (KNet No: 24467656)	T2DA	T2DA	M. SHORT	28.03.26

INDEX SHEET REFERENCE: 8011 SHEET 2871002

T2D TORRENS TO DARLINGTON

DESIGNER: A.SINGHAL / S.PODDAR
QUALIFICATION: MEng BEng
DATE: 25.03.26

REVIEWER: R.WOZNIAK
QUALIFICATION: FIEAust CPEng NER RPEV RPEQ MICE CEng
DATE: 25.03.26

INDEPENDENT DESIGN CERTIFIER (IF REQUIRED): T. JACKSON
QUALIFICATION: BEng (Hons), MEngSc
DATE: 27.03.26

Government of South Australia
Department for Infrastructure and Transport

PROJECT No.: 31921901 | FILE No.: 2020/10577
DESIGN No.: #15551401 | SURVEY No.: #15551401
PROJECT START ROAD RUNNING DISTANCE: NOT APPLICABLE
PROJECT END ROAD RUNNING DISTANCE: NOT APPLICABLE
SCALES: 200 0 100 200 300 400 1:10

ROAD No. 6200
RICHMOND ROAD
RAILWAY TERRACE - SOUTH ROAD; MILE END SOUTH
BRIDGE OVER KESWICK CREEK
BARRIER REINFORCEMENT - SHEET 02

DESIGNED: T2DA | DRAFTED: T2DA | ACCEPTED FOR USE: M. SHORT, TITLE: DIRECTOR OF ENG., DATE: 28.03.26 | ACCEPTANCE FORM KNET No.: 24467656, IN ACCORDANCE WITH DP013 | DRAWING No.: 8011 | SHEET No.: 2871122 | AMEND No.: 0

SHEET LATITUDE -34.94188 SHEET LONGITUDE 138.57397

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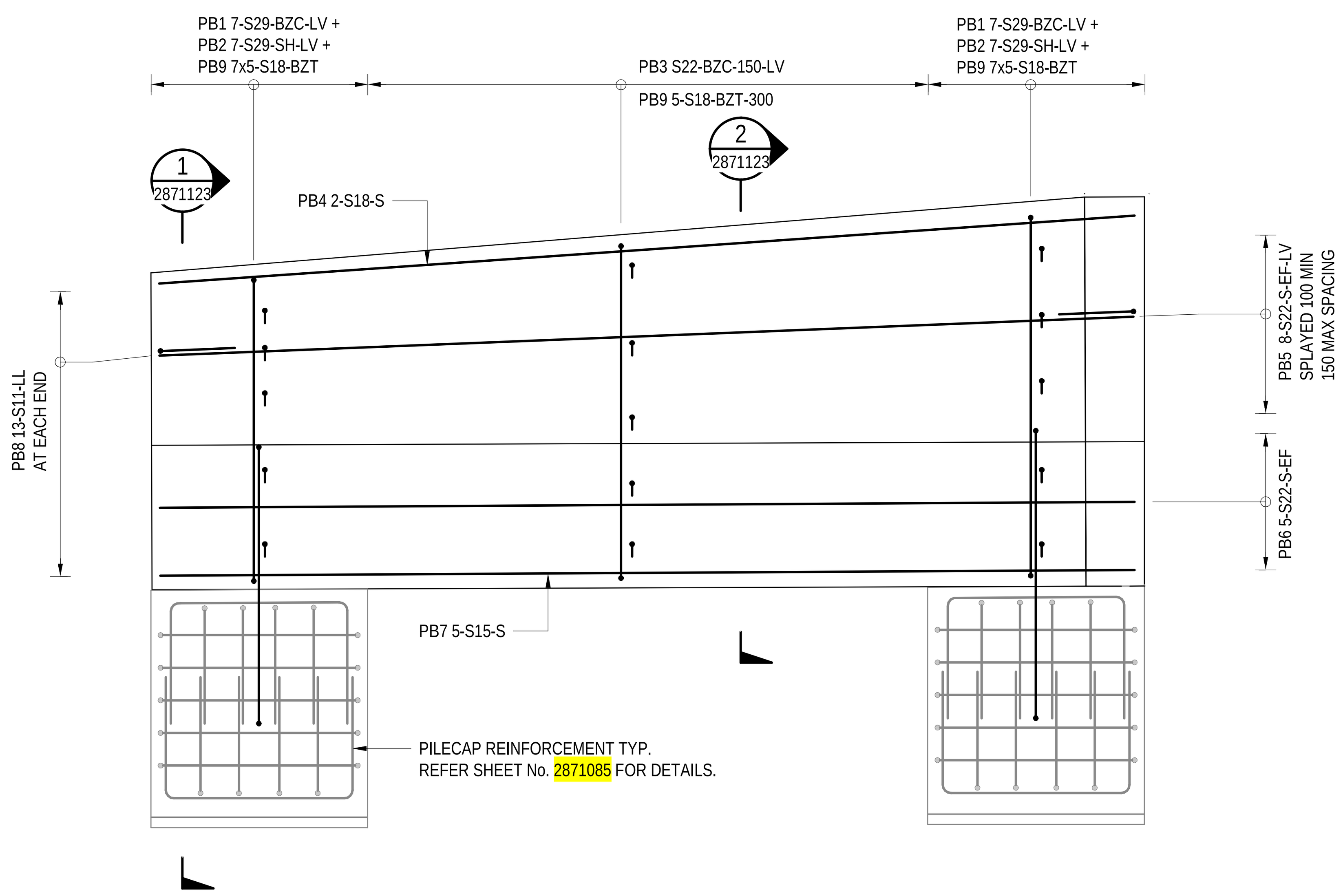
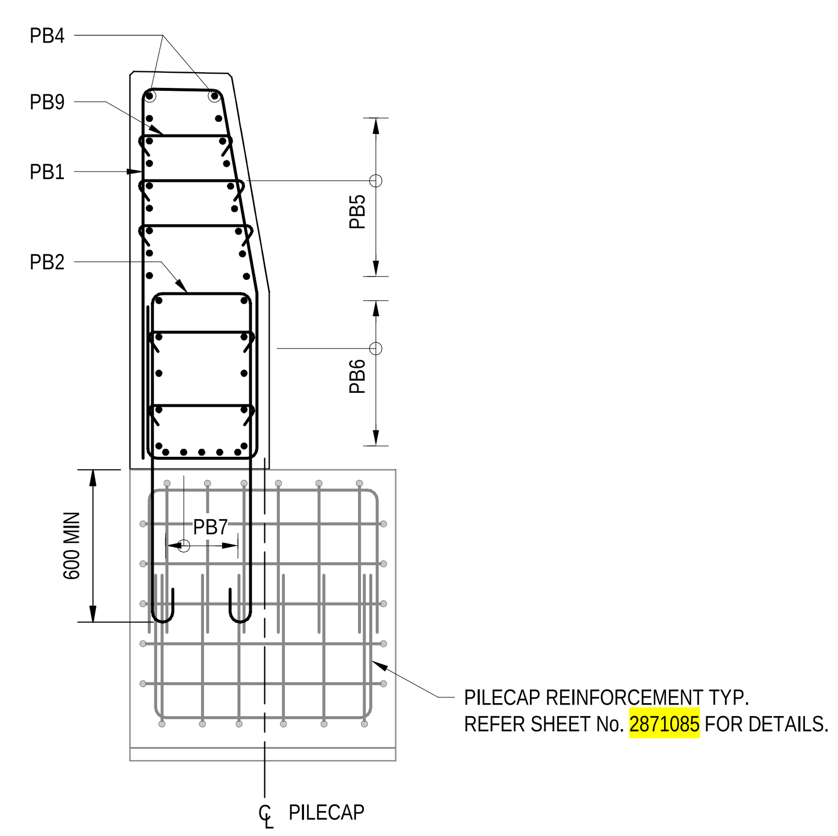
FILE NAME: Autodesk Docs://T2D - Torrens to Darlington /T2DPAA-T2D-C3S-BR-M3D-100302.rvt



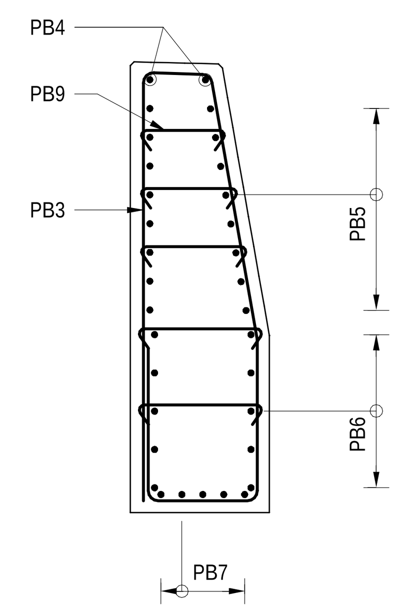
ELEVATION - TYPICAL OFF-STRUCTURE BARRIER REINFORCEMENT DETAILS

SCALE 1 : 25



SECTION 1 2871123

SCALE 1 : 25



SECTION 2 2871123

SCALE 1 : 25

NOTES

1. FOR GENERAL NOTES REFER TO SHEET No. 2871005 TO 2871008.

PROJECT DOCUMENT REFERENCE T2DPAA-T2D-C3S-BR-DRG-101123

FOR CONSTRUCTION

No.	AMENDMENT DESCRIPTION	BY	CHECK	ACCEPTANCE	DATE
0	ISSUED FOR CONSTRUCTION (KNet No: 24467656)	T2DA	T2DA	M. SHORT	28.03.26

INDEX SHEET REFERENCE: 8011 SHEET 2871002

T2D → TORRENS TO DARLINGTON

UNCONTROLLED COPY WHEN PRINTED ◄ 100 MILLIMETRES ON ORIGINAL DRAWING ► ALL DIMENSIONS ARE IN MILLIMETRES UNLESS SHOWN OTHERWISE

DESIGNER: A.SINGHAL / S.PODDAR
QUALIFICATION: MEng BEng
DATE: 25.03.26

REVIEWER: R.WOZNIAK
QUALIFICATION: FIEAust CPEng NER RPEV RPEQ MICE CEng
DATE: 25.03.26

INDEPENDENT DESIGN CERTIFIER (IF REQUIRED): T. JACKSON
QUALIFICATION: BEng (Hons), MEngSc
DATE: 27.03.26

Government of South Australia
Department for Infrastructure and Transport

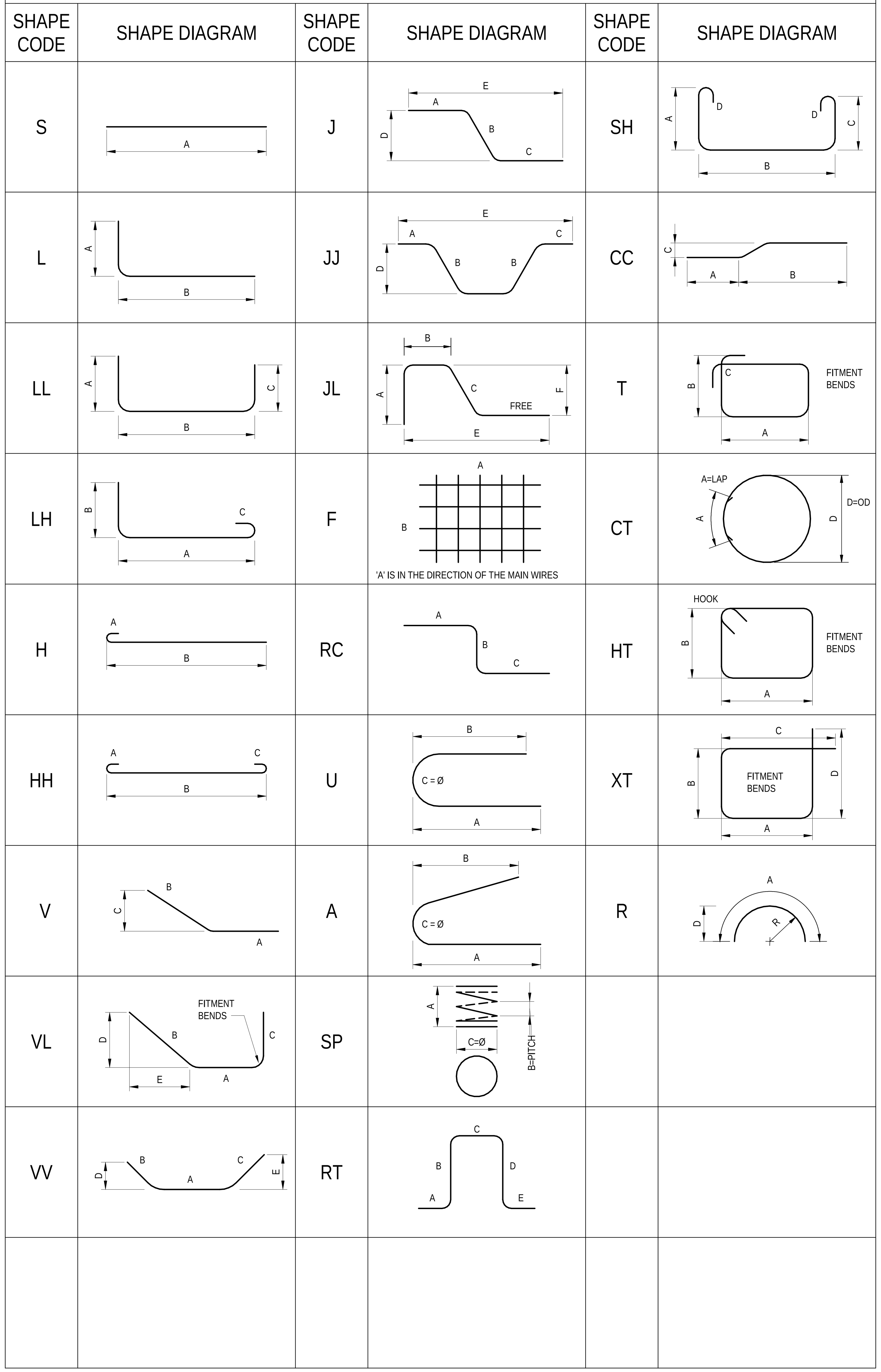
PROJECT No.: 31921901
FILE No.: 2020/10577
DESIGN No.: #15551401
SURVEY No.: #15551401
PROJECT START ROAD RUNNING DISTANCE: NOT APPLICABLE
PROJECT END ROAD RUNNING DISTANCE: NOT APPLICABLE
SCALES: 500 0 250 500 750 1000 1:25

ROAD No. 6200
RICHMOND ROAD
RAILWAY TERRACE - SOUTH ROAD; MILE END SOUTH
BRIDGE OVER KESWICK CREEK
BARRIER REINFORCEMENT - SHEET 03

DESIGNED: T2DA
DRAFTED: T2DA
ACCEPTED FOR USE: M. SHORT
TITLE: DIRECTOR OF ENG.
DATE: 28.03.26
ACCEPTANCE FORM KNET No.: 24467656
IN ACCORDANCE WITH DP013
DRAWING No.: 8011
SHEET No.: 2871123
AMEND No.: 0
SHEET LATITUDE -34.94188
SHEET LONGITUDE 138.57397

FILE NAME: Autodesk Docs://T2D - Torrens to Darlington /T2DPAA-T2D-C3S-BR-M3D-100302.rvt

AUSTRALIAN STANDARD BAR SHAPES TO AS/NZS 1100.501



BAR NOTATION

S8 10-N24-S-250-T

- STRUCTURE ELEMENT
- BAR MARK
- No. OF BARS
- DUCTILITY CLASS
- BAR Ø (mm)
- BAR BENDING CODE
- BAR SPACING
- LOCATION

LOCATION

T	TOP OF FACE
B	BOTTOM OF FACE
NF	NEAR FACE
FF	FAR FACE
EF	EACH FACE
C	PLACED CENTRALLY
EC	EACH CORNER
LV	VARIABLE LENGTH BAR
SF	SIDE FACE
NSOP	NOT SHOWN ON PLAN

MESH NOTATION

S7 SL81-F-B

- STRUCTURE ELEMENT
- BAR MARK
- MESH DESIGNATION TO AS/NZ 4671
- BAR BENDING CODE
- LOCATION

STRUCTURE ELEMENT PREFIXES

F	FOUNDATION (PILE / FOOTING)	B	BEAM
A	ABUTMENT	BI	BARRIER IN SITU
FW	FENDER WALL	PB	PRECAST BARRIER
W	WINGWALL	H	HEADWALL
P	PIER	S	APPROACH SLAB
D	DECK SLAB	G	PRECAST GIRDERS
BP	BEARING PLINTH / PEDESTAL	RB	RESTRAINT BLOCK

NOTES

1. AUSTRALIAN STANDARD BAR SHAPES ARE IN ACCORDANCE WITH AS 1100.501.
2. BAR SIZE IS THE NOMINAL DIAMETER IN MILLIMETRES, OR THE AS/NZS 4671 FABRIC NUMBER.
3. THE GRADE OF REINFORCEMENT, IF NOT STATED ON THE DRAWINGS, SHALL BE D500N TO AS/NZS 4671.
4. WHERE SHOWN ON THE DRAWINGS, 'W' SHALL DENOTE PLAIN ROUND REINFORCING BARS EQUIVALENT TO GRADE R500L TO AS/NZS 4671.
5. WHERE SHOWN ON DRAWINGS, 'RL' AND 'SL' SHALL DENOTE WELDED REINFORCING FABRIC (RECTANGULAR AND SQUARE), RESPECTIVELY.
6. DIMENSIONS SHOWN ON BAR SHAPE DIAGRAM ARE MEASURED FROM THE OUTSIDE FACE OF THE BARS AND ARE IN MILLIMETRES.
7. THE INCLUDED ANGLE OF ANY BEND SHALL BE A RIGHT ANGLE IF NO DIMENSION SHOWN.
8. BARS TO BE BENT SUCH THAT STRAIGHT LEGS LIE IN THE SAME HORIZONTAL PLANE.

PROJECT DOCUMENT REFERENCE
T2DPAA-T2D-C3S-BR-DRG-101151

FOR CONSTRUCTION

No.	AMENDMENT DESCRIPTION	BY	CHECK	ACCEPTANCE	DATE
0	ISSUED FOR CONSTRUCTION (KNet No: 24467656)	T2DA	T2DA	M. SHORT	28.03.26

INDEX SHEET REFERENCE: 8011 SHEET 2871002

T2D → TORRENS TO DARLINGTON

DESIGNER: A.SINGHAL / S.PODDAR
QUALIFICATION: MEng BEng
DATE: 25.03.26

REVIEWER: R.WOZNIAK
QUALIFICATION: FIEAust CPEng NER RPEV RPEQ MICE CEng
DATE: 25.03.26

INDEPENDENT DESIGN CERTIFIER (IF REQUIRED): T. JACKSON
QUALIFICATION: BEng (Hons), MEngSc
DATE: 27.03.26

Government of South Australia
Department for Infrastructure and Transport

PROJECT No.: 31921901 | FILE No.: 2020/10577
DESIGN No.: #15551401 | SURVEY No.: #15551401
PROJECT START ROAD RUNNING DISTANCE: NOT APPLICABLE
PROJECT END ROAD RUNNING DISTANCE: NOT APPLICABLE
SCALES: NOT TO SCALE

ROAD No. 6200
RICHMOND ROAD
RAILWAY TERRACE - SOUTH ROAD; MILE END SOUTH
BRIDGE OVER KESWICK CREEK
BAR SHAPES DIAGRAM - SHEET 01

DESIGNED: T2DA | DRAFTED: T2DA | ACCEPTED FOR USE: M. SHORT, TITLE: DIRECTOR OF ENG. DATE: 28.03.26 | ACCEPTANCE FORM KNET No.: 24467656, IN ACCORDANCE WITH DP013 | DRAWING No.: 8011 | SHEET No.: 2871151 | AMEND No.: 0

SHEET LATITUDE -34.94188 SHEET LONGITUDE 138.57397

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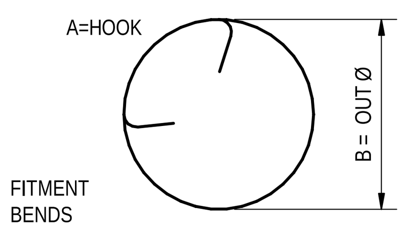
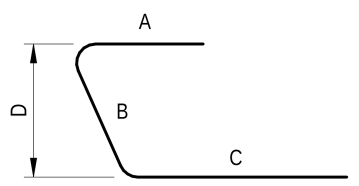
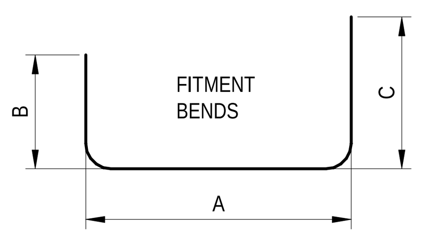
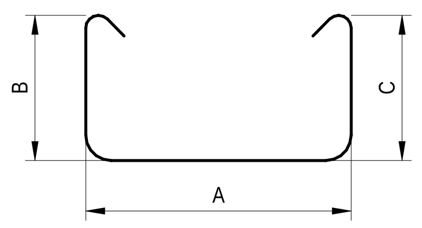
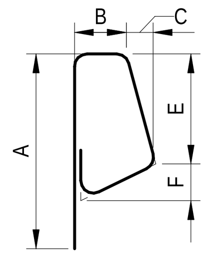
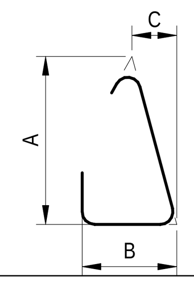
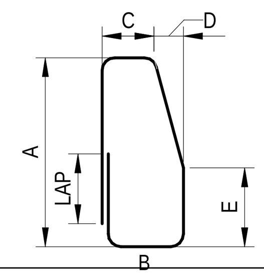
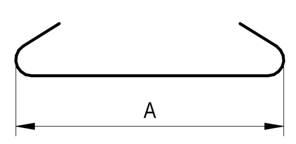
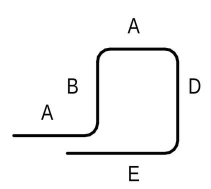
FILE NAME: Autodesk Docs://T2D - Torrens to Darlington /T2DPAA-T2D-C3S-BR-M3D-100302.rvt

NON STANDARD BAR SHAPES

SHAPE CODE	SHAPE DIAGRAM	SHAPE CODE	SHAPE DIAGRAM
CH			
VS			
UT			
AZT			
BZA			
BZB			
BZC			
BZT			
FZ			

PROJECT DOCUMENT REFERENCE
T2DPAA-T2D-C3S-BR-DRG-101152

FOR CONSTRUCTION

No.	AMENDMENT DESCRIPTION	BY	CHECK	ACCEPTANCE	DATE
0	ISSUED FOR CONSTRUCTION (KNet No: 24467656)	T2DA	T2DA	M. SHORT	28.03.26

INDEX SHEET REFERENCE: 8011 SHEET 2871002

T2D → TORRENS TO DARLINGTON

UNCONTROLLED COPY WHEN PRINTED ◄ 100 MILLIMETRES ON ORIGINAL DRAWING ► ALL DIMENSIONS ARE IN MILLIMETRES UNLESS SHOWN OTHERWISE

DESIGNER: A.SINGHAL / S.PODDAR
QUALIFICATION: MEng BEng
DATE: 25.03.26

REVIEWER: R.WOZNIAK
QUALIFICATION: FIEAust CPEng NER RPEV RPEQ MICE CEng
DATE: 25.03.26

INDEPENDENT DESIGN CERTIFIER (IF REQUIRED): T. JACKSON
QUALIFICATION: BEng (Hons), MEngSc
DATE: 27.03.26

Government of South Australia
Department for Infrastructure and Transport

PROJECT No.: 31921901
FILE No.: 2020/10577
DESIGN No.: #15551401
SURVEY No.: #15551401
PROJECT START ROAD RUNNING DISTANCE: NOT APPLICABLE
PROJECT END ROAD RUNNING DISTANCE: NOT APPLICABLE
SCALES: NOT TO SCALE

ROAD No. 6200
RICHMOND ROAD
RAILWAY TERRACE - SOUTH ROAD; MILE END SOUTH
BRIDGE OVER KESWICK CREEK
BAR SHAPES DIAGRAM - SHEET 02

DESIGNED: T2DA
DRAFTED: T2DA
ACCEPTED FOR USE: M. SHORT
TITLE: DIRECTOR OF ENG.
DATE: 28.03.26
ACCEPTANCE FORM KNET No.: 24467656
IN ACCORDANCE WITH DP013
DRAWING No.: 8011
SHEET No.: 2871152
AMEND No.: 0
SHEET LATITUDE -34.94188 SHEET LONGITUDE 138.57397

FILE NAME: Autodesk Docs://T2D - Torrens to Darlington /T2DPAA-T2D-C3S-BR-M3D-100302.rvt